DevOps Engineer – EKS Hybrid – DCS – 4/Mar/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Sr. Data Engineer/Sr. Clouds/Unix/Linux Administrator /Sr. Systems Engineer /Python/R Developer /AI/ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Sr. Data Engineer/Clouds/Linux Admin. /ML/BI/SPoint/D365 CRM Consultant /EDW
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4, Sora, Mistral, and Gemini 1.5.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements and cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Expert in Docker and Compose, including multi-stage builds and advanced networking. Manage S3 efficiently, including bucket configuration, security best practices, and cost-optimization through batch operations. Secure cloud infrastructure by implementing best practices in access control, identity and access management (IAM), and vulnerability management.
- Frequently implemented AWS cloud-native design patterns. Such as **Microservices Architecture** to break down a large application into smaller, independent services. **Serverless Computing** to run code without provisioning or managing servers, **API Gateway** to manage, secure, and monitor the APIs in a central location.
- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Utilized Ansible for automated configurations and deployments on the pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Unleash insights from diverse data: apply unsupervised ML to uncover patterns in unlabeled data, leverage RStudio for supervised learning, and expertly select algorithms (e.g., KNN, CNN, RNN) aided by AutoML tools like Amazon SageMaker Autopilot, Azure AutoML, H2O.ai, TPOT and KerasTuner.
- Streamlined the DevOps workflow on AWS using Atlassian tools (Jira, Confluence, Bitbucket, Bamboo), resulting in comprehensive lifecycle management.
- Executed implementation, documentation, integration, and support operations of a variety of REST API projects
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Designed and implemented a multi-tiered Data Vault Architecture supporting two or more cloud providers via Terraform, seamlessly integrating with their respective multi-layered analytics architectures through Snowflake
- Bridged the gap between development and operations by providing high-level options, conceptual approaches, and technical recommendations. Facilitated the smooth transition of system designs into operation through workshops and training on DevOps best practices for IT and business stakeholders.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or batch scheduling via AutoSys.

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer /Sr. Data Engineer/Clouds /Liunx Admin. /ML/BI/SharePoint Consultant /EDW
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.
Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Worked with DevOps Engineering team in Revenue Digital Branch to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Frequently implemented AWS cloud-native design patterns. Such as highly scalable and fault-tolerant **Amazon DynamoDB Amazon SQS** to send messages between microservices, or to queue tasks for later processing. **SNS** to send messages to a large number of subscribers or to send notifications to customers, or to trigger workflows in other applications.
Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Leverage AWS DevOps practices and tools, primarily Amazon EKS (Elastic Kubernetes Service), to streamline application development and deployment within the AWS cloud.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation and management of deployed machine learning models in production environments.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained AKS clusters for various microservices applications using Terraform and Helm Charts.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, multiple data transformation jobs or batch scheduling via AutoSys
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer /Clouds/Linux Administrator /ML/SharePoint /K2 Developer
Responsibilities:
Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Implemented AWS cloud-native design patterns, including leveraging Amazon CloudWatch for comprehensive monitoring and observability of AWS resources. This includes tracking metrics, logs, and events from both applications & infrastructure. Additionally, utilized Amazon Cognito for robust user identity and access management in web and mobile applications.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
Leverage AWS services for cloud-native design patterns:
Highly scalable and fault-tolerant monitoring: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
Unified user authentication and authorization: Implement Amazon Cognito for user identity and access management across web and mobile applications.
Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
Completed design documentation, data models, loading approaches and source to target mapping documents.
Leveraged technical expertise to consult on projects throughout their lifecycle. Conducted pre-project assessments to identify and mitigate risks, interdependencies, and exceptions. Developed solutions aligned with ICT strategy and architecture. Authored technical documentation to support new and existing applications.
Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.
Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
Migrated on-premise applications to the cloud using AWS Direct Connect, Storage Gateway and AWS Snowball Edge.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Industry: DevOps Engineer/Data Engineer/Clouds/Linux Administrator /SharePoint Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- **Provided technical consulting and subject matter expertise throughout project lifecycles**:
- **Pre-project**: Identified, mitigated, or escalated risks, interdependencies, and exceptions. Formulated solutions aligned with ICT strategy and architecture.
- **Development**: Delivered high-level options, conceptual approaches, technical recommendations, and estimates to support transitioning system designs into operation.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyze data sources, types, and formats to define transformation, loading, and data model requirements.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: DevOps Engineer/Data Engineer /Data Analyst /SharePoint Consultant
Responsibilities:

- Collaborating across the entire product lifecycle to build, release, maintain, and improve features. Ensure high-quality, fit-for-purpose, resilient, and reusable services that empower customer channels.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Actively contribute to and guide innovation in DevOps practices and incorporate agile delivery approaches and technologies that assist with the development and delivery of high-quality software
- Carry out and deliver efficient/repeatable platform engineering and application maintenance tasks through the orchestration and strategy-building tools to ensure optimal performance
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: DevOps Engineer/Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.

Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Machine Learning Engineer – Corelated Resources – 28/2/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Machine Learning Engineer /Senior DevOps Engineer / Senior Data Engineer /Python /R Developer /AI/ML/DL/BI/SharePoint/D365 CRM Consultant /EDW/DBA -Japan(7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior ML Engineer /Snr DevOps/Data Engineer /Python/R Developer/AI/ML/BI/SP/D365 CRM Consultant/EDW
Responsibilities:

- Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.
- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within Spark cluster using Spark Applications & Spark SQL
- Working as part of the ML Development team and closely collaborating with Data Scientists to designed and implement Big Data & ML platforms, pipelines, and tooling using various technologies like Hadoop, BigQuery, Spark, Redshift and AutoML.
- Utilized AI/ML technologies like CrewAI and MLOps to automate tasks, improve productivity, and reduce costs in sales, customer service, and model building. Leveraged TinyML to integrate AI with IoT/mobile devices for innovative solutions.
- **Diffusion Models and LLMs for Image and Content Generation**: Using diffusion models like Stable Diffusion or Dall-E 2 to create new products, generate realistic avatars for the metaverse. Using LLMs for image& content generation by Sora etc.
- As a Data and Software Engineer, design and implement ETL pipelines either using GCP's Dataflow or AWS Glue or AWS EMR (with Apatche Spark /Hive, etc) to transform and load data from various sources, optimizing data processing efficiency.
- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Utilized distributed file systems like HDFS (Hadoop Distributed File System) or GCS (Google Cloud Storage) to efficiently store and access terabytes of data across multiple machines.
- Engineer and productionise ML models for real-world impact, leveraging the Python, AWS stack and top-notch MLOps tools.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV for Imaging AI cancer detection, a different way than Dr. Cao Kei proposed a pancreatic cancer early detection model "PANDA" based on deep learning published a paper in "Nature Medicine" in Nov, 2023.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …

- Fine-tune pre-trained generative AI models (like Content generation, Chatbots, Data analysis) to build bespoke AI assistants based on clients' data to integrate into clients' business systems to automate / streamline / optimise existing processes.
- Develop medical-tailored LLMs (BERT, GPT-4, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- **Model Deployment and Monitoring**: Deploy machine learning models either using fully managed AWS SageMaker, or high-performance and scalable TensorFlow Serving or Open-source Kubeflow run on Kubenetes on any major clouds.
- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health. **API Development with Python**: Developed RESTful APIs for ML model deployment
- **Automation and CI/CD**: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX A100s system

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior ML Engineer /Senior DevOps/Data Engineer /AI//ML/DL/SP/BI /Python /R Developer/EDW
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Designed, developed, and implemented data models and algorithms to solve complex business problems.

Work closely & under the guidance of the Analytics Data Centre of Excellence Manager to develop & refine AI solutions and conduct pilots. Collaborate with the BI and IT teams to integrate production-ready AI solutions with business systems.

- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Executing training and retraining systems as needed, utilizing either traditional machine learning libraries like Scikit-learn, CNN, or RNN, or leveraging AutoML toolkits (e.g., Google Cloud AutoML, Azure Machine Learning) for automated feature engineering, model selection, and hyperparameter tuning.
- Successfully migrated ML models to TensorFlow Serving and TorchScript for efficient real-time predictions.

Championed the adoption of TinyML for resource-constrained edge devices, expanding data collection capabilities.

Working as part of the ML Development team and closely collaborating with Data Scientists to designed and implement Big Data & ML platforms, pipelines, and tooling using various technologies like Hadoop, BigQuery, Spark, Redshift and AutoML.

As a Data and Software Engineer, design and implement ETL pipelines either using GCP's Dataflow or AWS Glue or AWS EMR (with Apatche Spark /Hive, etc) to transform and load data from various sources, optimizing data processing efficiency.

Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.

Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.

Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).

Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.

Model Training and Optimization: Data sampling: Utilize techniques like reservoir sampling or stratified sampling to efficiently select representative subsets of data for training, reducing processing time and memory usage.

Docker Images: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.

API Development with Python: Developed RESTful APIs for ML model deployment.

Implemented RESTful APIs using Python and Flask to enable seamless data exchange between different components.
Optimized data pipelines utilizing Airflow and Luigi, reducing processing time by 30% and improving resource utilization.
Automated model versioning and rollbacks with MLflow, enabling seamless rollbacks and mitigating production issues.

- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.
- Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs
Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior ML Engineer /Senior DevOps/Data Engineer/AI/ML/DL Consultant/Python/SP/K2 Developer /EDW
Responsibilities:

Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
Working as part of the ML Development team and closely collaborating with Data Scientists to designed and implement Big Data & ML platforms, pipelines, and tooling using various technologies like Hadoop, BigQuery, Spark, Redshift and AutoML.
Utilized distributed file systems like HDFS (Hadoop Distributed File System) or object storage like AWS S3 or Azure Blob Storage to efficiently store and access terabytes of data across multiple machines.
Data Engineering & Software Engineering Foundations:
As a Data and Software Engineer, design and implement ETL pipelines either using Azure ADF or AWS Glue or AWS EMR (with Apatche Spark /Hive, etc) to transform and load data from various sources, optimizing data processing efficiency.
Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance. Employed best practices for ML and MLOps throughout the process, including feature engineering, data preprocessing, data versioning with dedicated tools like DVC or Pachyderm, and model training and evaluation on terabytes of data.
Model Training and Optimization: Data sampling: Utilize techniques like reservoir sampling or stratified sampling to efficiently select representative subsets of data for training, reducing processing time and memory usage.
Docker Images: Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

API Development with Python: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
Distributed computing frameworks for parallel processing. Leveraging cloud platforms (like GCP, AWS, or Azure) or on-premise solutions depending on specific needs, for scalable and cost-effective resource utilization.

- Proficient in ML frameworks: Skilled in Python, R, Java, Scikit-learn, H2O, TensorFlow, PyTorch, Keras for ML/DL tasks.
- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- **NLP, CV and RL**: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.

- **MLOps advocate**: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Machine Learning Engineer /DevOps/Data Engineer /AI/DL/ML /SharePoint /D365 CRM Consultant
Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Agile data scientist with experience in conducting research, creating datasets, and performing statistical analysis to build and train data science products (e.g., recommendation engines) and machine learning systems (e.g., fraud detection). Demonstrated proficiency in using iterative development, fostering collaboration with stakeholders, and applying automation for efficient and adaptable data science workflows.
- Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- MLOps advocate, championing data security through implementing tools like CI/CD pipelines, model versioning, and monitoring. Ensured data protection with encryption, access controls, and anonymization techniques.
- **Cloud migration & automation**: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher). Skilled in creating impactful visualizations using Tableau and Power BI.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Transform and train Data science products and ML Systems. Worked on developing and training ML models.
- Research and implement appropriate ML algorithms and tools according to business requirements
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Conducted regular system training and retraining activities using Python, scikit-learn, TensorFlow for ML/DL.
- **API Development with Python**: Developed ML models using Python frameworks (e.g., Flask, FastAPI) and secured RESTful APIs for web applications. Managed cloud infrastructure (AWS/Azure/GCP) and implemented MLOps/DevSecOps practices for secure and efficient deployment.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Machine Learning Engineer /DevOps Engineer /Data Engineer / SharePoint Consultant /DBA
Responsibilities:

- Built and deployed ML models using k-NN) and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy. Building and productionizing ML models with a strong focus on Python.
- Conducted research on appropriate ML algorithms and tools according to business requirements.

- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Built and deployed ML models using k-NN) and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy. Building and productionizing ML models with a strong focus on Python.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access

OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.

Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL

Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP

CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.

Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.

Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5

Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve

Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior AWS Cloud Engineer –AI ML – SustainRecruit Applied on 24/2/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AWS Cloud Engineer/Senior DevOps Engineer/Senior Data Engineer/ Data Analyst/Python/R Developer /Clouds/Linux Administrator /AI /ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AWS Cloud Engineer/Snr DevOps Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Designed, implemented, and maintained a highly scalable, serverless, and modular architecture on AWS, comprising Step Functions for orchestrating serverless workflows. S3 for secure and scalable object storage. DynamoDB for NoSQL database functionality. SageMaker for building, training, and deploying AI/ML models. AppSync for real-time GraphQL data access, integrated with Cognito for user authentication and fine-grained authorization based on roles and permissions.
- Developed cloud-native applications leveraging AWS services including S3, DynamoDB, Lambda, and API Gateway.

Leverage AWS Event Bridge, Beanstalk Lambdas, and API Gateway for AI/ML, IoT, AR/VR Development:
Events Trigger Actions: Data flows from IoT devices, streaming services, scheduled triggers, or API requests to Event Bridge. Event Bridge filters and routes data to relevant Beanstalk-managed Lambda functions.
Lambda Functions for Intelligence and Processing: Lambdas leverage AI/ML, IoT, AR/VR tools based on needs.
- Examples: AI sentiment analysis, IoT anomaly detection, AR/VR content generation, ML image recognition.
API Gateway Access & Control: API Gateway serves as the public endpoint for user interaction and triggers specific Lambdas. Implement authentication and authorization for secure access control.

- Designed and implemented CI/CD pipelines to automate the build, test, and deployment process.
- Collaborated with cross-functional teams to design and deploy cloud solutions.
- Orchestrating Complex Workflows with Step Functions:
- **Define**: Identify tasks, order, dependencies, & branching points. **Choose**:Standard, Express, or Parallel state machine type.
- **Build**: Use ASL to define states, transitions, and logic (Choice, Pass, Task, Wait). **Configure**: Specify details for each state (Task: service, input, errors). **Implement**: Set retries, backoff, and catch branches for error handling.
- **Deploy and Run**: Design and deploy scalable state machines in AWS Step Functions using console, CLI, SDKs (boto3), or AWS SAM (Serverless Application Model) for streamlined management.
- Trigger executions with input and monitor in real-time for serverless workflow orchestration.
- Implemented Infrastructure as Code practices using AWS CloudFormation to define and provision cloud resources.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Implemented automated deployments through GitLab CI/CD pipelines, integrating with Jenkins for specific tasks. This enabled efficient deployments, streamlined configuration changes, and facilitated rollback procedures when necessary.
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.
- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Applied software design patterns to enhance system reliability and performance.
- Integrated Machine Learning models into cloud applications using SageMaker and TensorFlow.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Automated infrastructure provisioning and configuration using Terraform and CloudFormation, adhering to IaC best practices to ensure consistency and efficiency.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, ANN analytics, etc.
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior AWS Cloud Engineer/Snr DevOps Engineer/Python/R Developer/AI/ML/SP Consultant/EDW
Responsibilities:

Built and secured a hybrid cloud architecture between AWS and GCP: Established encrypted tunnels for data transfer using Cloud VPN Gateways on both platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration across both clouds, ensuring consistency and efficiency.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Researched and implemented appropriate ML algorithms and tools based on business requirements.

Design and deploy a modern **AWS Lambda Architecture EDW**, comprising: **Batch layer**: Hadoop HDFS for distributed storage and MapReduce for batch processing of large datasets. **Speed layer**: Spark for in-memory processing of real-time data streams, utilizing RDDs for efficient data manipulation. **Service layer**: Spark Streaming for low-latency analytics on streaming data, generating SNS application notifications based on real-time insights. **Direct write queries**: Spark SQL for efficiently querying datasets within Spark cluster, enabling seamless integration with downstream applications & services.

Architectured, deployed, and managed resilient cloud infrastructure: Leveraged proven solutions like S3, SQS, DynamoDB, Lambdas, Step Functions, SageMaker, AppSync, and Cognito to build scalable and secure systems.

Embraced Infrastructure as Code: Automated infrastructure provisioning and configuration with best practices, ensuring cost-efficiency and consistency.

Streamlined development with CI/CD pipelines: Implemented automated testing, deployment, and rollback processes, enabling rapid and reliable updates.

Successfully designed and implemented highly available and scalable cloud infrastructure on AWS for various applications, utilizing S3 for object storage, SQS for message queuing, DynamoDB for NoSQL database, Lambdas for serverless functions, Step Functions for orchestration, AppSync for GraphQL API, and Cognito for user authentication.

Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.

Master containerization with Docker/Compose (multistage builds, advanced networking). Manage S3 efficiently: buckets, policies, security, batch optimizations. Secure cloud infrastructure through robust implementation of security guidelines.

Designed and implemented CI/CD pipelines for Python applications using poetry, pyproject.toml, and pytest.

Developed and optimized data pipelines leveraging Apache Flink and Kafka Streams for real-time processing.

Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Built serverless functions using AWS Lambda (Python/Go) for various purposes. Defined triggers, processed data, and generated outputs. Leveraged AWS Management Console for creation, configuration, testing, and deployment. Integrated functions with services like API Gateway for seamless execution.

Choreograph serverless workflows with AWS Step Functions & integrate data pipelines with MWAA.

Trigger tasks on events with CloudWatch & manage configurations across instances with SSM Run Command/Automation.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

- Leveraging TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, GitLab, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Employed Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis in Python/R programming & fine-tuning based on test results to improve model performance. Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed ML pipelines for large datasets using frameworks (Scikit-learn, TensorFlow, etc.) and distributed computing (Spark, Dask, etc.) on GPUs/TPUs. Expertise in feature engineering, model selection, hyperparameter tuning, and continuous monitoring.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior AWS Cloud Engineer/Snr DevOps Engineer/Python/R Developer/AI/ML/SharePoint /K2 Developer
Responsibilities:

Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Successfully designed and implemented highly available and scalable cloud infrastructure on AWS for various applications, utilizing S3 for object storage, SQS for message queuing, DynamoDB for NoSQL database, Lambdas for serverless functions, Step Functions for orchestration, AppSync for GraphQL API, and Cognito for user authentication.

Automated infrastructure provisioning and configuration using Terraform and CloudFormation, adhering to IaC best practices to ensure consistency and efficiency.

Established and maintained secure environments through IAM policies, CloudTrail logging, Security Groups, and AWS Config for continuous monitoring.

Implemented CI/CD pipelines using Jenkins and GitLab CI/CD to automate infrastructure deployments, configuration changes, and rollback procedures.

Optimized cloud resource utilization and costs through cost analysis tools and resource tagging.

Managed S3 buckets, policies, securities, applying batches, and implemented security guidelines on cloud infrastructure.

Configured secure S3 buckets with retention policies (Glacier, object expiry), access controls, and metadata management.

Developed IaC CI/CD pipelines for a variety of applications using Terraform, AWS/Azure DevOps, DevSecOps or AIOps.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

mplementing Infrastructure as Code (IAC) by creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.

Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources

Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.

Conducted and evaluations to measure model performance and accuracy using A/B testing and other methods.

Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3

Connected R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.

Used Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items

Master of AI/ML deployment: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.

Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).

NLP, CV and RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.

Conducted statistical analysis in Python/R programming & fine-tuning based on test results to improve model performance. Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: AWS Cloud Engineer/DevOps Engineer/Python/R Developer/AI/ML/SharePoint Consultant /DBA

Responsibilities:

- Designed, implemented, and maintained cloud infrastructure for pioneering software projects.
- Utilized AWS services including S3, SQS, DynamoDB,
- Implemented Infrastructure as Code methodologies and CICD pipelines to streamline deployment processes.
- Collaborated with cross-functional teams to integrate Machine Learning systems into cloud architecture.
- Designing and deploying a cloud-based data analytics platform on AWS for a large healthcare organization.
- Utilized Amazon Redshift and Athena for efficient data storage and analysis, enabling faster and more insightful reporting.
- efficiency. Secured the cloud environment by implementing best practices for IAM, encryption, and access control.
- Continuously monitored and optimized cloud resources to reduce costs and maximize performance.
- Engineered secure and scalable cloud architectures. Managed Amazon DynamoDB tables for high-throughput applications.
- Developed serverless functions using AWS Lambda. Implemented IAM policies for fine-grained access control.
- Conducted security audits and ensured compliance. Collaborated with data scientists on Machine Learning model deployment. Supported continuous delivery through automated pipelines.
- Designed, implemented, and maintained cloud infrastructure for diverse projects.
- Leveraged AWS services including S3, SQS, DynamoDB, Lambda, and Step Functions to optimize performance and scalability. Implemented AWS Infrastructure as Code methodologies to automate infrastructure provisioning and configuration management. Established CI/CD pipelines and processes to streamline application deployment and delivery.
- Contributed to the integration of Machine Learning systems into cloud environments, enhancing data processing capabilities. Applied various software design patterns to ensure efficient and scalable cloud solutions.
- Used AWS Migration Hub and Azure Migrate to plan and execute Azure to AWS migrations
- Leveraged AWS tools like EC2 Instance Migration, Serverless Migration Service, Storage Gateway, Database Migration Service, Management Console, CLI, CloudTrail, IAM and Azure tools like ExpressRoute, Virtual WAN, Migrate (Server Migration, Data Transfer Service), Azure portal, PowerShell, Monitor, Active Directory to ensure seamless migration.
- Managed S3 buckets, policies, securities, applying batches, and implemented security guidelines on cloud infrastructure.
- Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
- Designed and implemented private and public cloud solutions, leveraging AWS and Azure.
- Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
- Migrated on-premise applications to the cloud using AWS Direct Connect, Snow Family, Storage Gateway, S3TA, etc.
- Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
- Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
- Provided technical consulting and subject matter expertise to projects, ensuring alignment with ICT strategy and architecture.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Managed and enhanced infrastructure with Terraform, integrating development and test tools into CI/CD pipelines. Additionally, developed and maintained REST APIs, monitored AWS cloud infrastructure, and implemented large-scale logging, monitoring, and security measures.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.

- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: AWS Cloud Engineer /DevOps Engineer/Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented cloud infrastructure solutions using AWS services such as S3, SQS and DynamoDB.
- Designed and maintained Amazon S3 buckets for data storage. Ensured high availability using multi-region deployments.
- Managed cloud infrastructure, including AWS EC2, S3, and RDS (Relational Database Service) instances.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Managed Amazon EC2 instances for compute resources. Set up Amazon VPCs and configured network security groups.
- Conducted **performance optimization for cloud-based applications.
- Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
- Advised clients on AWS best practices and recommended cloud solutions to meet their business objectives.
- Implemented AWS services including S3, SQS to optimize performance and reliability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Systems Engineer /BI Developer /Data Engineer /Data Analyst /DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.

- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Cyber Security Engineer – Clicks IT Georgia – NSWPF - $1000 p.d include Super 22/2/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cyber Security Engineer/Senior DevOps Engineer/Senior Systems Engineer/Data Engineer/Python/R Developer /Clouds/Linux Administrator/ AI/ML/BI/SharePoint/D365 CRM Consultant/EDW -Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cyber Security Engineer/Senior DevOps/Data Engineer/Python/R/AI/ML/SP/D365 CRM Developer/EDW
Responsibilities:
- Designed and implemented a layered security approach for digital systems, ensuring data confidentiality, integrity, and data availability through integrated strategies:
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data. Such as McAfee ESM and Palo Alto Networks Cortex XDR for impact.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Federated Amazon Redshift access with Azure EntraID (aka AAD), AD FS, SSO, and MFA in Azure Security Centre.
- Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, Defender, Sentinel, and E5 suite, as well as AWS Security HUB, Guard Duty, Inspector, Detective.
- Ensured Cloud Security by leveraging AWS IAM for granular identity and access control, storing container images securely in AWS ECR, and deploying them on a chosen platform like AWS ECS or Fargate.

- **Network Security via SIEM:** Hybrid Cloud Security with Netskope SASE: Deploy Netskope Service Edge in data centers and cloud environments. Configure SASE functionalities including CASB, SWG, ZTNA, and DLP. Establish secure connections between on-premises and cloud (e.g., ExpressRoute, Direct Connect).
- Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- Configured a Netskope Cloud Private Access (NCPA) tenant within the Azure cloud environment and established secure connections between Azure and AWS environments to Netskope using either Azure ExpressRoute or AWS Direct Connect
- Defined granular policies for user authentication methods (SAML for Azure AD, LDAP for AWS IAM) to grant secure access to designated resources. Enabled comprehensive security features like cloud malware protection, data loss prevention (DLP), and web filtering across both cloud environments, creating a robust hybrid cloud security posture.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- Integrated security scanning into CI/CD pipelines in AWS using AWS CodePipeline with Amazon Inspector in .yaml file
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- **Industry Compliance (Healthcare):** Implemented key security controls aligned with ACSC Essential 8 mitigations, ISM 37 security controls and relevant NIST CSF categories to address healthcare-specific risks and regulations.
- **Compliance Adherence**: Ensured compliance with ISO27001/2's 114 security controls and NIST800-53 standards.
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- **Streamlined security automation**: Tight integration of security checks and gates within continuous integration and continuous delivery (CI/CD) pipelines to detect and address vulnerabilities early and efficiently.
- **Infrastructure as Code with Terraform**: Provisioning and managing infrastructure in a secure, scalable, and auditable manner using Terraform, guaranteeing consistent and auditable infrastructure configurations.
- **Multi-layered API security for data integrity and protection**: Protect sensitive data from unauthorized access and manipulation through comprehensive security measures including authentication, authorization, encryption, and input validation. Implement these measures using REST API, Azure API Management or AWS API Gateway, and leverage Azure Monitor/Log Analytics or AWS CloudWatch logs for monitoring and analysis.
- Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Golang, Bash and PowerShell, including Azure Container Registry (ACR) or AWS ECR.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.Utilizing Azure Event Hub and Functions for data processing, monitored by AppInsights. AWS Event Bridge triggers Lambdas managed by Beanstalk, with API Gateway serving Lambdas' APIs.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via ServiceNow.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Financial Bank: Senior Cyber Security Engineer/Senior DevOps/Data Engineer/AI/ML/SharePoint Consultant /EDW
Responsibilities:

Designed and implemented robust security architectures for cloud infrastructure, applications, and databases, ensuring data confidentiality through encryption, integrity through data backups and access controls, and availability through high availability architectures and disaster recovery plans.

Leveraged **AI-powered cloud security solutions** for anomaly detection, threat intelligence, & automated incident response

Implemented **network security measures with a SIEM** tool for log monitoring, intrusion detection, & incident investigation.

Configured a Netskope Cloud Private Access (NCPA) tenant within the Azure cloud environment and established secure connections between Azure and AWS environments to Netskope using either Azure ExpressRoute or AWS Direct Connect

Integrated DevSecOps practices with Infrastructure as Code (IaC) security principles to automate security configurations and ensure consistent security posture across deployments.

Industry Compliance (Central Bank): Implemented 5 key security controls aligned with ACSC Essential 8 (e.g., Application Control, MFA), ISM 37 (e.g., physical access control with security clearances), and relevant NIST CSF categories (e.g., ID.AM, PR.AC) for the Reserve Bank of Australia (RBA).

Mitigated Security Threats: Conducted in-depth analysis of intrusion attempts, proactively researched threats, and implemented countermeasures.

Code Review and Vulnerability Elimination: Reviewed codebases, scanned base images, automated patching, monitored and reported, continuous improvement.

Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.

Data Security and Management:

Securely stewarding data in AWS S3: Crafted robust buckets with multi-layered protection.

Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Ensured adherence to industry-specific compliance standards and regulations.
Utilized various security tools: Splunk, McAfee, Secret Server, CrowdStrike, Mimecast, Qualys, Tenable AD, Fortinet.
Managed security platforms: EDR and Vulnerability Management Platforms (CrowdStrike, Mimecast, Qualys, Tenable AD, Fortinet etc.). Performed day-to-day monitoring and triaged critical alerts. Triage phishing emails and took containment /remediation actions. Offered expert advice on emerging security issues and project priorities.
DevSecOps security: Integrated security throughout the development lifecycle using practices like static code analysis, DAST (Dynamic Application Security Testing), and IaC security.
Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
Leveraging serverless architectures for event-driven data processing and API exposure.
Implemented a secure and efficient multi-cloud development pipeline: Used **Lambda** for serverless execution of tasks, enhancing security. Leveraged **container orchestration** for efficient resource management across various cloud providers.
Secure Developer: Extensive code review and security best practices knowledge (code smells, SAST, DAST, RASP)
Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.

- Successfully deployed and managed microservices applications on diverse infrastructures (bare metal, virtual machines, public clouds) using OpenShift. Automated key Kubernetes management tasks like node provisioning and scaling, security & access control, image management & deployments, and network configuration & monitoring, ensuring consistent operations regardless of the underlying platform (AKS, EKS, GKE).
- Secure cloud infrastructure through robust implementation of security guidelines.

Incident Response Refinement: Enhanced incident response processes, minimizing impact and downtime during security incidents. Defined information security solutions to support business objectives.
Advanced Security Tool Management: Deployed and managed advanced security tools to identify and resolve security issues across systems and applications.

- **Threat modeling for proactive risk mitigation**: Proactive identification and assessment of potential threats and vulnerabilities through threat modeling techniques to implement preventive measures and reduce risk exposure.
- **Scan Regularly**: Regularly scan base images for vulnerabilities using dedicated tools like Snyk or Anchore. Set clear criteria for triggering patch upgrades based on severity and impact.
- **Automate Uplifts**: Integrate image building and deployment into the CI/CD pipeline. Automate patching for critical vulnerabilities based on established policy using tools like Skaffold.
- **Monitor and Report**: Visualize scan results and patching progress on a centralized dashboard. Set up alerts for critical vulnerabilities and successful/failed patching attempts.
- **Continuous Improvement**: Track patching time and continuously improve processes to minimize response time. Foster communication between security and development teams to address underlying code issues and prevent future vulnerabilities.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Cyber Security Engineer/Senior DevOps/Data Engineer/AI/ML/SharePoint /K2 Developer
Responsibilities:
Designed and Implemented Robust Security Architectures: Network security (SIEM, Netskope), DevSecOps practices, secure AWS landing zones, multi-cloud development pipeline, secure cloud infrastructure.
Cloud Security Design and Management: Designed secure cloud infrastructure with IAM policies, network segmentation, and encryption.
Network Security via SIEM: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
Configured a Netskope Cloud Private Access (NCPA) tenant within the Azure cloud environment and established secure connections between Azure and AWS environments to Netskope using either Azure ExpressRoute or AWS Direct Connect
Defined granular policies for user authentication methods (SAML for Azure AD, LDAP for AWS IAM) to grant secure access to designated resources. Enabled comprehensive security features like cloud malware protection, data loss prevention (DLP), and web filtering across both cloud environments, creating a robust hybrid cloud security posture.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
Integrated security scanning into CI/CD pipelines in AWS using AWS CodePipeline with Amazon Inspector in .yaml file
Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
Compliance Adherence: Ensured compliance with ISO27001/2's 114 security controls and NIST800-53 standards.
Industry Compliance: Ensured adherence to ACSC ISM, NIST CSF, ISO27001/2, and NIST800-53 standards.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock).
Incorporated **DevSecOps** practices for secure infrastructure provisioning, vulnerability scanning, & automated deployments.

Threat Mitigation: Monitored and mitigated threats using tools like Wireshark and Metasploit.
DevSecOps Collaboration: Integrated security into CI/CD pipelines.
Incident Response Planning: Developed incident response playbooks for cloud-based services.
Developing and refining incident response processes to ensure swift and effective action in the event of security incidents, minimising impact, and downtime.
Data Security and Management: Designed and implemented AI-driven data solutions, pipelines, and infrastructure components using secure practices.
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
Implementing Infrastructure as Code (IAC) by creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Identifying and mitigating potential security threats through in-depth analysis of intrusion attempts, proactive threat research, and the implementation of countermeasures.
Threat Intelligence Analysis: Investigated cyber threats, identified vulnerabilities, & recommended preventive measures.
Incident Handling: Responded swiftly to security incidents, minimizing business impact.
Code Review and Vulnerability Elimination: Reviewed code bases to eliminate common coding vulnerabilities, ensuring secure application development and deployment.
Secure Code Practices: Worked closely with developers to ensure secure coding practices.
Compliance Adherence: Ensured compliance with ISO27001/2 and NIST800-53 standards.
Setup and customized ServiceNow Enterprise-grade IT Service Management (ITSM) platform with robust features for incident, change, and problem management, pre-built integrations with IT infrastructure, security-focused design.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Cyber Security Analyst /Senior Systems Engineer /SharePoint Consultant /DBA
Responsibilities:

- **Designed and Implemented Robust Security Architectures**: Network security (SIEM, Netskope), DevSecOps practices, secure AWS landing zones, multi-cloud development pipeline, secure cloud infrastructure.
- **Security Architecture Enhancement**: Collaborated on enhancing security architectures, conducted threat intelligence analysis, responded to incidents swiftly, reviewed code for vulnerabilities, and ensured compliance with standards.

Network Security via SIEM: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
Configured a Netskope Cloud Private Access (NCPA) tenant and established secure connections between Azure and AWS environments to Netskope using virtual Private IPs or peering connections.
DevSecOps Security: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs of work.

- Ensuring adherence to industry-specific compliance standards and regulations in all security initiatives
- Developed and refined incident response processes, leading to swift and effective actions during security incidents, minimizing impact and downtime.
- **Threat Mitigation**: Monitored and mitigated threats using tools like Wireshark and Metasploit.
- **Incident Response Planning**: Developed incident response playbooks for cloud-based services.
- **DevSecOps Collaboration**: Integrated security into CI/CD pipelines.
- Scrutinized code bases to eliminate common coding vulnerabilities and ensure secure application development and deployment. Deployed & managed advanced security tools to systematically identify & resolve security issues across systems & applications.Ensured adherence to industry-specific compliance standards & regulations in all security initiatives.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Utilized Splunk leading platform for log management, analysis, and security information and event management (SIEM), perform real-time monitoring, and conduct forensic investigations, including IDS, IPS, IAM and related.
- Configure McAfee and McAfee VPN to encrypts users' data and protects it from potential snoopers, ensuring that users' online activities and personal information remain secure even on untrusted networks.
- Setup and configured Tenable, Tenable.sc (formerly SecurityCenter) vulnerability management platform that helps organizations identify and prioritize security vulnerabilities across their infrastructure.
- Deploying and managing advanced security tools to systematically identify and resolve security issues across a spectrum of systems and applications.

Design and implement AI-driven pipelines for Azure IaC pipeline creation, prediction, and optimization using Azure ML Service (AML) or AWS SegaMaker, Terraform, Azure DevOps Pipelines or AWS CodeBuild, & Azure SDKs or AWS SDKs.
Perform day-to-day monitoring and triaging critical alerts and coordinate with technical teams for incident response.
Triage phishing emails, taking appropriate containment and remediation actions to prevent security incidents

- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Security Engineer/Infrastructure Engineer /Systems Administrator/SharePoint Consultant
Responsibilities:

- **Security Architecture Enhancement**: Collaborated with cross-functional teams to enhance security architectures for critical projects.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- Configured a Netskope Cloud Private Access (NCPA) tenant and established secure connections between Azure and AWS environments to Netskope using virtual Private IPs or peering connections.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Static Code Analysis**: Conducted thorough code reviews to identify vulnerabilities.
- **Industry Compliance**: Ensured adherence to ACSC ISM and NIST CSF.
- **Security Testing**: Performed penetration testing and vulnerability scanning.
- **Secure SDLC Implementation**: Advised development teams on secure coding practices.
- **NIST Best Practices**: Followed NIST guidelines for secure software development.
- Securely stewarding data in AWS S3 involves crafting robust buckets with multi-layered protection: icy Glacier retention for long-term archives, precise object expiry for ephemeral data, granular metadata controls for organized access, and data access controls for airtight security. Implement security guidelines on cloud infrastructure, ensuring compliance.
- Collaborating with IT Security for the Standard Operating Environment (SOE) security hardening in line with PSPF, ISM, ACSC, CIS and DTA Protected pattern guidelines.Stayed updated with industry security trends and emerging threats.
- Leveraged McAfee, Splunk, and Secret Server to enhance the effectiveness of Cyber Security Management programs.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Cloud Security Engineer/Infrastructure Engineer/Systems Administrator /DBA
Responsibilities:

- **Cloud Security Design**: Designed secure cloud infrastructure, including IAM policies, network segmentation, and encryption.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- Configured a Netskope Cloud Private Access (NCPA) tenant and established secure connections between Azure and AWS environments to Netskope using virtual Private IPs or peering connections.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Network Security Enhancement**: Implemented network segmentation and intrusion detection systems.
- **Threat Hunting**: Proactively searched for vulnerabilities and potential threats.
- **Incident Handling**: Responded to security incidents, ensuring minimal impact.
- **Compliance Audits**: Conducted internal audits to meet industry standards.
- **Risk Assessment**: Evaluated emerging technologies for relevance and risk.
- Gained foundational experience in information security.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Systems Administrator /DWH
Responsibilities:

- Supported ongoing operational Cyber Security Management programs of work, contributing to a strong security framework.
- Managed hardware and server administration, ensuring reliable infrastructure performance.
- Oversaw AD, DFS, ADFS, AAD, DNS, and DHCP services, maintaining their stability and security.
- Provided level 3 reactive support for incident/service request management, monitoring, and event management.
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Systems Engineer /BI Developer /Data Engineer /Data Analyst /DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.

- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior Data Engineer – AKS – TRC Anne-Maria – Ubank 19/2/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Data Engineer/Senior Data Analyst /Senior DevOps Engineer /Clouds/Linux Administrator /Python /R Developer
AI /ML /BI /SharePoint /Dynamics 365 CRM Consultant /EDW -Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Data Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/BI/SPoint/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and implemented a multi-tier Data Vault architecture using Terraform to support data ingestion from multiple cloud providers and integrate with a Snowflake-based multi-layered analytics architecture.
- Developed and maintained data pipelines using Spark and Python for high-volume data processing and transformation, ensuring data quality and integrity.
- Implemented CI/CD practices with tools like Jenkins and Terraform Cloud for automated pipeline deployment and infrastructure provisioning.
- Utilized dbt to streamline data transformation and ensure code reusability.
- Collaborated with business stakeholders and analysts to translate data requirements into technical specifications.
- Monitored and optimized data pipelines for performance and efficiency.

- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- Utilised Azure ADF (Azure Data Factory) and Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, & deployment processes within the Azure ecosystem.
- Defined necessary build steps, like code compilation, running tests, and generating artifacts.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Collaborated with a central team on a large analytics project aimed at transforming customer experiences for a rapidly growing, digitally native business.
- Leveraged SQL and Python to develop robust data pipelines for ingestion, processing, and analysis of large datasets.
- Implemented distributed compute technologies: Spark & PySpark to optimize data processing performance and scalability.
- Applied CI/CD and IaC methodologies to automate deployment processes and ensure consistency across environments.
- Developed and maintained CI/CD pipelines for a variety of applications using Azure DevOps, DevSecOps towards AIOps.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Designed and implemented data models based on entity-relationship and relational modeling principles, following Kimball and Data Vault methodologies.
- Played a key role in multi-layered analytics architectures to support various analytical use cases and business requirements.
- Worked closely with stakeholders to understand requirements and deliver solutions that meet business needs effectively.
- Implemented Data Vault design principles, designing scalable solutions including hub, link, and satellite tables for improved data management and analytics.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Implemented Pydantic-Core for automated Data Validation, reducing overflow validation time. Pydantic defines data models, with specific types and validation rules, automating validation and minimizing manual error checking.
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Completed design documentation, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Implemented data transformation using Step Functions and Spark Notebooks, writing Spark SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
- Real-time streaming data processing using Amazon Kinesis Analytics to write SQL queries to process and analyze data from Amazon Kinesis Streams or other streaming sources. Configure jobs to ingest and process streaming data.
- Implemented ETL/ELT processes for Azure Synapse Analytics, leveraging tools like SSIS, PolyBase and Stream Analytics.
- Employed DAX Studio and DAX functional language for complex formula creation in Power BI.
- Designed and developed OLAP Cubes, Dashboard & Reports and visualized data either in Power BI, Tableau or QlikView

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Data Engineer/Sr. DevOps Engineer/Python/R Developer/AI/ML/SharePoint BI Developer /EDW
Responsibilities:

Designed and implemented a multi-tier Data Vault architecture using Terraform to support data ingestion from multiple cloud providers and integrate with a Snowflake-based multi-layered analytics architecture.

Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Developed and maintained data pipelines using Spark and Python for high-volume data processing and transformation, ensuring data quality and integrity.

Implemented CI/CD practices with tools like Jenkins and Terraform Cloud for automated pipeline deployment and infrastructure provisioning.

Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.

Development and implementation of a real-time data analytics platform using Kafka and Spark Streaming to provide live insights into customer behavior.
Designed and built robust data pipelines for ETL processes using technologies like Airflow and Luigi.
Developed data models using the Kimball methodology to support multi-dimensional data analysis.
Utilized SQL and Python for data manipulation and analysis.
Implemented data quality checks and monitoring mechanisms to ensure data reliability.
Spearheaded the integration of Docker and Kubernetes into the data engineering workflow, resulting in enhanced scalability and containerized data processing.
Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
Utilized message-based asynchronous integration tools like Kafka for real-time data streaming, enabling timely data updates and analytics.
Developed reusable and scalable data solutions for processing and managing various data formats (CSV, JSON, Parquet, Avro) using Azure Cloud Tools.
Designed and executed streaming processes using Spark Streaming, enhancing real-time data processing capabilities.
Utilized RESTful APIs and testing tools like Postman for efficient integration with external systems.
Implemented complex ETL/ELT processes either using Azure Synapse Analytics or Apache Spark, PySpark, AWS Glue.
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Complete End-to-End Azure Data Engineer project using Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard tools of Power BI, Looker Studio or Tableau, including write Spark code in Azure Databricks.
Using Azure Data Factory (ADF) to extra and move on-premises SQL Server Data into Azure Cloud
Build a data process flow to load raw data onto Data Lake storage for Databricks to write spark code and transform the data and load the data back to transform data Lake storage for synapse analytics to run the SQL queries on top of the transform
Create a Spark session using the SparkSession.builder and getOrCreate() methods.
Either manually configuring Apache Spark in PySpark by setting various configuration parameters to control the behavior of Spark application or configure Spark programmatically using the SparkConf class in PySpark.
Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
Designed Amazon Redshift data warehousing solutions to facilitate data-driven decision-making.
Offered data-driven recommendations to meet stakeholder needs and support decision-making.
Developed interactive Power BI dashboards for visualizing key performance indicators.
Designed and developed OLAP Cubes, Dashboard & Reports and visualized data either in Power BI, Tableau or QlikView Leveraged Azure Data Factory for data integration, enhancing insights for RBA services.
Automated data support processes through scripting in R, Python, and PowerShell 7.3.
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Established an automated training pipeline using Azure DevOps.
Created BI applications and visual analytics products using PowerQuery, Power Pivots, and more in Azure.
Designed Bank analysis dashboards with SharePoint Dashboard Designer or Azure/Power BI Pro.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Data Engineer/Senior DevOps Engineer/Python Developer/AI/ML/SharePoint /K2 Developer /EDW
Responsibilities:

- Designed and implemented a **Google Cloud Platform EDW** using a serverless Lambda architecture. Data is ingested via Pub/Sub and processed in real-time using Cloud Functions. Cloud Dataflow performs periodic batch processing to update the data warehouse stored in BigQuery, which serves as the single source of truth for analytics.
- Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
- Built-in data validation, reconciliation, and metadata management for quality assured information.

Embedded quality engineering with unit, functional, and non-functional testing to identify and fix issues at their roots.
Designed and implemented ETL pipelines using GCP's Dataflow to transform and load data from various sources, optimizing data processing efficiency.
Developed and maintained data pipelines for ingesting and processing data from various internal and external sources, including APIs, databases, and flat files.
Implemented data warehousing solutions using platforms like Amazon Redshift and Snowflake.
Utilized data modeling techniques like entity-relationship modeling and Data Vault to structure data for efficient querying and analysis.
Automated data tasks using Python scripts and scheduled jobs.

Documented data pipelines and processes for clarity and maintainability.
Utilized dbt to streamline data transformation and ensure code reusability.
Demonstrated expertise in cloud platforms and advanced scripting, contributing to the adoption of GCP Composer for orchestration and workflow management.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency
Built and maintained data pipelines using a combination of Python, Spark, and AWS Glue for data transformation and loading.
Played a pivotal role in building and maintaining data pipelines using technologies such as Apache Spark 3.3 and PySpark.
Contributed to the adoption of dbt for data modeling and orchestration, streamlining data delivery for analytics teams.
Proficient in modern data analytics and engineering tools such as dbt and Dataform.
Skilled in creating impactful visualizations using Tableau and Microsoft Power BI.
Synthesized and analyzed data from various sources to support data-driven decision-making.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Created interactive Power BI dashboards with DAX calculations, utilizing Databricks for complex analytics.
Implemented high-performance warehouse solutions with Amazon Redshift or Azure DevOps.
Executed ETL processes for traditional DWH and ELT for Azure Data Warehouse or Parallel Data Warehouse.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: Data Engineer/Data Analyst /DevOps Engineer/SharePoint /BI Developer /D365 CRM Consultant /DBA
Responsibilities:

- Designed and implemented data pipelines for a marketing analytics platform, enabling data-driven marketing campaigns and customer segmentation.
- Utilized SQL and Python for data analysis and reporting.
- Developed data visualizations using tools like Tableau and Power BI.
- Collaborated with marketing teams to understand their data needs and translate them into actionable insights.
- Actively participated in code reviews, providing valuable insights and feedback to enhance code quality and maintainability.
- Developed and maintained data pipelines, ensuring efficient data extraction, transformation, and loading (ETL) processes.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Created data models and ETL processes to enable efficient data integration into the data warehouse.
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Data Engineer /Data Analyst /DevOps Engineer/SharePoint Consultant /DBA
Responsibilities:

- Designed and built databases for various applications using relational database technologies like MySQL and PostgreSQL.
- Developed and maintained SQL queries for data retrieval and manipulation.
- Performed data analysis and reporting to support business decision-making.
- Contributed to the implementation of a data platform's architecture, focusing on data pipelines, ETL jobs, and ingestion patterns.
- Leveraged cloud platform capabilities, particularly GCP's BigQuery, to optimize data processing and analysis for better client service.

- Demonstrated expertise in version control using Git, and CI/CD processes using Jenkins/Bamboo, ensuring efficient and reliable code deployment.
- Actively participated in troubleshooting and resolving technical issues, ensuring the smooth operation of data pipelines and associated components.
- Developed Python scripts for data processing, supporting efficient ETL operations.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Utilized Jira and Confluence for effective project management and documentation.
- Designed and developed interactive dashboards using Power BI to visualize key performance indicators and data trends.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer /Data Analyst /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Responsible to develop and publish both scheduled and ad hoc data visualizations that align with the business's specific needs while adhering to best practices in visualization
- Conducted data extraction, modelling, and statistical analysis to enable data-driven decision-making.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Data Engineer /Data Analyst /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Lead the creation of context aligned aggregate data sets for use by Insights Analyst, Modelling & Visualization Specialists.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Data Engineer /Data Analyst /MS BI Developer /DWH /DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.

- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior Data Operations Engineer-Data & Integration – Bluefin –ASX 7/2/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Data Operations Engineer-Data & Integration/Sr. DevOps Engineer /Python/R Developer/Clouds/Linux Administrator AI/ML/BISP/Dynamics 365 CRM Consultant/-Japan (7.5yrs) & Aust.(24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Sr. Data Operations Engineer-Data & Integration/Sr. DevOps Engineer/AI/ML /SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and implemented a **Google Cloud Platform EDW** using a serverless Lambda architecture. Data is ingested via Pub/Sub and processed in real-time using Cloud Functions. Cloud Dataflow performs periodic batch processing to update the data warehouse stored in BigQuery, which serves as the single source of truth for analytics.
- Managed and operated production data pipelines on Google Cloud Platform (GCP) using BigQuery, DBT, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
- Built-in data validation, reconciliation, and metadata management for quality assured information.
- Embedded quality engineering with unit, functional, and non-functional testing to identify and fix issues at their roots.
- Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.
- Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs. Monitored and maintained system performance, identified and resolved performance bottlenecks.
- Operated in a product development team and used agile methodologies to deliver projects on time and within budget.
- Implemented automation for data ingestion, transformation, and loading (ETL) processes using Python and Bash scripting.
- Collaborated with cross-functional teams (business analysts, developers) to translate business requirements into technical solutions. Documented operational procedures and maintained high standards for data quality and compliance.
- Built robust & secure GCP Landing Zones with Terraform, ensuring compliance with HIPAA, GDPR, and PCI DSS standards, and cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Own and support the Commission's Data & Integration systems, ensuring high availability, monitoring, fault resolution, and seamless software/data pipeline releases using GCP DevOps tools: Terraform, Vault, Artifactory, and Gitlab.

- Interact with others outside own department (D&I) on relatively complex matters. Perform independently and guide, assist, and mentor others in analysing, debugging, testing, documenting, and installing solutions
- Developed and maintained CI/CD pipelines for a variety of applications using GCP DevOps, DevSecOps towards AIOps.
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, ANN analytics, etc.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Manage 24x7 on-call support roster, providing day-to-day support and incident management for Production and non-Production environments. Implement monitoring and logging solutions using Prometheus and Grafana.
- Plan and implement changes to enhance Data & Integration systems, adhering to Change Management policies.
- Utilizing either BMC Control-M 9.0 or Cloud Composer/Airflow for orchestrating workflows and advanced scheduling within GCP Dataflow. Employing Dataform for data pipeline orchestration, SQL code management with version control, data asset definition, testing and deployment environment creation, and pipeline documentation.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Employed Requirements.txt to list exact packages to install for the deployment and choose constraints.txt to control dependency versions without full installation.
- Implemented Pydantic-Core for automated Data Validation, reducing overflow validation time. Pydantic defines data models, with specific types and validation rules, automating validation and minimizing manual error checking.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Developed and maintained IaC scripts using Python and Terraform for provisioning and managing cloud resources, enhancing scalability and flexibility of applications. Developing Data APIs (REST, GraphQL, Graph API, gRPC)
- Prioritized and navigated multiple time-constrained objectives, ensuring efficient project delivery.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. Data Operations Engineer-Data & Integration/Sr. DevOps Engineer/AI/ML /BI/SP Consultant /EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Researched and implemented appropriate ML algorithms and tools based on business requirements.

Designed, developed, and implemented data pipelines using GCP services such as BigQuery, Cloud Composer/Airflow, DBT, Dataflow, Dataform, Cloud Build, and GKE.

Designed and implemented data integrations using SAP Data Services to connect various data sources and applications.

Developed data validation and reconciliation processes to ensure data accuracy and consistency.

Utilized SQL and analytics tools to query, extract, and analyze structured and unstructured data.

Maintained system availability and troubleshoot technical issues related to data integrations.

As a product development team member, utilized tools like Selenium, Cypress, and Appium to run automated tests seamlessly within GCP. Leveraging GCP's scalable infrastructure like Compute Engine or Kubernetes Engine ensures parallel test execution and effortless adaptation to growing business needs.

Worked closely with data architects & developers to implement data governance and information management strategies.

Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.

Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.

Developed and maintained CI/CD pipelines for a variety of applications using GCP DevOps, DevSecOps towards AIOps.

Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues

Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
- Manage 24x7 on-call support roster, providing day-to-day support and incident management for Production and non-Production environments.
- Seamlessly execute automated tests within GCP utilizing Selenium, Cypress, or Appium. Leverage GCP's scalable infrastructure like Compute Engine or Kubernetes Engine for parallel test execution, ensuring effortless adaptation to growing business needs.
- Conduct non-functional acceptance testing, develop test scripts, and execute test cases for system enhancements.
- Leveraged SAP Data Services and SAP Business Objects to deliver robust ETL, Reporting, and Analytics solutions, showcasing in-depth understanding of these processes.
- Applied technical expertise in monitoring, observability, performance management, risk management, troubleshooting, and service recoverability. Executed and reported on non-functional testing, contributing to system improvements.
- Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.
- Utilized either BMC Control-M 8.0 or Cloud Composer/Airflow for orchestrating and scheduling workflows within GCP, potentially including job/task scheduling on various platforms (Windows, Automic, SQL Server Agent) and providing operating system support (MS Windows Server, Enterprise Linux).
- Employing Dataform for data pipeline orchestration, SQL code management with version control, data asset definition, testing and deployment environment creation, and pipeline documentation.
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.
- Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.
- Integrate security, code analysis, and quality checks into the SDLC automation using SonarQube, Snyk, and OWASP ZAP.
- Monitor and optimize system and application performance using Stackdriver, Datadog, Splunk and New Relic.
- Leverage Kubespray with its community-maintained extensions and Ansible to automate Kubernetes cluster deployment on AWS and Azure clouds. Then, utilize Kustomize for managing Kubernetes applications deployed on Amazon EKS and Azure AKS, enabling consistent configurations and simplified deployments across both cloud providers.
- Engineered AI-driven ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. Data Operations Engineer-Data & Integration/Sr. DevOps Engineer /AI/ML/SharePoint /K2 Developer
Responsibilities:

- Design and deploy a modern **AWS Lambda Architecture EDW**, comprising: **Batch layer**: Hadoop HDFS for distributed storage and MapReduce for batch processing of large datasets. **Speed layer**: Spark for in-memory processing of real-time data streams, utilizing RDDs for efficient data manipulation. **Service layer**: Spark Streaming for low-latency analytics on streaming data, generating SNS application notifications based on real-time insights. **Direct write queries**: Spark SQL for efficiently querying datasets within Spark cluster, enabling seamless integration with downstream applications & services.
- Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.
- Engineered and operated data platforms on Google Cloud Platform, utilizing services such as BigQuery, Cloud Composer/Airflow, Dataflow, Dataform, Cloud Build, and GKE.
- Configure dbt for BigQuery by specifying the project ID, dataset, and other connection details in the profiles.yml file
- Automated data pipelines using Airflow and Dataform, ensuring efficient data processing and delivery.
- Implemented monitoring and alerting solutions for proactive issue identification and resolution.
- Collaborated with DevOps team to implement CI/CD pipelines for data platform deployments.
- As a product development team member, I utilize tools like Selenium, Cypress, and Appium to run automated tests seamlessly within GCP. Leveraging GCP's scalable infrastructure like Compute Engine or Kubernetes Engine ensures parallel test execution and effortless adaptation to growing business needs.
- Maintained operational documentation and best practices for GCP data management.
- Operated in a product development team, contributing to the development and enhancement of data platforms.
- Collaborated with cross-functional teams to identify and implement process improvements, emphasizing automation.
- Applied scripting experience with languages such as Python and Bash, streamlining operational activities.
- Leveraging Dataform for managing intricate data pipelines with dependency management, data quality checks, and data lineage tracking. Using Cloud Build to automate diverse CI/CD tasks such as building, testing, and deploying applications to various platforms like Kubernetes, Cloud Functions, and App Engine.
- Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.
- Conducted non-functional acceptance testing, developed test scripts, and executed non-functional test cases for Data and Integration systems including the deployment of application and data pipeline releases while ensuring operational process and procedural documentation are maintained professionally.

Completed design, data models, loading approaches and source to target mapping documents.
Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
Created a mapping document that defines how the source data fields correspond to the target data fields.
Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
Implemented data transformation using Cloud Workflows and Dataproc Notebooks, writing BigQuery SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases

- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Data Operations Engineer-Data & Integration/ DevOps Engineer/Clouds/Linux Admin,/SP Consultant
Responsibilities:

- Implemented a **3-tier lambda architecture EDW on Azure, combining batch and real-time processing**. An **ingestion layer** handles data intake, followed by a **batch processing layer** using HDInsight and PySpark for scalable data transformation. The **real-time analysis layer** leverages Azure Stream Analytics on Data Streams for near real-time insights. Additionally, Databricks SQL offers interactive querying on Data Lake Storage data.
- Migrated on-premises data platforms to GCP, leveraging services like BigQuery, Dataflow, and Cloud Storage.
- Deploying GKE as a managed Kubernetes service for running containerized applications on GCP, managing Kubernetes clusters with automatic scaling, load balancing, and health checks to ensure high availability and scalability of applications.
- Leveraging BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment.
- Created and implemented ETL processes to extract, transform, and load data into data warehouses and data lakes.
- Implement the infrastructure as code using Terraform and Google Cloud Deployment Manager.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Monitored and analyzed data platform performance metrics, identifying and resolving performance issues.
- Implemented scripting solutions (Python, Bash) to automate routine tasks and improve operational efficiency.
- Provided technical support to internal users and troubleshoot data access and quality problems.
- Documented data management processes and procedures for internal knowledge sharing.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Demonstrated proficiency in scripting languages such as Python/PowerShell, Ansible, Puppet and Go for automation.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Data Operations Engineer-Data & Integration/DevOps Engineer/ML/SharePoint Consultant
Responsibilities:

- Performed data analysis using SQL and reporting tools to support business intelligence initiatives.
- Developed and maintained data reports and dashboards for various stakeholders.
- Cleaned and transformed raw data to ensure data quality and consistency.
- Collaborated with data analysts and business users to define data requirements and metrics. Installed, monitored, and supported data systems. Troubleshot and resolved critical incidents.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet

- Proficient in scripting and programming languages, including Python and Ruby.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Senior Windows /Unix/Linux Systems Administrator /DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Installation and configuration of hardware and software solutions.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Systems Engineer /BI Developer /Data Engineer /Data Analyst /EDW/DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design Enterprise data warehouse architecture with hardware infrastructure landscape, define schemes, Fact & Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Cyber Security Engineer – DevSecOps – Larson Partners for Zip called back for recruit interview first
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cyber Security Engineer- DevSecOps /Senior DevOps /Senior Data Engineer/Clouds/Linux Administrator /Python/R Developer /AI/ML/BI/SharePoint /Dynamics 365 CRM Consultant /EDW/DBA -Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Sr. Cyber Security Engineer-DevSecOps/Sr. DevOps/Data Engineer/Py/R Dev./AI/ML/SPD365 CRM Consultant
Responsibilities:

- Ensured Cloud Security by leveraging AWS IAM for identity governance, storing container images in ECR within S3, deploying them on Azure AKS through ECS, and utilizing Lambda for serverless execution, establishes a secure multi-cloud development and container orchestration pipeline.
- To achieve a comprehensive and secure development lifecycle, we're embedding security practices throughout our development and operations processes by implementing a robust DevSecOps approach. **Key elements include**:
- **Streamlined security automation**: Tight integration of security checks and gates within continuous integration and continuous delivery (CI/CD) pipelines to detect and address vulnerabilities early and efficiently.
- **Infrastructure as Code with Terraform**: Provisioning and managing infrastructure in a secure, scalable, and auditable manner using Terraform, guaranteeing consistent and auditable infrastructure configurations.
- **API security for data integrity and protection**: Rigorous protection of application programming interfaces (APIs) through authentication, authorization, encryption, and input validation to safeguard sensitive data from unauthorized access and manipulation.
- **Threat modeling for proactive risk mitigation**: Proactive identification and assessment of potential threats and vulnerabilities through threat modeling techniques to implement preventive measures and reduce risk exposure.
- Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Bash and PowerShell.
- **Scan Regularly**: Regularly scan base images for vulnerabilities using dedicated tools like Snyk or Anchore. Set clear criteria for triggering patch upgrades based on severity and impact.

- **Automate Uplifts**: Integrate image building and deployment into the CI/CD pipeline. Automate patching for critical vulnerabilities based on established policy using tools like Skaffold.
- **Monitor and Report**: Visualize scan results and patching progress on a centralized dashboard. Set up alerts for critical vulnerabilities and successful/failed patching attempts.
- **Continuous Improvement**: Track patching time and continuously improve processes to minimize response time. Foster communication between security and development teams to address underlying code issues and prevent future vulnerabilities.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, SSO and VPN. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Developed and maintained CI/CD pipelines for a variety of applications using Amazon/Azure DevOps, DevSecOps or AIOps.
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Master containerization with Docker/Compose (multistage builds, advanced networking). Manage S3 efficiently: buckets, policies, security, batch optimizations. Secure cloud infrastructure through robust implementation of security guidelines.
- Integrated security scanning into CI/CD pipelines in AWS using AWS CodePipeline with Amazon Inspector in .yaml file
- Install and setup Splunk 9.1 leading platform for log management, analysis, and security information and event management (SIEM), perform real-time monitoring, and conduct forensic investigations.
- Strong understanding of the intelligence lifecycle and associated analytic methodologies (Cyber Kill Chain, Diamond Model, Mitre ATT&CK, etc.). Assisted in the implementation of E8 mitigation strategies.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu +NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics,
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Cyber Security Engineer-DevSecOps/Sr. DevOps/Data Engineer/SP/BI /Python/R Developer /EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
Designed and implemented secure architectures for cloud-based applications.
Conducted penetration testing and vulnerability assessments for web applications and APIs security.
Ensured Cloud Security by leveraging AWS IAM for identity governance, storing container images in ECR within S3, deploying them on Azure AKS through ECS, and utilizing Lambda for serverless execution, establishes a secure multi-cloud development and container orchestration pipeline.
Managed security incidents and coordinated response efforts with internal and external stakeholders.
Developed and maintained security policies and procedures.
Implemented a vulnerability management program for containerized applications, reducing unpatched vulnerabilities by 70%. Automated patching process for container base images using AWS ECR, ECS and Jenkins pipelines.
Integrated security scanning into CI/CD pipelines to shift security left and prevent vulnerabilities from reaching production.
Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Bash and PowerShell.
Collaborated with developers and DevOps teams to implement secure coding practices and DevSecOps principles.
Deploy Enterprise Identity and Access Management (EIAM) for internal users and Customer Identity and Access Management (CIAM) for external customers and project contractors.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Securely stewarding data in AWS S3 involves crafting robust buckets with multi-layered protection: icy Glacier retention for long-term archives, precise object expiry for ephemeral data, granular metadata controls for organized access, and data access controls for airtight security. Implement security guidelines on cloud infrastructure, ensuring compliance.
Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.
Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1.
Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.
Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs.
Conduct regular security reviews and coordinate remedial actions to mitigate risks.
Stay up-to-date with the latest IT and cybersecurity trends, providing recommendations for enhancing cybersecurity posture.
Discover, identify, and report regularly on significant cyber threats and risks to the Department, Government, and other related organizations' presented by cyber threat actors and groups originating in global regions.
Develop and produce strategic intelligence reports focused on industry-level analysis of developing cybersecurity threats, events, conditions, and impacts to mitigate any potential data threats and defend data.
Advanced knowledge in Cyber Threat Intelligence, threat management, vulnerability management, risk mitigation, compliance and related. Response and recovery to security incidents
Collaborated with clients to monitor, detect, respond, and report on cybersecurity vulnerabilities, incidents, and threats.
Familiarity with open-source intelligence (OSINT), social media intelligence (SOCMINT), human intelligence (HUMINT), and technical intelligence (TECHINT) gathering skills.
Conducted cybersecurity assessments, provided recommendations for enhancements, and supported incident management.
Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs of work.
Perform day-to-day monitoring and triaging critical alerts and coordinate with technical teams for incident response.
Triage phishing emails, taking appropriate containment and remediation actions to prevent security incidents
Manage and configure security platforms such as EDR and Vulnerability Management Platforms (CrowdStrike, Mimecast, Qualys, Tenable AD, Fortinet etc.)
Track and report on information security controls and their effectiveness, making recommendations for improvements
Configure McAfee and McAfee VPN to encrypts users' data and protects it from potential snoopers, ensuring that users' online activities and personal information remain secure even on untrusted networks.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. Cyber Security Engineer-DevSecOps /Sr. DevOps/Data Engineer /SharePoint /K2 Developer/DBA
Responsibilities:
Secured on-premises infrastructure by implementing network security controls and access management policies.
Monitored security logs and identified potential threats. Managed user accounts and access privileges.
Provided technical support for security-related issues.
Ensured Cloud Security by leveraging AWS IAM for identity governance, storing container images in ECR within S3, deploying them on Azure AKS through ECS, and utilizing Lambda for serverless execution, establishes a secure multi-cloud development and container orchestration pipeline.
Managed S3 buckets, policies, securities, applying batches, and implemented security guidelines on cloud infrastructure.
Configured secure S3 buckets with retention policies (Glacier, object expiry), access controls, and metadata management.
Managed AWS resources. Reviewed and guided other team members in Python development.
Developed IaC CI/CD pipelines for a variety of applications using Terraform, AWS/Azure DevOps, DevSecOps or AIOps.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
Implementing Infrastructure as Code (IAC) by creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Implementing DevSecOps by integrating CI/CD pipelines, Infrastructure as Code using Terraform, and ensuring and threat modeling for a comprehensive and secure development lifecycle.

Ensured robust container security involves leveraging technologies such as isolation, resource control, and access management, complemented by the implementation of container scanning tools like Snyk and Aqua Security using Scripting Languages: Python, Bash and PowerShell.
Implementing a comprehensive security strategy involves utilizing powerful security tools such as vulnerability scanners like Nessus and Qualys, alongside Security Information and Event Management (SIEM) solutions.
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Implementing a comprehensive security strategy involves utilizing powerful security tools such as vulnerability scanners like Nessus and Qualys, alongside Security Information and Event Management (SIEM) solutions.
Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.
Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
Conducted security assessments and provided recommendations to enhance security postures.
Supported the implementation of cybersecurity controls, including firewalls, intrusion detection systems, and access controls. Contributed to client awareness of information security best practices and compliance requirements.
Developed and maintained infrastructure security strategies, roadmaps, and artifacts to ensure the effective implementation of security capabilities.
Produced detailed component specifications and high-level designs, facilitating smooth implementation.
Install and setup Tenable, Tenable.sc (formerly SecurityCenter) vulnerability management platform that helps organizations identify and prioritize security vulnerabilities across their infrastructure.
Using Tenable for continuous monitoring, vulnerability assessment, and reporting capabilities to help organizations stay on top of potential security risks.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Cyber Security Engineer/DevSecOps/Data Engineer /Sr. Systems Engineer /SharePoint Consultant /DBA
Responsibilities:

- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Specialized in Cloud infrastructure security, DevOps, and DevSecOps practices.
- Work on challenging projects addressing current and emerging threats.
- Assisted in the development and maintenance of information security controls and standards.
- Conducted security assessments and audits to identify vulnerabilities and recommend improvements.
- Supported incident management processes, helping to reduce the impact of security incidents.
- Collaborated with teams to implement security solutions and tools.
- Contributed to the organization's Information Security Strategy development.
- Stayed informed about the evolving cybersecurity landscape, including threats and vulnerabilities.
- Developed and maintained relationships with internal and external stakeholders.
- Provided expertise in infrastructure security architecture, ensuring the development and implementation of effective security solutions.
- Worked closely with solutions architects to address IT infrastructure concerns and ensure a secure architecture.
- Produced detailed component specifications and high-level designs for successful implementation.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Cyber Security Engineer/DevSecOps/Data Analyst/Senior Systems Engineer /SharePoint Consultant
Responsibilities:

- Collaborating with IT Security for the Standard Operating Environment (SOE) security hardening in line with PSPF, ISM, ACSC, CIS and DTA Protected pattern guidelines
- Specialized in Cloud infrastructure security, DevOps, and DevSecOps practices.

- Contributed to the development and delivery of strategies protecting organizations from diverse cyber threats.
- Identified potential threats and risks in application code and platforms.
- Executed effective solutions with minimal disruption to wider technology business.
- Assisted in security audits and assessments, identifying security weaknesses.
- Collaborated with the IT team to ensure the organization's security policies were enforced.
- Stayed updated with industry security trends and emerging threats.
- Gained foundational experience in cybersecurity. Assisted in monitoring and analyzing security alerts.
- Contributed to the documentation of security procedures and policies.
- Supported ongoing operational Cyber Security Management programs of work, contributing to the overall security posture.
- Managed hardware and server administration tasks to ensure smooth operations.
- Administered AD, DFS, ADFS, AAD, DNS, and DHCP services, maintaining their integrity and availability.
- Leveraged McAfee, Tenable, Splunk, and Secret Server to enhance the effectiveness of Cyber Security Management programs.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Cyber Security Engineer/DevOps/Data Analyst/Senior Systems Engineer /DBA
Responsibilities:

- Install and setup Tenable, Tenable.sc (formerly SecurityCenter) vulnerability management platform that helps organizations identify and prioritize security vulnerabilities across their infrastructure.
- Using Tenable for continuous monitoring, vulnerability assessment, and reporting capabilities to help organizations stay on top of potential security risks.
- Specialized in Cloud infrastructure security, DevOps, and DevSecOps practices.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.
- Assisted senior security analysts in monitoring and analyzing security alerts.
- Install and configure Secret Server for a privileged access management (PAM) solution to secure and manage privileged credentials, such as administrator passwords, SSH keys, and API keys in addition to provides centralized storage, access controls, and auditing for sensitive credentials, helping organizations enforce security policies and mitigate the risks
- Developed infrastructure security strategies and roadmaps based on enterprise architecture best practices.
- Created and maintained infrastructure security architecture artifacts, establishing standards and procedures for enhanced security capabilities.
- Collaborated closely with solutions architects, addressing IT infrastructure concerns and aligning them with security architecture requirements.
- Produced detailed component specifications and high-level designs, ensuring successful implementation.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Supported ongoing operational Cyber Security Management programs of work, contributing to a strong security framework.
- Managed hardware and server administration, ensuring reliable infrastructure performance.
- Oversaw AD, DFS, ADFS, AAD, DNS, and DHCP services, maintaining their stability and security.
- Provided level 3 reactive support for incident/service request management, monitoring, and event management.
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Systems Engineer /BI Developer /Data Engineer /Data Analyst /DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.

- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior Cloud Engineer – EKS – Talent Insight 29/1/24 No resp again F.it
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Engineer /Senior DevOps Engineer/Senior Data Engineer/Sr. Data Analyst /Sr. Python /R Programmer /Clouds/Linux Administrator /AI /ML/BI/SharePoint /D365 CRM Consultant/EDW -Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior Cloud Engineer/Sr. DevOps/Data Engineer/Python/R Developer/AI/ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architected and implemented the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Serve as the main point of contact for all maintenance and implementation tasks. Liaise with stakeholders across engineering and DevOps. Act as a bridge between SecOps and TechOps, ensuring seamless collaboration.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Using Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker, CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Master containerization with Docker/Compose (multistage builds, advanced networking). Manage S3 efficiently: buckets, policies, security, batch optimizations. Secure cloud infrastructure through robust implementation of security guidelines.

- Securely stewarding data in AWS S3 involves crafting robust buckets with multi-layered protection: icy Glacier retention for long-term archives, precise object expiry for ephemeral data, granular metadata controls for organized access, and data access controls for airtight security. Implement security guidelines on cloud infrastructure, ensuring compliance.
- Hands-on management of AWS resources and guidance for Python development.
- Review and guide team members in structuring Python projects, CI, and package management.
- Either using CloudFormation templates for popular deployments like Microsoft Windows Server and SAP HANA or using Terraform/Ansible for multi-clouds via its declarative approach or Combine CloudFormation templates with Terraform or Ansible for hybrid deployments and leverage CD pipelines for efficient infrastructure automation.
- Configure secure buckets, manage retention policies, object metadata, and data access controls in AWS S3.
- Advanced Git skills, including multi-branching, rebase, merges, and Pull Request reviews.
- Proficiently in leveraging both self-hosted SonarQube and cloud-based SonarCloud to rigorously analyze Python projects, ensuring code quality, security, and GRC (Governance, Risk, and Compliance) alignment.
- Developed and maintained CI/CD pipelines for a variety of applications using Amazon/Azure DevOps, DevSecOps or AIOps.
- Used PyYAML Primary library for parsing & generating YAML & used ruamel.yaml Alternative library with extended features.
- Leveraged the flexibility of `poetry` to manage essential Python dependencies, streamlining project configuration through `pyproject.toml` for metadata and build instructions. This unified approach also facilitated virtual environment management using `pipenv`, ensuring a clean and isolated development environment.
- Further enhanced code quality by implementing `pytest` for comprehensive unit and functional testing. The seamless integration of `pytest` with `poetry` enabled configuring test settings within `pyproject.toml` and leveraged Poetry's automation capabilities for efficient test execution.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.
- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Employed Requirements.txt to list exact packages to install for the deployment and choose constraints.txt to control dependency versions without full installation.
- Implemented Pydantic-Core for automated Data Validation, reducing overflow validation time. Pydantic defines data models, with specific types and validation rules, automating validation and minimizing manual error checking.
- Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, ANN analytics, etc.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. Cloud Engineer/Sr. DevOps/Data Engineer/Python/R Developer/AI/ML/BI/SharePoint Consultant /EDW
Responsibilities:

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Serve as the main point of contact (POC) for all maintenance and implementation tasks. Liaise with stakeholders across engineering and DevOps. Act as a bridge between SecOps and TechOps, ensuring seamless collaboration.

Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.

Master containerization with Docker/Compose (multistage builds, advanced networking). Manage S3 efficiently: buckets, policies, security, batch optimizations. Secure cloud infrastructure through robust implementation of security guidelines.

Securely stewarding data in AWS S3 involves crafting robust buckets with multi-layered protection: icy Glacier retention for long-term archives, precise object expiry for ephemeral data, granular metadata controls for organized access, and data access controls for airtight security. Implement security guidelines on cloud infrastructure, ensuring compliance.

Federated Amazon RedShift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Leveraged the flexibility of `poetry` to manage essential Python dependencies, streamlining project configuration through `pyproject.toml` for metadata and build instructions. This unified approach also facilitated virtual environment management using `pipenv`, ensuring a clean and isolated development environment.

Further enhanced code quality by implementing `pytest` for comprehensive unit and functional testing. The seamless integration of `pytest` with `poetry` enabled configuring test settings within `pyproject.toml` and leveraged Poetry's automation capabilities for efficient test execution.

Used SonarQube for large enterprises with strict customizable security requirements, & on-premise control to review bugs, manage vulnerabilities, security hotspots, update python project, and work with GRC (Governance, Risk, and Compliance).
Designed and implemented CI/CD pipelines for Python applications using poetry, pyproject.toml, and pytest.
Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
Developed and optimized data pipelines leveraging Apache Flink and Kafka Streams for real-time processing.
Implemented robust data quality monitoring using Great Expectations and Metaflow.
Collaborated with SecOps and DevOps teams to improve overall security posture and automation.
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Carried out and delivered efficient/repeatable platform engineering and application maintenance tasks through the orchestration and strategy-building tools (AWS CloudFormation, CDK and Terraform etc.) to ensure optimal performance.
Choreograph serverless workflows with AWS Step Functions & integrate data pipelines with MWAA.
Trigger tasks on events with CloudWatch & manage configurations across instances with SSM Run Command/Automation.
Delivered technical consulting and subject matter expertise to projects, including pre-project to ensure risks, interdependencies and exceptions are identified, mitigated, or escalated and solutions are formulated in alignment with ICT strategy and architecture. Wrote technical documentation to support new and existing applications.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

- Leveraging TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Employed Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis in Python/R programming & fine-tuning based on test results to improve model performance.Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Cloud Engineer/Sr. DevOps/Data Engineer/Python/R Developer/AI/ML /SharePoint /K2 Developer
Responsibilities:
Serve as the main point of contact (POC) for all maintenance and implementation tasks. Liaise with stakeholders across engineering and DevOps. Act as a bridge between SecOps and TechOps, ensuring seamless collaboration.
Managed S3 buckets, policies, securities, applying batches, and implemented security guidelines on cloud infrastructure.
Configured secure S3 buckets with retention policies (Glacier, object expiry), access controls, and metadata management.
Managed AWS resources. Reviewed and guided other team members in Python development.
Developed IaC CI/CD pipelines for a variety of applications using Terraform, AWS/Azure DevOps, DevSecOps or AIOps.
Used Artificial Neural Networks (ANN) algorithms based on brain function to model complicated patterns & forecast issues
Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.

Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
Implementing Infrastructure as Code (IAC) by creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.

Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts.
Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Connected R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
Used Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
Master of AI/ML deployment: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.
Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
Internal security champion: Enhanced internal security through improved ML libraries & frameworks using TensorFlow, PyTorch, Keras, Docker and Git.
Cloud security expert: Managed & monitored cloud infra with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security & automation.
NLP, CV and RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Domain: Cloud Engineer/DevOps/Data Engineer /Clouds/Linux Administrator /SharePoint Consultant /DBA
Responsibilities:

- Serve as the main point of contact (POC) for all maintenance and implementation tasks. Liaise with stakeholders across engineering and DevOps. Act as a bridge between SecOps and TechOps, ensuring seamless collaboration.
- Managed S3 buckets, policies, securities, applying batches, and implemented security guidelines on cloud infrastructure.
- Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.
- Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
- Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
- Migrated on-premise applications to the cloud using AWS Direct Connect, Snow Family, Storage Gateway, S3TA, etc.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
- Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
- Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
- Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
- Developed AWS infrastructure components using Terraform, IAM and DevSecOps for enhanced security and automation.
- Provided technical consulting and subject matter expertise to projects, ensuring alignment with ICT strategy and architecture.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Managed and enhanced infrastructure with Terraform, integrating development and test tools into CI/CD pipelines. Additionally, developed and maintained REST APIs, monitored AWS cloud infrastructure, and implemented large-scale logging, monitoring, and security measures.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Engineer /DevOps Engineer/Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- Serve as the main point of contact (POC) for all maintenance and implementation tasks. Liaise with stakeholders across engineering and DevOps. Act as a bridge between SecOps and TechOps, ensuring seamless collaboration.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools: Terraform, Vault, Artifactory, and Gitlab, to streamline operations and improve productivity.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /Systems Engineer /BI Developer /Data Engineer /Data Analyst /DBA
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

R Programmer – SustainRecruit – 22/1/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Python /R Programmer /Senior AI/ML Engineer/Developer /Senior DevOps/Data Engineer /Senior Data Analyst / /BI/SharePoint/D365 CRM Consultant /DWH/DBA-Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health:Sr. Python/R Programmer/Sr. AI/ML Engineer/Sr.DevOps/Data Engineer/Analyst/SP/D365 CRM Consultant/EDW
Responsibilities:

- Developed statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Crafted a powerful ad reporting tool for our client's upcoming product, wielding R's statistical sorcery to tame complex data sets and deliver precise insights. Integrated ad reporting tool with Dynamics 365 CRM for efficient marking and follow up.
- Designing, developing code/scripts that leverage structured data from various sources using statistical tools of R, Git, etc.
- Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
- Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.

- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
- End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Using Python and R programming language for many statistical modeling and data science tasks.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like `powerbi`. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like `poweRbi` or `embedR`. Utilize RStudio Server on AWS instances for remote R development and execution.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Machine Learning with R for logistic regression user case, linear regression with user case
- Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision) for Imaging AI cancer detection, like pancreatic cancer, liver cancer...
- Implemented data augmentation, transfer learning and model compression techniques to improve model performance and efficiency. Worked with large-scale image datasets using SQL and BigQuery ML for Big Data.
- Seamless integration of Python code and libraries within R environments, enabling combined workflows.
- Deploy R models and scripts as web services or integrate with Azure Machine Learning services.
- Analyzed large-scale marketing campaign data to identify key performance indicators and optimize future campaigns.
- Pioneering AI workflow automation, working on various projects within the large Government healthcare organization that are used to streamline admin services for healthcare professionals by leveraging AI and other modern technologies.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
- Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney

Gov. Central Bank: Senior Python /R Programmer /Sr. DevOps/Data Engineer /AI/ML/BI /SPoint/Power BI Developer /EDW

Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Implemented complex trading algorithms, demonstrating a deep knowledge of Python and R with high-performance trading systems. Designed and optimized algorithms for high-frequency trading, achieving competitive advantages.

Instrumental in building a sophisticated ad reporting tool for RBA's upcoming product. Leveraging my expertise in Python and R programming to create robust statistical models that can handle complex data sets with precision.

Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.

Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.

Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.

Bridge the gap between R and low-level C++ for precise control and data manipulation.

Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R.

Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")

End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity

Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items

Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
Collaborated closely with Data Scientists and Engineers on Greenfield projects from concept to production deployment.
Analyse and process deep, complex data sets for accurate reporting, Proficiency in R programming, Linear modeling (statistical processing).
Perform function calls to the `c` function with multiple arguments: c(1,5, 6, 9) to combines these elements into a single vector
Developed & implemented statistical models for predicting customer churn rate, leading to a 15% reduction in customer loss. Built interactive dashboards and reports either using Power BI, Tableau or R Shiny for real-time data visualization &analysis.
Using Python and R programming language for many statistical modeling and data science tasks.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.
Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 4.x, Python 3.1x, PSPP, JMP, JASP, EViews, StatCrunch, etc.
Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
Applyed ML techniques to solve real-world business problems within a commercial setting.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and optimized ETL processes using SQL and PySpark, enabling seamless data integration and transformation.
Built and maintained R scripts for automated data processing and reporting tasks.
Analyzed financial data to identify fraudulent transactions and mitigate financial risks.
Developed forecasting models for predicting future sales trends, improving inventory management.
Contributed to a collaborative knowledge base to share data analysis best practices within the team.
Implementing the Reserve Bank's operations in domestic and foreign exchange markets, monitoring and analysing developments in financial markets and co-ordinating the Reserve Bank's relationship with international institutions.

- Design and implementation of Big Data and Machine Learning platforms, pipelines and tooling
- Performance tuning machine learning pipelines to be efficient, reliable, and scalable.
- Implemented API endpoints for data exchange using Python, Django, Flask and FastAPI.
- Utilized Pyspark and Pandas for data processing and transformation tasks.
- Worked with AWS core services, Athena, and Glue to integrate data pipelines and enhance platform capabilities.
- Defined, mapped, and modeled data flows from various sources, including transactional systems, client data, and external data providers. Utilized SQL and Python to ensure data accuracy, transformation, and integration.
- Utilizing either dbt (data build tool) or Dataform tools in the realm of modern data analytics and engineering.

Designed Bank analysis dashboards with SharePoint Dashboard Designer or Azure/Power BI Pro.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Python /R Programmer /Senior DevOps/Data Engineer/Analyst /AI/ML/BI /SharePoint /K2 Developer
Responsibilities:
Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.
Bridge the gap between R and low-level C++ for precise control and data manipulation.
Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R.
Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
Develop statistical models for ad reporting using R programming. Perform testing & quality assurance of statistical models.
Strong ability to manage and interpret complex data sets.
Performed data cleaning, transformation, and analysis using R for various research projects.
Developed linear regression models to predict student academic performance, increasing overall accuracy by 8%.
Created interactive data visualizations using ggplot2 to effectively communicate insights to stakeholders.
Documented data analysis workflows and methodologies for future reference and reproducibility.

Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to.
Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github actions

- Execute a comprehensive Azure Data Engineer project from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau. This involves crafting Spark code within the Azure Databricks environment.

Automated data support processes through scripting in R, Python, and PowerShell 7.3.
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Created BI applications and visual analytics products using PowerQuery, Power Pivots, and more in Azure.
Writing high quality and efficient python and R code with good software engineering practices.
Implemented RESTful APIs using Python and Flask to enable seamless data exchange between different components.
Integrated third-party APIs and services to enhance platform functionality.
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Either manually configuring Apache Spark in PySpark by setting various configuration parameters to control the behavior of Spark application or configure Spark programmatically using the SparkConf class in PySpark.
Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling.
Designed Amazon Redshift data warehousing solutions to facilitate data-driven decision-making.
Employed DAX Studio and DAX functional language for complex formula creation in Power BI.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Implemented high-performance warehouse solutions with Amazon Redshift or Azure DevOps.
Utilized Azure Open AI Service and Azure Open AI Studio for Large Language Model (LLM) training.
Executed ETL processes for traditional DWH and ELT for Azure Data Warehouse or Parallel Data Warehouse.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Python /R Programmer /DevOps/Data Engineer/Data Analyst /SP /BI Developer /D365 CRM Consultant
Responsibilities:

- Developed statistical models for ad reporting using R programming. Managed and interpreted complex data sets.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Worked on a sophisticated ad reporting tool for the client's upcoming product.
- Data visualization with R using Tidyverse and Modeldata libraries with ggplot package and View(crackets) datasets
- Utilized Python and multithreaded programming to achieve low-latency performance and high reliability.
- Proficiency in visualisation tools such as Tableau, Power BI, R-Shiny, Microstrategy, SAP Business Objects, etc.
- Developed API's and Dashboards to provide real-time insights and monitoring of trading activities.
- Applied Async programming using Python Asyncio to optimize system performance.
- Leveraged frameworks like Flask and Django to build robust and scalable components.
- Designed and implemented data visualizations and dashboards to facilitate data-driven decision-making across groups.
- Conducted statistical modeling and analysis using large and complex datasets for Employers Mutual workforce policies.
- Spearheaded the analysis of complex data sets to extract actionable insights and inform strategic initiatives.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Created data models and ETL processes to enable efficient data integration into the data warehouse.
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**

Government Council: Python /R Programmer /DevOps/Data Engineer /Data Analyst /BI Developer /SharePoint Consultant
Responsibilities:

- Developed statistical models for ad reporting using R programming.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Proficient in modern data analytics and engineering tools such as dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Microsoft Power BI.
- Utilized containerization and Kubernetes to enhance deployment and scalability of applications.
- Implemented unit testing and test-driven development practices, ensuring code quality and reliability.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Optimized query performance and database schema design to enhance overall data processing efficiency.
- Utilized Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis to uncover patterns and trends, supporting evidence-based decision-making.
- Designed and developed interactive dashboards using Power BI to visualize key performance indicators and data trends.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Python /R Programmer /Data Engineer /Data Analyst /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed statistical models for ad reporting using R programming.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Conducted data extraction, modelling, and statistical analysis to enable data-driven decision-making.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Lead the creation of context aligned aggregate data sets for use by Insights Analyst, Modelling & Visualization Specialists.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior DevOps Engineer – EKS- Aurec Gov 2x/1/24
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator /ML/BI/SharePoint /D365 CRM Consultant/EDW
-Japan (7.5yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior DevOps Engineer/Sr. Data Engineer/Clouds/Linux Admin. /ML/BI/SPoint/D365 CRM Consultant /EDW
Responsibilities:

- Architected and implemented the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.

- Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Automated tasks and scripts using Python/Lambda/ Terraform, Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Solved complex development problems and produced the highest quality code while striving to push the technical boundaries of online application. Troubleshot and resolved network and security issues.
- Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
- Carried out and delivered efficient/repeatable platform engineering and application maintenance tasks through the orchestration and strategy-building tools to ensure optimal performance.
- Delivered technical consulting and subject matter expertise to projects, including pre-project to ensure risks, interdependencies and exceptions are identified, mitigated, or escalated and solutions are formulated in alignment with ICT strategy and architecture. Wrote technical documentation to support new and existing applications.
- Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevOps best practices to IT and business stakeholders.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Implemented Pydantic-Core for Data Validation, significantly reducing overflow data validation time. Pydantic defines data models with specific types and validation rules, automating the validation and conversion process, minimizing the need for manual data cleansing and error checking.
- Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
- Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
- Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Used TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis in Python/R programming & fine-tuning based on test results to improve model performance.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer /Sr. Data Engineer/Clouds /Liunx Admin. /ML/BI/SharePoint Consultant /EDW
Responsibilities:

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.

Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.

Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.

Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Integrated development and test tools with CI/CD pipelines. Developed and maintained REST APIs.
Managed and monitored cloud infrastructure using AWS. Developed and maintained logging and monitoring systems at large scale. Developed and maintained security policies and procedures for cloud infrastructure.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Implemented Pydantic-Core for Data Validation, significantly reducing overflow data validation time. Pydantic defines data models with specific types and validation rules, automating the validation and conversion process, minimizing the need for manual data cleansing and error checking.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.
Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.

- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer /Clouds/Linux Administrator /ML/SharePoint /K2 Developer
Responsibilities:

Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Automated tasks and scripts using Python/Lambda/ Terraform, Javascript/Typescript, PowerShell, Bash, Ansible & Puppet
Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Solved complex development problems and produced the highest quality code while striving to push the technical boundaries of online application. Troubleshot and resolved network and security issues.

Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
Delivered technical consulting and subject matter expertise to projects, including pre-project to ensure risks, interdependencies and exceptions are identified, mitigated, or escalated and solutions are formulated in alignment with ICT strategy and architecture. Wrote technical documentation to support new and existing applications.
Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.
Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
Implemented logging and monitoring solutions to improve application performance and reliability.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Managed AWS CloudFormation, Ansible, and Terraform to automate infrastructure provisioning and configuration.
Migrated on-premise applications to the cloud using AWS Direct Connect, Snow Family, Storage Gateway, S3TA, etc.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Domain: Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator /SharePoint Consultant
Responsibilities:

- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Collaborated with the DevOps Engineering team to drive automation, establish software standards, and deploy/manage cloud-native services.
- Successfully solved complex development problems, pushing technical boundaries to ensure the highest quality code for a variety of online applications.
- Delivered efficient/repeatable platform engineering and application maintenance tasks, optimizing performance through orchestration and strategy-building tools.
- Provided technical consulting and subject matter expertise to projects, ensuring alignment with ICT strategy and architecture.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Worked with DevOps team to establish and maintain a continuous integration and continuous deployment (CI/CD) pipeline.
- Managed and enhanced infrastructure with Terraform, integrating development and test tools into CI/CD pipelines. Additionally, developed and maintained REST APIs , monitored AWS cloud infrastructure, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained network hardware and software. Troubleshot and resolved network and connectivity issues. Documented IT procedures and policies.
- Analyze data sources, types, and formats to define transformation, loading, and data model requirements.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.

- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Senior DevOps Engineer/Senior Data Engineer /SharePoint Consultant
Responsibilities:

- Worked with the DevOps team to establish and maintain a continuous integration and continuous deployment (CI/CD) pipeline.
- Managed infrastructure as code (IAC) with Terraform, integrated tools into CI/CD pipelines, developed REST APIs, monitored AWS cloud, and implemented large-scale logging, monitoring, and security measures.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools: Terraform, Vault, Artifactory, and Gitlab, to streamline operations and improve productivity.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Technical supported the installation, configuration, and testing of hardware and software solutions.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

DevOps Engineer – Taktalent – GCP AI
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Sr. DevOps Engineer/Sr.Data Engineer/Clouds/Linux Admin/AI/ML/SP/D365 CRM Consultant/-Japan (7.5yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer /Sr. Data Engineer /Clouds/Linux Admin./AI/ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Designed and implemented Serverless Lambda Architecture with Google Cloud Platform encompassing PubSub Layer (aka:Data Ingestion Layer), Stream Layer with Batch Layer and Serving Layer.
- Being responsible for the end-to-end DevOps function, focusing on building CI/CD pipelines for scalable growth. Instrumental in taking a platform from MVP to its first production launch.

- Proficient in technologies like React, GCP, FastAPI, Postgres, Kubernetes, and Next-js.
- Build and maintain CI/CD pipelines for Kubernetes applications. Develop & maintain infrastructure as code using Terraform.
- Implement monitoring and logging solutions using Prometheus and Grafana.
- Collaborate with development teams to ensure that applications are deployed in a timely and efficient manner.
- Played a pivotal role in shaping system architecture entire DevOps environment for an Enterprise AI technology business.
- Ensured the safety and security of the AI platform while building robust CI/CD pipelines for Kubernetes applications.
- Contributed over 7 years of experience in a DevOps/Platform Engineer role, ensuring the seamless operation of the development-to-production process. Expertise in Kubernetes, Terraform, & cloud platforms such as GCP, AWS and Azure.
- Provided technical leadership in a 100% hands-on DevOps role, ensuring the security and rapid scalability of AI platforms. Worked closely with the founding team to transition the platform from MVP to a successful production launch.
- Involved in the design and development of cloud-based solutions for various clients using AWS, Azure and GCP services.
- Implement the infrastructure as code using Terraform and Google Cloud Deployment Manager .
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage applications using Docker CRE and Kubernetes on GCP GKE clusters.
- Automated cloud infrastructure provisioning and configuration using tools such as Terraform, Ansible, Puppet, and Chef.
- Implemented containerisation concepts using Docker and Kubernetes for deploying microservices-based applications.
- Deploy and orchestrate microservices using Docker CRE and Kubernetes on AWS and GCP platforms.
- Integrate security, code analysis, and quality checks into the SDLC automation using SonarQube, Snyk, and OWASP ZAP.
- Monitor and optimize system and application performance using CloudWatch, Stackdriver, Datadog, Splunk and New Relic.
- Designed and implemented CI/CD pipelines using Jenkins, optimizing software build, test, and deployment processes, resulting in a 20% increase in efficiency.
- Developed and maintained IaC scripts using Terraform for provisioning and managing cloud resources, enhancing scalability and flexibility of applications. Developing Data APIs (REST, GraphQL, Graph API, gRPC)
- Implemented containerization and orchestration solutions using Docker and Kubernetes, enabling efficient application deployment and scaling. Utilized various DevOps & DevSecOps tools such as Gitlab, Nexus, Terraform, etc
- Comprehend optimal techniques for integrating applications and utilizing the Apigee platform to establish efficient internal and external API connections, creating reusable enterprise engineering models.
- Migrated on-premise applications to the GCP cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos and CloudEndure Migration.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365 CRM/Power BI/Power Automate etc.)
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Used TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**

Gov Central Bank: Senior DevOps Engineer /Sr. Data Engineer/Clouds /Liunx Administrator /ML/BI/SP Consultant /EDW

Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.

Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Design, implement, and maintain robust CI/CD pipelines to automate software build, test, and deployment processes for a cloud-native data management solution for large enterprise manufacturing companies.

Led the DevOps function in a complex environment, contributing to the success of an AI platform. Delivered DevOps tooling to support a robust software development life cycle (SDLC). Implemented CI/CD pipelines for Kubernetes applications.

Performed root cause analysis and resolved issues related to cloud infrastructure, Windows servers, Linux servers, and network connectivity
Managed and maintained the cloud and network infrastructure, optimizing performance and security.
Champion best practices and methodologies for DevOps and SDLC automation, staying current with industry trends.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Spearheaded the transition to containerization technologies (Docker and Kubernetes), enhancing application scalability and reliability. Implemented Ansible for configuration management, reducing configuration-related issues by 30%.
Resolved a critical security risk in a business application by identifying vulnerabilities and implementing security patches.
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Led the implementation of Docker and Kubernetes for containerization, improving application scalability and resilience.
Performed in-depth security assessments, identified vulnerabilities, and implemented necessary security patches to address significant risks.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
Managed Linux servers using Ansible, Chef and Puppet; Used Maven, ANT and Gradle for build tools
Used Docker and Kubernetes to containerize and orchestrate applications
Configured and managed Azure services such as App Service, Functions, Storage, SQL Database, etc.
Monitored and troubleshooted Azure cloud performance and issues using Azure Monitor, Application Insights, Log Analytics, etc. Collaborated with development and operations teams to implement DevOps best practices and principles
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Configured and managed Azure resources such as virtual machines, storage accounts, app services, load balancers, WAF and firewalls. Utilized Gitlab, GitHub and Bitbucket for source code management and version control
Automated the deployment and configuration of Windows servers and Linux servers using Ansible, Puppet and Chef
Collaborated with developers and testers to implement continuous integration and continuous testing using Azure DevOps and Selenium; Used Nagios, Grafana and ELK(Elasticsearch, Logstash, and Kibana), Prometheus for logging & monitoring
Developed and maintained PowerShell scripts and automation tools for efficient Azure infrastructure management.
Engineered AI-driven ML systems using Python and machine learning frameworks (PyTorch, scikit-learn)
Deploy ML solutions using IBM's Watson Studio platform
Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn
Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git
Showcase strong analytical and problem-solving skills in creating efficient self-learning applications.
Develop and optimize ETL processes using SQL and PySpark, ensuring seamless data integration and transformation.
Deploy machine learning solutions for various clients using IBM's Watson Studio platform.
Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer /Clouds/Linux Administrator /SharePoint /K2 Developer /DBA
Responsibilities:
Managed the company's Azure cloud infrastructure, ensuring high availability and disaster recovery capabilities.
Collaborated with development teams to integrate security, code analysis, and quality checks into the SDLC automation, ensuring software quality and compliance with industry standards.
Configured and managed observability components using Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Automated processes using Shell, Python, Lambda, Javascript, and TypeScript, enhancing efficiency and productivity.
Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
Provided technical guidance to junior team members and conducted training sessions on DevOps best practices.
Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus
Managed AWS CloudFormation, Ansible, and Terraform to automate infrastructure provisioning and configuration.
Maintained open-source tools for web and cloud application deployment, scaling, and monitoring.
Coordinated with development teams to ensure optimized DevOps practices for software delivery.
Actively promoted and enforced enterprise-wide DevOps standards and best practices.

Administered Atlassian suite of tools and facilitated Dynamics CRM 2016 integration.
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
Migrated on-premise applications to the cloud either using AWS Migration Hub or Azure Migrate, Azure Import/Export etc.
Collaborated with developers & testers to deliver high-quality products using Scrum and Kanban agile methodologies.
Implemented security best practices for Azure resources using Azure AD, Azure Key Vault and Azure Policies
Used Jenkins, Bamboo and Azure Pipelines to create CI/CD pipelines for .Net applications and APIs
Built and deployed Docker containers for web applications using Azure Container Registry and Azure Container Instances
Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Provided 3rd level desktop, systems, and network support, resolving complex technical issues to minimize downtime.
Conducted server and security audits, ensuring compliance with security policies and implementing necessary remediation measures. Used JIRA, Bugzilla and Remedy for bug tracking tools
Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Developed PowerShell scripts and automation tools for managing and maintaining Azure infrastructure.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator /SharePoint Consultant
Responsibilities:

- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Spearheaded the transition to CI/CD pipelines using Azure DevOps, streamlining the code delivery process.
- Conducted penetration testing and implemented security measures to address significant security risks.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Worked directly with critical customers to gather requirements and ensure the delivered solutions met their expectations.
- Demonstrated proficiency in scripting languages such as PowerShell and Go for automation.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Migrated on-premise applications to the cloud either using AWS Migration Hub or Azure Migrate, Azure SQL & Power BI
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- Implemented and managed networking solutions, including Cisco, Meraki SD-WAN, FortiGate Firewalls and WAF to enhance network performance and protect against cyber threats.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Automating systems administration by various automation scripting like Python, PowerShell, Bash, Puppet, etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Senior DevOps Engineer/Azure/M365 Administrator (AAD/Inutune /SCCM) /SharePoint Consultant

Responsibilities:

- Implemented container-based technologies and deployment tools like Docker, Kubernetes, and Helm Charts for enhanced project efficiency.
- Administered Atlassian suite of tools, ensuring optimal performance and providing support to development teams.
- Supported the integration of Dynamics 365, contributing to a more efficient CRM system.
- Collaborated on a DevOps Center of Excellence to promote automation and continuous delivery practices.
- Actively participated in Agile teams, adhering to Agile principles and practices.
- Managed configuration and automation using Ansible, enhancing system stability.
- Effectively managed infrastructure with Cisco CloudCenter, Ansible, Docker, and Kubernetes.
- Assisted in the integration of ServiceNow for incident management, improving response times.
- Actively participated in a DevOps Center of Excellence, emphasizing automation and continuous delivery practices.
- Proficient in scripting and programming languages, including Python and Ruby.
- Ensured systems and processes adhered to high standards of security, performance, and reliability.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized a range of DevOps tools, including Terraform, Vault, Artifactory, and Gitlab, to streamline operations and improve productivity.
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and IIS on Windows servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Senior Windows /Unix/Linux Systems Administrator /DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.

- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Performed installation and configuration of hardware, software, testing and networking equipment.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior DevOps Engineer – EKS, little GCP -DCS
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator /ML/BI/SharePoint /D365 CRM Consultant/EDW
-Japan (7.5yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Sr. Data Engineer/Clouds/Linux Admin. /ML/BI/SPoint/D365 CRM Consultant /EDW
Responsibilities:

- Architected and implemented the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.

- Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
- Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
- Automated tasks and scripts using Python/Lambda/ Terraform, Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Worked with DevOps Engineering team in Revenue Digital Branch to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
- Solved complex development problems and produced the highest quality code while striving to push the technical boundaries of online application. Troubleshot and resolved network and security issues.
- Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
- Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.
- Carried out and delivered efficient/repeatable platform engineering and application maintenance tasks through the orchestration and strategy-building tools to ensure optimal performance.
- Delivered technical consulting and subject matter expertise to projects, including pre-project to ensure risks, interdependencies and exceptions are identified, mitigated, or escalated and solutions are formulated in alignment with ICT strategy and architecture. Wrote technical documentation to support new and existing applications.
- Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevOps best practices to IT and business stakeholders.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
- Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
- Designed, developed, and implemented complex ETL/ELT processes and efficient data pipelines in AWS using Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3, resulting in improved system efficiency.
- Implemented Pydantic-Core for Data Validation, significantly reducing overflow data validation time. Pydantic defines data models with specific types and validation rules, automating the validation and conversion process, minimizing the need for manual data cleansing and error checking.
- Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
- Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
- Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Used TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer /Sr. Data Engineer/Clouds /Liunx Admin. /ML/BI/SharePoint Consultant /EDW
Responsibilities:

Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations. Documented IT procedures and policies.
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.
Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Automated tasks and scripts using Python/Lambda/ Terraform, Javascript/Typescript, PowerShell, Shell, Ansible & Puppet
Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing Git version control for tracking changes, and automating deployment through CI/CD tools.
Worked with the DevOps team to establish and maintain a continuous integration and continuous deployment (CI/CD) pipeline. Developed and maintained infrastructure as code using Terraform.
Integrated development and test tools with CI/CD pipelines. Developed and maintained REST APIs.
Managed and monitored cloud infrastructure using AWS. Developed and maintained logging and monitoring systems at large scale. Developed and maintained security policies and procedures for cloud infrastructure.
Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Implemented Pydantic-Core for Data Validation, significantly reducing overflow data validation time. Pydantic defines data models with specific types and validation rules, automating the validation and conversion process, minimizing the need for manual data cleansing and error checking.
Designed and developed data solutions using AWS EC2, RDS, S3, Python, PySpark/EMR, RedShift Spectrum, &Snowflake
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
Implemented data quality checks and monitoring using Amazon CloudWatch and Simple Notification Service (SNS)
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Delivered and managed multi-cloud networks, enhancing operational efficiency and network reliability.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts.
Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer /Sr. Data Engineer /Clouds/Linux Administrator /ML/SharePoint /K2 Developer
Responsibilities:

Designed & deployed modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS & MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Developed and maintained CI/CD pipelines for a variety of applications using Amazon DevOps, DevSecOps or AIOps.

Either using AWS CDK (Cloud Development Kit) for creating intricate pipelines with multiple stages and dependencies or using AWS CodePipeline GUI interface for creating straightforward CI/CD pipelines, simplifying the modeling, orchestration, and management of code flow through the release process.
Automated tasks and scripts using Python/Lambda/ Terraform, Javascript/Typescript, PowerShell, Bash, Ansible & Puppet
Implementing Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
Worked with DevOps Engineering team to drive automation, establish software standards, service modularity, testing standards, and deployment/management of cloud native services.
Solved complex development problems and produced the highest quality code while striving to push the technical boundaries of online application. Troubleshot and resolved network and security issues.
Demonstrated expertise in data architecture, employing various data modeling techniques such as 3NF models, star schemas, Data Vault 2.0, Erwin Data Modeler 2020 R2 and Pydantic for database modeling.
Actively contributed to and guided innovation in DevSecOps practices and incorporated agile delivery approaches and technologies that assisted with the development and delivery of high-quality software.
Carried out and delivered efficient/repeatable platform engineering and application maintenance tasks through the orchestration and strategy-building tools to ensure optimal performance.
Delivered technical consulting and subject matter expertise to projects, including pre-project to ensure risks, interdependencies and exceptions are identified, mitigated, or escalated and solutions are formulated in alignment with ICT strategy and architecture. Wrote technical documentation to support new and existing applications.
Delivered high level options, conceptual approaches, technical recommendations and estimates to support the transition of system designs into operation. Presented workshops on DevSecOps best practices to IT and business stakeholders.
Developed and deployed cloud-native applications using Amazon ECS, EKS, API Gateway, CodeDeploy, AWS CDK etc.
Implemented logging and monitoring solutions to improve application performance and reliability.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Assisted in the development and maintenance of support strategies, operational standard procedures, performance strategies, robust configuration practices and automated processes.
Managed AWS CloudFormation, Ansible, and Terraform to automate infrastructure provisioning and configuration.
Migrated on-premise applications to the cloud using AWS Direct Connect, Snow Family, Storage Gateway, S3TA, etc.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM2016/Power BI/Power Automate etc.)
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator /SharePoint Consultant
Responsibilities:

- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Collaborated with the DevOps Engineering team to drive automation, establish software standards, and deploy/manage cloud-native services.
- Successfully solved complex development problems, pushing technical boundaries to ensure the highest quality code for a variety of online applications.
- Actively contributed to and guided innovation in DevOps practices, incorporating agile delivery approaches and technologies.
- Delivered efficient/repeatable platform engineering and application maintenance tasks, optimizing performance through orchestration and strategy-building tools.
- Provided technical consulting and subject matter expertise to projects, ensuring alignment with ICT strategy and architecture.
- Solid background in CI/CD, automation, and tools like Bamboo, Bitbucket, Puppet, Chef, Ansible, Jenkins, virtualization, containers, and monitoring (Splunk, New Relic, etc).
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Shell, Ansible and Puppet
- Worked with DevOps team to establish and maintain a continuous integration and continuous deployment (CI/CD) pipeline.
- Managed and enhanced infrastructure with Terraform, integrating development and test tools into CI/CD pipelines. Additionally, developed and maintained REST APIs , monitored AWS cloud infrastructure, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained network hardware and software. Troubleshot and resolved network and connectivity issues. Documented IT procedures and policies.
- Analyze data sources, types, and formats to define transformation, loading, and data model requirements.
- Create a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.

- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Played a key role in container orchestration using Kubernetes or Nomad, enabling dynamic scaling of services.
- Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Senior DevOps Engineer/Senior Data Engineer /SharePoint Consultant
Responsibilities:

- Worked with the DevOps team to establish and maintain a continuous integration and continuous deployment (CI/CD) pipeline.
- Managed infrastructure as code (IAC) with Terraform, integrated tools into CI/CD pipelines, developed REST APIs, monitored AWS cloud, and implemented large-scale logging, monitoring, and security measures.
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Implemented and managed networking solutions, including Cisco, Meraki SD-WAN, FortiGate Firewalls and WAF to enhance network performance and protect against cyber threats.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools: Terraform, Vault, Artifactory, and Gitlab, to streamline operations and improve productivity.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Systems Engineer /Windows/Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**

Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Supported the installation, configuration, and testing of hardware and software solutions.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Machine Learning Engineer - Aurec
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Machine Learning Engineer /Senior DevOps/AI/DL/MLOps Engineer /Python Developer/Senior Data Engineer /AI/ML/DL/BI/SharePoint/D365 CRM Consultant /EDW/DBA -Japan(7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior ML Engineer /Senior DevOps/AI/DL/MLOps Engineer /Python Developer/SP/D365 CRM Consultant/EDW
Responsibilities:

- **Infrastructure Design and Deployment**: Design, implement, and maintain scalable and reliable infrastructure for machine learning workflows using AWS, Terraform, and Kubernetes.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV for Imaging AI cancer detection, a different way than Dr. Cao Kei proposed a pancreatic cancer early detection model "PANDA" based on deep learning published a paper in "Nature Medicine" in Nov, 2023.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Spearheaded the development and optimization of real-time data pipelines using Apache Flink and Kafka Streams, reducing latency by 25% and improving model training efficiency.
- Implemented Airflow micro-batching pipelines for real-time predictions, achieving 10x faster model response times.
- Collaborated with stakeholders to configure and deploy ML models using MLOps platforms like Kubeflow and MLflow, streamlining the ML lifecycle. Developed and managed feature stores with Feast and Hopsworks, promoting data reuse and consistency across diverse ML projects.
- Employed LIME and SHAP for Explainable AI, enhancing model interpretability and building trust with stakeholders.
- Successfully migrated ML models to TensorFlow Serving and TorchScript for efficient real-time predictions.
- Championed the adoption of TinyML for resource-constrained edge devices, expanding data collection capabilities.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- **Model Deployment and Monitoring**: Deploy machine learning models either using fully managed AWS SageMaker, or high-performance and scalable TensorFlow Serving or Open-source Kubeflow run on Kubenetes on any major clouds.
- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Utilized Great Expectations and Metaflow for data quality monitoring and lineage tracking, ensuring data integrity and pipeline health.
- Leveraged Prometheus and Grafana for proactive monitoring and anomaly detection, maintaining model performance and data drift control.
- **Automation and CI/CD**: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure. Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX A100s system
- **Security and Compliance**: Implement security best practices to safeguard machine learning infrastructure and data.
- Built robust and secure AWS landing zones with Terraform, ensuring compliance with relevant data protection, regulatory requirements & cross-platform compatibility for Windows & Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within Security/main.tf configuration file.
- Produced production-quality code deployed as Docker containers. Coded in a Linux environment using Git for source control. Implemented end-to-end solutions in a shared codebase.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior Machine Learning Engineer /Senior DevOps/AI/DL/MLOps Engineer /BI /Python Developer/EDW
Responsibilities:

Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Security-focused AI architect: Designed, developed and deployed secure and efficient AI/ML applications, deep learning models and systems. Based on bank business requirements, define most suitable ML Pipelines from Kubeflow, Mlflow, Dataflow MLOps , ZenML, Flyte, framework to create pipelines that orchestrate the ML workflow.

Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.

Master of AI/ML deployment: Built and deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure/GCP), secured pipelines with MLOps/DevSecOps.

Built and deployed production-ready ML models using TensorFlow and PyTorch for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.

Optimized data pipelines utilizing Airflow and Luigi, reducing processing time by 30% and improving resource utilization.

Introduced automated model versioning and rollbacks with MLflow, enabling seamless rollbacks and mitigating production issues. Collaborated with data scientists and software engineers to ensure effective data flow and model integration.

Contributed to the evolution of the cloud infrastructure on AWS and GCP, supporting scalability and resource management.

Actively participated in knowledge sharing and team collaboration, promoting a productive and efficient work environment.

Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.

- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).

Internal security champion: Enhanced internal security through improved ML libraries and frameworks using TensorFlow, PyTorch, Keras, Docker and Git.

Cloud security expert: Managed and monitored cloud infrastructure with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security and automation.

Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs

NLP, CV & RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.

Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.

Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.

Designed and implemented containerized and Kubernetes-native supervised ML inference pipelines utilizing Amazon SageMaker Studio for seamless deployment on Nvidia DGX-1 systems. This ensured real-time performance and elastic scalability to handle varying workloads.

Monitored model performance, conducted A/B testing, & proactively identified and addressed model drift or potential issues.

Leveraged BERT-based NLP models to analyze daily financial trading notes and assess potential adherence to financial outcome regulations, leading to a 5% improvement in compliance and a reduction in administrative costs.

Employed Deep Learning CNNs to automate error-prone manual processes in analyzing financial trading reports for specific accounts, resulting in a 20% increase in financial business processing efficiency.

Agile data scientist: Conducted research, created datasets, performed statistical analysis, built & trained data science products & ML systems.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.

Cloud migration & automation: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.

Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to

Migrated on-premises workloads to AWS Cloud using various tools, including AWS Server Migration Service (SMS) for migrate VMware workloads to AWS; AWS Database Migration Service (DMS) for migrating databases to AWS and AWS Database Gateway for connecting on-premises applications to cloud databases securely. Delivered CI/CD/DevOps in AWS

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior ML Engineer /Senior DevOps/AI/DL/MLOps Engineer /SharePoint /K2 /Python Developer /EDW
Responsibilities:

AI/ML architect: Enhanced international student engagement and sales through models, deployed and monitored with MLOps/DevSecOps for efficient, secure platform. Participated in the design and scoping of greenfield projects.
Designed, built, and operated software systems to transform petabytes of data into actionable insights.
Collaborated with a passionate team of Data Scientists, Machine Learning Engineers, and Data Engineers.
Worked on end-to-end technical solutions, from spike sessions to production deployment.
Spearheading large-scale Machine Learning data processing projects for an enterprise-level commercial brand.
Leading the optimization of data pipelines and liaising with stakeholders on pipeline configuration for ModelOps.
Driving the evolution of feature stores, including real-time applications and ML systems.
Utilizing the latest ML technologies: Stream Processing (Apache Flink, Kafka Streams) and Micro-batching (Airflow, Luigi).
Implementing commercial deployment of Machine Learning frameworks, including TensorFlow and Torch.
Building and productionizing ML models with a strong focus on Python.
Collaborating with cross-functional teams and maintaining a collaborative attitude.
Leveraging AWS and GCP for cloud services.Committed to software best practices, including IaC, GitOps, and CI/CD.
Implement and maintain monitoring solutions to track model performance, data drift, and system health.
Transformed and trained data science products and ML systems using Python and Pydantic on AWS EMR.
Conduct performance analysis and implement improvements to enhance the overall efficiency of machine learning systems.
Leverage Amazon SageMaker Studio to provide a collaborative environment for data scientists and ML engineers.
Data-driven problem solver: Developed and implemented ML models for key business challenges, optimizing processes & driving project success.
Security and automation expert: Secured and managed Linux systems, implemented best practices and automation, proactively mitigating threats and resolving complex issues.Developed and maintained AI /ML models to enhance international student's engagement and sales.

- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- **Master of AI/ML deployment**: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.
- Proficient in ML frameworks: Skilled in Python, R, Java, Scikit-learn, H2O, TensorFlow, PyTorch, Keras, MXNet, ImageAI, OpenCV for ML/DL tasks.
- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- **Internal security champion**: Enhanced internal security through improved ML libraries & frameworks using TensorFlow, PyTorch, Keras, Docker and Git.
- Cloud security expert: Managed & monitored cloud infra with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security & automation.
- **NLP, CV and RL**: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.
- **Agile data scientist**: Conducted research, created datasets, performed statistical analysis, built & trained data science products and ML systems.
- **MLOps advocate**: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Automated tasks and scripts using Python, PowerShell, Bash scripts, Ansible, Puppet, Lambda, JavaScript and Typescript
- **Cloud migration & automation**: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption. Employed Power BI, Tableau, or QlikView for real-time financial analysis.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Machine Learning Engineer /DevOps/AI/DL/MLOps Engineer /SharePoint /D365 CRM Consultant
Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Collaborated with data scientists and software engineers to develop, refine, and deploy ML solutions while optimizing data pipelines and infrastructure.

- Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Manage support requests. Perform maintenance tasks such as software upgrades
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher). Skilled in creating impactful visualizations using Tableau and Power BI.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Transform and train Data science products and ML Systems. Worked on developing and training ML models.
- Research and implement appropriate ML algorithms and tools according to business requirements
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Conducted regular system training and retraining activities using Python, scikit-learn, TensorFlow for ML/DL.
- Experienced in creating user interfaces with Flask, Django, FastAPI, utilizing scikit-learn for model evaluation.
- Proficient in Python, SQL, and containerization for deploying data applications, with additional expertise in Python application development and TypeScript. Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Platform Engineer-Azure / DevOps Engineer /ML Engineer/SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Built and deployed ML models using k-NN) and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy. Building and productionizing ML models with a strong focus on Python.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /ML /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Built and deployed ML models using k-NN) and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy. Building and productionizing ML models with a strong focus on Python.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /ML /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

R Programming – OsbornStaine
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior Python /R Programmer /Senior AI/ML Engineer/Developer /Senior DevOps/Data Engineer /Senior Data Analyst / /BI/SharePoint/D365 CRM Consultant /DWH/DBA-Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health:Sr. Python/R Programmer/Sr. AI/ML Engineer/Sr.DevOps/Data Engineer/Analyst/SP/D365 CRM Consultant/EDW
Responsibilities:

- Developed statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Crafted a powerful ad reporting tool for our client's upcoming product, wielding R's statistical sorcery to tame complex data sets and deliver precise insights. Integrated ad reporting tool with Dynamics 365 CRM for efficient marking and follow up.
- Designing, developing code/scripts that leverage structured data from various sources using statistical tools of R, Git, etc.
- Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
- Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.
- Bridge the gap between R and low-level C++ for precise control and data manipulation.
- Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R with RStudio and Rccp package.
- Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
- End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
- Using Python and R programming language for many statistical modeling and data science tasks.
- Connecting R to SQL Server and working with SQL Databases using RODBC and dplyr to querying SQL Databases.
- Programmatically interact with Power BI data and metadata using the Power BI REST API, accessible through R packages like `powerbi`. Embed Power BI reports and visuals in R Shiny apps using the Power BI Embedded API and dedicated R packages like `poweRbi` or `embedR`. Utilize RStudio Server on AWS instances for remote R development and execution.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Machine Learning with R for logistic regression user case, linear regression with user case
- Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision) for Imaging AI cancer detection, like pancreatic cancer, liver cancer...
- Implemented data augmentation, transfer learning and model compression techniques to improve model performance and efficiency. Worked with large-scale image datasets using SQL and BigQuery ML for Big Data.
- Seamless integration of Python code and libraries within R environments, enabling combined workflows.
- Deploy R models and scripts as web services or integrate with Azure Machine Learning services.
- Analyzed large-scale marketing campaign data to identify key performance indicators and optimize future campaigns.
- Pioneering AI workflow automation, working on various projects within the large Government healthcare organization that are used to streamline admin services for healthcare professionals by leveraging AI and other modern technologies.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
- Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**

Gov. Central Bank: Senior Python /R Programmer /Sr. DevOps/Data Engineer /AI/ML/BI /SPoint/Power BI Developer /EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Implemented complex trading algorithms, demonstrating a deep knowledge of Python and R with high-performance trading systems. Designed and optimized algorithms for high-frequency trading, achieving competitive advantages.
Instrumental in building a sophisticated ad reporting tool for RBA's upcoming product. Leveraging my expertise in Python and R programming to create robust statistical models that can handle complex data sets with precision.
Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.
Bridge the gap between R and low-level C++ for precise control and data manipulation.
Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R.
Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
Unleash insights from diverse data: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.
Collaborated closely with Data Scientists and Engineers on Greenfield projects from concept to production deployment.
Analyse and process deep, complex data sets for accurate reporting, Proficiency in R programming, Linear modeling (statistical processing).
Perform function calls to the `c` function with multiple arguments: c(1,5, 6, 9) to combines these elements into a single vector
Developed & implemented statistical models for predicting customer churn rate, leading to a 15% reduction in customer loss. Built interactive dashboards and reports either using Power BI, Tableau or R Shiny for real-time data visualization &analysis.
Using Python and R programming language for many statistical modeling and data science tasks.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.
Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 4.x, Python 3.1x, PSPP, JMP, JASP, EViews, StatCrunch, etc.
Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
Applyed ML techniques to solve real-world business problems within a commercial setting.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and optimized ETL processes using SQL and PySpark, enabling seamless data integration and transformation.
Built and maintained R scripts for automated data processing and reporting tasks.
Analyzed financial data to identify fraudulent transactions and mitigate financial risks.
Developed forecasting models for predicting future sales trends, improving inventory management.
Contributed to a collaborative knowledge base to share data analysis best practices within the team.
Implementing the Reserve Bank's operations in domestic and foreign exchange markets, monitoring and analysing developments in financial markets and co-ordinating the Reserve Bank's relationship with international institutions.

- Design and implementation of Big Data and Machine Learning platforms, pipelines and tooling
- Performance tuning machine learning pipelines to be efficient, reliable, and scalable.
- Implemented API endpoints for data exchange using Python, Django, Flask and FastAPI.
- Utilized Pyspark and Pandas for data processing and transformation tasks.
- Worked with AWS core services, Athena, and Glue to integrate data pipelines and enhance platform capabilities.
- Defined, mapped, and modeled data flows from various sources, including transactional systems, client data, and external data providers. Utilized SQL and Python to ensure data accuracy, transformation, and integration.
- Utilizing either dbt (data build tool) or Dataform tools in the realm of modern data analytics and engineering.

Designed Bank analysis dashboards with SharePoint Dashboard Designer or Azure/Power BI Pro.

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT/SYD (Working remotely from Sydney Home Office)
Gov. University: Senior Python /R Programmer /Senior DevOps/Data Engineer/Analyst /AI/ML/BI /SharePoint /K2 Developer

Responsibilities:

Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
Uncover data insights using R and linear modeling, ensuring model accuracy through rigorous testing and QA.
Tame complex, deep datasets for powerful reporting, wielding proficiency in testing and data interpretation.
Bridge the gap between R and low-level C++ for precise control and data manipulation.
Create a C++ file (e.g., `my_linear_function.cpp`) containing the C++ function for using in R.
Use the `sourceCpp()` function to compile and load the C++ code into R: sourceCpp("my_linear_function.cpp")
End-to-end data science execution including scoping, building, testing, implementation, maintenance, tracking and optimisation of predictive models. Setup ETL pipelines required to have data available, ensuring data quality and integrity
Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
Develop statistical models for ad reporting using R programming. Perform testing & quality assurance of statistical models.
Strong ability to manage and interpret complex data sets.
Performed data cleaning, transformation, and analysis using R for various research projects.
Developed linear regression models to predict student academic performance, increasing overall accuracy by 8%.
Created interactive data visualizations using ggplot2 to effectively communicate insights to stakeholders.
Documented data analysis workflows and methodologies for future reference and reproducibility.
Machine Learning with R programming language & associated packages (Tidyverse, Shiny, Core R) to evaluate, automate and enhance business processes ensuring they are robust and adaptable coding/scripting principles are adhered to.
Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github actions

- Execute a comprehensive Azure Data Engineer project from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau. This involves crafting Spark code within the Azure Databricks environment.

Automated data support processes through scripting in R, Python, and PowerShell 7.3.
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Created BI applications and visual analytics products using PowerQuery, Power Pivots, and more in Azure.
Writing high quality and efficient python and R code with good software engineering practices.
Implemented RESTful APIs using Python and Flask to enable seamless data exchange between different components.
Integrated third-party APIs and services to enhance platform functionality.
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Either manually configuring Apache Spark in PySpark by setting various configuration parameters to control the behavior of Spark application or configure Spark programmatically using the SparkConf class in PySpark.
Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling.
Designed Amazon Redshift data warehousing solutions to facilitate data-driven decision-making.
Employed DAX Studio and DAX functional language for complex formula creation in Power BI.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Implemented high-performance warehouse solutions with Amazon Redshift or Azure DevOps.
Utilized Azure Open AI Service and Azure Open AI Studio for Large Language Model (LLM) training.
Executed ETL processes for traditional DWH and ELT for Azure Data Warehouse or Parallel Data Warehouse.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: Python /R Programmer /DevOps/Data Engineer/Data Analyst /SP /BI Developer /D365 CRM Consultant
Responsibilities:

- Developed statistical models for ad reporting using R programming. Managed and interpreted complex data sets.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Worked on a sophisticated ad reporting tool for the client's upcoming product.
- Data visualization with R using Tidyverse and Modeldata libraries with ggplot package and View(crackets) datasets
- Utilized Python and multithreaded programming to achieve low-latency performance and high reliability.

- Proficiency in visualisation tools such as Tableau, Power BI, R-Shiny, Microstrategy, SAP Business Objects, etc.
- Developed API's and Dashboards to provide real-time insights and monitoring of trading activities.
- Applied Async programming using Python Asyncio to optimize system performance.
- Leveraged frameworks like Flask and Django to build robust and scalable components.
- Designed and implemented data visualizations and dashboards to facilitate data-driven decision-making across groups.
- Conducted statistical modeling and analysis using large and complex datasets for Employers Mutual workforce policies.
- Spearheaded the analysis of complex data sets to extract actionable insights and inform strategic initiatives.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Created data models and ETL processes to enable efficient data integration into the data warehouse.
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Python /R Programmer /DevOps/Data Engineer /Data Analyst /BI Developer /SharePoint Consultant
Responsibilities:

- Developed statistical models for ad reporting using R programming.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Proficient in modern data analytics and engineering tools such as dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Microsoft Power BI.
- Utilized containerization and Kubernetes to enhance deployment and scalability of applications.
- Implemented unit testing and test-driven development practices, ensuring code quality and reliability.
- Engineered end-to-end data pipelines to extract, transform, and load data from diverse sources to data warehouses.
- Optimized query performance and database schema design to enhance overall data processing efficiency.
- Utilized Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis to uncover patterns and trends, supporting evidence-based decision-making.
- Designed and developed interactive dashboards using Power BI to visualize key performance indicators and data trends.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Python /R Programmer /Data Engineer /Data Analyst /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed statistical models for ad reporting using R programming.
- Performed testing and quality assurance of statistical models.
- Analyzed and processed deep, complex data sets for accurate reporting.
- Conducted data extraction, modelling, and statistical analysis to enable data-driven decision-making.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Lead the creation of context aligned aggregate data sets for use by Insights Analyst, Modelling & Visualization Specialists.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.

- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH**
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior AWS Data Engineer – TRC for Synchrone Consulting
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AWS Data Engineer/Senior DevOps Engineer/Senior Data Analyst /ML/BI Developer /Clouds/Unix/Linux Administrator /SharePoint/D365 CRM Consultant /EDW -Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**

Gov. Health: Senior AWS Data Engineer/Sr. DevOps Engineer/Linux Administrator/BI/ML /SP/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
- Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services such as SageMaker, Glue, Airflow, Lambda, EC2, VPC, and IAM using Jenkins, Ansible, Chef, Puppet, Splunk, JIRA, confluence etc.
- Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing using Python, Pydantic, Java, R, SQL, git etc.
- Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
- Managed and optimized Redshift clusters to ensure high performance and scalability.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), Single Sign-On and MFA via VPN.
- Utilized Glue, a serverless ETL service, to collect, transforms, and load data into data warehouses and data lakes.
- Automated a wide range of data processing tasks, including ETL, machine learning, and data quality checks using Airflow.
- Used Lambda to run code without having to provision or manage servers, especially for users who need to run short-lived, event-driven tasks, such as processing data streams or responding to API requests.
- Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
- Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Monitored and troubleshoot data pipeline issues using Splunk and CloudWatch.
- Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
- Migrated on-premises workloads to AWS Cloud using various tools including AWS Server Migration Service (SMS) for mirate VMware workloads to AWS; AWS Database Migration Service (DMS) for migrating databases to AWS and AWS Database Gateway for connecting on-premises applications to cloud databases securely. Delivered CI/CD/DevOps in AWS Cloud.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Implemented data transformation using Step Functions and Spark Notebooks, writing Spark SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
- Real-time streaming data processing using Amazon Kinesis Analytics to write SQL queries to process and analyze data from Amazon Kinesis Streams or other streaming sources. Configure jobs to ingest and process streaming data.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal secure ML libraries and frameworks to enhance overall efficiency.
- Built, trained, deployed, and managed machine learning models using Sagemaker.
- Used Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior AWS Data Engineer /Sr. DevOps Engineer /Linux Admin/SharePoint/Power BI Developer /EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Led a team of data engineers in designing, developing, and implementing data pipelines and solutions on AWS.
Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services such as SageMaker, Glue, Airflow, Lambda, EC2, VPC, and IAM using Jenkins, Ansible, Chef, Puppet, Splunk, Jira, confluence etc.
Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing using Python, Pydantic, Java, R, SQL, git etc.
Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
Managed and optimized Redshift clusters to ensure high performance and scalability.
Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), Single Sign-On and MFA via VPN.
Developed and maintained data architecture documentation using Confluence.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Presented data findings and insights to senior management. Mentored and trained junior data engineers.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.

- Employ AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data.
- Define a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
- Executed the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
- Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Completed End-to-End Azure Data Engineer project using Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, Dashboard tools of Power BI, Looker Studio or Tableau, including write Spark code in Azure Databricks.
Build a data process flow to load raw data onto Data Lake storage for Databricks to write spark code and transform the data and load the data back to transform data Lake storage for synapse analytics to run the SQL queries on top of the transform
Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling. Employed DAX Studio and DAX functional language for complex formula creation in Power BI.
Automated tasks and scripts using Python, PowerShell, Bash, Ansible, Puppet, Lambda, JavaScript and Typescript
Linux administration, focusing on file system management, network configurations, and system performance optimization.
Implemented and maintained Amazon Elastic Kubernetes Service (Amazon EKS) clusters for various microservices applications using Terraform and Helm
Design and develop data solutions using AWS services (EC2, S3, EMR, RedShift Spectrum), Python and Snowflake
Built, trained, deployed, and managed machine learning models using SageMaker.
Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Implement MLOps for the ML application lifecycle and ModelOps for model management once deployed in a production environment. Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Utilized SageMaker for building, training, deploying, and managing machine learning models.
Interpreted trends and patterns, prepared data for prescriptive and predictive modeling.

- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Government Education: Senior AWS Data Engineer /Senior DevOps Engineer/BI Developer /SharePoint /K2 Developer /EDW
Responsibilities:
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing using Python, Pydantic, Java, R, SQL, git etc.
Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Used Lambda to run code without having to provision or manage servers, especially for users who need to run short-lived, event-driven tasks, such as processing data streams or responding to API requests.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Monitored and troubleshoot data pipeline issues using Splunk and CloudWatch.
Employ AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
Define a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
Designed and implemented data quality checks to ensure data integrity.
Developed and maintained data warehouses using Hive and Impala. Automated data processing tasks using airflow.
Created and maintained data visualizations using Tableau.
Built, trained, deployed, and managed machine learning models using Sagemaker.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Set up & managed AWS services such as Sagemaker, Glue, Airflow, Lambda, EC2, VPC, and IAM.
Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
Configured and managed observability components using Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Designed and developed fully automated CI/CD pipelines using Azure DevOps, GitHub Actions, and Jenkins
Delivered robust, scalable, and future-proof technical solutions aligning with industry best practices.
Swiftly handled incident management and BAU operations, maintaining a stable data platform.
Drove the implementation of best practices, processes, and methodologies across the data platform.
Built and maintained data pipelines using a combination of Python, Spark, and AWS Glue for data transformation and loading. Implemented ETL processes for data transformation, utilizing Python, Scala, and SQL for data processing.
Played a pivotal role in building and maintaining data pipelines using technologies such as Apache Spark 3.3 and PySpark.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Developed intricate SQL queries for data extraction, transformation, and analysis.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Created interactive Power BI dashboards with DAX calculations and utilized Databricks SQL for high performance analytics.
Utilized Azure Open AI /ML Service and Azure Open AI Studio for Large Language Model (LLM) training.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: **AWS Data Engineer /DevOps Engineer /SharePoint /BI Developer /D365 CRM Consultant /DBA**
Responsibilities:

- Successfully established the Batch, Speed, and Service layers of the Lambda Architecture on AWS Cloud, employing Hadoop for Batch processing, Spark with in-memory access and RDD for Speed layer, and Spark Streaming for Service layer to enable real-time data views and queries on the Spark cluster datasets.

- Demonstrated expertise in AWS developer analytics tools, including Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3.
- Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including install and configure Hadoop on a cluster. Set up Hadoop Distributed File System (HDFS) for storage. Install Apache Spark on the same cluster or a different one. Configure Spark to use Hadoop YARN for resource management.
- Set up Apache Hive for data warehousing and SQL-like queries. Create external tables or load data into Hive tables.
- Develop and implement MapReduce jobs for specific data processing tasks. Integrate MapReduce jobs into the workflow for distributed processing.
- Experienced with bash / PythonUtilized Python and SQL for development, working on Snowflake features like Snow Spark, DAG, Snowpipe, external functions, and API integration.
- Designing and implementing data ingestion processes using Azure Synapse, Azure Data Factory, and Power BI.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: AWS/Azure Data Engineer /DevOps Engineer /BI Developer /SharePoint Consultant /DBA
Responsibilities:

- Employed AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Defined a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
- Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
- Implemented ETL/ELT frameworks, orchestration, and configuration as code, contributing to efficient data processing and integration. Developed and deployed data solutions using AWS services, Python and SQL
- Demonstrated exceptional incident management and business-as-usual (BAU) support, ensuring a stable data platform.
- Collected, cleaned, and analyzed data from various sources. Developed and maintained data models using SQL and Python. Created and maintained data visualizations using Tableau. Presented data findings and insights to senior management. Developed Python scripts for data processing, supporting efficient ETL operations.
- Experienced in RESTful APIs and integration with external systems.
- Proficient in cloud data analytics technologies such as Databricks and Azure ADF.
- Solid experience using MS Stack: Microsoft SQL Server, SSIS, SSRS, SSAS, Power BI, and SQL.
- Familiarity with MongoDB, Graph databases, programming languages (R, Python, Spark ETL SQL VStudio), CI/tools (Bamboo, or similar), GitHub, Jira, and Confluence.
- Experienced in data modeling in enterprise data warehousing/lake environments and use of tools like ERWIN.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP,
- ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /Data Analyst /BI Developer /Senior Systems Engineer /EDW /DBA
Responsibilities:

- Align Architecture, Solution Design with business requirements. Design, Build and Maintain Physical data model in data platform layers. Map business glossary, data definition and data sets to physical data model
- Build ETL that collects, manage, and convert raw data into consumable tables applying technical logic based on business rules. Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Performed advanced data modeling and design, supporting data-driven decision-making processes within the organization.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Senior Data Engineer /MS BI Developer /Senior Database Administrator /EDW**
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, Control-M
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior MLOps Engineer /Senior DevOps/AI/DL/ML Engineer /Python Developer/Senior Data Engineer /AI/ML/DL/BI/SharePoint/D365 CRM Consultant /EDW/DBA-Japan(7.5yrs) and Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior MLOps Engineer /Sr. DevOps/AI/DL/ML Engineer /Python Developer/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.

Collaborated closely with Data Scientists and Engineers on Greenfield projects from concept to production deployment.

- Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Pioneering AI workflow automation, working on various projects within the large Government healthcare organization that are used to streamline admin services for healthcare professionals by leveraging AI and other modern technologies.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Produced production-quality code deployed as Docker containers. Coded in a Linux environment using Git for source control. Implemented end-to-end solutions in a shared codebase.
- Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
- Define ML Pipelines using the chosen MLOps framework to create pipelines that orchestrate the ML workflow.
- Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.
- Developed and automated infrastructure deployment using Terraform and Ansible, increasing developer velocity by 30%.
- Managed critical infrastructure upgrades and maintenance tasks, ensuring high availability and operational resilience.
- Owned the development and maintenance of custom Terraform modules for internal and venture consumption, fostering code reusability and standardization. Liaised with vendors as required on support issues.
- Managed support requests and performed maintenance tasks such as software upgrades.
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
- Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.
- Monitored model performance, conducted A/B testing, & proactively identified and addressed model drift or potential issues.
- Transformed and trained data science products and ML systems using Python and Pydantic on AWS EMR.
- Leverage Amazon SageMaker Studio to provide a collaborative environment for data scientists and ML engineers.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**

Gov. Central Bank: Senior MLOps Engineer /Senior DevOps /AI/DL/ML Engineer /BI /Python Developer/EDW

Responsibilities:

Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Designed and operated software systems to convert petabytes of data into actionable insights within a cloud-based infrastructure environment.

Collaborated closely with Data Scientists and Engineers on greenfield projects from concept to production deployment.

Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines.

Collaborated with a passionate team of Data Scientists, Machine Learning Engineers, and Data Engineers.

Participated in the design and scoping of greenfield projects.

Worked on end-to-end technical solutions, from spike sessions to production deployment.

Committed to software best practices, including infrastructure as code, GitOps, and CI/CD.

Developed and implemented machine learning models and algorithms, showcasing deep Python knowledge and high-performance trading system expertise. Quickly adapt to new technologies, enhancing the company's tech stack.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.

Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.

Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.

Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for financial trading devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
- Based on bank business requirements, define most suitable ML Pipelines from Kubeflow, Mlflow, Dataflow MLOps , ZenML, Flyte, framework to create pipelines that orchestrate the ML workflow.
- Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.
- Developed and automated infrastructure deployment using Terraform and Ansible, increasing developer velocity by 30%.
- Managed critical infrastructure upgrades and maintenance tasks, ensuring high availability and operational resilience.
- Owned the development and maintenance of custom Terraform modules for internal and venture consumption, fostering code reusability and standardization. Liaised with vendors as required on support issues.
- Streamlined financial claim processing through the deployment of 20+ production-ready models powered by Random Forest and K-Means Clustering, achieving a 15% reduction in processing errors and a 10% decrease in administrative costs.
- Leveraged BERT-based NLP models to analyze daily financial trading notes and assess potential adherence to financial outcome regulations, leading to a 5% improvement in compliance and a reduction in administrative costs.
- Employed Deep Learning CNNs to automate error-prone manual processes in analyzing financial trading reports for specific accounts, resulting in a 20% increase in financial business processing efficiency.
- Designed and implemented containerized and Kubernetes-native supervised ML inference pipelines utilizing Amazon SageMaker Studio for seamless deployment on Nvidia DGX-1 systems. This ensured real-time performance and elastic scalability to handle varying workloads.
- Conducted regular A/B testing to compare model performance and identify opportunities for improvement. This proactive approach ensured model stability and adherence to regulatory compliance on the Harrison Platform.

Implemented infrastructure stacks using Terraform and Ansible; refined Terraform modules for internal and venture use.

Demonstrated expertise in Kubernetes, Linux administration, TCP/IP networking, CI/CD pipelines, AWS, and Infrastructure as Code tooling including Terraform and Ansible.

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Use Python and machine learning frameworks (PyTorch, scikit-learn) to build scalable AI /ML solutions.

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
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Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior MLOps Engineer /Sr. DevOps/AI/DL/ML Engineer /SPoint /K2 /Python Developer /DBA
Responsibilities:

AI/ML architect: Enhanced international student engagement and sales through models, deployed and monitored with MLOps/DevSecOps for efficient, secure platform. Participated in the design and scoping of greenfield projects.
Designed, built, and operated software systems to transform petabytes of data into actionable insights.
Collaborated with a passionate team of Data Scientists, Machine Learning Engineers, and Data Engineers.
Worked on end-to-end technical solutions, from spike sessions to production deployment.
Committed to software best practices, including infrastructure as code, GitOps, and CI/CD.
Collaborated with data scientists and software engineers to develop, refine, and deploy ML solutions while optimizing data pipelines and infrastructure.
Data-driven problem solver: Developed and implemented ML models for key business challenges, optimizing processes & driving project success.
Security and automation expert: Secured and managed Linux systems, implemented best practices and automation, proactively mitigating threats and resolving complex issues.Developed and maintained AI /ML models to enhance international student's engagement and sales.

- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Master of AI/ML deployment: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.
- Proficient in ML frameworks: Skilled in Python, R, Java, Scikit-learn, H2O, TensorFlow, PyTorch, Keras, MXNet, ImageAI, OpenCV for ML/DL tasks.
- Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Internal security champion: Enhanced internal security through improved ML libraries & frameworks using TensorFlow, PyTorch, Keras, Docker and Git.
- Cloud security expert: Managed & monitored cloud infra with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security & automation.
- NLP, CV and RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.
- Agile data scientist: Conducted research, created datasets, performed statistical analysis, built & trained data science products and ML systems.
- MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.
- Automated tasks and scripts using Python, PowerShell, Bash scripts, Ansible, Puppet, Lambda, JavaScript and Typescript
- Cloud migration & automation: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Senior MLOps Engineer /Senior DevOps/AI/DL/ML Engineer /SharePoint /D365 CRM Consultant
Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Manage support requests. Perform maintenance tasks such as software upgrades
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.

- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Transform and train Data science products and ML Systems.
- Research and implement appropriate ML algorithms and tools according to business requirements
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Conducted regular system training and retraining activities using Python, scikit-learn, TensorFlow for ML/DL.
- Experienced in creating user interfaces with Flask, Django, FastAPI, utilizing scikit-learn for model evaluation.
- Proficient in Python, SQL, and containerization for deploying data applications, with additional expertise in Python application development and TypeScript.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Conducted statistical modeling and analysis for workforce policy insights.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Platform Engineer-Azure / DevOps Engineer /ML Engineer/SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Worked on data representation methods and contributed to effective data engineering processes.
- Proficient in Linux platform development, optimizing application performance.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.

- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Sr. Platform Engineer-K8s+MLOps – Harrison AI Pty Ltd applied on 7/Jan/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Platform Engineer-Kubenetes+MLOps/Sr. DevOps/AI/DL/MLOps Engineer /Python Developer/Senior Data Engineer /AI/ML/DL/BI/SharePoint/D365 CRM Consultant /EDW/DBA-Japan(7.5yrs) and Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. Platform Engineer-K8s+MLOps/Sr. DevOps/AI/DL/ML /Python Developer/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Pioneering AI workflow automation, working on various projects within the large Government healthcare organization that are used to streamline admin services for healthcare professionals by leveraging AI and other modern technologies.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows
- Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
- Define ML Pipelines using the chosen MLOps framework to create pipelines that orchestrate the ML workflow.
- Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.
- Developed and automated infrastructure deployment using Terraform and Ansible, increasing developer velocity by 30%.
- Managed critical infrastructure upgrades and maintenance tasks, ensuring high availability and operational resilience.
- Owned the development and maintenance of custom Terraform modules for internal and venture consumption, fostering code reusability and standardization. Liaised with vendors as required on support issues.
- Authored comprehensive technical documentation to guide platform users and facilitate knowledge transfer.
- Managed support requests and performed maintenance tasks such as software upgrades.
- Provided assistance in developing and deploying new features and improvements to the cluster. This covered both services in the physical datacenter and within AWS. Wrote end user technical documentation.
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
- Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.
- Collaborated with data scientists and software engineers to develop, refine, and deploy ML solutions while optimizing data pipelines and infrastructure.
- Monitored model performance, conducted A/B testing, & proactively identified and addressed model drift or potential issues.
- Transformed and trained data science products and ML systems using Python and Pydantic on AWS EMR.

- Leverage Amazon SageMaker Studio to provide a collaborative environment for data scientists and ML engineers.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney

Gov. Central Bank: Sr. Platform Engineer-Kubenetes+MLOps/Sr. DevOps/AI/MLOps Engineer /BI /Python Developer/EDW

Responsibilities:

Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Developed and implemented machine learning models and algorithms, showcasing deep Python knowledge and high-performance trading system expertise. Quickly adapt to new technologies, enhancing the company's tech stack.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.

Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.

Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for financial trading devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows
- Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.
- Based on bank business requirements, define most suitable ML Pipelines from Kubeflow, Mlflow, Dataflow MLOps , ZenML, Flyte, framework to create pipelines that orchestrate the ML workflow.
- Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Developed and automated infrastructure deployment using Terraform and Ansible, increasing developer velocity by 30%.
- Managed critical infrastructure upgrades and maintenance tasks, ensuring high availability and operational resilience.
- Owned the development and maintenance of custom Terraform modules for internal and venture consumption, fostering code reusability and standardization. Liaised with vendors as required on support issues.
- Streamlined financial claim processing through the deployment of 20+ production-ready models powered by Random Forest and K-Means Clustering, achieving a 15% reduction in processing errors and a 10% decrease in administrative costs.
- Leveraged BERT-based NLP models to analyze daily financial trading notes and assess potential adherence to financial outcome regulations, leading to a 5% improvement in compliance and a reduction in administrative costs.
- Employed Deep Learning CNNs to automate error-prone manual processes in analyzing financial trading reports for specific accounts, resulting in a 20% increase in financial business processing efficiency.
- Designed and implemented containerized and Kubernetes-native supervised ML inference pipelines utilizing Amazon SageMaker Studio for seamless deployment on Nvidia DGX-1 systems. This ensured real-time performance and elastic scalability to handle varying workloads.
- Collaborated with data scientists and software engineers to develop, refine, & deploy high-performance ML solutions while optimizing data ingestion pipelines & infrastructure on the Harrison Platform for efficient resource utilization & cost reduction.
- Established a robust MLOps monitoring framework for deployed models, leveraging tools like Amazon CloudWatch and Prometheus to track key performance metrics, data drift, and potential issues.
- Conducted regular A/B testing to compare model performance and identify opportunities for improvement. This proactive approach ensured model stability and adherence to regulatory compliance on the Harrison Platform.

Collaborated with ML engineers and data scientists to integrate MLOps pipelines into the Kubernetes environment, achieving seamless model deployment and monitoring.

Oversaw support requests, maintenance tasks, and liaised with vendors for critical support issues.

Developed and deployed new features and enhancements to the cluster, both in physical datacenters and AWS.

Authored comprehensive end-user technical documentation for various system components.

Implemented infrastructure stacks using Terraform and Ansible; refined Terraform modules for internal and venture use.

Demonstrated expertise in Kubernetes, Linux administration, TCP/IP networking, CI/CD pipelines, AWS, and Infrastructure as Code tooling including Terraform and Ansible.

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Use Python and machine learning frameworks (PyTorch, scikit-learn) to build scalable AI /ML solutions.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. Platform Engineer-Kubenetes+MLOps/Sr. DevOps/AI/ML Engineer /SPoint /K2 /Python Developer /DBA
Responsibilities:

AI/ML architect: Enhanced international student engagement and sales through models, deployed and monitored with MLOps/DevSecOps for efficient, secure platform.
Data-driven problem solver: Developed and implemented ML models for key business challenges, optimizing processes & driving project success.
Security and automation expert: Secured and managed Linux systems, implemented best practices and automation, proactively mitigating threats and resolving complex issues.Developed and maintained AI /ML models to enhance international student's engagement and sales.

- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Master of AI/ML deployment: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.
- Proficient in ML frameworks: Skilled in Python, R, Java, Scikit-learn, H2O, TensorFlow, PyTorch, Keras, MXNet, ImageAI, OpenCV for ML/DL tasks.
- Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Internal security champion: Enhanced internal security through improved ML libraries & frameworks using TensorFlow, PyTorch, Keras, Docker and Git.
- Cloud security expert: Managed & monitored cloud infra with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security & automation.
- NLP, CV & RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.
- Agile data scientist: Conducted research, created datasets, performed statistical analysis, built & trained data science products & ML systems.
- MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.
- Automated tasks and scripts using Python, PowerShell, Bash scripts, Ansible, Puppet, Lambda, JavaScript and Typescript
- Cloud migration & automation: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Senior Platform Engineer – AWS /Senior DevOps /AI/ML Engineer /SharePoint /D365 CRM Consultant
Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices.
- Manage support requests. Perform maintenance tasks such as software upgrades
- Liaise with vendors as required on support issues
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Transform and train Data science products and ML Systems.
- Research and implement appropriate ML algorithms and tools according to business requirements

- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Played a key role in the integration of AI technologies into business processes.
- Conducted regular system training and retraining activities.
- Contributed to the innovation of efficient AI tools, explored design libraries for new concepts, and developed an ML sentiment analysis prototype using Python, scikit-learn, and Azure OpenAI ML.
- Experienced in creating user interfaces with Flask, Django, FastAPI, utilizing scikit-learn for model evaluation.
- Proficient in Python, SQL, and containerization for deploying data applications, with additional expertise in Python application development and TypeScript.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Conducted statistical modeling and analysis for workforce policy insights.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Platform Engineer-Azure / DevOps Engineer /ML Engineer/SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Worked on data representation methods and contributed to effective data engineering processes.
- Designed and developed AI solutions driving automation and innovation.
- Proficient in Linux platform development, optimizing application performance.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform

- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior ML Operation Engineer /Senior DevOps/AI/DL/ML Engineer /Python Developer/Senior Data Engineer /AI/ML/DL/BI/SharePoint/D365 CRM Consultant /EDW/DBA -Japan(7.5yrs) and Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. ML Operation Engineer /Sr. DevOps/AI/DL/ML Engineer /Python Developer/SP/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- **Infrastructure Design and Deployment**: Design, implement, and maintain scalable and reliable infrastructure for machine learning workflows using AWS, Terraform, and Kubernetes.
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Collaborate with data scientists and software engineers to ensure seamless integration of machine learning models into production environments.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- **Model Deployment and Monitoring**: Deploy machine learning models either using fully managed AWS SageMaker, or high-performance and scalable TensorFlow Serving or Open-source Kubeflow run on Kubenetes on any major clouds.
- Implement and maintain monitoring solutions to track model performance, data drift, and system health.
- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- **Automation and CI/CD**: Develop and maintain automated deployment pipelines for machine learning models and related infrastructure.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Implement continuous integration and continuous deployment (CI/CD) best practices to streamline the development and deployment process.
- Championed software best practices, including infrastructure as code, GitOps, and CI/CD pipelines
- Pioneering AI workflow automation, working on various projects within the large Government healthcare organization that are used to streamline admin services for healthcare professionals by leveraging AI and other modern technologies.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Nvidia DGX A100s system
- Conduct performance analysis and implement improvements to enhance the overall efficiency of machine learning systems.
- **Security and Compliance**: Implement security best practices to safeguard machine learning infrastructure and data.
- Ensure compliance with relevant data protection and regulatory requirements.

- Produced production-quality code deployed as Docker containers. Coded in a Linux environment using Git for source control. Implemented end-to-end solutions in a shared codebase.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Sr. Machine Learning Operation Engineer /Sr. DevOps /AI/DL/ML Engineer /BI /Python Developer/EDW
Responsibilities:

Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Security-focused AI architect: Designed, developed and deployed secure and efficient AI/ML applications, deep learning models and systems.

Based on bank business requirements, define most suitable ML Pipelines from Kubeflow, Mlflow, Dataflow MLOps , ZenML, Flyte, framework to create pipelines that orchestrate the ML workflow.

Specify the sequence of steps, including data preparation, model training, evaluation, deployment, and monitoring.

Master of AI/ML deployment: Built and deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure/GCP), secured pipelines with MLOps/DevSecOps.

Designed and implemented automated inference pipelines using Amazon SageMaker Studio to ensure efficient model scalability and real-time performance.

- Designed, implemented, and optimized a high-performance, scalable physical ML Kubernetes cluster supporting machine learning workloads on Harrison Platform Nvidia DGX-1 systems.

Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.

- Leverage Amazon SageMaker Studio to provide a collaborative environment for data scientists and ML engineers.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).

Internal security champion: Enhanced internal security through improved ML libraries and frameworks using TensorFlow, PyTorch, Keras, Docker and Git.

Cloud security expert: Managed and monitored cloud infrastructure with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security and automation.

Leverage Bank-specific LLMs and tailored ML/DL algorithms (forecasting, trading, risk, compliance) for bespoke financial services analysis and execution, adhering to central bank regulations and proprietary needs

NLP, CV & RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.

Using the Harrison Platform to develop AI-powered financial trading devices which use ML algorithms to analyze data and make financial decisions to deliver better financial results for RBA.

Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.

Designed and implemented containerized and Kubernetes-native supervised ML inference pipelines utilizing Amazon SageMaker Studio for seamless deployment on Nvidia DGX-1 systems. This ensured real-time performance and elastic scalability to handle varying workloads.

Developed NLP models based on BERT to analyze patient notes and predict medication adherence, leading to a 5% improvement in patient compliance and a reduction in hospital readmission rates.

Monitored model performance, conducted A/B testing, & proactively identified and addressed model drift or potential issues.

Leveraged BERT-based NLP models to analyze daily financial trading notes and assess potential adherence to financial outcome regulations, leading to a 5% improvement in compliance and a reduction in administrative costs.

Employed Deep Learning CNNs to automate error-prone manual processes in analyzing financial trading reports for specific accounts, resulting in a 20% increase in financial business processing efficiency.

Transformed and trained data science products and ML systems using Python and Pydantic on AWS EMR.

Agile data scientist: Conducted research, created datasets, performed statistical analysis, built & trained data science products & ML systems.

Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.

Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.

Containerize ML Components by packaging the model training code, preprocessing scripts, and prediction logic as Docker containers. Use a Dockerfile to define the container environment and dependencies.

MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.

Cloud migration & automation: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.

Migrated on-premises workloads to AWS Cloud using various tools, including AWS Server Migration Service (SMS) for migrate VMware workloads to AWS; AWS Database Migration Service (DMS) for migrating databases to AWS and AWS Database Gateway for connecting on-premises applications to cloud databases securely. Delivered CI/CD/DevOps in AWS

Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Senior ML Operation Engineer /Sr. DevOps/AI/DL/ML Engineer /SPoint /K2 /Python Developer /DBA

Responsibilities:

AI/ML architect: Enhanced international student engagement and sales through models, deployed and monitored with MLOps/DevSecOps for efficient, secure platform. Participated in the design and scoping of greenfield projects.

Designed, built, and operated software systems to transform petabytes of data into actionable insights.

Collaborated with a passionate team of Data Scientists, Machine Learning Engineers, and Data Engineers.

Worked on end-to-end technical solutions, from spike sessions to production deployment.

Committed to software best practices, including infrastructure as code, GitOps, and CI/CD.

Collaborated with data scientists and software engineers to develop, refine, and deploy ML solutions while optimizing data pipelines and infrastructure.

Data-driven problem solver: Developed and implemented ML models for key business challenges, optimizing processes & driving project success.

Security and automation expert: Secured and managed Linux systems, implemented best practices and automation, proactively mitigating threats and resolving complex issues.Developed and maintained AI /ML models to enhance international student's engagement and sales.

- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Master of AI/ML deployment: Built & deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure), secured pipelines with MLOps/DevSecOps.
- Proficient in ML frameworks: Skilled in Python, R, Java, Scikit-learn, H2O, TensorFlow, PyTorch, Keras, MXNet, ImageAI, OpenCV for ML/DL tasks.
- Big data wrangling: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Internal security champion: Enhanced internal security through improved ML libraries & frameworks using TensorFlow, PyTorch, Keras, Docker and Git.
- Cloud security expert: Managed & monitored cloud infra with AWS SegeMaker, Config, CloudTrail, Shield, WAF, ensuring high security & automation.
- NLP, CV and RL: Applied NLP (text classification, named entity recognition), computer vision (image classification), and reinforcement learning to security problems.
- Agile data scientist: Conducted research, created datasets, performed statistical analysis, built & trained data science products and ML systems.
- MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.
- Automated tasks and scripts using Python, PowerShell, Bash scripts, Ansible, Puppet, Lambda, JavaScript and Typescript
- Cloud migration & automation: Migrated on-premise infrastructure to AWS, scripted automation routines, liaised with vendors, minimized operational disruption.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

Employed Power BI, Tableau, or QlikView for real-time financial analysis.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Health Sector: Machine Learning Operation Engineer /Senior DevOps/AI/DL/ML Engineer /SharePoint /D365 CRM Consultant

Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Manage support requests. Perform maintenance tasks such as software upgrades
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.

- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Transform and train Data science products and ML Systems.
- Research and implement appropriate ML algorithms and tools according to business requirements
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Conducted regular system training and retraining activities using Python, scikit-learn, TensorFlow for ML/DL.
- Experienced in creating user interfaces with Flask, Django, FastAPI, utilizing scikit-learn for model evaluation.
- Proficient in Python, SQL, and containerization for deploying data applications, with additional expertise in Python application development and TypeScript.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Conducted statistical modeling and analysis for workforce policy insights.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Platform Engineer-Azure / DevOps Engineer /ML Engineer/SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Worked on data representation methods and contributed to effective data engineering processes.
- Proficient in Linux platform development, optimizing application performance.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.

- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Engineer – AWS –Azure /Senior DevOps Engineer /Senior Data Engineer /AI /ML /BI Developer /EDW /Clouds/Unix/Linux Administrator /SharePoint/D365 CRM Consultant - Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cloud Engineer-AWS /Sr. DevOps Engineer/Linux Administrator /AI/ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Gps, IAM roles, etc. in Security/main.tf
- Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Migrated on-premises workloads to AWS Cloud using various tools including AWS Server Migration Service (SMS) for mirate VMware workloads to AWS; AWS Database Migration Service (DMS) for migrating databases to AWS and AWS Database Gateway for connecting on-premises applications to cloud databases securely. Delivered CI/CD/DevOps in AWS Cloud.
- Integrated monitoring solutions, including CloudWatch for real-time insights, CloudTrail for event management, and Dynatrace for comprehensive performance analysis in AWS environments.
- Utilized different APIs (FastAPI, RestAPI, etc.) to automate manual tasks for infrastructure provisioning and setup.
- Automating build, test and deployment using CI/CD pipeline with GitLab and DevOps workflows on AWS & IT infrastructure.
- Adopted DevOps best practices and AWS cloud infrastructure, IoC, modern cloud-native, and container platforms.
- Senior Linux administrator and networking engineer working in IT infrastructure supporting large corporate platforms.
- Leveraged AWS Compute, Storage, RDS Database, and Networking services collectively to build a robust and scalable cloud infrastructure.
- Utilized Terraform for provisioning, Ansible for configuration management, Packer for image creation, and Azure DevOps pipelines for seamless Infrastructure as Code deployment and management.
- Leveraging AWS Security Hub for centralized security insights, efficiently remediated security alerts using automated processes orchestrated through AWS Lambda.
- Implemented effective resource management practices, including tagging, and ensure system patching integrity using AWS Systems Manager (SSM) while streamlining data delivery through Kinesis Firehose for comprehensive data analysis.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services like EC2, S3, IAM, EKS, ELB, Cloud Formation, RDS, SageMaker, Glue, Lambda, VPC, VPS etc. using Jenkins, Ansible, Chef & Puppet.
- Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github actions
- Leveraged Arturo for setting up EKS, provisioning databases, and implementing platform monitoring dashboards and alerts.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Implemented data transformation using Step Functions and Spark Notebooks, writing Spark SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Leveraged ChatGPT Enterprise to develop AI solutions automating processes and fostering innovation in data security for an aged care business, employing well-trained LLMs (LlaMa, Mistrial, DeepSeek, etc.) on extensive enterprise datasets.
- Integrated LangChain with Azure OpenAI to create Private ChatGPTs for efficient data analysis by BAs and Data Analysts.
- Utilized R and Python in Jupyter Notebooks for statistical analysis and fine-tuning based on test results.

- Executed system training and retraining using Scikit-learn, CNN, RNN, or AutoML toolkits (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop for large-scale data processing in the machine learning pipelines.
- Built, trained, deployed, and managed ML models using Sagemaker, employed Spark MLlib for distributed machine learning on large-scale data sets, and used Git for code management.
- Implemented MLOps and ModelOps for the entire lifecycle and management of machine learning applications and models, including development, deployment, and maintenance in a production environment.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Cloud Engineer-AWS/Sr. DevOps Engineer/Linux Administrator/ML/SP/Power BI Developer /EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
Design and develop data solutions using AWS services (EC2, S3, EMR, RedShift Spectrum), Python and Snowflake
Leveraged Arturo for setting up EKS, provisioning databases, and implementing platform monitoring dashboards and alerts.
Managed and supported the infrastructure and platform including capacity management, performance monitoring, troubleshooting, & resolution of technical issues. Used various APIs to automate manual tasks for infra provisioning & setup.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
Real-time streaming data processing using Amazon Kinesis Analytics to write SQL queries to process and analyze data from Amazon Kinesis Streams or other streaming sources. Configure jobs to ingest and process streaming data.
Automating build, test and deployment using CI/CD pipeline and DevOps on AWS and IT infrastructure.
Adopted DevOps best practices and AWS cloud infrastructure, IoC, modern cloud-native, and container platforms.
Senior Linux administrator and networking engineer working in IT infrastructure supporting large corporate platforms.
Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services such as SageMaker, Glue, Airflow, Lambda, EC2, VPC, and IAM using Jenkins, Ansible, Chef, Puppet, Splunk, Jira, confluence etc.
Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github Actions
Developed and maintained CI/CD pipelines using Jenkins, GitLab, DevOps workflow and AWS CodePipeline.
Expertise in data integration, microservices, & backend platforms, coupled with the use of diverse APIs for automating manual tasks in infrastructure provisioning & setup. Manag & optimize Redshift clusters for high performance & scalability
Federated Amazon Redshift access with Azure AD, SSO and MFA in Azure Security Centre.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.

- Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling. Employed DAX Studio and DAX functional language for complex formula creation in Power BI.
Automated tasks and scripts using Python, PowerShell, Bash, Ansible, Puppet, Lambda, JavaScript and Typescript
Implemented and maintained Amazon Elastic Kubernetes Service (Amazon EKS) clusters for various microservices applications using Terraform and Helm
Built, trained, deployed, and managed machine learning models using SageMaker.
Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.
Implement MLOps for the ML application lifecycle and ModelOps for model management once deployed in a production environment. Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Utilized SageMaker for building, training, deploying, and managing machine learning models.
Interpreted trends and patterns, prepared data for prescriptive and predictive modeling.

- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Government Education: Senior Cloud Engineer-AWS /Sr. DevOps Engineer /ML/BI Developer/SPoint /K2 Developer /EDW
Responsibilities:

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing using Python, Pydantic, Java, R, SQL, git etc.
Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
Leveraged Arturo for setting up EKS, provisioning databases, and implementing platform monitoring dashboards and alerts.
Managed and supported the infrastructure and platform including capacity management, performance monitoring, troubleshooting, and resolution of technical issues.
Leveraged various APIs to automate manual tasks for infra provisioning and setup.
Play a key role in expanding capabilities for a critical transformation program that will build a new cloud-based HR platform on AWS from the ground up using the latest cloud technologies focusing on a global digital user's experience.
Automating build, test and deployment using CI/CD pipeline and DevOps on AWS and IT infrastructure.
Adopted DevOps best practices and AWS cloud infrastructure, IoC, modern cloud-native, and container platforms.
Senior Linux administrator and networking engineer working in IT infrastructure supporting large corporate platforms.
As an AWS Cloud Engineering SME with strong DevOps expertise, skilled in utilizing Node.js and Python, proficient in designing and optimizing cloud-based solutions for seamless and efficient operations.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Used Lambda to run code without having to provision or manage servers, especially for users who need to run short-lived, event-driven tasks, such as processing data streams or responding to API requests.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Monitored and troubleshoot data pipeline issues using Splunk and CloudWatch.
Employ AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
Developed and maintained data warehouses using Hive and Impala. Automated data processing tasks using airflow.
Created and maintained data visualizations using Tableau.
Built, trained, deployed, and managed machine learning models using Sagemaker.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Set up & managed AWS services such as Sagemaker, Glue, Airflow, Lambda, EC2, VPC, and IAM.
Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
Configured and managed observability components using Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Designed and developed fully automated CI/CD pipelines using GitLab, DevOps workflow, Jenkins, Ansible, Puppet etc.
Built and maintained data pipelines using a combination of Python, Spark, and AWS Glue for data transformation and loading. Implemented ETL processes for data transformation, utilizing Python, Scala, and SQL for data processing.
Played a pivotal role in building and maintaining data pipelines using technologies such as Apache Spark 3.3 and PySpark.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: Cloud Engineer-AWS/DevOps Engineer/Data Engineer/SharePoint/BI Developer /D365 CRM Consultant /DBA
Responsibilities:

- Successfully established the Batch, Speed, and Service layers of the Lambda Architecture on AWS Cloud, employing Hadoop for Batch processing, Spark with in-memory access and RDD for Speed layer, and Spark Streaming for Service layer to enable real-time data views and queries on the Spark cluster datasets.
- Demonstrated expertise in AWS developer analytics tools, including Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3.
- Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including install and configure Hadoop on a cluster. Set up Hadoop Distributed File System (HDFS) for storage. Install Apache Spark on the same cluster or a different one. Configure Spark to use Hadoop YARN for resource management.
- Set up Apache Hive for data warehousing and SQL-like queries. Create external tables or load data into Hive tables.
- Develop and implement MapReduce jobs for specific data processing tasks. Integrate MapReduce jobs into the workflow for distributed processing.
- Experienced with bash / PythonUtilized Python and SQL for development, working on Snowflake features like Snow Spark, DAG, Snowpipe, external functions, and API integration.
- Designing and implementing data ingestion processes using Azure Synapse, Azure Data Factory, and Power BI.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Cloud Engineer-AWS /DevOps Engineer /Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- Employed AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Defined a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
- Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
- Implemented ETL/ELT frameworks, orchestration, and configuration as code, contributing to efficient data processing and integration. Developed and deployed data solutions using AWS services, Python and SQL
- Demonstrated exceptional incident management and business-as-usual (BAU) support, ensuring a stable data platform.
- Collected, cleaned, and analyzed data from various sources. Developed and maintained data models using SQL and Python. Created and maintained data visualizations using Tableau. Presented data findings and insights to senior management. Developed Python scripts for data processing, supporting efficient ETL operations.
- Experienced in RESTful APIs and integration with external systems.
- Proficient in cloud data analytics technologies such as Databricks and Azure ADF.
- Solid experience using MS Stack: Microsoft SQL Server, SSIS, SSRS, SSAS, Power BI, and SQL.
- Familiarity with MongoDB, Graph databases, programming languages (R, Python, Spark ETL SQL VStudio), CI/tools (Bamboo, or similar), GitHub Actions, Jira, and Confluence.
- Experienced in data modeling in enterprise data warehousing/lake environments and use of tools like ERWIN.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP,
- ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: DevOps Engineer /Data Engineer/Data Analyst /BI Developer /Senior Systems Engineer /EDW
Responsibilities:

- Align Architecture, Solution Design with business requirements. Design, Build and Maintain Physical data model in data platform layers. Map business glossary, data definition and data sets to physical data model
- Build ETL that collects, manage, and convert raw data into consumable tables applying technical logic based on business rules. Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Performed advanced data modeling and design, supporting data-driven decision-making processes within the organization.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x

- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Global Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /EDW
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, Control-M
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.

Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer /Senior Data Engineer /Python Developer /Clouds/Linux Administrator /AI //ML /BI /SharePoint /D365 CRM Consultant /EDW/-Japan (7.5yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Sr. Azure DevOps Engineer/Sr. Data Engineer/Python Developer/AI/ML/SharePoint/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
- Designed and set up end-to-end enterprise CI/CD pipelines using Jenkins or GitHub Actions for optimizing deployment processes in diverse enterprise applications.
- Utilised Azure ADF (Azure Data Factory) and Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, and deployment processes within the Azure ecosystem.
- Defined necessary build steps, like code compilation, running tests, and generating artifacts.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
- Implemented and maintained AKS clusters, guaranteeing high availability and reliability of the production environment.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365 CRM/Power BI/Power Automate etc.)
- Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
- Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, etc.
- Automated systems administration using Python/Jenkins /Lambda/Javascript/ PowerShell, Bash, Ansible and Puppet
- Communicated effectively with business stakeholders and project managers to understand requirements and deliver solutions. Worked in a hybrid cloud environment with zero ClickOps.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Monitor and troubleshoot Azure services using Splunk, Data Dog and Azure Network Monitor
- Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v6.0/7.0
- Monitoring and troubleshooting AWS, Azure infrastructure and application issues, ensuring high availability and optimal performance, including configure load balance, troubleshoot FortiGate WAF, Firewall (Physical, VLAN & virtual interfaces)
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Interpreted trends and patterns; prepared data for prescriptive and predictive modeling.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

- Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevSecOps and Configuration-As-Code)
- Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
- Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
- Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Sr. AWS/Azure DevOps Engineer/Python Developer/Liunx Administrator/ML/SharePoint Consultant/EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Collaborated with cross-functional teams to design, implement, and optimize CI/CD pipelines using Jenkins and Bamboo,
Automated infrastructure provisioning & configuration using Python, Jenkins, Ansible, Terraform, Puppet/PowerShell etc.
Championed the adoption of best practice DevOps methodologies, enhancing continuous integration, deployment, and software delivery. Worked on multi-zone load balancing, redundancy, and system software updates.
Resolved complex issues, conducted post-release reviews, &contributed to centralized CI/CD suite using Atlassian products.
Led end-to-end ownership of secure CI/CD processes for various applications, including design, build, testing, and support.
Automated complex builds and deployments across AWS and Azure clouds, resulting in improved scalability and resource utilization. Managed infrastructure as code using Terraform to enabling seamless provisioning of resources
Collaborated with development teams to ensure continuous compliance with security policies and build secure applications.
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Conducted regular code reviews, identifying areas for improvement in the deployment process.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service), IaC tools: Terraform, CloudFormation, CDK (Cloud Developer Kit), etc., to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Configured and managed Azure resources such as virtual machines, storage accounts, app services, load balancers, WAF and firewalls. Utilized Gitlab, GitHub and Bitbucket for source code management and version control
Automated the deployment and configuration of Windows servers and Linux servers using Ansible, Puppet and Chef
Collaborated with developers and testers to implement continuous integration and continuous testing using Azure DevOps and Selenium; Used Nagios, ELK and Prometheus for logging and monitoring
Provided technical support for Windows and Linux servers, ensuring optimal functionality and performance.
High proficiency in Active Directory, Azure AD, Managed the installation, configuration, and testing of new and existing hardware, software, and WAN/LAN technologies using (IaC) tools: Azure Resource Manager(ARM) templates, Teraform or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code
High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate WAF, Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
Deploy vMX instance in Azure; Provision MX67W device in Meraki dashboard; Create a site-to-site VPN and configure it; Configure Azure VPN gateway in Azure portal; Establish VPN connection under “Security & SD-WAN” >”Site-to-site VPN”
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Developed and maintained PowerShell scripts and automation tools for efficient Azure infrastructure management.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau

Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Sr. Azure DevOps Engineer/Sr. Data Engineer/Python Developer /ML/SharePoint /K2 Developer /DBA
Responsibilities:

Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
Implemented DevSecOps security best practices and detection methods, ensuring a secure and compliant environment.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
Designed and set up end-to-end enterprise CI/CD pipelines using Jenkins or GitHub Actions for optimizing deployment processes in diverse enterprise applications.
Utilised Azure ADF (Azure Data Factory) and Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, and deployment processes within the Azure ecosystem.
Defined necessary build steps, like code compilation, running tests, and generating artifacts.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
Implemented and maintained AKS clusters, guaranteeing high availability and reliability of the production environment.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365 CRM/Power BI/Power Automate etc.)
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, etc.
Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus
Configured and optimized Azure storage accounts, SQL databases and Cosmos DB
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Provided 3rd level desktop, systems, and network support, resolving complex technical issues to minimize downtime.
Conducted server and security audits, ensuring compliance with security policies and implementing necessary remediation measures. Used JIRA, Bugzilla and Remedy for bug tracking tools
Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Developed PowerShell scripts and automation tools for managing and maintaining Azure infrastructure.

- Maintained and optimized Active Directory, Azure, MS Defender Suite, and M365 Environments for efficient user management and data security.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
Using Azure Open AI Service and Azure Open AI Studio to train Large Language Model with scikit-learn and TensorFlow.
Played a key role in delivering Cloud Native software solutions, including machine learning and artificial intelligence applications. Ensured the maturity of engineering practices across the business.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: Azure DevOps Engineer/Data Engineer/Python Developer /Linux Administrator /SharePoint Consultant
Responsibilities:

- Architected and implemented secure landing zones for cloud workloads, focusing on Windows-based systems and utilizing Azure Blueprints.

- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Automated systems administration using Python/Jenkins /Lambda/Javascript/ PowerShell, Bash, Ansible and Puppet
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Migrated on-premise applications to the cloud using AWS Migration Hub and Azure Migrate.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- Implemented and managed networking solutions, including Cisco, Meraki SD-WAN, FortiGate Firewalls and WAF to enhance network performance and protect against cyber threats.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Automating systems administration by various automation scripting like Python, PowerShell, Bash, Puppet, etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Azure DevOps Engineer/Azure/M365 Administrator (AAD/Inutune /SCCM) /SharePoint Consultant
Responsibilities:

- Led the setup of Azure-focused cloud landing zones, following best practices.
- Collaborated with Data Engineers to integrate DevOps practices into the newly formed team.
- Ensured automation, scalability, and reliability in the Cloud and DevOps ecosystem.
- Built and maintained cloud infrastructure using IaC in Terraform. Designed and set up enterprise CI/CD pipelines
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and IIS on Windows servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Senior Windows /Unix/Linux Systems Administrator /DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.

- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Assisted in the installation and configuration of hardware, software, and networking equipment.
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Supported the installation, configuration, and testing of hardware and software solutions.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Assisted in managing IT infrastructure and ensuring optimal performance.
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer /Senior Data Engineer /Python Developer /Clouds/Linux Administrator /AI //ML /BI /SharePoint /D365 CRM Consultant /EDW -Japan (7.5yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Sr. Data Engineer/Python Developer/AI/ML/SharePoint/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
- Employed Docker containers orchestrated by Kubernetes, utilizing Helm charts for optimized resource utilization and deployment processes in the Azure cloud environment.
- Integrated Azure ADF and Databricks as vital components to establish end-to-end CI/CD pipelines in the Azure environment, actively contributing to orchestration, automation, and deployment processes.
- Defined essential build steps, including code compilation, test execution, and artifact generation, while performing data transformation and integration tasks using Python in Azure DevOps to harness the capabilities of Azure.
- Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
- Implemented and managed AKS clusters to ensure the high availability and reliability of the production environment.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Monitor and troubleshoot Azure services using Splunk, Data Dog and Azure Network Monitor
- Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
- Leveraged Docker for containerization, Kubernetes for orchestration, and Helm for streamlined deployment of microservices on AWS, ensuring scalable and efficient cloud-native applications.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Automated infrastructure provisioning & configuration using Python, Jenkins, Ansible, Terraform, Puppet, GitLab, etc.
- Championed the adoption of best practice DevOps methodologies, enhancing continuous integration, deployment, and software delivery. Worked on multi-zone load balancing, redundancy, and system software updates.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Leveraged ChatGPT Enterprise to develop AI solutions automating processes and fostering innovation in data security for an aged care business, employing well-trained LLMs (LlaMa, Mistrial, DeepSeek, etc.) on extensive enterprise datasets.
- Integrated LangChain with Azure OpenAI to create Private ChatGPTs for efficient data analysis by BAs and Data Analysts.
- Utilized R and Python in Jupyter Notebooks for statistical analysis and fine-tuning based on test results.

- Executed system training and retraining using Scikit-learn, CNN, RNN, or AutoML toolkits (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop for large-scale data processing in the machine learning pipelines.
- Built, trained, deployed, and managed ML models using Sagemaker, employed Spark MLlib for distributed machine learning on large-scale data sets, and used Git for code management.
- Implemented MLOps and ModelOps for the entire lifecycle and management of machine learning applications and models, including development, deployment, and maintenance in a production environment.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer/Python Developer/Clouds/Liunx Admin. /ML /BI /SharePoint Consultant/EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Collaborated with cross-functional teams to design, implement, and optimize CI/CD pipelines using Jenkins and Bamboo,
Automated infrastructure provisioning & configuration using Python, Jenkins, Ansible, Terraform, Puppet/PowerShell etc.
Championed the adoption of best practice DevOps methodologies, enhancing continuous integration, deployment, and software delivery. Worked on multi-zone load balancing, redundancy, and system software updates.
Resolved complex issues, conducted post-release reviews, &contributed to centralized CI/CD suite using Atlassian products.
Automated complex builds and deployments across AWS and Azure clouds, resulting in improved scalability and resource utilization. Managed infrastructure as code using Terraform to enabling seamless provisioning of resources
Collaborated with development teams to ensure continuous compliance with security policies and build secure applications.
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Conducted regular code reviews, identifying areas for improvement in the deployment process.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service), IaC tools: Terraform, CloudFormation, CDK (Cloud Developer Kit), etc., to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Configured and managed Azure resources such as virtual machines, storage accounts, app services, load balancers, WAF and firewalls. Utilized Gitlab, GitHub and Bitbucket for source code management and version control
Automated the deployment and configuration of Windows servers and Linux servers using Ansible, Puppet and Chef
Collaborated with developers and testers to implement continuous integration and continuous testing using Azure DevOps and Selenium; Used Nagios, ELK and Prometheus for logging and monitoring
Provided technical support for Windows and Linux servers, ensuring optimal functionality and performance.
High proficiency in Active Directory, Azure AD, Managed the installation, configuration, and testing of new and existing hardware, software, and WAN/LAN technologies using (IaC) tools: Azure Resource Manager(ARM) templates, Teraform or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code
High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate WAF, Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
Deploy vMX instance in Azure; Provision MX67W device in Meraki dashboard; Create a site-to-site VPN and configure it; Configure Azure VPN gateway in Azure portal; Establish VPN connection under “Security & SD-WAN” >”Site-to-site VPN”
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Developed and maintained PowerShell scripts and automation tools for efficient Azure infrastructure management.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer/Senior Data Engineer/Python Developer /ML/SharePoint /K2 Developer /DBA
Responsibilities:

Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
Implemented DevSecOps security best practices and detection methods, ensuring a secure and compliant environment.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
Designed and set up end-to-end enterprise CI/CD pipelines using Jenkins or GitHub Actions for optimizing deployment processes in diverse enterprise applications.
Utilised Azure ADF (Azure Data Factory) and Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, and deployment processes within the Azure ecosystem.
Defined necessary build steps, like code compilation, running tests, and generating artifacts.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
Implemented and maintained AKS clusters, guaranteeing high availability and reliability of the production environment.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365 CRM/Power BI/Power Automate etc.)
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, etc.
Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus
Configured and optimized Azure storage accounts, SQL databases and Cosmos DB
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Provided 3rd level desktop, systems, and network support, resolving complex technical issues to minimize downtime.
Conducted server and security audits, ensuring compliance with security policies and implementing necessary remediation measures. Used JIRA, Bugzilla and Remedy for bug tracking tools
Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Developed PowerShell scripts and automation tools for managing and maintaining Azure infrastructure.

- Maintained and optimized Active Directory, Azure, MS Defender Suite, and M365 Environments for efficient user management and data security.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
Using Azure Open AI Service and Azure Open AI Studio to train Large Language Model with scikit-learn and TensorFlow.
Played a key role in delivering Cloud Native software solutions, including machine learning and artificial intelligence applications. Ensured the maturity of engineering practices across the business.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: DevOps Engineer/Data Engineer/Python Developer /Linux Administrator/SharePoint Consultant DBA
Responsibilities:

- Architected and implemented secure landing zones for cloud workloads, focusing on Windows-based systems and utilizing Azure Blueprints.

- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Automated systems administration using Python/Jenkins /Lambda/Javascript/ PowerShell, Bash, Ansible and Puppet
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Migrated on-premise applications to the cloud using AWS Migration Hub and Azure Migrate.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- Implemented and managed networking solutions, including Cisco, Meraki SD-WAN, FortiGate Firewalls and WAF to enhance network performance and protect against cyber threats.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Automating systems administration by various automation scripting like Python, PowerShell, Bash, Puppet, etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Azure/M365 Administrator (AAD/Inutune /SCCM) /SharePoint Consultant
Responsibilities:

- Led the setup of Azure-focused cloud landing zones, following best practices.
- Collaborated with Data Engineers to integrate DevOps practices into the newly formed team.
- Ensured automation, scalability, and reliability in the Cloud and DevOps ecosystem.
- Built and maintained cloud infrastructure using IaC in Terraform. Designed and set up enterprise CI/CD pipelines
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and IIS on Windows servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Senior Windows /Unix/Linux Systems Administrator /DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.

- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Assisted in the installation and configuration of hardware, software, and networking equipment.
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Supported the installation, configuration, and testing of hardware and software solutions.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Assisted in managing IT infrastructure and ensuring optimal performance.
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer /Senior Data Engineer/Senior Data Analyst /ML/BI Developer /Clouds/Unix/Linux Administrator /SharePoint/D365 CRM Consultant /EDW -Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AWS DevOps Engineer/Sr. Data Engineer/Linux Administrator/BI/ML /SP/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
- Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services such as SageMaker, Glue, Airflow, Lambda, EC2, VPC, and IAM using Jenkins, Ansible, Chef, Puppet, Splunk, JIRA, confluence etc.
- Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github actions
- Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
- Managed and optimized Redshift clusters to ensure high performance and scalability.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), Single Sign-On and MFA via VPN.
- Utilized Glue, a serverless ETL service, to collect, transforms, and load data into data warehouses and data lakes.
- Automated a wide range of data processing tasks, including ETL, machine learning, and data quality checks using Airflow.
- Used Lambda to run code without having to provision or manage servers, especially for users who need to run short-lived, event-driven tasks, such as processing data streams or responding to API requests.
- Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
- Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Monitored and troubleshoot data pipeline issues using Splunk and CloudWatch.
- Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
- Migrated on-premises workloads to AWS Cloud using various tools including AWS Server Migration Service (SMS) for mirate VMware workloads to AWS; AWS Database Migration Service (DMS) for migrating databases to AWS and AWS Database Gateway for connecting on-premises applications to cloud databases securely. Delivered CI/CD/DevOps in AWS Cloud.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Implemented data transformation using Step Functions and Spark Notebooks, writing Spark SQL scripts or utilizing PySpark for transformations, including data cleaning, enrichment, and aggregation.
- Real-time streaming data processing using Amazon Kinesis Analytics to write SQL queries to process and analyze data from Amazon Kinesis Streams or other streaming sources. Configure jobs to ingest and process streaming data.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration

- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, GitHub Actions, Docker, etc. to extend and improve existing internal secure ML libraries and frameworks to enhance overall efficiency.
- Built, trained, deployed, and managed machine learning models using Sagemaker.
- Used Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney

Gov. Central Bank: Senior AWS DevOps Engineer/Sr. Data Engineer/Clouds/Linux Admin./ML/SP/Power BI Developer /EDW

Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Led a team of data engineers in designing, developing, and implementing data pipelines and solutions on AWS.

Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services such as SageMaker, Glue, Airflow, Lambda, EC2, VPC, and IAM using Jenkins, Ansible, Chef, Puppet, Splunk, Jira, confluence etc.

Integrated Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing with Python, Pydantic, Java, R, SQL, Github Actions

Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.

Managed and optimized Redshift clusters to ensure high performance and scalability.

Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), Single Sign-On and MFA via VPN.

Developed and maintained data architecture documentation using Confluence.

Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.

Presented data findings and insights to senior management. Mentored and trained junior data engineers.

Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.

- Employ AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data.
- Define a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
- Executed the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
- Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.

Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.

Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Completed End-to-End Azure Data Engineer project using Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, Dashboard tools of Power BI, Looker Studio or Tableau, including write Spark code in Azure Databricks.

Build a data process flow to load raw data onto Data Lake storage for Databricks to write spark code and transform the data and load the data back to transform data Lake storage for synapse analytics to run the SQL queries on top of the transform

Utilized a range of statistical software tools including RStudio, R 4.0, Python, IBM SPSS, SAS, and more for in-depth data analysis and modeling. Employed DAX Studio and DAX functional language for complex formula creation in Power BI.

Automated tasks and scripts using Python, PowerShell, Bash, Ansible, Puppet, Lambda, JavaScript and Typescript

Linux administration, focusing on file system management, network configurations, and system performance optimization.

Implemented and maintained Amazon Elastic Kubernetes Service (Amazon EKS) clusters for various microservices applications using Terraform and Helm

Design and develop data solutions using AWS services (EC2, S3, EMR, RedShift Spectrum), Python and Snowflake

Built, trained, deployed, and managed machine learning models using SageMaker.

Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Applied statistical methods, particularly Predictive Analysis with Azure Machine Learning and AI Visuals.

Implement MLOps for the ML application lifecycle and ModelOps for model management once deployed in a production environment. Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)

Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Utilized SageMaker for building, training, deploying, and managing machine learning models.
Interpreted trends and patterns, prepared data for prescriptive and predictive modeling.

- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Government Education: Senior AWS DevOps Engineer /Sr. Data Engineer /ML/BI Developer /SharePoint /K2 Developer /EDW
Responsibilities:

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including Hadoop Setup, Spark Installation, Hive Configuration, MapReduce Integration, Data Pipeline Development and Incorporate Hive for structured data querying and analysis using PySpark (Python API for Spark) for data processing using Python, Pydantic, Java, R, SQL, git etc.
Developed and maintained CI/CD pipelines using Jenkins and AWS CodePipeline.
Utilized Glue, a serverless ETL service, to collect, transform, and load data into data warehouses and data lakes.
Implemented and maintained AWS secure infrastructure using IAM, Control Tower & Security Hub for enhanced security.
Used Lambda to run code without having to provision or manage servers, especially for users who need to run short-lived, event-driven tasks, such as processing data streams or responding to API requests.
Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources. Employed Amazon EMR(Elastic MapReduce) service with Apache Spark and Hadoop for processing extensive data on S3.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Monitored and troubleshoot data pipeline issues using Splunk and CloudWatch.
Employ AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
Define a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
Designed and implemented data quality checks to ensure data integrity.
Developed and maintained data warehouses using Hive and Impala. Automated data processing tasks using airflow.
Created and maintained data visualizations using Tableau.
Built, trained, deployed, and managed machine learning models using Sagemaker.
Utilized Amazon Redshift for running complex queries on large datasets and Amazon Athena for serverless SQL-based data analysis on Amazon S3. Set up & managed AWS services such as Sagemaker, Glue, Airflow, Lambda, EC2, VPC, and IAM.
Managed the Kafka platform, SQS cases messaging, and integration components, ensuring their development, support, administration, and configuration in a hybrid environment.
Configured and managed observability components using Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Designed and developed fully automated CI/CD pipelines using Azure DevOps, GitHub Actions, and Jenkins
Delivered robust, scalable, and future-proof technical solutions aligning with industry best practices.
Swiftly handled incident management and BAU operations, maintaining a stable data platform.
Drove the implementation of best practices, processes, and methodologies across the data platform.
Built and maintained data pipelines using a combination of Python, Spark, and AWS Glue for data transformation and loading. Implemented ETL processes for data transformation, utilizing Python, Scala, and SQL for data processing.
Played a pivotal role in building and maintaining data pipelines using technologies such as Apache Spark 3.3 and PySpark.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Developed intricate SQL queries for data extraction, transformation, and analysis.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Designed data models with Kimball or Data Vault methodologies for efficient data retrieval.
Created interactive Power BI dashboards with DAX calculations and utilized Databricks SQL for high performance analytics.
Utilized Azure Open AI /ML Service and Azure Open AI Studio for Large Language Model (LLM) training.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.

- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: AWS DevOps Engineer / Data Engineer SharePoint /BI Developer /D365 CRM Consultant /DBA
Responsibilities:

- Successfully established the Batch, Speed, and Service layers of the Lambda Architecture on AWS Cloud, employing Hadoop for Batch processing, Spark with in-memory access and RDD for Speed layer, and Spark Streaming for Service layer to enable real-time data views and queries on the Spark cluster datasets.
- Demonstrated expertise in AWS developer analytics tools, including Lambda, Step Functions, API Gateway, SQS, SNS, CloudWatch, CloudFormation, and S3.
- Integrating Hadoop, Spark, Hive, and MapReduce into a cohesive data pipeline, including install and configure Hadoop on a cluster. Set up Hadoop Distributed File System (HDFS) for storage. Install Apache Spark on the same cluster or a different one. Configure Spark to use Hadoop YARN for resource management.
- Set up Apache Hive for data warehousing and SQL-like queries. Create external tables or load data into Hive tables.
- Develop and implement MapReduce jobs for specific data processing tasks. Integrate MapReduce jobs into the workflow for distributed processing.
- Experienced with bash / PythonUtilized Python and SQL for development, working on Snowflake features like Snow Spark, DAG, Snowpipe, external functions, and API integration.
- Designing and implementing data ingestion processes using Azure Synapse, Azure Data Factory, and Power BI.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: AWS/Azure DevOps Engineer /Data Engineer /BI Developer /SharePoint Consultant /DBA
Responsibilities:

- Employed AWS Glue through the AWS Management Console to create a crawler for discovering and cataloging S3 bucket data. Defined a Data Lake Table in Glue, specifying tables, selecting the data source (S3), and setting table properties.
- Execute the crawler to populate the AWS Glue Data Catalog with metadata related to the data.
- Implemented ETL/ELT frameworks, orchestration, and configuration as code, contributing to efficient data processing and integration. Developed and deployed data solutions using AWS services, Python and SQL
- Demonstrated exceptional incident management and business-as-usual (BAU) support, ensuring a stable data platform.
- Collected, cleaned, and analyzed data from various sources. Developed and maintained data models using SQL and Python. Created and maintained data visualizations using Tableau. Presented data findings and insights to senior management. Developed Python scripts for data processing, supporting efficient ETL operations.
- Experienced in RESTful APIs and integration with external systems.
- Proficient in cloud data analytics technologies such as Databricks and Azure ADF.
- Solid experience using MS Stack: Microsoft SQL Server, SSIS, SSRS, SSAS, Power BI, and SQL.
- Familiarity with MongoDB, Graph databases, programming languages (R, Python, Spark ETL SQL VStudio), CI/tools (Bamboo, or similar), GitHub Actions, Jira, and Confluence.
- Experienced in data modeling in enterprise data warehousing/lake environments and use of tools like ERWIN.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP,
- ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer /Data Engineer/Data Analyst /BI Developer /Senior Systems Engineer /EDW
Responsibilities:

- Align Architecture, Solution Design with business requirements. Design, Build and Maintain Physical data model in data platform layers. Map business glossary, data definition and data sets to physical data model
- Build ETL that collects, manage, and convert raw data into consumable tables applying technical logic based on business rules. Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Performed advanced data modeling and design, supporting data-driven decision-making processes within the organization.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x

- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /EDW
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /EDW
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, Control-M
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.

Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI/ML Engineer /Python Developer/Senior DevOps/Senior Data Engineer /AI/ML/BI/SharePoint/D365 CRM Consultant /EDW/DBA-Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /ML/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based Csystems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- Utilized ChatGPT Enterprise to build AI solutions that automate processes, reduce carbon emissions, and drive innovation with a focus on data security for aged care business through trained LLMs using large enterprise datasets.
- Integrated LangChain with Azure OpenAI to build Private ChatGPTs for BAs efficient chat with word, pdf, text documents.
- Used Ollama to build Open-Source ChatGPTs from scratch (FULLY Local and secure) for marketing and sales teams
- Implemented latest multimodal GPT-4 for Decision Makers to upload and analyze various document types.
- Utilized Multi-GPTs to build Synapse GPT for clinical research for those who are Alzheimer and dementia aged patients.
- Analyze data sources, types, and formats to define security auto detect requirements against stakeholders' requirements.
- Load the dataset, considering features related to security and environmental factors like carbon emissions.
- Model Training: Use a RandomForestClassifier for security prediction and a simple neural network using TensorFlow for automation, innovation, and carbon emissions reduction.
- Evaluation: Assess the security model's accuracy on a test set and train the TensorFlow model for multi-objective tasks.
- Integration: Combine models to create a comprehensive solution addressing security concerns, automation, innovation, and carbon emissions reduction, including machine learning, deep learning, and natural language processing (NLP)
- Deployment: Implement the solution in real-world scenarios, continuously improving and adapting based on performance feedback. Create a Think Library of reusable AI models and automations
- Interrogate extensive libraries of existing designs to develop concept designs for new bridges
- Collaborate with engineering experts to refine AI solutions and ensure they meet client requirements
- Maintain strong communication with clients and stakeholders throughout the project lifecycle
- Demonstrated skill in designing secure AI solutions and utilizing relevant tools and frameworks, coupled with strong problem-solving, analytical, and effective communication abilities.
- Proficiency in programming languages such as Python, R, or Java, and experienced with machine learning frameworks such as Scikit-learn, H20; DL frameworks: TensorFlow, PyTorch, Keras, Apache MXNet; Python libraries: ImageAI, OpenCV etc.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal security related ML libraries and frameworks to enhance overall efficiency.
- Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
- Hands on experience building AI tools with ML methods, DL NLP either on multi-GPU Azure AI Studio, Azure AI Content Safety, GCP's Cloud AI Platform, Cloud AutoML or AWS SageMaker, DeepLens, DeepComposer or well scaled Kaggle standalone platforms. Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation. Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance. Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.

- Strong understanding of the intelligence lifecycle and associated analytic methodologies (Cyber Kill Chain, Diamond Model, Mitre ATT&CK, etc.). Assisted in the implementation of E8 mitigation strategies.
- Applied NLP, computer vision, recommender systems and anomaly detection techniques to various domains
- Trained and deployed models on the cloud using AWS SageMaker, Azure Machine Learning and Google Cloud AI Platform
- Applied natural language processing (text classification, named entity recognition), computer vision (image classification) and reinforcement learning (optimal control) techniques to defense problems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SP /Power BI Developer /EDW
Responsibilities:

Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
Implement advanced trading algorithms, showcasing deep Python knowledge and high-performance trading system expertise. Quickly adapt to new technologies, enhancing the company's tech stack.
Implemented AI tools prioritizing data security through comprehensive measures, including data encryption, access control, secure storage, regular audits, data anonymization, secure model deployment, API security, secure development practices, user education, compliance adherence, incident response planning, and ongoing security training.
Designing and developing AI solutions that automate processes, reduce carbon emissions, and drive innovation with a focus on data security by implementing robust measures to protect sensitive information.
Contributed to the development of a Think Library for efficient model and automation reuse.
Maintained clear communication with clients and stakeholders throughout project lifecycles.
Collaborate with engineering experts to refine AI solutions and ensure they meet client requirements
Designed and built AI-powered ChatBots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
Use Python and machine learning frameworks (PyTorch, scikit-learn) to build scalable AI /ML solutions.
Developed and deployed end-to-end machine learning solutions for various clients using DataRobot's automated machine learning platform with best practice development methodologies and practices of Agile, DevOps & Configuration-As-Code
Used MLOps for the entire lifecycle of machine learning applications and predictive analytics to use data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data as well as to predict customer behavior, identify fraud, and optimize business processes.
Used ModelOps for operations and management of machine learning models once they are deployed in a production env.
Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.
Clearly articulate the problem that clients want to solve with AI against analyzed data sources, types, and formats
Gather relevant data for the problem. Ensure data quality and cleanliness.
Data Preprocessing: Clean, normalize, and transform the data to make it suitable for the AI training model.
Feature Engineering: Select or create features that are most relevant to the problem.
Split the Data: Divide the dataset into training and testing sets.
Choose a Model: Select a machine learning or deep learning model based on the nature of the problem.
Train the Model: Use the training dataset to train the chosen model.
Model Evaluation: Assess the performance of your model using the testing dataset.
Hyperparameter Tuning: Optimize the hyperparameters of the model for better performance.
Deployment: If satisfied with the model, deploy it for real-world use.
Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn.
Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
Deploy machine learning solutions for various clients using IBM's Watson Studio platform.
Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Spearhead the development of Python-based applications, showcasing proficiency in Python and Typescript development.
Contribute to the development of full-stack applications, emphasizing both front-end and backend components.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SharePoint /K2 Developer /DBA
Responsibilities:

Developed and maintained AI /ML models to enhance customer engagement and sales.
Proficient in Python, Java, and R, leveraging them for software development.
Built software components for data processing, forming the foundation for AI /ML applications.
Optimized code for AI /ML projects using data structures, algorithms, and software architecture knowledge.
Solid understanding of machine learning, deep learning, and NLP.
Either using Python, R with SQL 2022 Machine Learning feature on Azure or run Machine Learning on PyTorch based the applications on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Demonstrated problem-solving and analytical skills in AI development.
Tailored effective presentations for diverse technical audiences, showcasing strong communication skills.
Worked on client applications using Java, SQL, and Oracle Database, focusing on quality and performance.
Utilized Agile methodology and Git for task planning, execution, and code management.
Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
Proficient in technologies like React, GCP, FastAPI, Postgres, Kubernetes, and Next-js.
Designed and optimized ML platforms in collaboration with data scientists and engineers.
Proficient in modern data analytics and engineering tools like dbt and Dataform.
Skilled in creating impactful visualizations using Tableau and Power BI.
Utilized SQL for data extraction, transformation, and integration, employing various platforms.
Designed data models for efficient data retrieval using recognized methodologies.
Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
Used Agile methodology to plan and execute tasks and used Git to manage code
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: Mid Machine Learning Engineer Python Developer /Data Engineer /SharePoint /D365 CRM Consultant
Responsibilities:

- Assisted in ML prototype development and testing, gaining valuable experience.
- Conducted research on advanced ML technologies for defense applications.
- Collaborated with senior ML engineers to understand and apply ML algorithms to the ML models to promote the sales.
- Contributed to the creation of innovative AI tools with a focus on efficiency.
- Engaged in the interrogation of extensive design libraries for new concept developments.
- Ensured seamless communication with stakeholders during project execution.
- Developed ML sentiment analysis prototype using Python, scikit-learn, and Azure ML.
- Utilized Flask for creating a user interface and scikit-learn for model evaluation.
- Proficient in Python, SQL, Kubernetes, and containerization for data app deployment.
- Engaged in Python application development and TypeScript experience.
- Involved in DB management and SQL database administration tasks.
- Contributed to DevOps, configuring CI/CD pipelines and Git version control.

- Utilized Python for low-latency, reliable multithreaded programming.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Leveraged Flask, Django for scalable component development.
- Conducted statistical modeling and analysis for workforce policy insights.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Used Flask to create a web interface for the application
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Senior Machine Learning Engineer /Python Developer /BI Developer /SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked as a research assistant in AI and machine learning.
- Designed and developed AI solutions driving automation and innovation.
- Collaborated with cross-functional teams to refine and enhance AI implementations.
- Maintained strong communication channels with clients to align solutions with their needs.
- Contributed to data-related features in software projects.
- Designed and implemented data-driven solutions using Python and SQL.
- Proficient in Linux platform development, optimizing application performance.
- Aided in data collection, preprocessing, and AI research analysis.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Optimized query performance and database schema design.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, optimizing IT costs and integrating with Sales Force Automation (SFA).
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: OpenAI ChatGPT Enterprise, Mutimodal GPT-4, Multi-GPTs, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI/ML Engineer /Python Developer/Senior DevOps/Senior Data Engineer /AI/ML/BI/SharePoint/D365 CRM Consultant /EDW/DBA-Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer AI/ML/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
- Dedicated and accomplished Machine Learning Engineer with a proven track record in developing and implementing cutting-edge AI solutions. Adept at designing, optimizing training and inference pipelines for efficient model deployment.
- Collaborating with cross-functional teams, exploring & implementing state-of-the-art techniques to enhance AI capabilities.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI
- Design and develop ML models using Natural Language Processing (NLP), ML algrithirm, OpenCV, Q-Learning(Qstar), etc.
- Transformed and trained data science products and ML systems using Python and Pydantic on AWS EMR.
- Utilized AWS Machine Learning Tools: ECS, EKS, to manage and scale containerized ML workloads.
- Leverage Amazon SageMaker Studio to provide a collaborative environment for data scientists and ML engineers.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Deployed well trained and tuned ML model into production for web applications using Flask or FastAPI
- Worked with ML team to develop & train ML models using tools such as Erwin, Pydantic, LlaMa 2, MakerSuite, or X.AI Grok.
- Developed and maintained ML libraries & frameworks. Managed and monitored cloud infrastructure either on AWS or Azure
- Either using Python, R with SQL 2022 Machine Learning feature on Azure or run Machine Learning on PyTorch based the applications on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
- Developed and maintained logging and monitoring systems at large scale.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- Experienced in creating innovative solutions for real-world problems.
- Utilized ChatGPT Enterprise to build AI solutions that automate processes, reduce carbon emissions, and drive innovation with a focus on data security for aged care business through trained LLMs using large enterprise datasets.
- Integrated LangChain with Azure OpenAI to build Private ChatGPTs for BAs efficient chat with word, pdf, text documents.
- Used Ollama to build Open-Source ChatGPTs from scratch (FULLY Local and secure) for marketing and sales teams
- Implemented latest multimodal GPT-4 for Decision Makers to upload and analyze various document types.
- Utilized Multi-GPTs to build Synapse GPT for clinical research for those who are Alzheimer and dementia aged patients.
- Analyze data sources, types, and formats to define security auto detect requirements against stakeholders' requirements.
- Proficiency in programming languages such as Python, R, or Java, and experienced with machine learning frameworks such as Scikit-learn, H20; DL frameworks: TensorFlow, PyTorch, Keras, Apache MXNet; Python libraries: ImageAI, OpenCV etc.
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal security related ML libraries and frameworks to enhance overall efficiency.
- Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
- Hands on experience building AI tools with ML methods, DL NLP either on multi-GPU Azure AI Studio, Azure AI Content Safety, GCP's Cloud AI Platform, Cloud AutoML or AWS SageMaker, DeepLens, DeepComposer or well scaled Kaggle standalone platforms. Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Applied NLP, computer vision, recommender systems and anomaly detection techniques to various domains

- Applied natural language processing (text classification, named entity recognition), computer vision (image classification) and reinforcement learning (optimal control) techniques to security problems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney

Gov. Central Bank: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SP /Power BI Developer /EDW

Responsibilities:

Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).

Developed and implemented machine learning models and algorithms, showcasing deep Python knowledge and high-performance trading system expertise. Quickly adapt to new technologies, enhancing the company's tech stack.

Collaborate with cross-functional teams to understand business requirements and translate them into technical solutions.

Explore and implement state-of-the-art techniques in machine learning to enhance the AI capabilities.

Led the design and optimization of training and inference pipelines, resulting in a significant improvement in model deployment efficiency. Build and optimise training and inference pipelines for efficient model deployment.

Conducted extensive research and implementation of state-of-the-art techniques in machine learning, enhancing the organization's overall AI capabilities. Applied expertise in data structures, data modeling, and software architecture.

Developed and trained ML models, contributing to the scalability of a profitable product.

Researched and implemented appropriate ML algorithms and tools based on business requirements.

Collaborated with cross-functional teams to integrate ML solutions into existing software systems.

Utilized Python and Pytorch for model development and extended ML libraries.

Conducted statistical analysis and created datasets for effective model training.

Implemented ML tools prioritizing data security through comprehensive measures, including data encryption, access control, secure storage, regular audits, data anonymization, secure model deployment, API security, etc.

Designing and developing AI solutions that automate processes, and drive innovation with a focus on data security by implementing robust measures to protect sensitive information.

Collaborate with engineering experts to refine AI solutions and ensure they meet client requirements

Designed and built AI-powered ChatBots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.

Use Python and machine learning frameworks (PyTorch, scikit-learn) to build scalable AI /ML solutions.

Developed and deployed end-to-end machine learning solutions for various clients using DataRobot's automated machine learning platform with best practice development methodologies and practices of Agile, DevOps & Configuration-As-Code

Used MLOps for the entire lifecycle of machine learning applications and predictive analytics to use data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data as well as to predict customer behavior, identify fraud, and optimize business processes.

Used ModelOps for operations and management of machine learning models once they are deployed in a production env.

Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.

Clearly articulate the problem that clients want to solve with ML against analyzed data sources, types, and formats

Collect pertinent data for the problem, ensuring data quality and cleanliness for testing AI/ML models.

Data preprocessing by cleaning, normalizing, and transforming the data to make it suitable for the ML training model.

Feature Engineering through Select or create features that are most relevant to the problem.

Split the Data by dividing the dataset into training and testing sets. Use the training dataset to train the chosen model.

Select a machine learning or deep learning model based on the nature of the problem.

Assess the performance of the model using the testing dataset for model evaluation:

Optimize model hyperparameters for improved performance and deploy it for real-world use if satisfied.

Worked on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn.

Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.

Deploy machine learning solutions for various clients using IBM's Watson Studio platform.

Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.

Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.

Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.

Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.

Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.

Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet

Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: **AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SharePoint /K2 Developer /DBA**
Responsibilities:

Developed and maintained AI /ML models to enhance customer engagement and sales.
Collaborated with cross-functional teams, translating business requirements into technical solutions and ensuring seamless integration of machine learning solutions.
Developed and implemented machine learning models to address complex business challenges, contributing to the success of key projects. Implemented MLOps and DevOps practices to enhance model deployment efficiency.
Worked closely with domain experts to identify opportunities for the application of machine learning solutions and delivered innovative results. Contributed to the extension of existing ML libraries and frameworks.
Developed appropriate datasets and data representation methods and performed statistical analysis.
Worked with the ML team to develop and train ML models. Developed and maintained ML libraries and frameworks.
Managed and monitored cloud infrastructure using AWS SegeMaker, AWS Config, AWS CloudTrail, AWS Shield, WAF etc.
Developed and maintained logging and monitoring systems at large scale using AWS Security Hub ,AWS CloudWatch Logs
Proficient in Python, Java, and R, leveraging them for software development.
Built software components for data processing, forming the foundation for AI /ML applications.
Optimized code for AI /ML projects using data structures, algorithms, and software architecture knowledge.
Demonstrated expertise as a Machine Learning Engineer with a strong background in designing and optimizing training and inference pipelines, proficient in AI pipeline development, and possessing a solid understanding of machine learning frameworks and libraries.
Either using Python, R with SQL 2019 Machine Learning feature on Azure or run Machine Learning on PyTorch based the applications on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Demonstrated problem-solving and analytical skills in AI development.
Tailored effective presentations for diverse technical audiences, showcasing strong communication skills.
Worked on client applications using Java, SQL, and Oracle Database, focusing on quality and performance.
Utilized agile methodology and Git for task planning, execution, and code management.
Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
Proficient in technologies like React, GCP, FastAPI, Postgres, Kubernetes, and Next-js.
Designed and optimized ML platforms in collaboration with data scientists and engineers.
Proficient in modern data analytics and engineering tools like dbt and Dataform.
Skilled in creating impactful visualizations using Tableau and Power BI.
Utilized SQL for data extraction, transformation, and integration, employing various platforms.
Designed data models for efficient data retrieval using recognized methodologies.
Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
Used Agile methodology to plan and execute tasks and used Git to manage code
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: **Machine Learning Engineer /Python Developer /Data Engineer /SharePoint /D365 CRM Consultant**
Responsibilities:

- Played a pivotal role in building and optimizing training pipelines, ensuring the seamless integration of machine learning models into existing systems.
- Explored and implemented novel techniques, staying abreast of the latest developments in machine learning to maintain a competitive edge. Conducted research on advanced ML technologies for security applications
- Demonstrated strong Python programming skills in developing and optimizing ML algorithms.
- Played a key role in the integration of AI technologies into business processes.

- Conducted regular system training and retraining activities.
- Contributed to the innovation of efficient AI tools, explored design libraries for new concepts, and developed an ML sentiment analysis prototype using Python, scikit-learn, and Azure OpenAI ML.
- Experienced in creating user interfaces with Flask, Django, FastAPI, utilizing scikit-learn for model evaluation.
- Proficient in Python, SQL, Kubernetes, and containerization for deploying data applications, with additional expertise in Python application development and TypeScript.
- Contributed to DevOps, configuring CI/CD pipelines and Git version control.
- Utilized Python for low-latency, reliable multithreaded programming.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Conducted statistical modeling and analysis for workforce policy insights.
- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Machine Learning Engineer /Python Developer /BI Developer /SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Gained hands-on experience in machine learning development, providing valuable support to the team in implementing algorithms and models.
- Assisted in optimizing training pipelines and contributed to ongoing projects, gaining exposure to real-world AI challenges.
- Worked on data representation methods and contributed to effective data engineering processes.
- Collaborated with cross-functional teams to enhance software architecture for ML applications.
- Demonstrated expertise in version control software for maintaining ML codebase.
- Worked as a research assistant in AI and machine learning.
- Designed and developed AI solutions driving automation and innovation.
- Proficient in Linux platform development, optimizing application performance.
- Aided in data collection, preprocessing, and AI research analysis.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.

- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: OpenAI ChatGPT Enterprise, Mutimodal GPT-4, Multi-GPTs, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Seems for NSWTF long time ago based on yellow highted ones?

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Operation Support Engineer /Sr. DevOps/Data Engineer /Sr. Systems Administrator/Clouds/Linux Admin./ /ML/BI/SharePoint/D365 CRM Consultant /EDW –Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Cloud Operation Support Engineer/DevOps/Data Engineer/Systems Admin./ML/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and implemented infrastructure solutions for security, performance, and operations.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
- Administered hybrid Azure, GCP and AWS cloud environments services, cloud security and governance best practices
- Supported cloud solutions for various clients using AWS, Azure, and GCP platforms
- Managed large-scale & complex deployments & changes to cloud systems & applications using CI/CD pipelines and GitHub
- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, ADF, ARM, Docker, GitHub, Jenkins etc.
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Using Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
- Developed, reviewed, and maintained cloud support documentation and best practices
- Escalated and resolved performance issues with cloud providers and vendors
- Managed the integration of cloud solutions across disparate platforms and technologies
- Responsible for change implementation planning and change management for cloud systems and applications
- Managed the integration of system solutions across various platforms and technologies, supporting DevOps and Cloud.
- Oversaw change implementation planning and change management, ensuring minimal risk to business and data.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
- Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with Databricks, ADF, AKS, Ansible, and Jenkins.
- Automated tasks and scripts using GitHub/Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal DSTG ML libraries and frameworks to enhance overall efficiency.
- Used Spark MLlib to perform distributed machine learning on large-scale data sets and used GitHub to manage code
- Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

- Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Cloud Operation Support Engineer/DevOps/Data Engineer/Systems Admin./SP/BI Developer/EDW
Responsibilities:

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9 /J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
Administered hybrid Azure, GCP and AWS cloud environments services, cloud security and governance best practices
Designed and implemented cloud solutions for various projects using AWS and Azure platforms
Managed large-scale & complex deployments & changes to cloud systems & applications using CI/CD pipelines and GitHub
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services like EC2, RDS, S3, RedShift, DynamoDB, and Kinesis to optimize platform performance.
Leveraging AWS DevOps best practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Using Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
Supported and maintained hybrid cloud systems and applications using DevOps tools and methodologies
Scripted and automated cloud tasks and processes using Python, PowerShell, and Bash
Ensured that cloud installations were completed on time, cost, and quality
Assisted in the development of business cases and proposals for cloud solutions
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
Managed the operation's infrastructure across the globe and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with ADF, AKS, Ansible etc.
Designed and implemented infrastructure solutions for security, performance, and operations.
Implemented security baselines and security hardening in response to emerging threats.
Designed, implemented, configured, and supported Azure and Office 365 technology at an expert level, including services such as Azure Files and others. Minimized implementation risk through change control methodologies.
Managed and supported Standard Operating Environment (SOE) for RBA, including application packaging, SCCM/Intune administration, Group Policy modification, SharePoint and Dynamics 365 CRM online etc.
Conducted root cause analysis and implemented system changes to prevent incident recurrence.
End-to-end ownership of secure CI/CD processes for various applications, including design, build, testing, and support Coordinated SOE rollouts and desktop environment changes following Change Management procedures.
Develop & enhance automated processes to release new solutions into the live environment in a timely, repeatable fashion
High proficiency in Active Directory, Azure AD, Managed the installation, configuration, and testing of new and existing hardware, software, and WAN/LAN technologies using (IaC) tools: Azure Resource Manager(ARM) templates, Teraform or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code
Managed Intune/SCCM for SOE image delivery, operating system deployment, and patch management.
Ensure all requests (Service, Tasks and Incidents) are managed in a timely manner and using the service management tool
Management and coordination of SOE rollouts and upcoming desktop environment changes through an approve Change Management process. Administer and modification of Group Policy objects
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Implemented data augmentation, transfer learning and model compression techniques to improve model performance and efficiency. Worked with large-scale image datasets using SQL and BigQuery ML for Big Data.

Oct. 2013 – Nov.2015 University of Canberra (UC), ACT/SYD (Working remotely from Sydney Home Office)
Gov. University: Cloud Operation Support Engineer/DevOps/Data Engineer/Systems Administrator /SP/K2 Developer/DBA
Responsibilities:

Implemented and supported cloud solutions for various clients using AWS, Azure, and GCP platforms
Collaborated with cross-functional teams to implement application and platform solutions aligning with architecture standards, enhancing system and service support.
Managed large-scale deployments and changes to meet operational requirements, contributing to solution delivery.

Conducted performance analysis of assigned systems and applications, providing valuable insights for business case development and issue mitigation.
Ensured timely, cost-effective, and high-quality installations, reducing operational and project risks while containing costs.
Developed, reviewed, and maintained comprehensive support documentation for applications and platforms, aiding in service restoration during unplanned outages.
Effectively escalated performance issues to minimize adverse impacts on business continuity.
Spearheaded change implementation planning and change management, meeting business needs with minimal risk.
Provided technical support and troubleshooting for cloud systems and applications using AWS and Azure platforms
Supported the implementation of hybrid cloud solutions to enhance system and service capabilities.
Collaborated with cross-functional teams to ensure successful system deployments.
Conducted performance analysis and reported findings to support system improvement decisions.
Managed installations to meet project deadlines while minimizing operational risks.
Maintained comprehensive support documentation to aid in service restoration.
Worked with providers to address performance issues and minimize system downtime.
Assisted in change management and planning for applications and platforms.
Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
Automated systems administration using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Designed and implemented infrastructure solutions for security, performance, and operations.
Using Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Conducted regular infrastructure assessments to identify configuration issues, security vulnerabilities, and potential risks.
Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Sr. Cloud Operation Support Engineer/DevOps/Data Engineer/Systems Admin./SP Consultant /DBA
Responsibilities:

- Designed and implemented cloud infrastructure solutions in adherence to current architecture standards.
- Managed complex deployments and changes to meet operational requirements, ensuring solution delivery.
- Analyzed and reported on system performance, aiding in the development of business cases for system upgrades.
- Ensured timely, cost-effective, and high-quality installations, reducing operational and project risks.
- Developed, reviewed, and maintained support documentation, facilitating service restoration during unplanned outages.
- Collaborated with internal and external providers to resolve performance issues and minimize system downtime.
- Provided expert-level support for Azure and Office 365 technologies, addressing configuration and performance issues.
- Managed and supported Azure cloud infrastructure for a large enterprise customer
- Implemented and maintained cloud infrastructure automation and orchestration solutions using Terraform and Ansible
- Monitored and analyzed cloud infrastructure performance and usage. Troubleshooted and resolved cloud-related incidents
- Collaborated with other engineering teams to develop and implement new cloud-based solutions
- Administered and maintained on-premise and cloud servers, networks, and applications
- Installed and configured hardware and software. Performed system updates, patches, and upgrades
- Using Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
- Conducted system backup and recovery activities, ensuring data integrity and business continuity.
- Designed and implemented infrastructure solutions for security, performance, and operations.
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps with ADF, AKS, Ansible, Puppet etc.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Minimized implementation risk through change control methodologies.
- Successfully managed and maintained the Standard Operating Environment (SOE) for over 800 workstations, ensuring system stability and optimal performance.

- Executed SOE rollouts, adhering to Change Management processes and maintaining high system availability.
- Using Azure Monitor to monitored and troubleshooted Azure infrastructure and applications issues, ensuring high availability
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Centre etc.
- Automated systems administration using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deployment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Operation Support Engineer/Sr. DevOps/Data Engineer/Sr. Systems Admin./SP Consultant
Responsibilities:

- Designed and implemented infrastructure solutions for security, performance, and operations.
- Administered and maintained Microsoft operating systems and domain administration.
- Designed, implemented, and managed AWS cloud infrastructure for a variety of customers
- Developed and maintained CI/CD pipelines using GitHub and Jenkins
- Worked with customers to understand their requirements and design and implement cloud solutions that met their needs
- Provided technical support to customers for AWS-related issues
- Provided IT support and assistance to users and clients. Resolved hardware and software issues
- Installed and configured desktops, laptops, and mobile devices. Performed data backup and recovery
- Maintained IT inventory and documentation
- Assisted in the management and optimization of multi-domain and multi-forest systems across diverse environments.
- Contributed to Azure hosting maintenance, including system configuration, security enhancement, and workload migration.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Using Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
- Provided 3rd level desktop, systems, and network support, diagnosing and resolving complex technical issues.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Azure Administrator (AD/AAD/InTune/SCCM/SCOM)/DBA
Responsibilities:

- Installed, configured, and maintained on-premises servers and networks. Supported users with technical issues
- Monitored and analyzed system performance. Troubleshooted and resolved system-related incidents
- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Assisted in managing email and backup servers and their associated operating systems.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /ICT System /Linux Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided 3rd level technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planned, designed, and managed Active Directory with high-level support for AD migrations within and between AD forests.
- Installed and configured hardware, software, and networking equipment.
- Supported Office Online Specialist functions, resolving technical issues promptly.
- Managed and maintained the operation's infrastructure across the globe.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI/ML Engineer /Python Developer/Senior DevOps/Senior Data Engineer /AI/ML/BI/SharePoint/D365 CRM Consultant /EDW/DBA-Japan(7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney
Gov. Health: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /ML/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based Csystems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- Utilized ChatGPT Enterprise to build AI solutions that automate processes, reduce carbon emissions, and drive innovation with a focus on data security for aged care business through trained LLMs using large enterprise datasets.
- Integrated LangChain with Azure OpenAI to build Private ChatGPTs for BAs efficient chat with word, pdf, text documents.
- Used Ollama to build Open-Source ChatGPTs from scratch (FULLY Local and secure) for marketing and sales teams
- Implemented latest multimodal GPT-4 for Decision Makers to upload and analyze various document types.
- Utilized Multi-GPTs to build Synapse GPT for clinical research for those who are Alzheimer and dementia aged patients.
- Analyze data sources, types, and formats to define security auto detect requirements against stakeholders' requirements.
- Load the dataset, considering features related to security and environmental factors like carbon emissions.
- Model Training: Use a RandomForestClassifier for security prediction and a simple neural network using TensorFlow for automation, innovation, and carbon emissions reduction.
- Evaluation: Assess the security model's accuracy on a test set and train the TensorFlow model for multi-objective tasks.
- Integration: Combine models to create a comprehensive solution addressing security concerns, automation, innovation, and carbon emissions reduction, including machine learning, deep learning, and natural language processing (NLP)
- Deployment: Implement the solution in real-world scenarios, continuously improving and adapting based on performance feedback. Create a Think Library of reusable AI models and automations
- Interrogate extensive libraries of existing designs to develop concept designs for new bridges
- Collaborate with engineering experts to refine AI solutions and ensure they meet client requirements
- Maintain strong communication with clients and stakeholders throughout the project lifecycle
- Demonstrated skill in designing secure AI solutions and utilizing relevant tools and frameworks, coupled with strong problem-solving, analytical, and effective communication abilities.
- Proficiency in programming languages such as Python, R, or Java, and experienced with machine learning frameworks such as Scikit-learn, H20; DL frameworks: TensorFlow, PyTorch, Keras, Apache MXNet; Python libraries: ImageAI, OpenCV etc.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal security related ML libraries and frameworks to enhance overall efficiency.
- Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
- Hands on experience building AI tools with ML methods, DL NLP either on multi-GPU Azure AI Studio, Azure AI Content Safety, GCP's Cloud AI Platform, Cloud AutoML or AWS SageMaker, DeepLens, DeepComposer or well scaled Kaggle standalone platforms. Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Utilized Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code

Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Strong understanding of the intelligence lifecycle and associated analytic methodologies (Cyber Kill Chain, Diamond Model, Mitre ATT&CK, etc.). Assisted in the implementation of E8 mitigation strategies.
- Applied NLP, computer vision, recommender systems and anomaly detection techniques to various domains
- Trained and deployed models on the cloud using AWS SageMaker, Azure Machine Learning and Google Cloud AI Platform
- Applied natural language processing (text classification, named entity recognition), computer vision (image classification) and reinforcement learning (optimal control) techniques to defense problems

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SP /Power BI Developer /EDW
Responsibilities:

- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Implement advanced trading algorithms, showcasing deep Python knowledge and high-performance trading system expertise. Quickly adapt to new technologies, enhancing the company's tech stack.
- Implemented AI tools prioritizing data security through comprehensive measures, including data encryption, access control, secure storage, regular audits, data anonymization, secure model deployment, API security, secure development practices, user education, compliance adherence, incident response planning, and ongoing security training.
- Designing and developing AI solutions that automate processes, reduce carbon emissions, and drive innovation with a focus on data security by implementing robust measures to protect sensitive information.
- Contributed to the development of a Think Library for efficient model and automation reuse.
- Maintained clear communication with clients and stakeholders throughout project lifecycles.
- Collaborate with engineering experts to refine AI solutions and ensure they meet client requirements
- Designed and built AI-powered ChatBots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
- Use Python and machine learning frameworks (PyTorch, scikit-learn) to build scalable AI /ML solutions.
- Developed and deployed end-to-end machine learning solutions for various clients using DataRobot's automated machine learning platform with best practice development methodologies and practices of Agile, DevOps & Configuration-As-Code
- Used MLOps for the entire lifecycle of machine learning applications and predictive analytics to use data, statistical algorithms, and machine learning techniques to identify the likelihood of future outcomes based on historical data as well as to predict customer behavior, identify fraud, and optimize business processes.
- Used ModelOps for operations and management of machine learning models once they are deployed in a production env.
- Utilized Python, Java, R, SQL, and various tools for data science prototypes and ML model development.
- Clearly articulate the problem that clients want to solve with AI against analyzed data sources, types, and formats
- Gather relevant data for the problem. Ensure data quality and cleanliness.
- Data Preprocessing: Clean, normalize, and transform the data to make it suitable for the AI training model.
- Feature Engineering: Select or create features that are most relevant to the problem.
- Split the Data: Divide the dataset into training and testing sets.
- Choose a Model: Select a machine learning or deep learning model based on the nature of the problem.
- Train the Model: Use the training dataset to train the chosen model.
- Model Evaluation: Assess the performance of your model using the testing dataset.
- Hyperparameter Tuning: Optimize the hyperparameters of the model for better performance.
- Deployment: If satisfied with the model, deploy it for real-world use.
- Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn.
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Deploy machine learning solutions for various clients using IBM's Watson Studio platform.
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
- Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Spearhead the development of Python-based applications, showcasing proficiency in Python and Typescript development.
- Contribute to the development of full-stack applications, emphasizing both front-end and backend components.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: AI /ML Engineer/Python Developer/Sr. DevOps/Data Engineer /SharePoint /K2 Developer /DBA
Responsibilities:

- Developed and maintained AI /ML models to enhance customer engagement and sales.

Proficient in Python, Java, and R, leveraging them for software development.
Built software components for data processing, forming the foundation for AI /ML applications.
Optimized code for AI /ML projects using data structures, algorithms, and software architecture knowledge.
Solid understanding of machine learning, deep learning, and NLP.
Either using Python, R with SQL 2022 Machine Learning feature on Azure or run Machine Learning on PyTorch based the applications on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Demonstrated problem-solving and analytical skills in AI development.
Tailored effective presentations for diverse technical audiences, showcasing strong communication skills.
Worked on client applications using Java, SQL, and Oracle Database, focusing on quality and performance.
Utilized Agile methodology and Git for task planning, execution, and code management.
Designed and built AI-powered Chatbots tools that can simulate human conversation for frequently asked aged care Questions and Answers to automate customer services, and provide personalized recommendations.
Proficient in technologies like React, GCP, FastAPI, Postgres, Kubernetes, and Next-js.
Designed and optimized ML platforms in collaboration with data scientists and engineers.
Proficient in modern data analytics and engineering tools like dbt and Dataform.
Skilled in creating impactful visualizations using Tableau and Power BI.
Utilized SQL for data extraction, transformation, and integration, employing various platforms.
Designed data models for efficient data retrieval using recognized methodologies.
Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
Used Agile methodology to plan and execute tasks and used Git to manage code
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
Employed Power BI, Tableau, or QlikView for real-time financial analysis.
Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Health Sector: **Mid Machine Learning Engineer Python Developer /Data Engineer /SharePoint /D365 CRM Consultant**
Responsibilities:

- Assisted in ML prototype development and testing, gaining valuable experience.
- Conducted research on advanced ML technologies for defense applications.
- Collaborated with senior ML engineers to understand and apply ML algorithms to the ML models to promote the sales.
- Contributed to the creation of innovative AI tools with a focus on efficiency.
- Engaged in the interrogation of extensive design libraries for new concept developments.
- Ensured seamless communication with stakeholders during project execution.
- Developed ML sentiment analysis prototype using Python, scikit-learn, and Azure ML.
- Utilized Flask for creating a user interface and scikit-learn for model evaluation.
- Proficient in Python, SQL, Kubernetes, and containerization for data app deployment.
- Engaged in Python application development and TypeScript experience.
- Involved in DB management and SQL database administration tasks.
- Contributed to DevOps, configuring CI/CD pipelines and Git version control.
- Utilized Python for low-latency, reliable multithreaded programming.
- Developed APIs and Dashboards for real-time trading insights.
- Applied Async/await in Python for non-blocking, concurrent request handling.
- Leveraged Flask, Django for scalable component development.
- Conducted statistical modeling and analysis for workforce policy insights.

- Spearheaded complex data analysis for actionable insights and strategies.
- Managed ETL processes, ensuring consistent and accurate data.
- Utilized AWS services for enhanced data storage, processing, and visualization.
- Profiled, tested, and modeled data from various sources, identifying trends and outliers.
- Designed data models and ETL processes for efficient data integration.
- Implemented data automation solutions using R and Python.
- Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Used Flask to create a web interface for the application
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Senior Machine Learning Engineer /Python Developer /BI Developer /SharePoint Consultant
Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked as a research assistant in AI and machine learning.
- Designed and developed AI solutions driving automation and innovation.
- Collaborated with cross-functional teams to refine and enhance AI implementations.
- Maintained strong communication channels with clients to align solutions with their needs.
- Contributed to data-related features in software projects.
- Designed and implemented data-driven solutions using Python and SQL.
- Proficient in Linux platform development, optimizing application performance.
- Aided in data collection, preprocessing, and AI research analysis.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Optimized query performance and database schema design.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, optimizing IT costs and integrating with Sales Force Automation (SFA).

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Senior Data Engineer /MS BI Developer /Senior Database Administrator /DWH**
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: OpenAI ChatGPT Enterprise, Mutimodal GPT-4, Multi-GPTs, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Support Engineer /Windows/Linux Administrator/Sr. DevOps Engineer/Sr. Data Engineer /ML/BI/SharePoint /D365 CRM Consultant /EDW –Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cloud Support Engineer/Senior DevOps/Data Engineer/ML/BI/SP/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP or AWS Cloud via Terraform.
- Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
- Design and build cloud infrastructure solutions. Ensure group data platforms are well maintained, secure, and stable
- Support various teams during design, development, and delivery of platforms and data analytics
- Provided Level 3 technical support, maintenance and advanced troubleshooting across a wide variety of technologies.
- Engaged in the design and construction of cloud infrastructure solutions.
- Ensured the security and stability of group data platforms.
- Provided Level 3 support, maintenance, and advanced troubleshooting across diverse technologies.
- Maintained, optimized, secured, and stabilized the group's data platforms. Provided support to teams to ensure the delivery of supportable, scalable, and robust data analytics solutions during design, development, and delivery.
- Design and develop data solutions using AWS web services (EC2, S3, EMR, RedShift Spectrum), Python and Snowflake
- Build scalable and reliable data pipelines using Airflow and Lambda to ingest and process data from various sources
- Implemented and maintained Amazon Elastic Kubernetes Service (Amazon EKS) clusters for various microservices applications using Terraform and Helm. Implement data quality checks and monitoring using CloudWatch and SNS
- Collaborated with cross-functional teams to design, implement, and optimize CI/CD pipelines using Jenkins and Bamboo,
- Automated infrastructure provisioning & configuration using Ansible, Terraform, leading to cost-effective & scalable solutions.
- Championed the adoption of best practice DevOps methodologies, enhancing continuous integration, deployment, and software delivery. Worked on multi-zone load balancing, redundancy, and system software updates.
- Resolved complex issues, conducted post-release reviews, &contributed to centralized CI/CD suite using Atlassian products.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- Efficiently managed Azure cloud infrastructure, including virtual machines, storage, & networking components by DevOps.
- Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with Azure Databricks, ADF, Azure Kubenete Service (AKS), Ansible, and Jenkins.
- Implemented DevOps security controls, detection methods, and maintained firewalls/NSGs in alignment with Cyber security standards, ensuring a secure and compliant environment, including distributed file system namespace (DFSN) and DFSR
- Managed ExpressRoute setup, network design, implementation, and support a variety of enterprise applications
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Completed design, data models, loading approaches and source to target mapping documents.
- Developed ELT for Azure Synapse Analytics using Azure Data Factory (ADF)/Synapse Pipelines, Spark Notebooks, SSIS, enhanced PolyBase on SQL Server 2019 /2022 and Stream Analytics sitting on Windows Server 2019/2022.
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics 365 CRM, Power BI, Power Apps and SharePoint online under M365
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Developed and deployed AI-driven machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
- Implemented data augmentation, transfer learning and model compression techniques to improve model performance and efficiency. Worked with large-scale image datasets using SQL and BigQuery ML for Big Data.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior Cloud Support Engineer/Windows/Linux Admin./Sr. DevOps/Data Engineer/SP/BI Developer/EDW
Responsibilities:

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9 /J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to GCP Cloud via Terraform.
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Designed and built cloud infrastructure solutions. Ensured group data platforms were well maintained, secure, and stable
Supported various teams during design, development, and delivery of platforms and data analytics
Provided Level 3 technical support, maintenance and advanced troubleshooting across a wide variety of technologies.
Collaborated with various teams during the design, development, and delivery of platforms and data analytics.
Demonstrated proficiency in supporting projects from inception to completion.
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Proficiently managed Azure Cloud Platform deployment, design and implementation for government clients.
Using Azure Monitor in CSP, Nagios etc. to monitor and ensure the systems' security, integrity, & operational performance.
Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, and Azure SQL.
Effectively administered Azure cloud infrastructure, encompassing virtual machines, storage, and networking components.
Implemented security controls and maintained firewalls/NSGs in alignment with Cyber security standards
Delivered extensive Azure cloud-related support to users, resolving issues and enhancing system performance.
Implemented DevOps security controls, detection methods, and maintained firewalls/NSGs in alignment with Cyber security standards, ensuring a secure and compliant environment, including distributed file system namespace (DFSN) and DFSR
Performed post Go-Live support to ensure optimal functioning of systems and applications.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Designed and implemented infrastructure solutions for security, performance, and operations.
Implemented security baselines and security hardening in response to emerging threats.
Administered and managed Active Directory, Azure, and Endpoint Management.
Managed and supported Standard Operating Environment (SOE) for RBA, including application packaging, SCCM/Intune administration, Group Policy modification, SharePoint and Dynamics 365 CRM online etc.
Conducted root cause analysis and implemented system changes to prevent incident recurrence.
End-to-end ownership of secure CI/CD processes for various applications, including design, build, testing, and support Coordinated SOE rollouts and desktop environment changes following Change Management procedures.
Develop & enhance automated processes to release new solutions into the live environment in a timely, repeatable fashion
Managed Intune/SCCM for SOE image delivery, operating system deployment, and patch management.
Configured profiles to manage and configure the settings from centralized console using Intune.
Administered Group Policy objects, ensuring consistent configuration and compliance across the organization.
Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint online under M365
Ensure all requests (Service, Tasks and Incidents) are managed in a timely manner and using the service management tool
Management and administration of Intune/SCCM for the delivery of SOE images, operating system, application patching and process automations via PowerShell scripts. Application packaging and deployment with the endorsed by IT Security
Collaborating with IT Security for the SOE security hardening in line with PSPF, ISM, ACSC, CIS and DTA Protected pattern guidelines.
High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
Deploy vMX instance in Azure; Provision MX67W device in Meraki dashboard; Create a site-to-site VPN and configure it; Configure Azure VPN gateway in Azure portal; Establish VPN connection under "Security & SD-WAN" >"Site-to-site VPN"

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Cloud Support Engineer /Senior DevOps/Data Engineer/SharePoint/K2 Developer/DBA
Responsibilities:
Designed, implemented, and supported Azure cloud solutions for a variety of enterprise applications

Managed ExpressRoute setup and network design
Implemented security controls and maintained firewalls/NSGs in alignment with cybersecurity standards
Managed vendors to ensure delivery of cloud related services on time and within budget
Provided Azure cloud related support on application hosting/migration into Azure
Contributed to the successful delivery of new services in a technical capacity. Demonstrated expertise in AWS and Azure environments. Stayed updated on contemporary ICT management and governance frameworks.
Performed post-go-live support to ensure smooth operation of Azure cloud solutions
Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, and Azure SQL.
Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
Automated systems administration using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Implemented Azure Active Directory solutions to manage identities, enabling single sign-on and multifactor authentication.
Designed, implemented, & configured Azure and Microsoft 365 services, contributing to enhanced operational performance.
Contributed to change control processes to minimize implementation risks associated with system changes.
Managed the operation's infrastructure across Asia-Pacific and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with ADF, AKS, ansible etc.
Designed and implemented infrastructure solutions for security, performance, and operations.
Implemented security baselines and security hardening in response to emerging threats.
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0
Designed, implemented, and configured Azure and Microsoft 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint online under M365
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Conducted regular infrastructure assessments to identify configuration issues, security vulnerabilities, and potential risks.
Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Senior Cloud Administrator/Win/Linux Admin./DevOps/Data Engineer/SharePoint Consultant/DBA
Responsibilities:

- Provided expert-level support for Azure and Office 365 technologies, addressing configuration and performance issues.
- Assisted in automating administrative tasks using PowerShell scripts, improving operational efficiency.
- Defined successful new product requirements and implementation strategies.
- Applied proven ability to align cloud solutions with business objectives.
- Brought hands-on experience in managing and developing cloud platform services.
- Designed and implemented infrastructure solutions for security, performance, and operations.
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps with ADF, AKS, Ansible, Puppet etc.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2013, Power BI, Microsoft Flow and SharePoint 2013 online under M365
- Implemented security baselines and security hardening in response to emerging threats.
- Minimized implementation risk through change control methodologies.
- Successfully managed and maintained the Standard Operating Environment (SOE) for over 800 workstations, ensuring system stability and optimal performance.
- Implemented application packaging and deployment strategies, enhancing user experience and productivity.
- Executed SOE rollouts, adhering to Change Management processes and maintaining high system availability.
- Collaborated with cross-functional teams to identify root causes of incidents, contributing to proactive problem resolution.
- Designed and architected SCCM solutions for enterprise deployment and update management.
- Using Azure Monitor to monitored and troubleshooted Azure infrastructure and applications issues, ensuring high availability
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Centre etc.
- Automated systems administration using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.

- Update Active Directory Schema for Exchange Hybrid Deployment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council (SCC), QLD Australia**
Government Council: Clouds Administrator /Senior DevOps/Data Engineer/SharePoint Consultant
Responsibilities:

- Designed and implemented infrastructure solutions for security, performance, and operations.
- Administered and maintained Microsoft operating systems and domain administration.
- Played a crucial role in ensuring the stability and security of cloud infrastructure. Provided support to various teams, facilitating seamless project delivery. Demonstrated adaptability and advanced troubleshooting skills across technologies.
- Assisted in the management and optimization of multi-domain and multi-forest systems across diverse environments.
- Contributed to Azure hosting maintenance, including system configuration, security enhancement, and workload migration.
- Utilized PowerShell for automation, reducing manual administrative efforts.
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery and Azure SQL.
- Supported the implementation of Azure and Office 365 technologies, improving operational efficiency.
- Collaborated with the team to ensure compliance with security baselines and best practices.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2010, Power BI, Microsoft Flow and SharePoint 2010online under M365
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with ADF, AKS, ansible etc.
- Developed PowerShell scripts to automate routine tasks, resulting in time savings and increased operational efficiency.
- Supported and contributed to IT-related projects, ensuring technical excellence and alignment with strategic goals.
- Provided 3rd level desktop, systems, and network support, diagnosing and resolving complex technical issues.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Cloud Administrator (AD/AAD/InTune/SCCM/SCOM)/Senior Systems Engineer /DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Assisted in managing email and backup servers and their associated operating systems.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /ICT System /Linux Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided 3rd level technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor: Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planned, designed, and managed Active Directory with high-level support for AD migrations within and between AD forests.
- Installed and configured hardware, software, and networking equipment.
- Supported Office Online Specialist functions, resolving technical issues promptly.
- Managed and maintained the operation's infrastructure across the globe.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Infrastructure Engineer /Senior DevOps Engineer /Senior Systems Engineer /Senior Data Engineer /Clouds/Unix/Linux Administrator /ML/BI/ SharePoint/ Dynamics 365 CRM Consultant /EDW –Japan (7.5yrs) & Aust. (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Cloud Infrastructure Engineer/Sr. DevOps/Data Engineer/ML/BI/SharePoint/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- Managed migration and deployment projects for a Government Health business across the Country within the IT team.
- Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity. Overseeing, monitoring, and generating reports on Azure environments
- Ensuring the availability, security, and optimal performance of the Commission's environmental and cloud platforms
- Handling a mix of small-scale projects and day-to-day operational tasks with a specific focus on Azure and O365
- Configured, administered, migrated, and provided ongoing support for Azure and Microsoft 365 environments.
- Deliver Level 2/3 technical and systems support for Microsoft infrastructures on Azure (PaaS / IaaS), covering AAD, DNS, DHCP, Exchange 2019, Windows Server 2022, and Microsoft 365 applications such as Office 365, Power BI, MS Teams, SharePoint 2019, Power Apps, Power Automate and Dynamics 365 CRM online, including Networking and Monitoring
- Proactively identifying opportunities for improvement, incorporating new methods and best practices into their environments
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics 365 CRM, Power BI, Power Platform and SharePoint 2019 online
- Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
- Designed and implemented a unified application portal allowing users to access local-hosted, remote-hosted, and cloud-based systems. Developed and integrated custom reporting utilizing Python ReST to enable business analysis
- Performed and automated storage benchmark using Python and FIO
- Helped Operations teams by providing them with automated scripts for deployment activities using Ksh, Bash, and Python
- Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
- Implemented security baselines and security hardening measures, and ensured compliance with industry standards.
- Implemented end-to-end solutions for data ingestion and processing using Azure Synapse, ADF and Power BI.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using ADF/Synapse Pipelines, Spark Notebooks, SSIS, integrated PolyBase to query external data resources, and Copilots with M365 to increase Teams' productivities.
- Analyze data sources, types, and formats to define transformation, loading, and data model requirements.
- Implemented data transformation by utilizing Synapse Pipelines and Spark Notebooks. Write Spark SQL scripts or use PySpark for transformations. Implemented data cleaning, enrichment, and aggregation.
- Created & managed entities, business rules, & strategies in SQL Server 2022 Master Data Service Configuration Manager
- Configured and managed observability components using SCOM, Graphite, Grafana and ELK (Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Used Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senior Cloud Infrastructure Engineer/Sr. DevOps/Data Engineer /ML/SPoint/Power BI Developer /EDW
Responsibilities:

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9 /J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
Created FTP, SFTP scripts to automate Windows administrative tasks in Python and functions of Windows and Unix/Linux Operating Systems. Developed automation and reporting processes in Perl, Python, and Java
Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.
Managed migration and deployment projects for a Government Central Bank business across the globe within the IT team.
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity. Overseeing, monitoring, and generating reports on Azure environments
Ensuring the availability, security, and optimal performance of the Commission's environmental and cloud platforms
Handling a mix of small-scale projects and day-to-day operational tasks with a specific focus on Azure and O365
Configured, administered, migrated, and provided ongoing support for Azure and Microsoft 365 environments.
Conducted technical support and maintenance for their environments and applications.
Provide Level 2/3 technical and systems support for Microsoft infrastructures on Azure (PaaS / IaaS), covering AAD, DNS, DHCP, Exchange 2016, Windows Server 2016, and Microsoft 365 applications such as Office 365, Power BI, MS Teams, SharePoint 2016, Power Apps, Power Automate and Dynamics CRM 2016 online, including Networking and Monitoring.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
Perform operational security tasks, including audits and security testing, to enhance system protection.
Successfully managed projects, meeting scope, schedule, quality, and cost requirements.
Proactively identifying opportunities for improvement, incorporating new methods and best practices into their environments
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Maintain cloud infrastructure across AWS and Azure, optimizing resources and availability.
Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
Used Compliance Manager to implement security baselines and security hardening measures with compliance timelines. Worked to compliance timelines with internal and external auditors to conduct IT compliance testing, evidence gathering
Managed capacity management and extended SQL capacity by using Graph API to request specific data from the server.
End-to-end ownership of secure CI/CD processes for various applications, including design, build, testing, and support Coordinated SOE rollouts and desktop environment changes following Change Management procedures.
Managed Intune/SCCM/SCOM for SOE image delivery, operating system deployment, and patch management.
Using RBA's detection and response tools, primarily Splunk Security Information and Event Management (SIEM) platform, develop reporting and analytics that automate analysis of threat detection gaps.
Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
Interpreted trends and patterns; prepared data for prescriptive and predictive modeling.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Cloud Infrastructure Engineer/Sr. DevOps/Data/Systems Engineer /SharePoint/K2 Developer/DBA
Responsibilities:

Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint 2016 online
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
Managed migration and deployment projects for a Government Health business across the Country within the IT team.
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity. Overseeing, monitoring, and generating reports on Azure environments
Ensuring the availability, security, and optimal performance of the Commission's environmental and cloud platforms
Handling a mix of small-scale projects and day-to-day operational tasks with a specific focus on Azure and O365
Configured, administered, migrated, and provided ongoing support for Azure and Microsoft 365 environments.
Deliver Level 2/3 technical and systems support for Microsoft infrastructures on Azure (PaaS / IaaS), covering AAD, DNS, DHCP, Exchange 2019, Windows Server 2022, and Microsoft 365 applications such as Office 365, Power BI, MS Teams, SharePoint 2019, Microsoft Flow and Dynamics 365 CRM online, including networking and monitoring.
Proactively identifying opportunities for improvement, incorporating new methods and best practices into their environments
Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics 365 CRM, Power BI, Power Platform and SharePoint 2019 online
Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
Maintain cloud infrastructure across AWS and Azure, optimizing resources and availability.
Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
Automated systems administration using Python/Lambda/PowerShell, Bash, Javascript/Typescript, Ansible and Puppet
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0
Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Insurance Sector: Cloud Infrastructure Engineer/DevOps/Systems Engineer/Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity. Overseeing, monitoring, and generating reports on Azure environments
- Ensuring the availability, security, and optimal performance of the Commission's environmental and cloud platforms
- Handling a mix of small-scale projects and day-to-day operational tasks with a specific focus on Azure and O365
- Oversaw, monitored, and generated reports on Azure environments.
- Configured, administered, migrated, and provided ongoing support for Azure and Office 365 environments.
- Conducted technical support and maintenance for systems and applications.
- Identified and implemented opportunities for improvement and best practices.
- Configured, administered, migrated, and provided ongoing support for Azure and Microsoft 365 environments.

- Deliver Level 2/3 technical and systems support for Microsoft infrastructures on Azure (PaaS / IaaS), covering AAD, DNS, DHCP, Exchange 2010, Windows Server 2012R2, and Microsoft 365 applications such as Office 365, Power BI, MS Teams, SharePoint 2013, and Dynamics CRM 2013 online, including networking and monitoring.
- Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps with ADF, AKS, Ansible, Puppet etc.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Executed SOE rollouts, adhering to Change Management processes and maintaining high system availability.
- Designed and architected InTune/SCCM/SCOM solutions for enterprise deployment and update management.
- Using Azure Monitor to monitored and troubleshooted Azure infrastructure and applications issues, ensuring high availability
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Centre etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deployment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Infrastructure Engineer/DevOps/Systems Engineer/Data Engineer /SharePoint Consultant
Responsibilities:

- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2010, Power BI, Power Apps and SharePoint 2010 online
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
- Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
- Maintain cloud infrastructure across AWS and Azure, optimizing resources and availability.
- Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.
- Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
- Designed, implemented, configured, and supported Azure and Office 365 technology at an expert level, including services such as Azure Files and others, including Dynamics CRM 2010, Power BI, Power Apps and SharePoint 2010 online
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Cloud Infrastructure Engineer /Azure Administrator (AD/AAD/InTune/SCCM/SCOM)/DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.

- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Global Insurance: Project Manager /Infrastructure Engineer /Systems Engineer /Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Managed migration and deployment projects for a Global Insurance business within the IT team.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided 3rd level technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Infrastructure Engineer /Systems Engineer/Systems Administrator /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided Level 2/3 technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planned, designed, and managed Active Directory with high-level support for AD migrations within and between AD forests.
- Managed and maintained the operation's infrastructure across the globe.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Windows Administrator /Senior Systems Engineer/Senior DevOps Engineer/Senior Data Engineer /Clouds/Linux Administrator /ML/BI/ SharePoint/ Dynamics 365 CRM Consultant /EDW –Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Windows Administrator/Sr. Systems Engineer/Sr. DevOps/ML /SharePoint/D365 CRM Consultant /EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or Cloud via Terraform.
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics 365 CRM, Power BI, Power Platform and SharePoint 2019 online
- Managed Windows systems, conducted OS installations, configurations, and upgrades. Implemented group policies and security policies. Collaborated on technical solutions and supported ongoing projects.
- Conducted VMWare, Windows systems administration including Operating System installation, configuration, integration, hardening, performing OS upgrades, software and patch management and relevant documentation.
- Diagnosed issues with Active Directory and other identity and authentication services (eg Okta, Kerberos).
- Advised on the configuration of group policies and security policies for Windows servers, workstations, and applications.
- Automated and scripted tasks, implemented and managed storage, performed server password recovery.
- Developed and implemented technical solutions (MS Windows Implementations, Upgrades, Migrations and Decommission) and supported ongoing projects.
- Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
- Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.

- Implemented data transformation by utilizing Synapse Pipelines and Spark Notebooks. Write Spark SQL scripts or use PySpark for transformations. Implemented data cleaning, enrichment, and aggregation.
- Created & managed entities, business rules, & strategies in SQL Server 2022 Master Data Service Configuration Manager
- Configured and managed observability components using SCOM, Graphite, Grafana and ELK (Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
- Conducted thorough testing and implemented monitoring technologies, including SCOM and ELK stack, to monitor server health and activity. Deployed configuration changes and development releases, and provided support thereof.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Migrated on-premise applications & databases to Azure using Azure Migrate, Azure SQL, Power BI & Azure Import/Export
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Use R and Python in Jupyter Notebooks for performance of statistical analysis and fine-tuning using test results
- Executing train and retrain systems as necessary either use Scikit-learn, CNN, RNN or use AutoML toolkit (Google AutoML, Azure AutoML, H2O.ai) with Spark and Hadoop to process large-scale data sets for the machine learning pipeline, including feature engineering, model selection, and hyperparameter tuning. Distributed Computing and NVIDIA for GPU acceleration
- Use TensorFlow, PyTorch, or Keras with Python or PyCharm, Jupyter nootbook, Git, Docker, etc. to extend and improve existing internal ML libraries and frameworks to enhance overall efficiency.
- Used Spark MLlib to perform distributed machine learning on large-scale data sets and used Git to manage code
- Conducted statistical analysis and fine-tuning based on test results to improve model performance.
- Use MLOps for the entire lifecycle of machine learning applications, including development, deployment, and maintenance; Use ModelOps for the operations and management of machine learning models once they are deployed in a production env.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Windows Administrator/Sr. Systems Engineer/DevOps /ML/SharePoint/Power BI Developer /EDW
Responsibilities:

Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9 /J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
Administered and monitored both Windows and Linux-based systems, ensuring smooth operation of critical applications.
Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint 2016 online
Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
Led Windows Server implementations, upgrades, and migrations. Diagnosed issues with Active Directory and identity services. Provided automation and scripting expertise for server maintenance.
Conducted VMWare, Windows systems administration including Operating System installation, configuration, integration, hardening, performing OS upgrades, software and patch management and relevant documentation.
Diagnosed issues with Active Directory and other identity and authentication services (eg Okta, Kerberos).
Advised on the configuration of group policies and security policies for Windows servers, workstations, and applications.
Automated and scripted tasks, implemented and managed storage, performed server password recovery.
Developed and implemented technical solutions (MS Windows Implementations, Upgrades, Migrations and Decommission) and supported ongoing projects.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.
Setup and configure production environments for banking systems, ensuring minimal interruptions to products and services.
Deployed configuration changes, provided support to software development teams, & handled escalations &system support.
Managed capacity management and extended SQL capacity by using Graph API to request specific data from the server.
Configured and managed observability components using SCOM, Graphite, Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service) and other AWS services, to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Designed and developed an end-to-end data pipeline on Amazon S3 buckets, integrated with Snowflake, and orchestrated using AWS Lambda functions.
Developed pipelines in data lake EDW from scratch-built on S3 buckets, Snowflake, Lambda functions, using Python, Glue.
Created data pipelines between various source systems such as SAS, Oracle, Salesforce Guidewire and SQL Server to an enterprise Data Lake / warehouse complying with a STAR schema with multiple SCD's, CDC's and FACT tables
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
End-to-end ownership of secure CI/CD processes for various applications, including design, build, testing, and support Coordinated SOE rollouts and desktop environment changes following Change Management procedures.
Managed Intune/SCCM/SCOM for SOE image delivery, operating system deployment, and patch management.
Using RBA's detection and response tools, primarily Splunk Security Information and Event Management (SIEM) platform, develop reporting and analytics that automate analysis of threat detection gaps.
Interpreted trends and patterns; prepared data for prescriptive and predictive modeling.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Windows Administrator/Sr. Systems Engineer/Sr. DevOps/SharePoint/K2 Developer/DBA
Responsibilities:
Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint 2016 online
Executed VMWare and Windows systems administration. Oversaw OS integration, hardening, and patch management. Implemented storage solutions and managed server password recovery.
Diagnosed issues with Active Directory and other identity and authentication services (eg Okta, Kerberos).
Advised on the configuration of group policies and security policies for Windows servers, workstations, and applications.
Automated and scripted tasks, implemented and managed storage, performed server password recovery.
Developed and implemented technical solutions (MS Windows Implementations, Upgrades, Migrations and Decommission) and supported ongoing projects.
Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
Maintain cloud infrastructure across AWS and Azure, optimizing resources and availability.
Successfully managed projects, meeting scope, schedule, quality, and cost requirements.
Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.
Configured and managed observability components using SCOM, Graphite, Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
Automated systems administration using Python/Lambda/PowerShell, Bash, Javascript/Typescript, Ansible and Puppet
Designed, implemented, configured, and supported Azure and Office 365 technology at an expert level, including services such as Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint 2016 online
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Managed the IT infrastructure, including email, print, and backup servers and their associated operating systems, network and links either using Commvault Data Protection or Veeam Backup for Microsoft 365 v4.0
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.

Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.

Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.

- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Senior Windows Administrator/Clouds/Linux Admin./Sr. DevOps/SharePoint Consultant /DBA
Responsibilities:

- Provided expert-level support for Azure and Office 365 technologies, addressing configuration and performance issues.
- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2016, Power BI, Power Apps and SharePoint 2016 online
- Conducted VMWare, Windows systems administration including Operating System installation, configuration, integration, hardening, performing OS upgrades, software and patch management and relevant documentation.
- Contributed to complex IT landscapes, ensuring seamless operations across the enterprise. Provided technical planning for Windows OS deployments, upgrades, and migrations.
- Performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using Azure DevOps engineering with Azure SQL, Key Vault, Databricks, ADF, AKS, Ansible, and Jenkins.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Maintain cloud infrastructure across AWS and Azure, optimizing resources and availability.
- Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.
- Configured and managed observability components using SCOM, Graphite, Grafana and ELK(Elasticsearch, Logstash, and Kibana), enabling operational reporting and capacity management for the streaming and integration solutions.
- Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps with ADF, AKS, Ansible, Puppet etc.
- Configured Azure Active Directory to manage and secure identities including SSO, MFA, and Conditional Access.
- Designed, implemented, configured, and supported Azure and Office 365 technology at an expert level, including services such as Azure Files and others, including Dynamics CRM 2013, Power BI, Power Apps and SharePoint 2013 online
- Implemented application packaging and deployment strategies, enhancing user experience and productivity.
- Executed SOE rollouts, adhering to Change Management processes and maintaining high system availability.
- Designed and architected InTune/SCCM/SCOM solutions for enterprise deployment and update management.
- Using Azure Monitor to monitored and troubleshooted Azure infrastructure and applications issues, ensuring high availability
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Centre etc.
- Automated systems administration using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deployment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Senior Systems Engineer/Azure Administrator/Senior DevOps Engineer/SharePoint Consultant
Responsibilities:

- Designed, implemented, and configured Azure and Office 365 technology at an expert level, encompassing services like Azure Files and others, including Dynamics CRM 2010, Power BI, Power Apps and SharePoint 2010 online
- Demonstrated expertise in cloud platforms, including Azure and AWS. Managed stakeholder relationships, emphasizing effective communication and issue management. Presented technical solutions for MS Windows implementations.
- Automated tasks and scripts using Python/Lambda/Javascript/Typescript, PowerShell, Bash, Ansible and Puppet
- Using Azure Monitor in CSP, Nagios etc. to monitor and maintain infrastructure and vendors to meet changing growth, security, performance and availability requirements and ensure the systems' security, integrity, & operational performance.
- Provide expert-level technical support to end-users, resolving system performance, software, and hardware issues.
- Implemented security baselines and security hardening measures, and ensured compliance with industry standards.
- Managed and optimized Azure AD for identity management and security, including SSO, MFA, and Conditional Access.

- Designed, implemented, configured, and supported Azure and Office 365 technology at an expert level, including services such as Azure Files and others, including Dynamics CRM 2010, Power BI, Power Apps and SharePoint 2010 online
- Managed the operation's infrastructure across the country and performed hands-on maintenance, configuration, security, backup, and migration of workloads on self-managed Azure hosting using DevOps engineering with ADF, AKS, ansible etc.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Azure Administrator (AD/AAD/InTune/SCCM/SCOM)/DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /ICT System /Linux Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided 3rd level technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planned, designed, and managed Active Directory with high-level support for AD migrations within and between AD forests.
- Managed and maintained the operation's infrastructure across the globe.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.

- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Infrastructure Azure Consultant /Senior DevOps Engineer /Senior Data Engineer /Python Developer /EDW /Clouds/Linux Administrator /AI //ML /BI /SharePoint /Dynamics 365 CRM Consultant /EDW/-Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior Infrastructure Azure Consultant/Sr. DevOps/Data Engineer/AI/ML/SPoint/D365 CRM Consultant/EDW
Responsibilities:

- Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
- Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
- Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
- As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
- Lead continuous improvement, development, and refinement of Cloud Service catalogue requirements.
- Collaborate with technical & non-technical stakeholders to design & implement Cloud Services, producing relevant patterns.
- Showcase proficiency in Cloud Architecture and technology, emphasizing strong automation capabilities across multiple Cloud platforms. Utilize coding skills in writing and modifying Terraform templates for automation purposes.
- Leverage World of Warcraft (WoW) as the main provisioning technology, particularly focusing on Terraform integration.
- Designed and deployed a highly scalable and secure Azure environment for a large e-commerce platform.
- Automated infrastructure provisioning and configuration using Terraform and Ansible, achieving a 50% reduction in deployment time. Created and maintained documentation for cloud infrastructure and automation scripts.
- Managed Azure resources, including VMs, storage accounts, & network configurations, ensuring optimal performance and security. Collaborated with developers and DevOps engineers to integrate Azure services into CI/CD pipelines.
- Utilised Azure ADF (Azure Data Factory) and Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation & deployment processes within the Azure ecosystem.
- Defined necessary build steps, like code compilation, running tests, and generating artifacts.
- World of Warcraft (WoW) and Terraform synergize seamlessly to streamline infrastructure provisioning, combining the immersive world of gaming with the efficiency of Terraform's infrastructure-as-code capabilities.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Migrated on-premises workloads to Azure using Azure Migrate and Azure Hybrid Cloud Manager.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Implemented and maintained AKS clusters, guaranteeing high availability and reliability of the production environment.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, with a focus on M365 environments (O365, MS Teams, SharePoint 2019, Dynamics 365 CRM, Power BI, Power Automate, etc.).
- Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, etc.
- Automated systems administration using Python/Jenkins /Lambda/Javascript/ PowerShell, Bash, Ansible and Puppet
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Monitor and troubleshoot Azure services using Splunk, Data Dog and Azure Network Monitor
- Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot for national users reported issues via Manage Engine Ticketing.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Interpreted trends and patterns; prepared data for prescriptive and predictive modeling.
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV (Open Source Computer Vision).
- Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevSecOps and Configuration-As-Code)
- Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
- Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau

- Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior Infrastructure Azure Consultant/Sr. DevOps/Data Engineer/Liunx Admin./ML/SP Consultant/EDW
Responsibilities:

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL
Collaborated with cross-functional teams to design, implement, and optimize CI/CD pipelines using Jenkins and Bamboo,
Demonstrate expertise in stakeholder management, especially with both technical and non-technical audiences.
Automated infrastructure provisioning & configuration using Python, Jenkins, Ansible, Terraform, Puppet/PowerShell etc.
Championed the adoption of best practice DevOps methodologies, enhancing continuous integration, deployment, and software delivery. Worked on multi-zone load balancing, redundancy, and system software updates.
Resolved complex issues, conducted post-release reviews, &contributed to centralized CI/CD suite using Atlassian products
Migrated on-premises workloads to Azure using Azure Migrate and Azure Hybrid Cloud Manager.
Implemented Azure cost optimization strategies, resulting in a 20% reduction in cloud spending.
Managed Azure Active Directory and Azure AD Connect for identity and access management.
Monitored and troubleshoot Azure resources to ensure high availability and performance.
Provided technical support and guidance to internal teams on Azure services and best practices.
Automated complex builds and deployments across AWS and Azure clouds, resulting in improved scalability and resource utilization. Managed infrastructure as code using Terraform to enabling seamless provisioning of resources
Collaborated with development teams to ensure continuous compliance with security policies and build secure applications.
Extensive expertise in programming languages such as Java, Python, Go, C# and PowerShell/Shell scripting languages,
Leveraging AWS DevOps practices and tools, including Amazon EKS (Elastic Kubernetes Service), IaC tools: Terraform, CloudFormation, CDK (Cloud Developer Kit), etc., to enable the seamless development and deployment of applications within a cloud environment while automating and optimizing software development and IT operations.
Developed and maintained AWS infrastructure components using Terraform and IAM for enhanced security and automation.
Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
Configured and administered Azure resources, including virtual machines, storage accounts, app services, load balancers, WAF, and firewalls. Employed GitLab, GitHub Actions, & Bitbucket for effective source code management & version control.
Automated the deployment and configuration of Windows servers and Linux servers using Ansible, Puppet and Chef
Collaborated with developers and testers to implement continuous integration and continuous testing using Azure DevOps and Selenium; Used Nagios, ELK and Prometheus for logging and monitoring
Provided technical support for Windows and Linux servers, ensuring optimal functionality and performance.
High proficiency in Active Directory, Azure AD, Managed the installation, configuration, and testing of new and existing hardware, software, and WAN/LAN technologies using (IaC) tools: Azure Resource Manager(ARM) templates, Teraform or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code
High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate WAF, Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
Deploy vMX instance in Azure; Provision MX67W device in Meraki dashboard; Create a site-to-site VPN and configure it; Configure Azure VPN gateway in Azure portal; Establish VPN connection under “Security & SD-WAN” >”Site-to-site VPN”
Either using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
Developed and maintained PowerShell scripts and automation tools for efficient Azure infrastructure management.
Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).
Developed and deployed end-to-end ML solutions for clients using DataRobot's automated ML platform and best practice methodologies. (Agile, DevOps and Configuration-As-Code)
Conducted experiments and evaluations to measure model performance and accuracy using A/B testing and other methods.
Presented results and insights to stakeholders using data visualization tools like Matplotlib, Seaborn, Power BI or Tableau
Delivered optimal ML solutions either using Python, R with SQL 2022 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Execute a comprehensive Azure Data Engineer projects from start to finish, employing tools such as Data Factory, Data Lake Gen2, Azure Databricks, Azure Synapse Analytics, and Dashboard utilities like Power BI, Looker Studio, or Tableau.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Infrastructure Azure Consultant/Sr. DevOps/Data Engineer /ML/SharePoint /K2 Developer /DBA
Responsibilities:

Architect and implement the 3 Layers (Batch, Speed and Service) Lambda Architectures either in AWS, GCP or Azure.
Designed and deployed secure landing zones for cloud workloads using Azure Blueprints in both Windows-based systems and Linux based trading systems using Azure Ubuntu Container where can connect to AWS or GCP Cloud via Terraform.
Implemented DevSecOps security best practices and detection methods, ensuring a secure and compliant environment.
Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
As an experienced Azure DevOps engineer, I crafted cloud infrastructure using Infrastructure as Code (IaC) in Terraform. Provisioned Azure resources in Terraform, defining IaC with DDL language, and developed Terraform Modules using Python programming for streamlined resource provisioning. Producing efficient, high-quality code/solutions for projects.
Designed and set up end-to-end enterprise CI/CD pipelines using Jenkins or GitHub Actions for optimizing deployment processes in diverse enterprise applications.
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
Utilised Azure ADF (Azure Data Factory) & Databricks as integral components for setting up end-to-end CI/CD pipelines in the Azure environment. Contributing to the orchestration, automation, & deployment processes within the Azure ecosystem.
Defined necessary build steps, like code compilation, running tests, and generating artifacts.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Successfully migrated on-premises workloads to Azure Cloud using various methods, including Azure Migrate and Azure Site Recovery for infrastructure and application migration, Azure SQL for database migration, and Power BI SQL Server Data Connector for seamless data connectivity.
Implemented and maintained AKS clusters, guaranteeing high availability and reliability of the production environment.
Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365 CRM/Power BI/Power Automate etc.)
Azure/M365 administration either using (IaC) tools (Azure Resource Manager(ARM) or PowerShell Desired State Configuration (DSC) to define the Infrastructure as configuration code or manually by PowerShell or CML scripts.
Utilized bicep and ARM templates to provision and configure Azure resources such as App Service, Functions, Logic Apps, Storage, Key Vault, etc.
Communicated effectively with business stakeholders and project managers to understand requirements and deliver solutions. Worked in a hybrid cloud environment with zero ClickOps.
Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus
Configured and optimized Azure storage accounts, SQL databases and Cosmos DB
Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
Provided 3rd level desktop, systems, and network support, resolving complex technical issues to minimize downtime.
Conducted server and security audits, ensuring compliance with security policies and implementing necessary remediation measures. Used JIRA, Bugzilla and Remedy for bug tracking tools
Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including Oracle, MySQL and MongoDB for databases
Designed and implemented Windows-based cloud solutions using Azure services, including Azure Resource Manager (ARM), Azure Policy, Azure Security Center, Azure Active Directory, Azure SQL and Azure Networking.
Developed PowerShell scripts and automation tools for managing and maintaining Azure infrastructure.

- Maintained and optimized Active Directory, Azure, MS Defender Suite, and M365 Environments for efficient user management and data security.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Setting up SharePoint 2013 and its development environment by installing and configuring WFE-Tier two clustered web servers, APS-Tier one application server & SQL-Tier two clustered database servers, Office/Napa Developer Tools etc.
Using Azure Open AI Service and Azure Open AI Studio to train Large Language Model with scikit-learn and TensorFlow.
Played a key role in delivering Cloud Native software solutions, including machine learning and artificial intelligence applications. Ensured the maturity of engineering practices across the business.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: Infrastructure Azure Consultant/DevOps/Data Engineer/Linux Administrator /SharePoint Consultant
Responsibilities:

- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, with a focus on M365 environments (O365, MS Teams, SharePoint 2013, Dynamics CRM 2013, Power BI, Power Automate, etc.).
- Designed and built cloud-based infrastructure solutions for various clients using AWS and Azure services.
- Implemented secure CI/CD pipelines using Terraform, Vault, Artifactory, and GitLab for various applications.
- Automated systems administration using Python/Jenkins /Lambda/Javascript/ PowerShell, Bash, Ansible and Puppet
- Deployed and managed container-based applications using Docker, Kubernetes, and Helm Charts.
- Migrated on-premise applications to the cloud using AWS Migration Hub and Azure Migrate.
- Installed, configured, and maintained Windows servers and Linux servers for various business applications
- Administered Windows servers using PowerShell and IIS; Used Vagrant, VMware and Hyper-V for virtualization platforms
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Troubleshot and resolved issues related to hardware, software, network, security, and performance
- Implemented and managed networking solutions, including Cisco, Meraki SD-WAN, FortiGate Firewalls and WAF to enhance network performance and protect against cyber threats.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Configuring Office 365 DirSync and Active Directory/LDAP using Microsoft Software Deployment Toolkit (MDT2013)
- Install and configure Exchange 2013 Mail Servers with High Availability (HA) in Data Access Group (DAG) features, including configure Send/Receive connector, Malware Protection Settings in Administrator Center etc.
- Automating systems administration by various automation scripting like Python, PowerShell, Bash, Puppet, etc.
- Configure Automatic Updates either by local Group Policy or manually by group policies settings with GPUpdate /force
- Install and configure SC Orchestrator2012 for Exchange Automation by Orchestration Console and PowerShell scripts.
- Update Active Directory Schema for Exchange Hybrid Deplolyment by PowerShell cmd>setup.exe /prepareAD etc.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Azure DevOps Engineer/Azure/M365 Administrator (AAD/Inutune /SCCM) /SharePoint Consultant
Responsibilities:

- Led the setup of Azure-focused cloud landing zones, following best practices.
- Collaborated with Data Engineers to integrate DevOps practices into the newly formed team.
- Ensured automation, scalability, and reliability in the Cloud and DevOps ecosystem.
- Built and maintained cloud infrastructure using IaC in Terraform. Designed and set up enterprise CI/CD pipelines
- Configured and managed DNS, DHCP, Active Directory, Exchange Server, SharePoint Server, SQL Server, and IIS on Windows servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Configuring Office 365 DirSync and SSO with AD FS using federated Identity Microsoft Software Deployment Toolkit 2012
- Installing and configuring SC Data Protection Manager (DPM2012) for Exchange 2010 SP2
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /Senior Windows /Unix/Linux Systems Administrator /DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Design, implement, configure and administer Infrastructure solutions (Iaas, Saas, PaaS etc.)

- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008/BDD2007Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Windows/Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Assisted in the installation and configuration of hardware and software solutions.
- Conducted testing and troubleshooting of IT systems to ensure optimal performance.
- Assisted in managing network infrastructure and security measures.
- Provided technical support to end-users, resolving issues in a timely manner.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Technical supported for the installation, configuration, and testing of hardware and software solutions.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA

Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

- Design, implementation, and optimization of a high-performance, scalable physical ML Kubernetes cluster supporting ML workloads on Nvidia DGX A100 systems, includes a pre-installed with DGX OS, a customized Ubuntu distribution with additional NVIDIA Software Stack specifically optimized for running AI and HPC workloads via Containers for DL training & inference, HPC, analytics, and industry-specific applications.
- vs
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.

- Designed and deployed secure landing zones for AWS cloud workloads using Terraform. Ensured compatibility with both Windows and Linux-based systems. Implemented security measures: Security Groups, IAM roles, etc. in Security/main.tf
- vs
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.

- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine tasks on the physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Using Harrison platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- vs
- 1. Harrison Platform: Medical AI dev made easy. Automate with robots building robots, from AWS to Azure and GCP.
- 2. Pre-built tools & collab flow: Streamline the AI factory for med devices. From data to docs, deploy & monitor with ease.
- 3. Common processes, shared success: Harrison unifies your medical AI journey. Prep, train, validate, deploy & watch - all in one place.

- Design and develop Embodied AI for DGX A100s systems so that it equipped with limbs and senses to interact with patients directly to pick up riche data from the patients via sensors for diagnosis and making medical decisions.
- Built secure and scalable Azure landing zones using Azure Blueprints and Terraform, ensuring compatibility for Windows and Linux workloads through proactive implementation of robust security measures, including Security Groups, IAM roles, and strategic configuration within the Security/main.tf file.

Westpac - **DevOps Engineer – Lanson – 10/6/2024**

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior DevOps Engineer/Java/Python Developer/Snr Data Engineer/Integration Solution Architect/Snr Systems Engineer/ Salesforce/Mulesoft/AI/ML/DL/BI/SharePoint /D365 CRM Developer /EDW -Japan (7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Java/Python /AI/ML/DL/BI/SharePoint/D365 CRM Developer /Snr Data Engineer /EDW
Responsibilities:

- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Design and develop software solutions, ensuring adherence to relevant standards and processes.
- Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.
- Full-stack engineer skilled in React/TypeScript for frontend development, Java/APIs for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Collaborate closely with development and QA teams to ensure efficient application development and deployment.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation templates or Terraform. Utilized AWS Control Tower for centralized management of multiple AWS accounts, enabling parallel processing for various projects.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Design and implement secure and scalable CI/CD pipelines using AWS CodeBuild, RDS & S3 with Bitbucket repositories.
- Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Utilized Terraform's merge function to consolidate configuration data across various maps and objects, promoting efficient and maintainable infrastructure code. Experienced in Agile and able to promote Agile practices across IT teams.
- **Ensured high-quality, secure code**: Utilized SonarQube for code analysis, JFrog and Artifactory for artifact management, and Fortify Scan for security vulnerability detection, throughout the development process.
- **Integration Solution Design with AWS CodeBuild (CI/CD)**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Lambda functions, or Hybrid).
- Leverage AWS CodePipeline and AWS CodeDeploy to identify bottlenecks and inefficiencies in the CI/CD pipeline, and proactively implement improvements with AWS Lambda functions.
- **Supercharge automation!** Combine custom logic in Python or Go with Workato's pre-built integrations to streamline even the most complex business workflows. Maintain DevOps Maturity Plan, Standardised Process across squads.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, or RNN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior DevOps Engineer/Mulesoft/Java/Python/R/AI/ML/DL/BI/SharePoint Developer /Snr Data Engineer
Responsibilities:

Hybrid Cloud Architect: Established secure connections between AWS, GCP, and Azure cloud environments for data transfer. Utilized infrastructure-as-code tools (Terraform/Ansible, or Puppet) for automated and consistent provisioning.

Built secure and scalable Azure Landing Zones using Infrastructure as Code (IaC) with Azure Blueprints, ARM templates, or Terraform. Utilized Azure Lighthouse for centralized governance and management of multiple Azure subscriptions, enabling efficient parallel processing for various projects.

Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with RBA's enterprise services layer, databases, and host systems.

Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets

Developed Mulesoft Anypoint APIs and Dashboards to provide real-time insights and monitoring of trading activities.

Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.

Implemented automated security scans (SAST, DAST) into the GitLab CI pipeline, enforced granular access controls using roles, and managed secrets securely. Regular audits ensured compliance with industry standards like PCI-DSS or SOC 2.

- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
  - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.

Automation and CI/CD: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.

Streamlined Terraform configuration management through the strategic use of the merge function to consolidate settings from diverse sources.

Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.

Data Pipeline Automation: Built and orchestrated data pipelines with Azure Data Factory, Azure Event Hubs, Azure Data Factory (Orchestration) for efficient data movement & transformation. Normalized data in Azure Cosmos DB & Azure SQL Database for optimal performance.

Cloud-Agnostic Data Platform: Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.

Automated Kubernetes Deployments: Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.

CI/CD for Integration Solutions: Designed integration flows with GCP Cloud Build (CI/CD), defining data flow, transformations, and choosing orchestration platforms (MuleSoft, Cloud Functions, or Hybrid).

Data Transformation Expertise: Employed MuleSoft Anypoint Platform (API, Exchange, DataWeave) for complex data transformations and DBT for standardization and basic transformations.

Design and implement secure and scalable CI/CD pipelines using Azure DevOps, Azure SQL Database & Azure Blob Storage with Bitbucket repositories. Maintain DevOps Maturity Plan, Standardised Process across squads.

Implement effective branching strategies utilizing Gitflow for seamless deployments and continuous application monitoring.

Optimized CI/CD pipelines with DevOps tools: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.

Production-Ready Machine Learning Models: Engineered and deployed production-ready ML models for real-world impact using Python and MLOps tools.

CI/CD and DevSecOps Automation: Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.

AI-Powered Pattern Recognition: Used Artificial Neural Networks to model complex patterns and forecast potential issues.

High-Performance Kubernetes Cluster: Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS for running containerized AI, HPC, and industry-specific workloads.

Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Utilize MLOps for the entire machine learning application lifecycle, encompassing development, deployment, and maintenance. Leverage ModelOps specifically for the operation & management of deployed ML models in production env.

Delivered optimal ML solutions either using Python, R with SQL 2016 Machine Learning feature on Azure or run PyTorch based ML applications with BigQuery on GCP or doing PyTorch programming with TensorFlow and scikit-learn on AWS.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Integration Solution Architect/Snr DevOps Engineer /Java/Salesforce/Mulesoft/SharePoint /K2 Developer
Responsibilities:

Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.

Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.

Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.

Designed and maintained robust CI/CD pipelines using platform-specific tools:

36. **Azure: Azure DevOps**
37. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
38. **GCP: Cloud Build/Cloud Deploy**

Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:

Python with Pandas and Airflow to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or

Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.

Developed and deployed secure, RESTful Python APIs for web applications using frameworks like Flask or FastAPI. These APIs can integrate with MLOps pipelines for data access, model predictions, or other machine learning tasks.

Applied Async programming using Python Asyncio to optimize system performance.

Leveraged frameworks like Flask and Django to build robust and scalable components.

Implemented API endpoints for data exchange using Python, Django, Flask and FastAPI

Collaborate closely with development and QA teams to ensure efficient application development and deployment.

Develop Mule Flows: Design flows in Anypoint Studio to act as intermediaries between systems.

Utilize Connectors: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.

Configure Connections: Provide access credentials, URLs, and other details for each system within the connectors.

Design Integration Logic: Transform data, handle errors, and implement security measures within the Mule flow.

Deploy and Monitor: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.

Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

Integration Solution Design with Azure DevOps CI/CD: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).

Develop and Build: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

Deployment Stages: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).

Developed and deployed a MuleSoft application: Used Anypoint Studio to design, build, and package the application. Configured deployment settings for different environments (Dev, UAT, Production) and managed resource allocation.

Secured API Access for Student Enrollment: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of student data.

CI/CD and DevSecOps Automation (AWS): Built and maintained CI/CD pipelines for various applications, employing DevSecOps practices using industry-standard tools (CodeBuild/GitHub) & following best practices for secure development

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Data Transformation Expertise (Azure):** Employed Azure Logic Apps (Connectors, Logic Flows) for complex data transformations, leveraging Azure Data Factory (ADF) for standardization and basic transformations.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or on-premises environment.

Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: DevOps Engineer/Data Engineer /Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:

- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- **Automation:** Created and maintained automated scripts using Python or Perl programming, along with Bash or Shell scripting, to streamline development and operational processes.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- **Collaboration:** Work closely with software developers, QA engineers, and system administrators to ensure smooth deployment and operation of applications.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within an AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- **Troubleshooting:** Diagnose and resolve technical issues related to CI/CD pipelines, containerization, and infrastructure.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack, and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon ECR and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Design and develop cloud-native applications using AWS services and Python
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using API Gateway
- Deploy and maintain applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.

- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Data Engineer -

Blow CV is for Macquarie bank via Lewis in Bluefin $850d

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter

Senior Data Engineer (DATA ANALYTICS - Data Engineer - STL Data Engineer) /AI/ML/DL/SP/D365 CRM Consultant/EDW

-Japan (7.5yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality Safety and Commission (ACQSC), Sydney

Gov. Health: Sr. Data Engineer(DATA ANALYTICS-Data Engineer-STL Data Engineer)/AI/ML/SP/D365 CRM Consultant/EDW

Responsibilities:

- Designed and developed big data solutions from scratch within both data lake and Lambda architectures, integrating S3, Snowflake, Lambda functions, Redshift, Python, PySpark, Pydantic, and Glue on AWS EMR.
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Designed and implemented real-time analytics pipelines using Spark (Python) to optimize operational efficiency across the organization or using the 3 Layers (Batch, Speed and Service) Lambda DWH Architectures either in AWS, GCP or Azure.
- Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
- Leveraged the distributed SQL power of Presto or Impala seamlessly within AWS architecture, crafted lightning-fast queries across S3 data lakes and Redshift data warehouses, empowering real-time insights and agile decision-making
- Developed and maintained data pipelines and ETL processes using Redshift and AWS services to ensure data quality and timely delivery for stakeholders. Leveraged big data technologies such as Spark & Hive to handle large datasets effectively.
- Experienced with AWS including any or all of EC2, S3, IAM, EKS, RDS, ELB, Cloud Formation, SageMaker, VPC & VPS.
- Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.
- Championed the adoption of CI/CD tools for DevOps practices, automating deployments and reducing time-to-market.
- Leveraged mastery over SQL, Python, Java, and Node.JS to develop robust data engineering solutions.
- Ensured the highest commitment to quality and customer value, resulting in improved data analytics approaches and tools.
- Either using orchestration tool of Airflow to author, schedule, and monitor workflows or using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

- Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services like EC2, S3, IAM, EKS, ELB, Cloud Formation, RDS, SageMaker, Glue, Lambda, VPC, VPS etc. using Jenkins, Ansible, Chef & Puppet.
- Proficiently queried data from various sources, both structured and semi-structured, to support data-driven decision-making.
- Design and implement data models and reporting structures using Snowflake and Tableau.
- Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 4.3, Python 3.11, PSPP, JMP, JASP, EViews, StatCrunch, etc.
- Developed interactive dashboards and reports using Microsoft BI stacks (Power BI, SQL Server) to provide actionable insights to various departments, providing valuable insights to support decision-making.
- Drawing meaningful conclusions and provide actionable insights, making weekly Data Integrity Check Report for the Operations and Decision-Making groups, carefully deleting the decisions from duplicated /cancelled activities in D365 CRM.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Either develop Python scripts to call APIs or use PowerShell or Git for automated products development, release & support.
- Conducted data extraction, modeling, and statistical analysis to facilitate data-driven decision-making for ongoing and future projects. Integrated the pipeline with GitHub or Azure Repository source code either by Python, Scalar, or R to train LLM
- Providing Technical support for Better Business Program applications covering SharePoint/Dynamics 365 CRM, TIBCO servers and associated SQL Databases troubleshooting for national users reported issues via Manage Engine Ticketing.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney

Gov. Central Bank: Sr. Data Engineer(DATA ANALYTICS-STL Data Engineer)/AI/ML/DL/SP/Power BI Developer /EDW

Responsibilities:

Designed and developed big data solutions from scratch within both data lake and Lambda architectures, integrating S3, Snowflake, Lambda functions, Redshift, Python, PySpark, Pydantic, and Glue on AWS EMR.

Built and maintained data pipelines for on-premise data warehouse, leveraging Presto/Impala for ad-hoc querying and reporting. Successfully migrated on-premise data platform to AWS, ensuring seamless data availability and scalability.

Design and deploy modern AWS Lambda Architecture comprising Batch layer with Hadoop HDFS and MapReduce, Speed layer with Spark in-memory access and RDD, and Service layer employing Spark Streaming for high-speed real-time SNS application notifications via direct write queries on datasets within the Spark cluster using Spark Applications and Spark SQL

Leveraged the distributed SQL power of Presto or Impala seamlessly within AWS architecture, crafted lightning-fast queries across S3 data lakes and Redshift data warehouses, empowering real-time insights and agile decision-making

Developed and maintained data pipelines and ETL processes using Redshift and AWS services to ensure data quality and timely delivery for stakeholders. Leveraged big data technologies such as Spark & Hive to handle large datasets effectively.

Experienced with AWS including any or all of EC2, S3, IAM, EKS, RDS, ELB, Cloud Formation, SageMaker, VPC & VPS.

Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL processes, achieving streamlined data integration.

Championed the adoption of CI/CD tools for DevOps practices, automating deployments and reducing time-to-market.

- Leveraged mastery over SQL, Python, Java, and Node.JS to develop robust data engineering solutions.
- Ensured the highest commitment to quality and customer value, resulting in improved data analytics approaches and tools.
- Either using orchestration tool of Airflow to author, schedule, and monitor workflows or using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 systems with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, & analytics.
- Using the Harrison Platform to develop AI-powered medical devices which use machine learning algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.

Performing Machine Building the Machine(MBM) tasks on physical ML Training Cluster and the Kubernetes based software stack it hosts, just like an automated factory where robots build other robots driven by automation, MLOps capabilities with Pre-built components & Collaboration tools to streamline & optimize AI development process either on AWS, Azure or GCP

Successfully orchestrated and managed AWS data services including Redshift, EMR, Athena, and S3, while designing and implementing efficient data pipelines using Airflow. Additionally, adeptly configured and managed AWS services like EC2, S3, IAM, EKS, ELB, Cloud Formation, RDS, SageMaker, Glue, Lambda, VPC, VPS using Jenkins, Ansible, Chef & Puppet.

Developed and implemented data quality checks and monitoring systems to guarantee data integrity and accuracy.

Collaborated with cross-functional teams (analysts, developers, IT) to support data-related projects and initiatives.

Architect and implement MS Analytics Platform System(APS) to combine Massively Parallel Processing (MPP) SQL Server DWH with Hadoop Distribution Insight (HDInsight) to support Azure /Power BI stream analytics for digital transformation.

Merged the marketing data environments of multiple separate banks, leading to a more unified, efficient data system.

Implemented ETL solutions using Informatica, optimizing data processing efficiency.

Either using orchestration tool of Airflow to author, schedule, and monitor workflows or using BMC Control-M v8.0 to manage, schedule multiple data transformation jobs or using Teradata Data Mover

Developed and executed source-to-target mappings between systems.

Migrated IBM TM1, PostgreSQL, Tableau and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.

Collaborated with cross-functional teams to develop a highly efficient Data Lake/Data Warehouse environment.

Leveraged AWS Services to enhance cloud native software delivery and data engineering processes.

Developed and maintained ETL processes using SQL Server, optimizing data extraction and transformation procedures.

Played a pivotal role in migrating data infrastructure to the Google Cloud Platform, leading to improved scalability and performance. Developed and optimized SQL queries for data extraction and analysis.

Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 4.0, Python 3.7, IBM SPSS , SAS, Stata, MATLAB, ATLAS.ti, etc

Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define continuous integration & continuous deployment (CI/CD) workflows or in Snowflake's Data Cloud.

Automated data product support processes through R, Python, PowerShell 7.3 scripting, improving efficiency in access requests, data refreshes, and releases.

Conducted data extraction, modeling, and statistical analysis to facilitate data-driven decision-making for ongoing and future projects using Power BI, Databricks, Data Factory, Data Lake Analytics, Synapse Pipeline and Analytics in Azure

- Applied 7 statistical methods, especially Predictive Analysis with Azure Machine Learning (AML) , AI Visuals to define the likelihood of future outcome, behavior & trend through analyzing data and deriving well-reasoned & defensible conclusions.

Create a training pipeline, Setup an Azure DevOps pipeline to automate the training process. Define a YAML pipeline that includes the necessary steps to deploy infrastructure, prepare data, launch the training job, and monitor its progress.

- Design and develop Bank analysis dashboards either using SharePoint Dashboard Designer or Azure /Power BI Pro with clickable drilldown on charts feature for visually analysis live database to deliver daily, weekly & monthly dashboards.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)

Gov. University: Senior Data Engineer (DATA ANALYTICS - Data Engineer - STL Data Engineer) /AI/ML/DL /K2 Developer
Responsibilities:

Architected and implemented 3 layers modern lambda architecture on Azure comprising batch layer with HDInsight, speed layer with PySpark and unified service layer with MS Analytics Platform System (APS) to combine Massively Parallel Processing (MPP) by Databricks SQL native query support or data views for Azure Stream Analytics real-time analysis.
Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
Implemented and maintained Azure Kubernetes Service (AKS) clusters for various microservices applications using Terraform and Helm Charts. Implemented big data technologies like Hadoop to process and analyze large datasets.
Designed and implemented real-time analytics pipelines using Spark (Python) to optimize operational efficiency across the organization or using the 3 Layers (Batch, Speed and Service) Lambda DWH Architectures either in AWS, GCP or Azure.
Built robust and secure AWS landing zones with Terraform, ensuring cross-platform compatibility for Windows and Linux workloads. Proactively implemented comprehensive security measures, including Security Groups and IAM roles, strategically defined within the Security/main.tf configuration file.
Leveraged the distributed SQL power of Presto or Impala seamlessly within AWS architecture, crafted lightning-fast queries across S3 data lakes and Redshift data warehouses, empowering real-time insights and agile decision-making
Developed and maintained data pipelines and ETL processes using Redshift and AWS services to ensure data quality and timely delivery for stakeholders. Leveraged big data technologies such as Spark & Hive to handle large datasets effectively.
Experienced with AWS including any or all of EC2, S3, IAM, EKS, RDS, ELB, Cloud Formation, SageMaker, VPC & VPS.
Collaborated with business analysts and product owners to translate data requirements into actionable solutions, driving data-driven decision-making. Utilized Talend for ETL/ELT processes, achieving streamlined data integration.
Championed the adoption of CI/CD tools for DevOps practices, automating deployments and reducing time-to-market.
Leveraged mastery over SQL, Python, Java, and Node.JS to develop robust data engineering solutions.
Developed and implemented data security policies to safeguard sensitive information and ensured compliance with evolving privacy guidelines. Implemented security measures, managing user access, roles, and permissions.
Manage and support existing and new platforms, emphasizing CI/CD pipelines, release pipelines, and DevOps practices.
Collaborate with key stakeholders, including SMEs, product owners, business analysts, and users across different locations and time zones to define and create solutions.
Contribute to maintaining and continuously improving the stability of platforms, supporting a DevOps culture and practices.
Demonstrate a strong awareness of risk, identifying, escalating, and remediating issues to ensure a robust data engineering environment. Supported the development and implementation of data warehousing solutions.
Performed data analysis and reporting using SQL, Excel, and Tableau, uncovering insights to inform business decisions.
Designed and implemented data quality checks and data cleaning procedures to improve data accuracy and usability.
Applied advanced Python programming to analyze large datasets, resulting in a significant improvement on identification.
Managed end-to-end data integration projects, from requirements gathering to deployment.
Utilized DataStage for ETL processes, achieving high performance and scalability.
Implemented big data technologies like Hadoop to process and analyze large datasets.
Developed expertise in querying data from a range of data sources, effectively handling structured and semi-structured data.
Worked closely with the team to improve data modeling techniques & practices for Data Warehouse/Data Mart applications.
Demonstrated proficiency in Python programming and experience with ORMs like SQLAlchemy and Django ORM.
Created reports and visualizations using Tableau to support business decisions.
Conducted in-depth data analysis and modeling to provide evidence-based insights supporting workforce policies and project initiatives using statistical software of RStudio, R 3.9, Python 3.5, SPSS, SAS, Stata, MATLAB, Gretl, etc.
Developed advanced analytics solutions to identify trends and patterns in large datasets via scikit-learn and TensorFlow
Developed complex SQL queries to extract, transform, and analyze data from multiple sources for business reporting needs.
Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
Design and implement data models either use Kimball or Data Vault methodologies to support efficient data retrieval.

- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: Senior Data Engineer /Senior Data Analyst /BI Developer /SharePoint Consultant
Responsibilities:

- Conducted extensive data analysis and provided actionable insights to support business decisions.
- Developed and documented data-related processes and procedures.
- Leveraged Databricks Lakehouse Platform to optimize data storage and retrieval processes, reducing data latency by 20%.

- Utilized data analytics approaches and tools to troubleshoot and problem-solve, consistently delivering effective solutions.
- Developed a keen understanding of Source to Target mapping between systems.
- Contributed to ETL development, data cleansing, and transformation efforts.
- Assisted with technical evaluation of requirements from users, product owners, and stakeholders.
- End-to-end ownership for solution design, developing, deploying and supporting the solutions.
- Worked as part of a team managing and supporting existing and new platforms, with a focus on CI/CD pipelines, release pipelines, and DevOps practices.
- Collaborated and communicated with key business SME's, product owners, business analysts and other key stakeholders across different locations and time zones to define and create solutions.
- Maintained and continuously improved the stability of the platform(s) and driving exceptional client experience and service.
- Demonstrated strong awareness of risk and equally strong accountability in identifying, escalating and remediating risk.
- Collaborated with cross-functional teams to troubleshoot and resolve data issues.
- Handling large scale, data processing systems in a Data Lake/Data Warehouse environment.
- Designed and implemented data visualizations and dashboards to facilitate data-driven decision-making across groups.
- Prepared and published technical reports and briefings for senior management and government entities.
- Spearheaded the analysis of complex data sets to extract actionable insights and inform strategic initiatives.
- Developed and maintained data pipelines, ensuring efficient data extraction, transformation, and loading (ETL) processes.
- Profile, test and understand data extracted from multiple data sources to identify data trends, outliers and modelling outputs
- Created data models and ETL processes to enable efficient data integration into the data warehouse.
- Implemented data automation solutions using R, Python, streamlining data product development and support processes.
- Design and develop OLAP Layer Cubes using Analysis Services (SSAS) either in SSDT2012 or Visual Studio 2010 Shell
- Data Profiling, Data Quality Analysis by Master Data Services and Data Cleansing by Data Quality Client 2012
- Optimisation and performance tuning either by optimising SQL query structures or from infrastructure perspective by creating one data file per CPU core in the tempdb file in System Database of SQL Server 2012 for optimal performance.
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: **Senior Data Engineer /Senior Data Analyst /BI Developer /SharePoint Consultant**
Responsibilities:

- Developed and maintained complex SQL queries and stored procedures, facilitating data extraction for trend analysis, financial forecasting.
- Maintained and updated data sets to support various business functions.
- Collaborated with the development team to create interactive and user-friendly Tableau dashboards, enhancing accessibility to insights.
- Assisted in the development and revision of data migration approaches to meet new data requirements.
- Supported application development activities by using data conversion and related techniques to acquire necessary data.
- Conducted thorough data audits, ensuring data integrity and compliance with data governance policies.
- Designed and developed interactive dashboards using Power BI to visualize key performance indicators and data trends.
- Developed and optimized SQL queries to extract, transform, and load data from various sources into the data warehouse.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: **Senior Data Engineer /Senior Data Analyst /BI Developer /DWH /DBA**
Responsibilities:

- Responsible to develop and publish both scheduled and ad hoc data visualizations that align with the business's specific needs while adhering to best practices in visualization
- Conducted data extraction, modelling, and statistical analysis to enable data-driven decision-making.
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Toad Data Modeler 6.x
- Develop OLAP Cubes either using SSAS or Cognos TM1 and visualize data either using Power BI, Tableau 10 or QlikView
- Data Vault modelling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**

Banking/Insurance Domain: Project Manager / Principle Data Engineer /Senior Data Analyst /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Responsible to prepare and mine large, diverse data sets to identify patterns and outliers.
- Lead the creation of context aligned aggregate data sets for use by Insights Analyst, Modelling & Visualization Specialists.
- Collaborated with business stakeholders to understand data requirements and provided timely data extracts for analytical purposes. Assisted in data analysis and predictive modeling using Python and machine learning techniques.
- Use HP BCA, System Center 2010 and SPLUNK to manage the Bank's internally managed infrastructure and applications,
- Based on clients' needs, recommend suitable BI solutions from the cause-seeking BI systems, the action- scenario type BI system and select its associated BI tools -Hyperion based Planning (PI), Budget Management (BM) for the clients.)
- Develop complex SQL Queries, stored procedures and functions using SQL / T-SQL code/ Stored Procedures/Triggers.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Systems demo for selected vendors and response to vendors' technical queries regarding the project

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor: Senior Data Engineer /Senior Data Analyst /MS BI Developer /DWH
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) and being responsible for delivering the products to market.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Maintain data models for analytic and reporting, including input to the enhancements required for the logical model, development and staging areas and source systems
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Provided technical support to end-users for database-related issues.
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio (BIDS2008) for DTS package development in Agile /RAD development environment.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
AI, ML, DL, BI IDE: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
Database: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
Clouds: MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
CMS: SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality and Safety Commission (ACQSC), Sydney**
Health Care: Senior SharePoint Administrator /Senior SharePoint /D365 CRM /Nintex K2 Counsultant /DBA
Responsibilities:

- Architect, develop and implement clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2019/IIS10
- Integrate Dynamics 365 CRM application with SharePoint 2019 using Office365 either on-premise**s** or in Hybrid /Private Azure Cloud by configuring MS Windows Server Active Directory infrastructure and Single Sign On (SSO) via ADFS.
- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks using K2 Configuration Analysis, K2 Designer, K2 Package and Deployment, K2 Server (Service account), K2 Management, Managing application administrators list, Workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, MS Power Automate workflows, SharePoint 2019 either on AWS or on Nintex K2 Cloud.
- Implemented Elastic Load Balancing (ELB) for Nintext K2, SharePoint 2019 on AWS' EC2 instance and EBS storage to distribute traffic among SharePoint front-end servers for high availability and load distribution.
- Create and maintain SmartObjects using Nintex K2 Designer and create Nintex K2 packages in Nintex K2 Five or K2 Cloud
- Collaborated with cross-functional teams to successfully migrate SharePoint 2013 to SharePoint 2019 on AWS' EC2, EBS.
- Provide technical support for hybrid aged care business applications (SharePoint, Dynamics CRM &TIBCO BPM workflows)
- Managing the upgrade from Sharepoint 2013 to Sharepoint 2016/19 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Providing incident, service request and problem management support to users in their daily use of SharePoint applications
- Upgraded Better Business Program system to Dynamics 365 CRM to manage business activities by upload various eForms to SharePoint 2019 Doc List triggering TIBCO AMX BPM Workflows for multi-parallel workflows via TIBCO iProcessor.
- Developing, customising and implementing MS Dynamics 365 /CRM 2016 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Monitor and resolve issues on Office 365, AWS, Azure and SharePoint / SharePoint online platforms
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Financial Bank: Senior SharePoint Administrator /Senior SharePoint /Nintex K2 Consultant /Power BI Developer
Responsibilities:

- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks: Managing application administrators list, Inviting users to use the Apps, Enabling link for favorite Appls workspace, App seetings, Managing workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, SharePoint either on AWS or on Nintex K2 Cloud.
- Provided technical operational support for mission-critical hosted applications, ensuring high availability and optimal performance.
- Played a key role in troubleshooting and resolving connectivity and compatibility issues with third-party systems like Nintex.
- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using KingswaySoft XrmSvcToolkit lightweight open source library to execute CRUD operations and remove all the complications from working with the REST Web API interface to offer an easier method of working with CRM Data.
- Designing, developing and maintaining BI and Reporting solutions either using SSRS, Power BI, Tableau 10.x or QlikView;
- As Senior Systems Administrator using LAMP Stack/AWS to help the RBA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP

- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Using wsl screenfetch etc. distro command line to switch between Windows 10 and Linux for B2B website development
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Design and develop enterprise Dynamics 365 CRM solutions using Dynamics CRM2016 Developer Tookit for VS2017
- Setup SharePoint 2016 integration with Dynamics 365 CRM, including OneNotes integration in Dynmaics 365 CRM
- Developing a CI/CD strategy to deliver web and App based e-commerce solutions using Sitecore8.2U3 for end-customers and partners, including both traditional e-commerce models as well as Accelerated Mobile Pages(AMP) mobile applications
- Manage SharePoint system and content permissions, manage system changes, and enforcing policies to conform to firm operational standards. Enhancing and providing Technical support (up to 3rd level) on all IT applications
- Administration and technical support of SharePoint 2013/2016, Dynamics 365 CRM and .NET Applications
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Client-Side development using App for SharePoint 2016 in VS2015TFS /VS2017 to develop SharePoint 2016 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- SQL 2014 /SQL 2016 development ecifically using Microsoft BI stack, including MDX and DAX in Power BI
- SharePoint Content migration (team sites) from SharePoint 2013 to 2016 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on “MS SharePoint 2013 Upgrade Process” best practice.
- Design and develop Digital Workflows using Nintex for SharePoint 2016, including Nintex Forms 2016 with Nintex Mobile.
- Develop and modify process flows by SharePoint 2016- Import Reusable (SharePoint) 2010 workflow in VS2017
- Administration/configuration of SharePoint 2013/2016, Dynamics 365, CRM 2016 and SQL Server 2014/2016 databases.
- Deep Learning with Python Tensorflow to train business data to predict future business trend on Jupyter notebook either in AWS or Google Cloud or Azure Cloud and disply well-trained data on Power BI with clickable drilldown on charts feature.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 10.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Senior SharePoint Administrator /Senior SharePoint /K2 Developer /Dynamics CRM Consultant
Responsibilities:

- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using SmartForms in K2 Designer to grab multiple line-of-business systems’ databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Managed SharePoint farm with a focus on maintaining web applications, site collections, and content databases.
- Collaborated with development teams to design and implement custom workflows using SharePoint Designer and K2.
- Conducted regular security audits and implemented enhancements to align with industry best practices.
- Developing, customising and implementing MS Dynamics CRM 2015 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Management of technologies supporting organisation systems such as: VMWARE, Azure,Office 365 and Palo Alto NGFW
- SharePoint document managemen development using Total Record Information Management (TRIM 8.1).
- Continuous improvement of existing technologies, processes and adoption of new technology based on CI/CD principles.
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Using VMware SRM /ESX servers with HA via vSphere6.5 Web Client to manage production and data recovery data centres
- Design and develop SharePoint pages, web parts, templates and libraries using Design Manager, SPD2013 and VS2015
- Deployment & configuration of custom SharePoint solutions by Install-SPSolution -Identity SolName.wsp -GACDeployment
- Maintain documentation relating to applications including development, troubleshoot and deployment
- Administration of multiple SharePoint Farm configurations either by PowerShell 3.0 scripts, SharePoint Central Administration or SharePoint Designer 2013, including Website configuration, customise backups and documentation.
- Using Adobe Photoshop, Dreamweaver to develop templates for BAU use, Prepare correspondence and reports as directed
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Develop templates either by SharePoint Design Manager, SPD2013 or VS2015, including HTML5/CSS3, JavaScript/jQuery, item templates, onet.xml(for the definition of list templates available for creating new lists), basesitestempling.xml
- Managing SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Migrator /HarePoint or using Collab365 to migrate data, sites and documents from on-premises to hybrid or online.
- Client-Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Integrated BA Insight Enterprise Search solution for SharePoint 2013 using BA Insight Longitude Connectors to optimise and enable UC to rapidly implement powerful search-driven applications at a fraction of the cost, time and risk of alternatives.
- Setup RecordPoint for SharePoint 2013 ECM by RecordPoint for SharePoint 2013 Configure Pack Settings v3.x; Select Term Store, Group and Set used for RecordPoint. Pre-configure Managed Metadata Service Application in SharePoint CA etc.
- Integrated Newsgator with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location. Develop simple document management

- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Migrated SP2010 WF to SP2016WF either by HarePoint or SP2016- Import Reusable (SP) 2010 workflow in VS2017
- Microsoft CRM 2015 Mutli-Server Deployment on Windows Server 2012R2/SQL Server2014/SharePoint 2013/IIS8.5 platforms, including Install and Configure the Outlook Client, Import Multi Users & CRM Online portal on Azure Cloud.
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client or using DreamFactory API automation to turn any database into API for analystics
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Assisted in the migration of SharePoint 2010 to SharePoint 2013, ensuring seamless data transition and minimal downtime.
- Provided technical support to end users, resolving issues related to permissions, access, and usability.
- Contributed to the development of documentation and training materials for SharePoint users.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Administering SharePoint 2010/2013, Dynamics CRM 2011/2013 and Microsoft SQL Server 2012R2 databases.
- Design and develop enterprise CRM2013 solutions using Microsoft Dynamics CRM2013 Developer Tookit for VS2013
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- Support the manager with other tasks as required. Ensure transfer of knowledge to the rest of the team
- SharePoint 2013 Profile Management and Synchronisation either by deploy Calligo or harmon.ie for SharePoint2013.
- Configuring and customising SharePoint implementations with AutoSPInstaller powershell scripts and SharePoint templates
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Leading customer SharePoint engagements for designing innovative SharePoint /Cloud solutions for EML customers.
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Integrating custom HTML5/CSS3 to SharePoint master page and page layouts using Adobe Photoshop, SPD2015&VS2015
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Using SharePoint 2013 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or visually in new Rest API's: http://spserver/_api/web or http://spserver/_vti_bin/client.svc/web
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Developing SharePoint Web Templates for on-permises and the Cloud via Office365, SharePoint Design Manager&VS2013
- SQL 2014 development ecifically Microsoft SQL Server 2008 R2 and 2012, including stored procedure, trigger, etc.
- Using test-driven development (TDD) and refactoring to support the focus on automation and continuous improvement
- Configure Metadata Term Store Management on for SharePoint 2013 landing page multi-navigate menus
- Changing the look of Search Results using Display Templates and Client Side Rendering (CSR).
- Implemented Microsoft EPM/PPM bespoke Solution via install MS Project Professional 2013, MS Project Server 2013 on top of SharePoint2013 Server Farm platform with ensuring solutions do not impose security risk to the Business.
- Mentoring and coaching other team members for customer-focused approach to control Agile/Scrum delivery process.
- Hands on experience with .Net application master page/SharePoint corp landing pages design,like skinning/re-branding.
- Integrated HP TRIM 7.2 with SharePoint 2010 to deliver Enterprise Content Management solutions, Document Management solutions, Nintex workflow solutions and custom solutions on the SharePoint platform.
- Service and application integration via Microsoft BPM's Tools like BizTalk2013's BAM, SharePoint workflow with InforPath form, Scan, approve and store invoices electronically into Dynamics GP or CRM by CRM Adapter or Nintex Workflow etc.

- Implement a high traffic website using Umbaraco 7, Sitecore CEP 7 according to OOD, DDD, IoC and other principles
- Perform unit (automated) testing; Support both QA testing (system & integration) and level 2 support for Production systems
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux 9.3 - 10.1 or Red Hat Enterprise Linux 7.2 for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Gathering requirements from stakeholders and translate them into technical solution design for SharePoint/CRM online.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Supported the administration and maintenance of SharePoint 2013 environments, gaining foundational experience in SharePoint infrastructure.
- Assisted in user training sessions to improve SharePoint adoption and usage among employees.
- Collaborated with senior administrators to troubleshoot and resolve configuration issues.
- Perform technical business analytical work and suggest how to improve AS-IS process and develop TO-BE process enhancements through business operation automation by implementing SharePoint /Nintex workflow solutions.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior SharePoint Administrator /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to develop effective Hybrid and Private Cloud solutions for new customers from architecture design to and implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on AWS Cloud.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Install and configure AD FS Server Farm with load balancing clusters in managing a mid to large AD environment.
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Add an Exchange Edge Server to Lync Server 2010, create Edge pool and publish the topology on SCSM2007 Dashboard.
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.
- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.
- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.
- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior SharePoint Administrator /Full Stack Developer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Design DWH concepts from Bill Inmon & Ralph Kimball and build data models via Toad Data Modeler 10.x &Informatica 8.x;
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and Agile /Scrum delivery methodology
- Develop and customise Dynamics CRM solutions using C#, VB, ASP.Net in Visual Studio 2008TFS.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective-C formulas for insurance premium calculations
- Create financial reports either by SSRS or SAP Crystal Reports for VS2008 or Business Object BOXIR2 on HP-UX platform
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both Jave FileNet workflows and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Software Engineer /Senior SharePoint Administrator / Developer/Full Stack .NET Developer
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) with Azure cloud and being responsible for delivering the products to market.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Install and configure ADFS Server Farm with load balancing clusters in managing a mid to large AD environment, including AD migrations, cross-forest migrations and interoperability scenarios
- Plan, design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for delivering the products to market.
- Standardising and automating software delivery processes such as: Version control by TFS/Git; Consistent eInvironment configuration, build and packaging by SCCM2007; Deployment by -Altiris and release management by SCSM and SCOM
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions. Advanced data query via LINQ-to-SQL
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OODesign patterns(Singleton, Factory etc.) based on business requirements.
- Maintaining and auditing real time security of SQL Server databases via customizable SQL Compliance Manager.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-Tier web applications with C#3.5, ASP.Net, MVC3/4, Service-Oriented Applications (Workflow Application, Servcie Library) using WCF to GetData and create righ GUI applications for UI/UX using WPF with data-bound controls.
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Developing N-tier & multi-threaded applications using C#.NET4, MVC3/4, ADO.Net, WPF, WCF, Rest, Linq, WinForms, etc.
- BizTalk2006(R2)/2009, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: Microsoft Azure Cloud, Amazon Cloud (AWS), Google Cloud Platform (GCP)
CMS: SharePoint 2007~2022, CRM3.0~D365 CRM, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net2.0~5/Core, NodeJs, AngularJS/2/4/5, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices

Workflows: K2 Workflow, Nintex Workflow 2007~2019, SharePoint 2007~2019/O365, MS Flow~Power Automate workflow, SharePoint Workflow Manager for SharePoint Server, TibCo/SharePoint/CRM Workflow, BizTalk Server 2006R2~2020 workflow

Systems Administrator-Application Support SME (SharePoint-Power Automate) –PRA-25/9/2023
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English /Japanese Interpreter
Senior Systems Administrator/Application Support SME (SharePoint/Power Automate) /D365 CRM Consultant /DWH /DBA – Japan (7.5+yrs) & Australia (23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality and Safety Commission (ACQSC), Sydney**
NSW Gov.: Sr. Systems Administrator/Application Support SME (SharePoint/Power Automate) /D365 CRM Counsultant
Responsibilities:

- Architect, develop and implement clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2019/IIS10
- Integrate Dynamics 365 CRM application with SharePoint 2019 using Office365 either on-premise**s** or in Hybrid /Private Azure Cloud by configuring MS Windows Server Active Directory infrastructure and Single Sign On (SSO) via ADFS.
- Application support SME for Aged Care Quality business applications covering SharePoint 2019, Power Automate, TIBCO, Dynamics 365 CRM, and associated SQL Servers.
- Provide technical support, maintenance, and configuration of SharePoint and Power Automate solutions. Troubleshootings for nationwide end users reported issues via Manage Engine ticketing system.
- Troubleshoot and improve SharePoint and Power Automate workflow designs.
- Perform data analytics and predictive maintenance tasks using SharePoint and Power Automate.
- Engage with stakeholders and super-users to understand their requirements and provide solutions.
- Document and report on the performance and issues of SharePoint and Power Automate solutions.
- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks using K2 Configuration Analysis, K2 Designer, K2 Package and Deployment, K2 Server (Service account), K2 Management, Managing application administrators list, Workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, MS Power Automate workflows, SharePoint 2019 either on AWS or on Nintex K2 Cloud.
- Implemented Elastic Load Balancing (ELB) for Nintext K2, SharePoint 2019 on AWS' EC2 instance and EBS storage to distribute traffic among SharePoint front-end servers for high availability and load distribution.
- Create and maintain SmartObjects using Nintex K2 Designer and create Nintex K2 packages in Nintex K2 Five or K2 Cloud
- Collaborated with cross-functional teams to successfully migrate SharePoint 2013 to SharePoint 2019 on AWS' EC2, EBS.
- Managing the upgrade from Sharepoint 2013 to Sharepoint 2016/19 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Providing incident, service request and problem management support to users in their daily use of SharePoint applications
- Upgraded Better Business Program system to Dynamics 365 CRM to manage business activities by upload various eForms to SharePoint 2019 Doc List triggering TIBCO AMX BPM Workflows for multi-parallel workflows via TIBCO iProcessor.
- Developing, customising and implementing MS Dynamics 365 /CRM 2016 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Monitor and resolve issues on Office 365, AWS, Azure and SharePoint / SharePoint online platforms
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Central Bank-Gov.: Sr. Systems Administrator/Application Support SME(SharePoint/Power Automate/Power BI Developer)
Responsibilities:

- Developed and deployed Power Automate flows for various business processes and scenarios.
- Used Power Automate connectors to integrate with various data sources and services.
- Tested and debugged Power Automate flows using Power Automate Monitor and other tools.
- Optimized Power Automate flows for performance, reliability, and scalability.
- Documented Power Automate flows and best practices.
- Administered and maintained SharePoint sites and applications for the bank's internal and external users.
- Implemented security policies and permissions for SharePoint users and groups.
- Managed SharePoint backups, restores, migrations, and updates.
- Provided training and support to SharePoint users and developers.
- Collaborated with IT teams to integrate SharePoint with other systems and applications.
- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks: Managing application administrators list, Inviting users to use the Apps, Enabling link for favorite Appls workspace, App seetings, Managing workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.

- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, SharePoint either on AWS or on Nintex K2 Cloud.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, MS Power Automate workflows, SharePoint 2016 either on AWS or on Nintex K2 Cloud.
- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using KingswaySoft XrmSvcToolkit lightweight open source library to execute CRUD operations and remove all the complications from working with the REST Web API interface to offer an easier method of working with CRM Data.
- Designing, developing and maintaining BI and Reporting solutions either using SSRS, Power BI, Tableau 10.x or QlikView;
- As Senior Systems Administrator using LAMP Stack/AWS to help the RBA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP
- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Using wsl screenfetch etc. distro command line to switch between Windows 10 and Linux for B2B website development
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Design and develop enterprise Dynamics 365 CRM solutions using Dynamics CRM2016 Developer Tookit for VS2017
- Setup SharePoint 2016 integration with Dynamics 365 CRM, including OneNotes integration in Dynmaics 365 CRM
- Manage SharePoint system and content permissions, manage system changes, and enforcing policies to conform to firm operational standards. Enhancing and providing Technical support (up to 3rd level) on all IT applications
- Administration and technical support of SharePoint 2013/2016, Dynamics 365 CRM and .NET Applications
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Client-Side development using App for SharePoint 2016 in VS2015TFS /VS2017 to develop SharePoint 2016 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- SQL 2014 /SQL 2016 development specifically using Microsoft BI stack, including MDX and DAX in Power BI
- SharePoint Content migration (team sites) from SharePoint 2013 to 2016 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on “MS SharePoint 2013 Upgrade Process” best practice.
- Design and develop Digital Workflows using Nintex for SharePoint 2016, including Nintex Forms 2016 with Nintex Mobile.
- Develop and modify process flows by SharePoint 2016- Import Reusable (SharePoint) 2010 workflow in VS2017
- Administration/configuration of SharePoint 2013/2016, Dynamics 365, CRM 2016 and SQL Server 2014/2016 databases.
- Deep Learning with Python Tensorflow to train business data to predict future business trend on Jupyter notebook either in AWS or Google Cloud or Azure Cloud and disply well-trained data on Power BI with clickable drilldown on charts feature.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 10.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education-Gov.: Senior SharePoint Administrator /Senior SharePoint /K2 Developer /Dynamics CRM Consultant
Responsibilities:

- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using SmartForms in K2 Designer to grab multiple line-of-business systems’ databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Managed SharePoint farm with a focus on maintaining web applications, site collections, and content databases.
- Developing, customising and implementing MS Dynamics CRM 2015 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- SharePoint document managemen development using Total Record Information Management (TRIM 8.1).
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Using VMware SRM /ESX servers with HA via vSphere6.5 Web Client to manage production and data recovery data centres
- Design and develop SharePoint pages, web parts, templates and libraries using Design Manager, SPD2013 and VS2015
- Deployment & configuration of custom SharePoint solutions by Install-SPSolution -Identity SolName.wsp -GACDeployment
- Maintain documentation relating to applications including development, troubleshoot and deployment
- Administration of multiple SharePoint Farm configurations either by PowerShell 3.0 scripts, SharePoint Central Administration or SharePoint Designer 2013, including Website configuration, customise backups and documentation.
- Using Adobe Photoshop, Dreamweaver to develop templates for BAU use, Prepare correspondence and reports as directed
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Develop templates either by SharePoint Design Manager, SPD2013 or VS2015, including HTML5/CSS3, JavaScript/jQuery, item templates, onet.xml(for the definition of list templates available for creating new lists), basesitestempling.xml
- Managing SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Migrator /HarePoint or using Collab365 to migrate data, sites and documents from on-premises to hybrid or online.
- Client-Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Setup RecordPoint for SharePoint 2013 ECM by RecordPoint for SharePoint 2013 Configure Pack Settings v3.x; Select Term Store, Group and Set used for RecordPoint. Pre-configure Managed Metadata Service Application in SharePoint CA etc.

- Integrated Newsgator with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location. Develop simple document management
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Migrated SP2010 WF to SP2016WF either by HarePoint or SP2016- Import Reusable (SP) 2010 workflow in VS2017
- Microsoft CRM 2015 Mutli-Server Deployment on Windows Server 2012R2/SQL Server2014/SharePoint 2013/IIS8.5 platforms, including Install and Configure the Outlook Client, Import Multi Users & CRM Online portal on Azure Cloud.
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client or using DreamFactory API automation to turn any database into API for analystics
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Sr SharePoint Administrator /Sr. SharePoint /Power Automate Developer /MS Dynamics CRM Developer

Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Installed, configured, and maintained Windows servers, workstations, and network devices.
- Monitored system performance, availability, and security using various tools.
- Resolved system issues and incidents using ITIL best practices.
- Performed system backups, restores, patches, and upgrades.
- Assisted with system audits, compliance, and documentation.
- Provided technical support to end users, resolving issues related to permissions, access, and usability.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Administering SharePoint 2010/2013, Dynamics CRM 2011/2013 and Microsoft SQL Server 2012R2 databases.
- Design and develop enterprise CRM2013 solutions using Microsoft Dynamics CRM2013 Developer Tookit for VS2013
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- SharePoint 2013 Profile Management and Synchronisation either by deploy Calligo or harmon.ie for SharePoint2013.
- Configuring and customising SharePoint implementations with AutoSPInstaller powershell scripts and SharePoint templates
- Leading customer SharePoint engagements for designing innovative SharePoint /Cloud solutions for EML customers.
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Integrating custom HTML5/CSS3 to SharePoint master page and page layouts using Adobe Photoshop, SPD2015&VS2015
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Using SharePoint 2013 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or visually in new Rest API's: http://spserver/_api/web or http://spserver/_vti_bin/client.svc/web
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Developing SharePoint Web Templates for on-permises and the Cloud via Office365, SharePoint Design Manager&VS2013
- SQL 2014 development ecifically Microsoft SQL Server 2008 R2 and 2012, including stored procedure, trigger, etc.
- Using test-driven development (TDD) and refactoring to support the focus on automation and continuous improvement
- Configure Metadata Term Store Management on for SharePoint 2013 landing page multi-navigate menus
- Changing the look of Search Results using Display Templates and Client Side Rendering (CSR).
- Implemented Microsoft EPM/PPM bespoke Solution via install MS Project Professional 2013, MS Project Server 2013 on top of SharePoint2013 Server Farm platform with ensuring solutions do not impose security risk to the Business.
- Hands on experience with .Net application master page/SharePoint corp landing pages design,like skinning/re-branding.
- Integrated HP TRIM 7.2 with SharePoint 2010 to deliver Enterprise Content Management solutions, Document Management solutions, Nintex workflow solutions and custom solutions on the SharePoint platform.

- Service and application integration via Microsoft BPM's Tools like BizTalk2013's BAM, SharePoint workflow with InforPath form, Scan, approve and store invoices electronically into Dynamics GP or CRM by CRM Adapter or Nintex Workflow etc.
- Implement a high traffic website using Umbaraco 7, Sitecore CEP 7 according to OOD, DDD, IoC and other principles
- Perform unit (automated) testing; Support both QA testing (system & integration) and level 2 support for Production systems
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux 9.3 - 10.1 or Red Hat Enterprise Linux 7.2 for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Provided technical support to end-users via phone, email, chat, or remote access.
- Troubleshot hardware, software, network, and application issues.
- Escalated complex issues to senior IT staff or vendors as needed.
- Installed, configured, and updated hardware, software, drivers, and applications.
- Maintained inventory of IT assets and supplies.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior SharePoint Administrator /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to develop effective Hybrid and Private Cloud solutions for new customers from architecture design to and implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on AWS Cloud.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Install and configure AD FS Server Farm with load balancing clusters in managing a mid to large AD environment.
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Add an Exchange Edge Server to Lync Server 2010, create Edge pool and publish the topology on SCSM2007 Dashboard.
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.
- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.
- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.
- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior SharePoint Administrator /Full Stack Developer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Design DWH concepts from Bill Inmon & Ralph Kimball and build data models via Toad Data Modeler 10.x &Informatica 8.x;

- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and Agile /Scrum delivery methodology
- Develop and customise Dynamics CRM solutions using C#, VB, ASP.Net in Visual Studio 2008TFS.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective-C formulas for insurance premium calculations
- Create financial reports either by SSRS or SAP Crystal Reports for VS2008 or Business Object BOXIR2 on HP-UX platform
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both Jave FileNet workflows and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Software Engineer /Senior SharePoint Administrator / Developer/Full Stack .NET Developer
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) with Azure cloud and being responsible for delivering the products to market.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Install and configure ADFS Server Farm with load balancing clusters in managing a mid to large AD environment, including AD migrations, cross-forest migrations and interoperability scenarios
- Plan, design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for delivering the products to market.
- Standardising and automating software delivery processes such as: Version control by TFS/Git; Consistent eInvironment configuration, build and packaging by SCCM2007; Deployment by -Altiris and release management by SCSM and SCOM
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions. Advanced data query via LINQ-to-SQL
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OODesign patterns(Singleton, Factory etc.) based on business requirements.
- Maintaining and auditing real time security of SQL Server databases via customizable SQL Compliance Manager.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-Tier web applications with C#3.5, ASP.Net, MVC3/4, Service-Oriented Applications (Workflow Application, Servcie Library) using WCF to GetData and create righ GUI applications for UI/UX using WPF with data-bound controls.
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Developing N-tier & multi-threaded applications using C#.NET4, MVC3/4, ADO.Net, WPF, WCF, Rest, Linq, WinForms, etc.
- BizTalk2006(R2)/2009, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: Microsoft Azure Cloud, Amazon Cloud (AWS), Google Cloud Platform (GCP)
CMS: SharePoint 2007~2022, CRM3.0~D365 CRM, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net2.0~5/Core, NodeJs, AngularJS/2/4/5, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
Workflows: K2 Workflow, Nintex Workflow 2007~2019, SharePoint 2007~2019/O365, MS Flow~Power Automate workflow, SharePoint Workflow Manager for SharePoint Server, TibCo/SharePoint/CRM Workflow, BizTalk Server 2006R2~2020 workflow

Senior SharePoint Administrator–Genesis–16/8/2023 @2-12 Macquarie St. Parramatta Alex mobile voice full, no resp at all
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English /Japanese Interpreter
Senior SharePoint Administrator /Sr SharePoint /D365 CRM /Nintex K2 Consultant/DWH/DBA –Japan (7.5+yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Aged Care Quality and Safety Commission (ACQSC), Sydney
Health Care: Senior SharePoint Administrator /Senior SharePoint /D365 CRM /Nintex K2 Counsultant /DBA
Responsibilities:

- Architect, develop and implement clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2019/IIS10
- Integrate Dynamics 365 CRM application with SharePoint 2019 using Office365 either on-premise**s** or in Hybrid /Private Azure Cloud by configuring MS Windows Server Active Directory infrastructure and Single Sign On (SSO) via ADFS.
- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks using K2 Configuration Analysis, K2 Designer, K2 Package and Deployment, K2 Server (Service account), K2 Management, Managing application administrators list, Workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, MS Power Automate workflows, SharePoint 2019 either on AWS or on Nintex K2 Cloud.
- Implemented Elastic Load Balancing (ELB) for Nintext K2, SharePoint 2019 on AWS' EC2 instance and EBS storage to distribute traffic among SharePoint front-end servers for high availability and load distribution.
- Create and maintain SmartObjects using Nintex K2 Designer and create Nintex K2 packages in Nintex K2 Five or K2 Cloud
- Collaborated with cross-functional teams to successfully migrate SharePoint 2013 to SharePoint 2019 on AWS' EC2, EBS.
- Provide technical support for hybrid aged care business applications (SharePoint, Dynamics CRM &TIBCO BPM workflows)
- Managing the upgrade from Sharepoint 2013 to Sharepoint 2016/19 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Providing incident, service request and problem management support to users in their daily use of SharePoint applications
- Upgraded Better Business Program system to Dynamics 365 CRM to manage business activities by upload various eForms to SharePoint 2019 Doc List triggering TIBCO AMX BPM Workflows for multi-parallel workflows via TIBCO iProcessor.
- Developing, customising and implementing MS Dynamics 365 /CRM 2016 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Monitor and resolve issues on Office 365, AWS, Azure and SharePoint / SharePoint online platforms
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Financial Bank: Senior SharePoint Administrator /Senior SharePoint /Nintex K2 Consultant /Power BI Developer
Responsibilities:

- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks: Managing application administrators list, Inviting users to use the Apps, Enabling link for favorite Appls workspace, App seetings, Managing workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, SharePoint either on AWS or on Nintex K2 Cloud.
- Provided technical operational support for mission-critical hosted applications, ensuring high availability and optimal performance.
- Played a key role in troubleshooting and resolving connectivity and compatibility issues with third-party systems like Nintex.
- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using KingswaySoft XrmSvcToolkit lightweight open source library to execute CRUD operations and remove all the complications from working with the REST Web API interface to offer an easier method of working with CRM Data.

- Designing, developing and maintaining BI and Reporting solutions either using SSRS, Power BI, Tableau 10.x or QlikView;
- As Senior Systems Administrator using LAMP Stack/AWS to help the RBA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP
- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Using wsl screenfetch etc. distro command line to switch between Windows 10 and Linux for B2B website development
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Design and develop enterprise Dynamics 365 CRM solutions using Dynamics CRM2016 Developer Tookit for VS2017
- Setup SharePoint 2016 integration with Dynamics 365 CRM, including OneNotes integration in Dynmaics 365 CRM
- Developing a CI/CD strategy to deliver web and App based e-commerce solutions using Sitecore8.2U3 for end-customers and partners, including both traditional e-commerce models as well as Accelerated Mobile Pages(AMP) mobile applications
- Manage SharePoint system and content permissions, manage system changes, and enforcing policies to conform to firm operational standards. Enhancing and providing Technical support (up to 3rd level) on all IT applications
- Administration and technical support of SharePoint 2013/2016, Dynamics 365 CRM and .NET Applications
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Client-Side development using App for SharePoint 2016 in VS2015TFS /VS2017 to develop SharePoint 2016 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- SQL 2014 /SQL 2016 development ecifically using Microsoft BI stack, including MDX and DAX in Power BI
- SharePoint Content migration (team sites) from SharePoint 2013 to 2016 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on “MS SharePoint 2013 Upgrade Process” best practice.
- Design and develop Digital Workflows using Nintex for SharePoint 2016, including Nintex Forms 2016 with Nintex Mobile.
- Develop and modify process flows by SharePoint 2016- Import Reusable (SharePoint) 2010 workflow in VS2017
- Administration/configuration of SharePoint 2013/2016, Dynamics 365, CRM 2016 and SQL Server 2014/2016 databases.
- Deep Learning with Python Tensorflow to train business data to predict future business trend on Jupyter notebook either in AWS or Google Cloud or Azure Cloud and disply well-trained data on Power BI with clickable drilldown on charts feature.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 10.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Senior SharePoint Administrator /Senior SharePoint /K2 Developer /Dynamics CRM Consultant
Responsibilities:

- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using SmartForms in K2 Designer to grab multiple line-of-business systems’ databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Managed SharePoint farm with a focus on maintaining web applications, site collections, and content databases.
- Collaborated with development teams to design and implement custom workflows using SharePoint Designer and K2.
- Conducted regular security audits and implemented enhancements to align with industry best practices.
- Developing, customising and implementing MS Dynamics CRM 2015 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Management of technologies supporting organisation systems such as: VMWARE, Azure,Office 365 and Palo Alto NGFW
- SharePoint document managemen development using Total Record Information Management (TRIM 8.1).
- Continuous improvement of existing technologies, processes and adoption of new technology based on CI/CD principles.
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Using VMware SRM /ESX servers with HA via vSphere6.5 Web Client to manage production and data recovery data centres
- Design and develop SharePoint pages, web parts, templates and libraries using Design Manager, SPD2013 and VS2015
- Deployment & configuration of custom SharePoint solutions by Install-SPSolution -Identity SolName.wsp -GACDeployment
- Maintain documentation relating to applications including development, troubleshoot and deployment
- Administration of multiple SharePoint Farm configurations either by PowerShell 3.0 scripts, SharePoint Central Administration or SharePoint Designer 2013, including Website configuration, customise backups and documentation.
- Using Adobe Photoshop, Dreamweaver to develop templates for BAU use, Prepare correspondence and reports as directed
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Develop templates either by SharePoint Design Manager, SPD2013 or VS2015, including HTML5/CSS3, JavaScript/jQuery, item templates, onet.xml(for the definition of list templates available for creating new lists), basesitestempling.xml
- Managing SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Migrator /HarePoint or using Collab365 to migrate data, sites and documents from on-premises to hybrid or online.
- Client-Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Integrated BA Insight Enterprise Search solution for SharePoint 2013 using BA Insight Longitude Connectors to optimise and enable UC to rapidly implement powerful search-driven applications at a fraction of the cost, time and risk of alternatives.
- Setup RecordPoint for SharePoint 2013 ECM by RecordPoint for SharePoint 2013 Configure Pack Settings v3.x; Select Term Store, Group and Set used for RecordPoint. Pre-configure Managed Metadata Service Application in SharePoint CA etc.

- Integrated Newsgator with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location. Develop simple document management
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Migrated SP2010 WF to SP2016WF either by HarePoint or SP2016- Import Reusable (SP) 2010 workflow in VS2017
- Microsoft CRM 2015 Mutli-Server Deployment on Windows Server 2012R2/SQL Server2014/SharePoint 2013/IIS8.5 platforms, including Install and Configure the Outlook Client, Import Multi Users & CRM Online portal on Azure Cloud.
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client or using DreamFactory API automation to turn any database into API for analystics
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Assisted in the migration of SharePoint 2010 to SharePoint 2013, ensuring seamless data transition and minimal downtime.
- Provided technical support to end users, resolving issues related to permissions, access, and usability.
- Contributed to the development of documentation and training materials for SharePoint users.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Administering SharePoint 2010/2013, Dynamics CRM 2011/2013 and Microsoft SQL Server 2012R2 databases.
- Design and develop enterprise CRM2013 solutions using Microsoft Dynamics CRM2013 Developer Tookit for VS2013
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- Support the manager with other tasks as required. Ensure transfer of knowledge to the rest of the team
- SharePoint 2013 Profile Management and Synchronisation either by deploy Calligo or harmon.ie for SharePoint2013.
- Configuring and customising SharePoint implementations with AutoSPInstaller powershell scripts and SharePoint templates
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Leading customer SharePoint engagements for designing innovative SharePoint /Cloud solutions for EML customers.
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Integrating custom HTML5/CSS3 to SharePoint master page and page layouts using Adobe Photoshop, SPD2015&VS2015
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Using SharePoint 2013 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or visually in new Rest API's: http://spserver/_api/web or http://spserver/_vti_bin/client.svc/web
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Developing SharePoint Web Templates for on-permises and the Cloud via Office365, SharePoint Design Manager&VS2013
- SQL 2014 development ecifically Microsoft SQL Server 2008 R2 and 2012, including stored procedure, trigger, etc.
- Using test-driven development (TDD) and refactoring to support the focus on automation and continuous improvement
- Configure Metadata Term Store Management on for SharePoint 2013 landing page multi-navigate menus
- Changing the look of Search Results using Display Templates and Client Side Rendering (CSR).
- Implemented Microsoft EPM/PPM bespoke Solution via install MS Project Professional 2013, MS Project Server 2013 on top of SharePoint2013 Server Farm platform with ensuring solutions do not impose security risk to the Business.
- Mentoring and coaching other team members for customer-focused approach to control Agile/Scrum delivery process.
- Hands on experience with .Net application master page/SharePoint corp landing pages design,like skinning/re-branding.
- Integrated HP TRIM 7.2 with SharePoint 2010 to deliver Enterprise Content Management solutions, Document Management solutions, Nintex workflow solutions and custom solutions on the SharePoint platform.

- Service and application integration via Microsoft BPM's Tools like BizTalk2013's BAM, SharePoint workflow with InforPath form, Scan, approve and store invoices electronically into Dynamics GP or CRM by CRM Adapter or Nintex Workflow etc.
- Implement a high traffic website using Umbaraco 7, Sitecore CEP 7 according to OOD, DDD, IoC and other principles
- Perform unit (automated) testing; Support both QA testing (system & integration) and level 2 support for Production systems
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux 9.3 - 10.1 or Red Hat Enterprise Linux 7.2 for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Gathering requirements from stakeholders and translate them into technical solution design for SharePoint/CRM online.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Supported the administration and maintenance of SharePoint 2013 environments, gaining foundational experience in SharePoint infrastructure.
- Assisted in user training sessions to improve SharePoint adoption and usage among employees.
- Collaborated with senior administrators to troubleshoot and resolve configuration issues.
- Perform technical business analytical work and suggest how to improve AS-IS process and develop TO-BE process enhancements through business operation automation by implementing SharePoint /Nintex workflow solutions.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior SharePoint Administrator /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to develop effective Hybrid and Private Cloud solutions for new customers from architecture design to and implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on AWS Cloud.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Install and configure AD FS Server Farm with load balancing clusters in managing a mid to large AD environment.
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Add an Exchange Edge Server to Lync Server 2010, create Edge pool and publish the topology on SCSM2007 Dashboard.
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.
- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.
- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.
- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior SharePoint Administrator /Full Stack Developer /DWH

Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Design DWH concepts from Bill Inmon & Ralph Kimball and build data models via Toad Data Modeler 10.x &Informatica 8.x;
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and Agile /Scrum delivery methodology
- Develop and customise Dynamics CRM solutions using C#, VB, ASP.Net in Visual Studio 2008TFS.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective-C formulas for insurance premium calculations
- Create financial reports either by SSRS or SAP Crystal Reports for VS2008 or Business Object BOXIR2 on HP-UX platform
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both Jave FileNet workflows and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Software Engineer /Senior SharePoint Administrator / Developer/Full Stack .NET Developer
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) with Azure cloud and being responsible for delivering the products to market.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Install and configure ADFS Server Farm with load balancing clusters in managing a mid to large AD environment, including AD migrations, cross-forest migrations and interoperability scenarios
- Plan, design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for delivering the products to market.
- Standardising and automating software delivery processes such as: Version control by TFS/Git; Consistent eInvironment configuration, build and packaging by SCCM2007; Deployment by -Altiris and release management by SCSM and SCOM
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions. Advanced data query via LINQ-to-SQL
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OODesign patterns(Singleton, Factory etc.) based on business requirements.
- Maintaining and auditing real time security of SQL Server databases via customizable SQL Compliance Manager.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-Tier web applications with C#3.5, ASP.Net, MVC3/4, Service-Oriented Applications (Workflow Application, Servcie Library) using WCF to GetData and create righ GUI applications for UI/UX using WPF with data-bound controls.
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Developing N-tier & multi-threaded applications using C#.NET4, MVC3/4, ADO.Net, WPF, WCF, Rest, Linq, WinForms, etc.
- BizTalk2006(R2)/2009, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions

EDUCATION / CERTIFICATION

Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: Microsoft Azure Cloud, Amazon Cloud (AWS), Google Cloud Platform (GCP)
CMS: SharePoint 2007~2022, CRM3.0~D365 CRM, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5

Framework: .Net2.0~5/Core, NodeJs, AngularJS/2/4/5, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
Workflows: K2 Workflow, Nintex Workflow 2007~2019, SharePoint 2007~2019/O365, MS Flow~Power Automate workflow, SharePoint Workflow Manager for SharePoint Server, TibCo/SharePoint/CRM Workflow, BizTalk Server 2006R2~2020 workflow

Microsoft Dynamics 365 Developer-Eagna Consulting – 17/6/2018 -
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English /Japanese Interpreter
MS Dynamics CRM Developer/Consultant /Senior SharePoint /BI Developer/DW Consultant–Japan (7+yrs) & Aust.(21yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Australia Aged Care Quality Agent (AACQA), Sydney**
Health Care: Microsoft Dynamics CRM Developer /Consultant /Senior SharePoint /BI Developer
Responsibilities:

- Architect, develop and integrate Dynamics 365 CRM application with SharePoint 2016 using Office365 either on-premise**s** or in Hybrid /Private Azure Cloud by configuring MS Windows Server Active Directory infrastructure and SSO via ADFS.
- BBP(Better Business Program) system upgrade using Dynamics 365 to manage business activities by upload various eForms to SharePoint 2016 Doc List triggering TIBCO AMX BPM Workflows for multi-parallel workflows via iProcessor.
- Developing, customising and implementing MS Dynamics 365 /CRM 2016 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Using Records365 for Office 365 integration and using RecordPoint for records management solution for on-premises, hybrid, and cloud content records management.
- Technical Support and Project Officer to manage day-to-day BBP issues and enhancement requests from business users and help facilitate resolution of the same. Administration of Microsoft Dynamics CRM 2016 and support maintenance
- Manage data migration & systems configuration from previous systems database, SQL DB maintenance and administration
- 3rd party and vendor management using JIRA Confluence, FTS / Git source control and VS2017 build systems, Jenkins
- Gather and analyse data either using MS SQL 2016 (SSIS, SSAS & SSRS) or visually by Power BI for streaming analysis.
- Either using Power BI, SQL2016 SSIS/SSRS or SSDT for VS2017 to develop reporting capability for Dynamics CRM users
- Rich Internet application (RIA) for B2B website development either using Angular5/6 or React-Redux on .NET Core 2.1

- Next generation web app development either using Sails.Js or Koa expressive middleware with ES2017 or Server side backend framework Hapi or in the combination of SailsJs with Angular, or Koa with Redux and React.
- Assist project managers/solutions architect in projects that touch upon Salesforce.com/Pardot.
- Setup Pardot B2B Marketing Automation in Salesforce so that marketing team is completely aligned with their sales team.
- Develop all needed on Salesforce (Visualforce pages, Apex classes & triggers, Apex test classes, Community development, Lightning components, Flows, Workflows and Business Process Builder and other admin level configuration)
- As DevOps & Operations System Administrator to provide Infrastructure, networking and support services to the organisation in outsource service capacity. Contribute to the broader strategic direction of company activities and IT environment
- Support the ICT teams and business units in effective and efficient use of the companies collaboration and communications applications including Microsoft based cloud hosted and on-premise services, Office 365, SharePoint 2013/2016 Online, Dynamics CRM 365, VSO, Records Management System, RecordPoint and related bespoke solutions
- Senior Systems Administrator using LAMP Stack/AWS to help the AACQA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP
- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Install Windows Subsystem for Linux (WSL) to setup and run multiple distro's on Windows 10 for B2B website development HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Monitor and resolve issues on Office 365, AWS, Azure and SharePoint / SharePoint online platforms
- Provide support and solution towards building large scale on premise and cloud-based infrastructure for the Company future.
- Develop plan and implement architecture / infrastructure for systems such as SharePoint, Dynamics CRM, Skype for business, Web technologies and Cloud based Virtualisation technologies (Azure, Hyper V, VMware, vSphere etc)
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Development and scripting experiences either in PowerShell, .net, bash, ruby or Python with Scikit-learn Machine Learning

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Financial Bank: Microsoft Dynamics CRM Developer /DW Consultant /Senior SharePoint /BI Developer
Responsibilities:

- Architect and implement MS Analytics Platform System(APS) to combine Massively Parallel Processing (MPP) SQL Server DWH with Hadoop Distribution Insight (HDInsight) to support Azure /Power BI stream analytics for digital transformation.
- Link Azure account to Docker Cloud account to deploy nodes and node clusters using Docker Cloud's Dashboard, API /CLI
- Configured SQL2016 PolyBase new feature for easy access to Hadoop and Azure Blob Storage data from anywhere.
- Performing Power BI stream analytics in Azure to build-in Power BI/Blob storage/Event Hub /SQL Database jobs via by Add an output & Authorize Now API with advanced SQL statement (T-SQL, Stored Procedures) &SQL2016 PolyBase optimized.
- Using KingswaySoft XrmSvcToolkit lightweight open source library to execute CRUD operations and remove all the complications from working with the REST Web API interface to offer an easier method of working with CRM Data.
- Develop continuous release, continuous build and continuous deployment environment using TFS, VSO and MCCM2016
- Designing, developing and maintaining BI and Reporting solutions either using SSRS, Power BI, Tableau 10.x or QlikView;
- Undertake Master Data work using the departments Golden Record Matching solution &solve solution implementation issues
- Design DWH concepts from Bill Inmon to Ralph Kimball and build data models either by Power BI or Informatica 8.x
- Connecting HDInsight Data with Azure Machine Learning for international students' Analytics Platform Service (APS)
- Deploy and execute ETL for traditional DWH and ELT either for Azure Data Warehouse OR Parallel Data Warehouse (PDW)
- Create Power BI tile, Ping a tile to Power BI dashboard either from a report or from Excel
- Using DAX functional language to create complex DAX formula and execute flows with function calls
- Nesting row contexts on the same or on differnct tables; Conditional formatting with the same number in gradient color
- Pin slicers to the dashboard or Pin Report Pages to Dashboards in Office 365 /Power BI or Azure /Power BI platforms.
- Using Power BI to produce Json formated datasets from live websites & using pandas and scikit-learn in Python to classify & training data on Jupyter notebook, computing business trend results to predict and display them on Power BI or notebook
- Consulting and development on BPM products like TIBCO AMX BPM, Biztalk, Oracle BPM Suite11gR2, Pega BPM, TM1 etc
- Integrate web and B2B portal and Power BI portal; Develop reporting capability for Dynamics CRM users
- Administration of Dynamics CRM2016 & support maintenance; Manage data migration and systems configuration.
- Manage day-to-day issues and enhancement requests from business users and help facilitate resolution of the same.
- Support the Sales/Marketing team by partnering with the IT department admins to solve user's problems.
- Assist project managers/solutions architect in projects that touch upon Salesforce.com/Pardot.
- Develop all needed on Salesforce (Visualforce pages, Apex classes & triggers, Apex test classes, Community development, Lightning components, Flows, Workflows and Business Process Builder and other admin level configuration)
- 3rd party and vendor management using JIRA Confluence, FTS / Git source control and VS2017 build systems, Jenkins
- Install and configure Windows Server 2016 to deploy a Docker-managed Hyper-V VM directly from the Azure Marketplace using new Container technologies for next generation Docker Cloud needed for big data digitized solutions.
- As Senior Systems Administrator using LAMP Stack/AWS to help the RBA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP
- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Using wsl screenfetch etc. distro command line to swith between Windows 10 and Linux for B2B website development

- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Design and develop enterprise Dynamics 365 CRM solutions using Dynamics CRM2016 Developer Tookit for VS2017
- Setup SharePoint 2016 integration with Dynamics 365 CRM, including OneNotes integration in Dynmaics 365 CRM
- Developing a CI/CD strategy to deliver web and App based e-commerce solutions using Sitecore8.2U3 for end-customers and partners, including both traditional e-commerce models as well as Accelerated Mobile Pages(AMP) mobile applications
- Manage SharePoint system and content permissions, manage system changes, and enforcing policies to conform to firm operational standards. Enhancing and providing Technical support (up to 3rd level) on all IT applications
- Administration and technical support of SharePoint 2013/2016, Dynamics 365 CRM and .NET Applications
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Rich Internet application (RIA) for B2B website development either using Angular4/5 or React-Redux on .NET Core 2.0
- Next generation web app development either using Sails.Js or Koa expressive middleware with ES2017 or Server side backend framework Hapi or in the combination of SailsJs with Angular, or Koa with Redux and React.
- Using sails new clientapi to build custom, enterprise-grade Node.js Apps with realtime MVC and safe/alter/drop database options; Run sails generate api client to create model and controller with REST API in a minutes.
- High performance Mobile App development using Progressive Web App(PWA) with React/Redux and Accelerated Mobile Pages(AMP) on AWS either using Xamarin or ionic with Async JS & Cordova or React Native with JS and React Library.
- Client-Side development using App for SharePoint 2016 in VS2015TFS /VS2017 to develop SharePoint 2016 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- SQL 2014 /SQL 2016 development ecifically using Microsoft BI stack, including MDX and DAX in Power BI
- SharePoint Content migration (team sites) from SharePoint 2013 to 2016 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on “MS SharePoint 2013 Upgrade Process” best practice.
- Design and develop Digital Workflows using Nintex for SharePoint 2016, including Nintex Forms 2016 with Nintex Mobile.
- Develop and modify process flows by SharePoint 2016- Import Reusable (SharePoint) 2010 workflow in VS2017
- Administration/configuration of SharePoint 2013/2016, Dynamics 365, CRM 2016 and SQL Server 2014/2016 databases.
- Deep Learning with Python Tensorflow to train business data to predict future business trend on Jupyter notebook either in AWS or Google Cloud or Azure Cloud and disply well-trained data on Power BI with clickable drilldown on charts feature.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 10.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Microsoft Dynamics CRM Developer /Consultant /Senior SharePoint /BI /K2 Developer
Responsibilities:

- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using SmartForms in K2 Designer to grab multiple line-of-business systems’ databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Developing, customising and implementing MS Dynamics CRM 2015 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Management of technologies supporting organisation systems such as: VMWARE, Azure,Office 365 and Palo Alto NGFW
- SharePoint document managemen development using Total Record Information Management (TRIM 8.1).
- Continuous improvement of existing technologies, processes and adoption of new technology based on CI/CD principles.
- Proficient in B2B website development or web application programming either using Ruby on Rails (RoR), PHP in LAMP stack, .NET in WISA stack, Java in Eclipse or the combination programming of RoR, Django and NodeJS in Python, like:
- Implementing and using GraphQL in Python platform by combination of using Django, Rails and Node web frameworks by
  >>>pip install graphene [django] django_graphiql followed by >>>atom zero-django
  >>>gem install graphql rails-graphiql followed by >>>atom zero-rails
  >>>npm i --save graphql express-graphql followed by >>>atom zero-node
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Using VMware SRM /ESX servers with HA via vSphere6.5 Web Client to manage production and data recovery data centres
- Design and develop SharePoint pages, web parts, templates and libraries using Design Manager, SPD2013 and VS2015
- Deployment & configuration of custom SharePoint solutions by Install-SPSolution -Identity SolName.wsp -GACDeployment
- Maintain documentation relating to applications including development, troubleshoot and deployment
- Administration of multiple SharePoint Farm configurations either by PowerShell 3.0 scripts, SharePoint Central Administration or SharePoint Designer 2013, including Website configuration, customise backups and documentation.
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Senior .NET developer to deliver on a large and complex intranet and internet with SharePoint2013, CRM2015，Sitecore8.2 using Helix Project, feature & foundation architecture/MVC design patterns, including HTML5/CSS3/SCSS and JavaScript.
- Develop templates either by SharePoint Design Manager, SPD2013 or VS2015, including HTML5/CSS3, JavaScript/jQuery, item templates, onet.xml(for the definition of list templates available for creating new lists), basesitestempling.xml
- Managing SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Migrator /HarePoint or using Collab365 to migrate data, sites and documents from on-premises to hybrid or online.

- Client-Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Integrated BA Insight Enterprise Search solution for SharePoint 2013 using BA Insight Longitude Connectors to optimise and enable UC to rapidly implement powerful search-driven applications at a fraction of the cost, time and risk of alternatives.
- Setup RecordPoint for SharePoint 2013 ECM by RecordPoint for SharePoint 2013 Configure Pack Settings v3.x; Select Term Store, Group and Set used for RecordPoint. Pre-configure Managed Metadata Service Application in SharePoint CA etc.
- Integrated Newsgator with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location. Develop simple document management
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Migrated SP2010 WF to SP2016WF either by HarePoint or SP2016- Import Reusable (SP) 2010 workflow in VS2017
- Microsoft CRM 2015 Mutli-Server Deployment on Windows Server 2012R2/SQL Server2014/SharePoint 2013/IIS8.5 platforms, including Install and Configure the Outlook Client, Import Multi Users & CRM Online portal on Azure Cloud.
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client or using DreamFactory API automation to turn any database into API for analystics
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Microsoft Dynamics-CRM Developer /Consultant /Senior SharePoint /BI Developer
Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Administering SharePoint 2010/2013, Dynamics CRM 2011/2013 and Microsoft SQL Server 2012R2 databases.
- Design and develop enterprise CRM2013 solutions using Microsoft Dynamics CRM2013 Developer Tookit for VS2013
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- Troubleshooting and support of the environment across SharePoint 2010, SharePoint 2013 and Sharepoint Online
- SharePoint 2013 Profile Management and Synchronisation either by deploy Calligo or harmon.ie for SharePoint2013.
- Configuring and customising SharePoint implementations with AutoSPInstaller powershell scripts and SharePoint templates
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Leading customer SharePoint engagements for designing innovative SharePoint /Cloud solutions for EML customers.
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Integrating custom HTML5/CSS3 to SharePoint master page and page layouts using Adobe Photoshop, SPD2015&VS2015
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Using SharePoint 2013 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or visually in new Rest API's: http://spserver/_api/web or http://spserver/_vti_bin/client.svc/web
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Developing SharePoint Web Templates for on-permises and the Cloud via Office365, SharePoint Design Manager&VS2013
- SQL 2014 development ecifically Microsoft SQL Server 2008 R2 and 2012, including stored procedure, trigger, etc.
- Using test-driven development (TDD) and refactoring to support the focus on automation and continuous improvement
- Configure Metadata Term Store Management on for SharePoint 2013 landing page multi-navigate menus
- Changing the look of Search Results using Display Templates and Client Side Rendering (CSR).
- Implemented Microsoft EPM/PPM bespoke Solution via install MS Project Professional 2013, MS Project Server 2013 on top of SharePoint2013 Server Farm platform with ensuring solutions do not impose security risk to the Business.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.

- Using Ninject DI container for Context Variables with Entity Framework for decoupling the classes on the server.
- Mentoring and coaching other team members for customer-focused approach to control Agile/Scrum delivery process.
- Hands on experience with .Net application master page/SharePoint corp landing pages design,like skinning/re-branding.
- Integrated HP TRIM 7.2 with SharePoint 2010 to deliver Enterprise Content Management solutions, Document Management solutions, Nintex workflow solutions and custom solutions on the SharePoint platform.
- Service and application integration via Microsoft BPM's Tools like BizTalk2013's BAM, SharePoint workflow with InforPath form, Scan, approve and store invoices electronically into Dynamics GP or CRM by CRM Adapter or Nintex Workflow etc.
- Implement a high traffic website using Umbaraco 7, Sitecore CEP 7 according to OOD, DDD, IoC and other principles
- Perform unit (automated) testing; Support both QA testing (system & integration) and level 2 support for Production systems
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux 9.3 - 10.1 or Red Hat Enterprise Linux 3-4 for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Microsoft Dynamics CRM Developer /Consultant /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Gathering requirements from stakeholders and translate them into technical solution design for SharePoint/CRM online.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Perform technical business analytical work and suggest how to improve AS-IS process and develop TO-BE process enhancements through business operation automation by implementing SharePoint /Nintex solutions.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Integrated HP Trim 7.3 with SharePoint 2010 ECM, ERM and eDiscovery to supports SCC Electronic Document and Records Management System (EDRMS) Strategic Plan and articulates the policy adopted within DFAT.
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Client side development by SharePoint Desinger 2010 with jquery, HTML5/CSS3 and App for SharePoint in VS2012
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Microsoft Dynamics-CRM Developer /Consultant /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to develop effective Hybrid and Private Cloud solutions for new customers from architecture design to and implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on AWS Cloud.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Install and configure AD FS Server Farm with load balancing clusters in managing a mid to large AD environment.
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Add an Exchange Edge Server to Lync Server 2010, create Edge pool and publish the topology on SCSM2007 Dashboard.
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.
- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.
- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.

- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior SharePoint /Office Online Administrator /Full Stack Developer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Design DWH concepts from Bill Inmon & Ralph Kimball and build data models via Toad Data Modeler 10.x &Informatica 8.x;
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and Agile /Scrum delivery methodology
- Develop and customise Dynamics CRM solutions using C#, VB, ASP.Net in Visual Studio 2008TFS.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective-C formulas for insurance premium calculations
- Create financial reports either by SSRS or SAP Crystal Reports for VS2008 or Business Object BOXIR2 on HP-UX platform
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both Jave FileNet workflows and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Software Engineer /Senior SharePoint /Full Stack .NET Developer / DW Consultant
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) with Azure cloud and being responsible for delivering the products to market.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Install and configure ADFS Server Farm with load balancing clusters in managing a mid to large AD environment, including AD migrations, cross-forest migrations and interoperability scenarios
- Plan, design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for delivering the products to market.
- Setup SCCM Site Server, Site DB Server, WSUS DB, Reporting Services DB with Management Points & Distribution Points to provide IaaS, Saas, PaaS services for corporate clients.
- Standardising and automating software delivery processes such as: Version control by TFS/Git; Consistent eInvironment configuration, build and packaging by SCCM2007; Deployment by -Altiris and release management by SCSM and SCOM
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions. Advanced data query via LINQ-to-SQL
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OODesign patterns(Singleton, Factory etc.) based on business requirements.
- Maintaining and auditing real time security of SQL Server databases via customizable SQL Compliance Manager.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- Deliver quality Corporate Portal implementation services that include both training and consulting
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-Tier web applications with C#3.5, ASP.Net, MVC3/4, Service-Oriented Applications (Workflow Application, Servcie Library) using WCF to GetData and create righ GUI applications for UI/UX using WPF with data-bound controls.
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Developing N-tier & multi-threaded applications using C#.NET4, MVC3/4, ADO.Net, WPF, WCF, Rest, Linq, WinForms, etc.
- BizTalk2006(R2)/2209, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC3/5), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Solaris8-12, HP-UX8-12c, RHEL5-7, SLES7-12, CenOS 5.0-6.5, AIX5.3/6.1, Distro, etc
Database: Oracle DB9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005-2016, Sybase, DB2, MySQL
CMS: SharePoint 2007-2016, MS Dynamics CRM3/4/2011/2013/2016/365, Sitecore, Kentico, Umbraco4.x-7.x, HP Trim 7.2~8.x

Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Tomcat, IIS 6~8.5, LAMP, WISA, TIBCO AMX BPM
Virtualization: Citrix Xen 5.0~7.x, VMWare ESX 3.5~5.x / ESXi, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS, etc.
Security: SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Encryption with US Data Encryption Standard (DES), RSA Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
BI Tools: SQL Server 2005~2016, Visual Studio 2005-2015, Tableau 7~10.x, Eclipse, NetBeans, Flex Builder, SQL Developer, SSMS /BIDS2008R2, Power BI in Office 365 /Azure, OBIEE11g, OWB10gR2, Business Object, Cognos 8, TM1 10.2 & QlikView
Framework: .Net2.0~5/Core, NodeJs, AngularJS/2/4/5, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
Workflows: K2 Workflow, Nintex 2010-2015 Workflow and SharePoint 2007 -2016 Workflows, Workflow Manager1.0

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English /Japanese Interpreter
Senior SharePoint Administrator /Sr SharePoint /D365 CRM /Nintex K2 Consultant/DWH/DBA –Japan (7.5+yrs) & Aust.(23yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality and Safety Commission (ACQSC), Sydney**
Health Care: Senior SharePoint Administrator /Senior SharePoint /D365 CRM /Nintex K2 Counsultant /DBA
Responsibilities:

- Architect, develop and implement clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2019/IIS10
- Integrate Dynamics 365 CRM application with SharePoint 2019 using Office365 either on-premise**s** or in Hybrid /Private Azure Cloud by configuring MS Windows Server Active Directory infrastructure and Single Sign On (SSO) via ADFS.
- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks using K2 Configuration Analysis, K2 Designer, K2 Package and Deployment, K2 Server (Service account), K2 Management, Managing application administrators list, Workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.
- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, MS Power Automate workflows, SharePoint 2019 either on AWS or on Nintex K2 Cloud.
- Implemented Elastic Load Balancing (ELB) for Nintext K2, SharePoint 2019 on AWS' EC2 instance and EBS storage to distribute traffic among SharePoint front-end servers for high availability and load distribution.
- Create and maintain SmartObjects using Nintex K2 Designer and create Nintex K2 packages in Nintex K2 Five or K2 Cloud
- Collaborated with cross-functional teams to successfully migrate SharePoint 2013 to SharePoint 2019 on AWS' EC2, EBS.
- Provide technical support for hybrid aged care business applications (SharePoint, Dynamics CRM &TIBCO BPM workflows)
- Managing the upgrade from Sharepoint 2013 to Sharepoint 2016/19 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Providing incident, service request and problem management support to users in their daily use of SharePoint applications
- Upgraded Better Business Program system to Dynamics 365 CRM to manage business activities by upload various eForms to SharePoint 2019 Doc List triggering TIBCO AMX BPM Workflows for multi-parallel workflows via TIBCO iProcessor.
- Developing, customising and implementing MS Dynamics 365 /CRM 2016 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Monitor and resolve issues on Office 365, AWS, Azure and SharePoint / SharePoint online platforms
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Financial Bank: Senior SharePoint Administrator /Senior SharePoint /Nintex K2 Consultant /Power BI Developer
Responsibilities:

- Administered SharePoint environments, managing installation, configuration, and maintenance of SharePoint servers and services, including administering Nintex K2 workflows published onto SharePoint.
- Migrated on-premise SharePoint 2013/2019 with Nintex K2 forms/workflows from Macquarie Government DC to AWS
- Performing Nintex K2 administrative tasks: Managing application administrators list, Inviting users to use the Apps, Enabling link for favorite Appls workspace, App seetings, Managing workflow task approvers lists etc.
- Implemented and enforced SharePoint security concepts, ensuring user permissions, groups, and roles align with best practices, including roles in Nintex K2 workflows.

- Diagnosed and resolved complex issues related to SharePoint configuration, performance, and compatibility with external systems, such as Nintex K2, SharePoint either on AWS or on Nintex K2 Cloud.
- Provided technical operational support for mission-critical hosted applications, ensuring high availability and optimal performance.
- Played a key role in troubleshooting and resolving connectivity and compatibility issues with third-party systems like Nintex.
- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using KingswaySoft XrmSvcToolkit lightweight open source library to execute CRUD operations and remove all the complications from working with the REST Web API interface to offer an easier method of working with CRM Data.
- Designing, developing and maintaining BI and Reporting solutions either using SSRS, Power BI, Tableau 10.x or QlikView;
- As Senior Systems Administrator using LAMP Stack/AWS to help the RBA team deliver excellent, high quality service using the latest Conteners within Docker Cloud infrastructure and web development technologies either in **WISA,** MEAN or LAMP
- Manage EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Using wsl screenfetch etc. distro command line to switch between Windows 10 and Linux for B2B website development
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Design and develop enterprise Dynamics 365 CRM solutions using Dynamics CRM2016 Developer Tookit for VS2017
- Setup SharePoint 2016 integration with Dynamics 365 CRM, including OneNotes integration in Dynmaics 365 CRM
- Developing a CI/CD strategy to deliver web and App based e-commerce solutions using Sitecore8.2U3 for end-customers and partners, including both traditional e-commerce models as well as Accelerated Mobile Pages(AMP) mobile applications
- Manage SharePoint system and content permissions, manage system changes, and enforcing policies to conform to firm operational standards. Enhancing and providing Technical support (up to 3rd level) on all IT applications
- Administration and technical support of SharePoint 2013/2016, Dynamics 365 CRM and .NET Applications
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Client-Side development using App for SharePoint 2016 in VS2015TFS /VS2017 to develop SharePoint 2016 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- SQL 2014 /SQL 2016 development ecifically using Microsoft BI stack, including MDX and DAX in Power BI
- SharePoint Content migration (team sites) from SharePoint 2013 to 2016 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on “MS SharePoint 2013 Upgrade Process” best practice.
- Design and develop Digital Workflows using Nintex for SharePoint 2016, including Nintex Forms 2016 with Nintex Mobile.
- Develop and modify process flows by SharePoint 2016- Import Reusable (SharePoint) 2010 workflow in VS2017
- Administration/configuration of SharePoint 2013/2016, Dynamics 365, CRM 2016 and SQL Server 2014/2016 databases.
- Deep Learning with Python Tensorflow to train business data to predict future business trend on Jupyter notebook either in AWS or Google Cloud or Azure Cloud and disply well-trained data on Power BI with clickable drilldown on charts feature.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 10.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Senior SharePoint Administrator /Senior SharePoint /K2 Developer /Dynamics CRM Consultant
Responsibilities:

- Administration/configuration of clustered SharePoint WFE-Tier Web servers either with F5 or Netscaler load balancer, APS-Tier Application Server Farm and SQL-Tier mirrored SQL Database Servers sitting on Windows Server 2012R2/IIS8.5
- Using SmartForms in K2 Designer to grab multiple line-of-business systems’ databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Managed SharePoint farm with a focus on maintaining web applications, site collections, and content databases.
- Collaborated with development teams to design and implement custom workflows using SharePoint Designer and K2.
- Conducted regular security audits and implemented enhancements to align with industry best practices.
- Developing, customising and implementing MS Dynamics CRM 2015 solutions and plugIns either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics Developer Toolkit with VSO(Visual Studoi Online) / VS Code, TFS or Git.
- Management of technologies supporting organisation systems such as: VMWARE, Azure,Office 365 and Palo Alto NGFW
- SharePoint document managemen development using Total Record Information Management (TRIM 8.1).
- Continuous improvement of existing technologies, processes and adoption of new technology based on CI/CD principles.
- Automation and Scripting either using PowerShell, PnP-PowerShell or using Git-Bash, Chef, Puppet, or Ansible tools
- Using VMware SRM /ESX servers with HA via vSphere6.5 Web Client to manage production and data recovery data centres
- Design and develop SharePoint pages, web parts, templates and libraries using Design Manager, SPD2013 and VS2015
- Deployment & configuration of custom SharePoint solutions by Install-SPSolution -Identity SolName.wsp -GACDeployment
- Maintain documentation relating to applications including development, troubleshoot and deployment
- Administration of multiple SharePoint Farm configurations either by PowerShell 3.0 scripts, SharePoint Central Administration or SharePoint Designer 2013, including Website configuration, customise backups and documentation.
- Using Adobe Photoshop, Dreamweaver to develop templates for BAU use, Prepare correspondence and reports as directed
- Implemented SharePoint 2013 Electronic Document and Records Management System (EDRMS) with HP Trim 8.1
- Develop templates either by SharePoint Design Manager, SPD2013 or VS2015, including HTML5/CSS3, JavaScript/jQuery, item templates, onet.xml(for the definition of list templates available for creating new lists), basesitestempling.xml

- Managing SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Migrator /HarePoint or using Collab365 to migrate data, sites and documents from on-premises to hybrid or online.
- Client-Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Integrated BA Insight Enterprise Search solution for SharePoint 2013 using BA Insight Longitude Connectors to optimise and enable UC to rapidly implement powerful search-driven applications at a fraction of the cost, time and risk of alternatives.
- Setup RecordPoint for SharePoint 2013 ECM by RecordPoint for SharePoint 2013 Configure Pack Settings v3.x; Select Term Store, Group and Set used for RecordPoint. Pre-configure Managed Metadata Service Application in SharePoint CA etc.
- Integrated Newsgator with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location. Develop simple document management
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Migrated SP2010 WF to SP2016WF either by HarePoint or SP2016- Import Reusable (SP) 2010 workflow in VS2017
- Microsoft CRM 2015 Mutli-Server Deployment on Windows Server 2012R2/SQL Server2014/SharePoint 2013/IIS8.5 platforms, including Install and Configure the Outlook Client, Import Multi Users & CRM Online portal on Azure Cloud.
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client or using DreamFactory API automation to turn any database into API for analystics
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution and Feature design and implementation across multiple Web Applications with EML Work Cover Assessment Nintex Workflow in SharePoint 2013 Online with Office 365 /Nintex2013/CRM2013/SQL Server 2012/SSRS platforms.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Assisted in the migration of SharePoint 2010 to SharePoint 2013, ensuring seamless data transition and minimal downtime.
- Provided technical support to end users, resolving issues related to permissions, access, and usability.
- Contributed to the development of documentation and training materials for SharePoint users.
- Expand Dynamics CRM2013 functionality by customising Dynamics CRM to better fit the company and customers' needs
- Developing Dynamics CRM plugins either using KingswaySoft XrmSvcToolkit or Microsoft Dynamics 365 Developer Toolkit
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Administering SharePoint 2010/2013, Dynamics CRM 2011/2013 and Microsoft SQL Server 2012R2 databases.
- Design and develop enterprise CRM2013 solutions using Microsoft Dynamics CRM2013 Developer Tookit for VS2013
- Developing custom farm solutions in SharePoint 2013 using Design Manager, SP2013 Designer and TFS2015/VS2015
- Support the manager with other tasks as required. Ensure transfer of knowledge to the rest of the team
- SharePoint 2013 Profile Management and Synchronisation either by deploy Calligo or harmon.ie for SharePoint2013.
- Configuring and customising SharePoint implementations with AutoSPInstaller powershell scripts and SharePoint templates
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Leading customer SharePoint engagements for designing innovative SharePoint /Cloud solutions for EML customers.
- SharePoint Content migration (team sites) from SharePoint 2010 to 2013 either quickly by AvePoint DocAve Quickr Migrator /HarePoint/MetaLogix/Sharegate or manually migrate based on "MS SharePoint 2013 Upgrade Process" best practice.
- Integrating custom HTML5/CSS3 to SharePoint master page and page layouts using Adobe Photoshop, SPD2015&VS2015
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Using SharePoint 2013 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or visually in new Rest API's: http://spserver/_api/web or http://spserver/_vti_bin/client.svc/web
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Developing SharePoint Web Templates for on-permises and the Cloud via Office365, SharePoint Design Manager&VS2013
- SQL 2014 development ecifically Microsoft SQL Server 2008 R2 and 2012, including stored procedure, trigger, etc.
- Using test-driven development (TDD) and refactoring to support the focus on automation and continuous improvement

- Configure Metadata Term Store Management on for SharePoint 2013 landing page multi-navigate menus
- Changing the look of Search Results using Display Templates and Client Side Rendering (CSR).
- Implemented Microsoft EPM/PPM bespoke Solution via install MS Project Professional 2013, MS Project Server 2013 on top of SharePoint2013 Server Farm platform with ensuring solutions do not impose security risk to the Business.
- Mentoring and coaching other team members for customer-focused approach to control Agile/Scrum delivery process.
- Hands on experience with .Net application master page/SharePoint corp landing pages design,like skinning/re-branding.
- Integrated HP TRIM 7.2 with SharePoint 2010 to deliver Enterprise Content Management solutions, Document Management solutions, Nintex workflow solutions and custom solutions on the SharePoint platform.
- Service and application integration via Microsoft BPM's Tools like BizTalk2013's BAM, SharePoint workflow with InforPath form, Scan, approve and store invoices electronically into Dynamics GP or CRM by CRM Adapter or Nintex Workflow etc.
- Implement a high traffic website using Umbaraco 7, Sitecore CEP 7 according to OOD, DDD, IoC and other principles
- Perform unit (automated) testing; Support both QA testing (system & integration) and level 2 support for Production systems
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- HP-UX 12c, SLES 12, RHEL 7 systems administratoring via HP LVM and RedHat LVM, Linux LVM and hpaculi utilities
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux 9.3 - 10.1 or Red Hat Enterprise Linux 7.2 for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Senior SharePoint Administrator /Senior SharePoint /BI Developer /MS Dynamics CRM Developer
Responsibilities:

- Solution Architect to provide SharePoint /Nintex Solution designs and technical architecture for SCC SharePoint 2010 farm with F5 application load balancing in front-end web servers and Clustering Failover for back-end SQL database servers.
- Gathering requirements from stakeholders and translate them into technical solution design for SharePoint/CRM online.
- Consulting and development on BPM products like K2, Nintex, Biztalk, Oracle BPM Suite11gR2, Pega BPM, IBM TM1 etc
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Supported the administration and maintenance of SharePoint 2013 environments, gaining foundational experience in SharePoint infrastructure.
- Assisted in user training sessions to improve SharePoint adoption and usage among employees.
- Collaborated with senior administrators to troubleshoot and resolve configuration issues.
- Perform technical business analytical work and suggest how to improve AS-IS process and develop TO-BE process enhancements through business operation automation by implementing SharePoint /Nintex workflow solutions.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Determing the record type of managed SharePoint content and how SharePoint Metadata is to be captured (Mapping columns to HP TRIM fields, Configuring Custom Mappings, maintenance of mappings and mapped values)
- Apply records management to SharePoint Content; Records management options(RMO);Enabling the use of RMOs, Site Level RMOs, List level RMOs and Self Service BI with SharePoint 2010
- Using SharePoint 2010 SSOM -SPSite, SPweb, SPList, SPListItem Objects to create items, updae and query items either in ASP.Net C# or PRC Call(owssrc.dll), SPServices, ASMX Web Services, Rest API's: http://spserver/_vti_bin/listdata.svc
- Using C#, MVC5, App for SharePoint in VS2010 to deliver SharePoint 2010 /Nintex solutions in an enterprise environment.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- Integrated Yammer with on-premises SharePoint 2013 to provide a richer enterprise social experience and bring employees, content, conversations and business data in a single location.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.
- Developing web parts, event receivers, web portals, web service via VS2010TFS,CAML in SharePoint 2010 platform

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior SharePoint Administrator /Senior SharePoint /BI Developer
Responsibilities:

- Solution Architect to develop effective Hybrid and Private Cloud solutions for new customers from architecture design to and implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on AWS Cloud.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- System Design and Development using Office Developer Tools & Dynamics CRM 2011 Developer Toolkit for VS2012
- Install and configure AD FS Server Farm with load balancing clusters in managing a mid to large AD environment.
- DevOps deployment across multiple cloud platforms (AWS, Salesforce, Windows Azure, Oracle Exa Cloud).
- Install and configure CAS Server interfacing centric Active Directory to setup Exchange 2010 HA Network Load Balancing
- Add an Exchange Edge Server to Lync Server 2010, create Edge pool and publish the topology on SCSM2007 Dashboard.
- Create and develop Content Query Webpart with XSLT, Search Results Webpart for Enterprise Search Center.
- Branding SharePoint2010 by Yeoman, SharePoint Designer2010 and deploy MegaMenu for corp landing pages in VS2010.

- Design and developing Dynamic Data Entry User Interfaces, ASP.Net Dynamic Data Scaffolding and Page Templates.
- Performance Tuning of Applications via SSIS on BIDS2008R2 platform with capacity to produce high level technical documentations and results via SSRS; Develop, monitor and review migration of ETL packages using SSIS.
- Integrate AJAX into a Spring MVC application using JQuery on a client side for sending websphere JSM queries.
- Create financial reports either by SSRS or SAP Crystal Reports for VS2011 or Business Object BOXIR2 on HP-UX platform

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior SharePoint Administrator /Full Stack Developer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Design DWH concepts from Bill Inmon & Ralph Kimball and build data models via Toad Data Modeler 10.x &Informatica 8.x;
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and Agile /Scrum delivery methodology
- Develop and customise Dynamics CRM solutions using C#, VB, ASP.Net in Visual Studio 2008TFS.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective-C formulas for insurance premium calculations
- Create financial reports either by SSRS or SAP Crystal Reports for VS2008 or Business Object BOXIR2 on HP-UX platform
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both Jave FileNet workflows and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Software Engineer /Senior SharePoint Administrator / Developer/Full Stack .NET Developer
Responsibilities:

- Team leader to plan, design and develop large and complex enterprise level packaged software for BI web solutions across Microsoft BI Suite (SQL/SSIS/SSAS/SSRS) with Azure cloud and being responsible for delivering the products to market.
- Client interaction and **c**onsulting with an array of stakeholders to elicit requirements for SharePoint online with Office365.
- Install and configure ADFS Server Farm with load balancing clusters in managing a mid to large AD environment, including AD migrations, cross-forest migrations and interoperability scenarios
- Plan, design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for delivering the products to market.
- Standardising and automating software delivery processes such as: Version control by TFS/Git; Consistent eInvironment configuration, build and packaging by SCCM2007; Deployment by -Altiris and release management by SCSM and SCOM
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions. Advanced data query via LINQ-to-SQL
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OODesign patterns(Singleton, Factory etc.) based on business requirements.
- Maintaining and auditing real time security of SQL Server databases via customizable SQL Compliance Manager.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-Tier web applications with C#3.5, ASP.Net, MVC3/4, Service-Oriented Applications (Workflow Application, Servcie Library) using WCF to GetData and create righ GUI applications for UI/UX using WPF with data-bound controls.
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Developing N-tier & multi-threaded applications using C#.NET4, MVC3/4, ADO.Net, WPF, WCF, Rest, Linq, WinForms, etc.
- BizTalk2006(R2)/2009, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions

EDUCATION / CERTIFICATION
Bachelor of Automobile Technology, Shanghai University, China [1987]
Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS
Programming Language : Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
OS: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

BI, AI, ML, DL, IDE: SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8, TM1 10.2; Azure OpenAI, Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, ATLAS.ti, etc.
Database: Oracle9i/10gR2/11gR2, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, DB2, MySQL
Clouds: Microsoft Azure Cloud, Amazon Cloud (AWS), Google Cloud Platform (GCP)
CMS: SharePoint 2007~2022, CRM3.0~D365 CRM, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
Server: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
Virtualization: Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
Security: MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
Framework: .Net2.0~5/Core, NodeJs, AngularJS/2/4/5, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
Workflows: K2 Workflow, Nintex Workflow 2007~2019, SharePoint 2007~2019/O365, MS Flow~Power Automate workflow, SharePoint Workflow Manager for SharePoint Server, TibCo/SharePoint/CRM Workflow, BizTalk Server 2006R2~2020 workflow

8/29/2023
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Full Stack Software Engineer/MT4 Specialist/Java /.NET Developer /SharePoint /CRM Developer -Japan(7+yrs)& Aust.(18yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present Australia Aged Care Quality Agent (AACQA)/Aged Care Quality Safety and Commission, **Sydney**
Health Care: **Full Stack Software Engineer/Senior Golang Developer /JavaScript Developer**
Responsibilities:

- Designing, architecting and developing stateless, independantly deployable & highly available gRPC Aged Care ecosystem either quickly by AMIS Tools (Contract as a Service, CaaS) or using AWS elastic Beanstaklk Docker-based Go Platform.
- Collaborated with the Hardware team to design, develop, and test real-time signal generation and trade execution applications for high-frequency trading.
- Utilized Python and multithreaded programming to achieve low-latency performance and high reliability.
- Developed API's and Dashboards to provide real-time insights and monitoring of trading activities.
- Applied Async programming using Python Asyncio to optimize system performance.
- Leveraged frameworks like Flask and Django to build robust and scalable components.
- Implemented complex trading algorithms, demonstrating a deep knowledge of Python and high-performance trading systems.
- Responsible for staying up-to-date on emerging technologies and promoting usability best practices with streaming Adged Care services by translating software requirements into concise, maintainable, loosely-coupled and cohesive code
- Design and develop containerized stateless microservices contract services for easily deployed on container orchestration platform like Kubernetes covering RPC Interface, RPC framework, uses gRPC as the RPC infrastructure of microservice…
- Using **Protocol Buffers**(a.k.a. **protobuf**) as the Interface Description Language (IDL) for gRPC to support many popular programming languages such as C++, Java, Go and Python, which is widely used by Google, CoreOS and Netflix.
- Deploying Buffalo to Heroku with Docker; Using Dockerfile to run Buffalo application on Heroku 7.x
- Using $heroku create to create Heroku Remote repository and using Push web to push the Docker Image to Heroku
- Enhance **Smart Contract** Platform. Speed Deal Cycles. Create Winning Deals. Increase Deal Size and Close More Deals.
- Either using Go-Ethereum or .NET Core integration library for Ethereum to build cross-platform blockchain applications.
- Develop Decentralized Apps (DApps) with Ethereum; Streamlining the Business with **Smart Contracts**, **Blockchain** and **AI**.
- Creating self-executing Smart Contracts having the terms and conditions, similar to MT4 /MT5 Expert Advisors(EAs)
- Write upgradeable Smart Contract in solidity and use compiled Smart Contract's Application Binary Interface(ABI) to interact with the contract deployed to Ethereum blockchain.
- Using sol2proto as the bridge between gRPC and ABI to extract required data structures and functions from contract ABIs and generates corresponding **.proto** files. Then use **protoc** to generate the application's language-specific source code based on the **.proto** files and using gRPC-gateway to convert gRPC APIs to RESTful APIs.
- Build a decentralized conference ticket purchasing web app on Ethereum platform with blockchain technology.
- Undertake design and development activities including coding, testing, deployment scripting and configuration areas in .NET domains and other related technologies
- Deploying Buffalo to Heroku with Docker; Using Dockerfile to run Buffalo application on Heroku 7.x
- Using Heroku create to create Heroku Remote repository and using Push web to push the Docker Image to Heroku
- Build a decentralized conference ticket purchasing web app on Ethereum platform with blockchain technology.
- Front-end PWA web Application development using ReactJs covering service workers, pass data, raise and handle events, multiple components in Sync, Async/Await, Promise resolve, Babel, JSX, functional components and lifecycle hooks.
- Building components and applications using complex approval workflows, visualizing loan data, messaging, document generation, etc. either on AWS or Azure Cloud, includes mobile applications.

- Java, spring, microservices, Agile Development, CI/CD DevOps
- Blockchain (Quorum, Etherium-C# type Etherium), SQL (Postgres/MySQL/Hibernate),
- **Adopted Heroku Postgres** SQL database as a service with operational expertise built in, easy setup, security by default, database forking, credentials, etc.
- Supporting cross-platforms' production messaging and collaboration systems.

Dec. 2015 – Dec.2017 **Reserve Bank of Australia (RBA), Sydney**
Equity / FX Dealing Room: Murex IT Trade Support Analyst /MT4 /MT5 Specialist /Senior Software Engineer
Responsibilities:

- Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Contributed to the design and architecture of real-time trading systems, ensuring efficiency and reliability.
- Collaborated with cross-functional teams to optimize trading algorithms, improving execution speed and accuracy.
- Worked in a real-time Linux environment, coding solutions to meet low-latency requirements.
- Utilized Pandas and NumPy for data manipulation and analysis in trading strategies.
- Played a key role in identifying and addressing performance bottlenecks, enhancing overall system efficiency.
- Online Trading Platform provider offers an exceptional trading experience in forex, indices, commodities and more.
- Being responsible for the continuous operation and running of the MT4 and MT5 systems.
- Providing operational and trade support for the Murex MX3 /MT4/MT5 platforms for desks trading Interest Rate Derivatives, Credit Derivatives, Commodities, Fixed Income, FX Options, Fixed Income products and Group Treasury services.
- Integrated within the platform is the ability to perform pricing of credit valuation adjustments and collateral management.
- Supports the end to end services required for trading including Risk Management, Product Control and Market Operations.
- Works closely with the business as well as development and project teams delivering into the Murex /MT4/MT5 platforms.
- Follow service management processes to trouble shoot and resolve issues raised by customers within the context of the Murex MX3 /MT4/MT5 platforms
- Design, document, configure and test new configuration on the Murex /MT4 /MT5 platforms to meet business specifications
- Propose, design, document and implement approaches to improve/automate processes for configuring and testing the end to end Murex environment
- Gather and document business requirements, and design and develop solutions within an agreed timeframe, adhering to the Banks Solution Delivery Life Cycle (SDLC) process and documentation framework
- Provide 1st, 2nd and 3rd level support where appropriate as well as escalate and manage issue resolution with the Murex /MT4/MT5 vendors
- Experienced Murex Analyst with front office experience and particular experience working with the Murex VaR module including configuration of reports and scenario containers.
- Using Murex VaR module to calculate Prob{delta Pie <<-VaR}=0.05
- Either using Murex VaR module Linear Valuation: The delta normal valuation methods or Full Valuation: Monte Carlo an Historic Simulation method to setup VaR=3% to 5% based on each profilio clients requests.
- Possess significant exposure to financial markets environments and products (Interest Rate Derivatives, Commodities, Options, Equities, FX Options) and detailed knowledge of various derivative and traded instruments including different valuation and risk methodologies. Expertise on the datamart module in Murex and ability to independently troubleshoot datamart report issues would be a significant advantage. Your detailed knowledge of Murex simulation views and configuration combined with your ability to engage directly with business staff (Trades, Quants, Managers, Product Control, Market Operations etc.) will help you succeed.
- Develop Decentralized Apps (DApps) with Ethereum; Streamlining the Business with **Smart Contracts**, **Blockchain** and **AI**.
- Creating self-executing Smart Contracts having the terms and conditions, similar to MT4 /MT5 Expert Advisors(EAs)
- Write upgradeable Smart Contract in solidity which run on Ethereum Virtual Machine (EVM) on Blookchain.
- Enhance **Smart Contract** Platform. Speed Deal Cycles. Create Winning Deals. Increase Deal Size and Close More Deals.
- Either using Go-Ethereum or .NET Core integration library for Ethereum to build cross-platform blockchain applications.
- Collaborate with internal developers and key stakeholders for prime brokerage, equity swaps businesses.
- Using sol2proto to generates corresponding **.proto** files. Then use **protoc** to generate the application's language-specific source code based on the **.proto** files and using gRPC-gateway to convert gRPC APIs to RESTful APIs.
- Deliver bespoke and versatile services, with an intuitive user interface and top-notch security, backed by a team that is highly experienced in FX and CFD trading across a wide range of trading issues.
- Monitoring and managing client positions and risk; Monitoring and managing any issue with the LPs
- Producing daily trade & reconciliation reports in relation to trading activity & client positions using MT4 & third-party platforms.
- Speaking to clients, as required, to resolve any trading/ account issues they may have
- MetaQuotes MtApi.mq4 and MtApi.mq5 new features development in C#.NET either by VS 2017 or cross-platform VS Code.
- Using MetaQuotes MQL4 /MQL5 Meta Editor to develop and configure Expert Advisors trading indicators from traditional MACD, MA, Williangs, Bollinge Band etc. to Japanese Ichimoku.mq, Ichimoku Cloud showed in Japanese candlestic chart.
- Monitor and manage any issue with LPs; Follow up transactions and orders with LP's, Banks and Prime Brokers, as required

- Dealing with Equities (both CASH and CFD), corporate actions and system & IT implementations
- Back office allocations and accounts setup for trading Shares, CFDs and FX, and be open to learning about new products
- Handle any quotes issue with the real time server, such as freezing, gaps, filtrations
- Training company employees to use MetaTrader4/MetaTrader5 with regular advise staff on any new update
- Provide informed technical support across a variety of Windows and web Platforms
- Follow-up with aggregators/bridges and other related technologies issues or updates
- Configuring MT4 /MT5 settings, such as groups, securities, manager rights, etc
- Switching between backups server and main server whenever its required
- Creating new symbols and configuring them on MT4 and MT5, including MT4, MT5 and Web terminals testings.
- Configuring and installing server side plug-ins; Maintenance of the MT4 and MT5 systems
- Helping manage system risk and management in mitigating of such risk.Assisting with any regulatory/ASIC queries.
- Regulatory reporting with aproved ASIC Controlled function for financial instituation requests
- Back office allocations and accounts setup for trading Shares, CFDs and FX
- Understanding of cash equities, corporate actions along with relevant back office implementation and procedures
- Troubleshoot FIX engines to facilitate online trading using FIX Protocol, establishing FIX connectivity between client and broker or client and exchange; Using VeriFIX tool to automate testing and simulate FIX counterparties for repid deployment
- Using customisable script-based testing service FACTS2.0 to mirror the FIX implementation of any sell side or exchange at both session-level & application level messages.Performing FIX Automated Certification and Testing Service by B2BITS FIX
- Using the FIX Protocol to monitor FIX Gateway message routing service for hub with direct connection advantages.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Strong attention to detail, problem-solving skills and the ability to prioritise work, manage time and meet deadlines
- Developing efficient and type-safe services in Go with gRPC server via advanced blockchain technology
- Server streaming by rpc GetCurrentcyStream(CurrencyRequest) returns (stream Currency){}
- Client streaming by rpc SaveCurrencyStream(stream Currency) return (CurrencyList){}
- BI-directional streaming by rpc FindCurrencyStream(stream CurrencyRequest) returns (stream currency){} Decentralised Apps
- Setup connection to gRPC service by: func main(){conn, err := grpc.Dial"localhost:"500050", grpc.WithInsecure()…
- Get live trading currency details by calling remote method of func printAUD(client pb.CurrencyServiceClient){curList,err:=client.GetCurrencyList(context.Background(),curReq)
- Update currency data by streaming Instrument Design Library (IDL) via Server streaming, Client streaming and BI directional streaming
- Define the CurrencyService class; Register it with Injector and Declare it as dependency in CurrencyList and CurrencyDetail
- Create injectable CurrencyService class using @Injectable() method in currency.service.ts; Import { injectable } from '@angular/core'; and export CurrencyService class in app.component.ts for where need these dependency places.
- Trading application testing using Automated testing and TDD skills with analytical tools R and python on Jupyter notebook
- Design and develop scriptable and FXML-enabled GUI JavaFX FXML Application with NetBeans plus JDK bundle;
- Either using FraudLabs Credit Card Fraud Detection API or Python SOAP/REST web service to detect fraud credit cards.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 9.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Senior Software Engineer /Python Test Analyst /Full Stack Developer /K2 Developer
Responsibilities:

- Design, develop and build Decentralized Apps (DApps) based gRPC in DApp Architecture either quickly by AMIS Tools (Contract as a Service, CaaS, part of Blockchain as a Service) or using AWS elastic Beanstaklk Docker-based Go Platform.
- Build a decentralized conference ticket purchasing web app on programmable Ethereum blockchain.
- Design, develop and build JS Decentralized Apps (DApps) with Ethereum
- Developed and maintained Python-based trading systems, focusing on real-time signal generation and execution.
- Collaborated with traders and analysts to implement algorithmic trading strategies.
- Designed and optimized algorithms for high-frequency trading, achieving competitive advantages.
- Contributed to the development of monitoring tools and dashboards for real-time performance analysis.
- Engaged in code reviews and knowledge sharing to ensure code quality and best practices.
- Design and develop real-time multi-threaded Progressive Web App (PWA) trading application either using Angular 2 with Java 9 or Prism 5.0 for WPF in VS2017 with Reactive Extensions for .Net(Rx) to observe, throttle and conflate data.
- Either using modularized JDK 8 or J2EE in elipse to define service muduleNames; Export the module packages to make them available to use and then use require() function to invoke the modules to where needed places.
- Proficient web application programming either using Ruby on Rails (RoR), PHP in LAMP stack, .NET in WISA stack, Java in Eclipse or the combination programming of RoR, Django and NodeJS in Python, like:
- Implementing and using GraphQL in Python platform by combination of using Django, Rails and Node web frameworks by >>>pip install graphene [django] django_graphiql followed by >>>atom zero-django

>>>gem install graphql rails-graphiql followed by >>>atom zero-rails
>>>npm i --save graphql express-graphql followed by >>>atom zero-node

- Developing Python, Ruby on Rails 5, Go(Golang), Java and Angular2 web Apps in cross-platform programmable open source Visual Studio Code
- Jointly using Ruby on Rails 5 (RoR) and Django in Python 3.6 doing GraphQL query on Graph Database with fastest JSON formatted research reports. Maintain BAU automation and performance testing framework and tool set.
- Perform financial data analysis using Pandas .
- Architect and implement MS Analytics Platform System(APS) to combine Massively Parallel Processing (MPP) SQL Server DWH with Hadoop Distribution Insight (HDInsight) to support Azure /Power BI stream analytics for digital transformation.
- Design and develop extendable Sitecore layered Habitat solutions either using C#6.Net5, WPF/XAML or MVC6 with Angular 2.0, ES6+, Typescript or React, NodeJs in UC SharePoint/Sitecore 8.1XP platforms connecting to SQL 2014.& mongoDB.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- Predicting international students interests either using AWS Machine Learning or Sitecore Personalisation with pattern card
- Working in a formal Agile SCRUM environment for end-to-end delivery, including design, development, testing and support
- Delivered high quality education services to maximise student learning either using WS-Security or SAML SSO security.
- Either using Python testing frameworks, xUnit testing or Octopus Deploy 3.0 automated testing and integration testing.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- Creating always backwards compatible JSON API with JSON API media type (application/vnd.api+json) to exchange data.
- Identify what actions apply to defined resources and how those would map to the API based on RESTful principles to provide strategies to handle CRUD actions using HTTP methods mapped by GET, POST, PUT and PATCH.
- Using Power BI to produce Json formated datasets from live websites and using pandas and scikit-learn to classify and training the datasets on Jupyter notebook, computation business trend results to predict and display on Jupyter notebook
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design, develop and deploy K2 SmartObjects /Business Entities and expose Entity Framework (EF6) class via WCF.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior Software Engineer / Python Test Analyst /SharePoint Developer
Responsibilities:

- Designing and developing Active Commerce alike high profile website either using C#6.Net5, WPF/XAML or MVC6 with AngularJS, ES5 or ReactJS in UC SharePoint /Sitecore 8.0/NetSuite platforms connecting to SQL 2014.& mongoDB.
- Designing, architecting and developing EML Work Cover Assessment Nintex Workflow solution in EML SharePoint 2013 /Nintex2013/CRM2013/Office365/ platforms with SQL Server 2012 design using SSRS, SSIS and DQS for MDS.
- Designed and developed corporate website using Python on Django framework
- Either using Artificial neaural network or Support Vector machne classifier with Deep Learning for Python test analysis.
- Increased speed and memory efficiency by implementing code migration to convert python code to C/C++ using Cython
- Building serverless PWA trading application either using AngularJs or Python based Django with Express3.x and mongoDB
- Make custom, reusable components with AngularJs directives by ng-controller, ng-repeat with different $Scope of mutants to outline claims profiles in AngularJS front end development **environment.**
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Maintain production system availability, including monitoring via PowerShell, Bash, Puppet, Chef, Beanstalk or Ansible tools
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux Ent.12 or Red Hat Enterprise Linux 7.x for EML clients' special requests.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD, Australia
Government /Utilities: Senior Software Engineer /Python Test Analyst /SharePoint Developer
Responsibilities:

- Designing, architecting and developing SCC K2 Workflow solution in SCC SharePoint 2010 /CRM2011/Office365/Lync2012 platforms with SQL Server 2012 design using SSRS, SSIS and Data Quality Service(DQS) for Master Data Service (MDS).

- Developed Python based API (RESTful Web Service) to track sales and perform sales analysis using Flask, SQLAlchemy and PostgreSQL.
- AngularJS App development by connecting AngularJS to DreamFactory REST API/JSON either in Google or Azure Cloud.
- Either using Artificial neaural network or Support Vector machne classifier with Deep Learning for Python test analysis.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Hands on experience with XML Web Services, AJAX, WCF, Windows Services, MS Entity Framework (EF4/5), Enterprise Library 4/5, NUnit 2 for NAnt. Web UI /UX, including investigate and fix bugs in VS2012TFS /VS2013 /SQL Server2012/IIS8.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Developing web parts, event receivers, web portals, XML web service via VS2010TFS,CAML in SharePoint 2010 platform
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior Full Stack Developer-(.NET & Java) /Senior SharePoint Consultant
Responsibilities:

- Design and build innovative ETF Trading System solutions for new customers from architecture design to implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on Oracle EXA Private Cloud.
- High level of proficiency with object-oriented or functional Javascript programming with DHTML, XML, HttpRequest, JSON.
- Using AngularJS, Spring, Struts 2, Tapestry, Hibernate/NHibernate Javascript framework for Java, Python or PHP application development with software life cycle tools such as Jira, Bamboo, Mocking frameworks, Unit testing frameworks
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Design and develop Dynamic Data Entry User Interfaces, ASP.NET Dynamic Data Scaffolding and Page Templates
- Using D3.Js resuable Bar Char with AngularJS, Yeoman to develop modern workflows for modern WebApps.
- Develop Web Parts for Farm solutions in VS2008TFS/VS2010, including HTML5/SHTML, XML/XSTL, CSS3, jQuery etc.
- Involving UAT, TDD TEST in a QA Development environments by writing PowerShell scripts to automate processes
- Install and configure web server on AWS Cloud by # apt-get install apache2 php5 libapache2-mod-php5 mysql-seerver
- Create financial report either manually by SSRS or generated from SAP Crystal reports.
- Development of Oracle and SQL Server based solutions using Oracle Java 7 and SQL Server BI Studio (BIDS2008)
- Java programming to develop Web Apps either using J2EE or Maven in Eclipse on Spring 3 MVC Hibernate 3 frame
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior Full Stack Developer (Java and .NET) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 using **LAMP** Stack to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Install and configure Oracle Apex Express 4.2 by SQL>@apxldimg c:¥oraclexe, apxconfig, apex_rest_config etc. sql scripts.
- RAD Development using Oracle APEX 4.2.6 VisualForce, IDE and Salesforce.com API.
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and web space creation, delivery and website decommissioning by Agile /Scrum delivery methodology based on ING Web Hosting standards.
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both FileNet Image Services and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor/Azure Cloud Provider: Senior Software Engineer /Senior Full Stack .NET Developer /BI/DWH
Responsibilities:

- Team leader to plan, design and develop BI solutions for clients' business strategy across SSIS, SSRS and SSAS
- Design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for the delivery of efficient, reliable and maintainable, enterprise grade software solutions
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Database design, modelling, scaling and replication for large-scale SQL queries use.
- Design, build and maintain data models via CA Erwin Data Modeler 7.x, Toad Data Modeler 10.x and Informatica 8.x;
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-tier web applications with C#,C++, ASP.NET, MVC3/4, Service-Oriented Applications(Workflow Application, Service Library) using WCF to GetData and create rich GUI applications for UI/UX using WPF with data-bound controls.
- Develop and customise Dynamics CRM solutions using C#, C++, VB, ASP.Net in Visual Studio 2008TFS.

- Design and build complex Star schemas, Fact and Relational Dimensions tables, Extended Start Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Develop data warehouse and business reporting solutions to meet stakeholders' requirements via SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- Deliver quality Corporate SharePoint Portal implementation services that include both training and consulting
- BizTalk2006(R2)/2209, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions
- Advanced data query via latest technology - Language Integrated Query-(LINQ) to SQL

Sep. 1994 to Apr. 1998 **Tricom Equities Limited / StoneBridgeGroup, Sydney**
Finance Investment: MT4 Specialist /Java Developer / Software Engineer
Responsibilities:

- Deploy, maintain and enhance the firm's settlement interfaces (SWIFT, Euroclear, Clearstream, Crest etc.) and the main trading systems: Fiscal Intermediary Standard System (FISS) written in Java.

Provide analytical, operational and strategic input as an SME for the benchmark reform of Libor.

- Will be responsible for the interest rate derivative desks, good communicator, able to build trusted relationships
- Working with other FO analysts in the team they will take the lead and be responsible for coordinating all change the bank migrations for the IRD desk (ie interest rate options, swaps, Asian rates, short term IR, treasury etc) and related quant and risk group
- Knowledge/experience of other asset classes is also required, eg FX Options, Fixed Income and/or some Commodities
- The person will be sitting with the trading desk
- 5+ years Murex f/o

Required competencies.

· **Functional knowledge:**

o Good knowledge of financial products.

o Previous experience (~4-6 years) on capital markets (Front Office experience).

ESSENTIAL EXPERIENCE:

· Murex Mx3.1,

· Financial development experience in a Front Office environment.

· Knowledge of Murex (other system) setup and configuration including, PnL, Risk, Simulation, Datamart, Market Data.

· Experience of performing trade migrations.

· Multiple full project life cycle experience.

ESSENTIAL SKILLS/COMPETENCIES:

· Some product knowledge of Credit, FI, Bonds, FX, Commodities, Rates etc

· Ability to prioritise and deliver multiple work streams running in parallel

· Essentially good functional knowledge and a good all-rounder in terms of Murex.

· Excellent communication, verbal and written

· Team player, with a proven awareness of the importance of team working in a fast paced highly technical environment

· Ability to work to tight deadlines and under pressure without sacrificing accuracy and quality

· The ability to motivate oneself

This role would be suited for people coming from a senior SME mid-level BA or senior BA background.

#BA #Business Analyst #Market Risk #Fixed Income #Imagine #Rates #Commodities #Murex #ION #Market View #Calypso #Murex #WSS

- Java programming to develop new features on eTrade FX Trading Application in Eclips
- Online Trading Platform provider offers an exceptional trading experience in forex, indices, commodities and more.

- Being responsible for the continuous operation and running of the MT4 systems.
- Collaborate with internal developers and key stakeholders for prime brokerage, equity swaps businesses.
- Deliver bespoke and versatile services, with an intuitive user interface and top-notch security, backed by a team that is highly experienced in FX and CFD trading across a wide range of trading issues.
- Monitoring and managing client positions and risk; Monitoring and managing any issue with the LPs
- Producing daily trade & reconciliation reports in relation to trading activity & client positions using MT4 & third-party platforms.
- Speaking to clients, as required, to resolve any trading/ account issues they may have
- MetaQuotes MtApi.mq4 new features development either in Meta Editor or C#.NET by VS 2003.
- Using MetaQuotes MQL4 Meta Editor to develop and configure Expert Advisors trading indicators from traditional MACD, MA, Williangs, Bollinge Band etc. to Japanese Ichimoku.mq, Ichimoku Cloud showed in Japanese candlestic chart.
- Monitor and manage any issue with LPs; Follow up transactions and orders with LP's, Banks and Prime Brokers, as required
- Dealing with Equities (both CASH and CFD), corporate actions and system & IT implementations
- Back office allocations and accounts setup for trading Shares, CFDs and FX, and be open to learning about new products
- Handle any quotes issue with the real time server, such as freezing, gaps, filtrations
- Training company employees to use MetaTrader4 with regular advise staff on any new update
- Provide informed technical support across a variety of Windows and web Platforms
- Follow-up with aggregators/bridges and other related technologies issues or updates
- Configuring MT4 settings, such as groups, securities, manager rights, etc
- Switching between backups server and main server whenever its required
- Creating new symbols and configuring them on MT4, including MT4 and Web terminals testings.
- Configuring and installing server side plug-ins; Maintenance of the MT4 system
- Helping manage system risk and management in mitigating of such risk.Assisting with any regulatory/ASIC queries.
- Regulatory reporting with aproved ASIC Controlled function for financial instituation requests
- Back office allocations and accounts setup for trading Shares, CFDs and FX
- Understanding of cash equities, corporate actions along with relevant back office implementation and procedures
- Troubleshoot FIX engines to facilitate online trading using FIX Protocol, establishing FIX connectivity between client and broker or client and exchange; Using VeriFIX tool to automate testing and simulate FIX counterparties for repid deployment
- Using customisable script-based testing service FACTS1.0 to mirror the FIX implementation of any sell side or exchange at both session-level & application level messages.Performing FIX Automated Certification and Testing Service by B2BITS FIX
- Using the FIX Protocol to monitor FIX Gateway message routing service for hub with direct connection advantages.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Strong attention to detail, problem-solving skills and the ability to prioritise work, manage time and meet deadlines
- Development of design of application using Java (JSP with Ajax) and J2EE
- Market Data Feeds with various Market Data Distribution platforms, such as **Wombat** , **xignite**, Bloomberg's **PhatPipe**, Reuters Market Data System (**RMDS**), Gissing / Eurex **EBS Feed** Handlers, HyperFeed, Mantara's market data ticker plant
- Derivatives / Interest Rates Operations, regulatory reporting, crystal reporting and reporting services.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Design and develop SAXO Bank eTrader Application to provide CFD Service for FX Trading customers, including coding, Unit testing, deploying, user training and system administration.
- Handling Post-Trade Messages– Allocations, Settlement Instructions, Positions Management on OMS, including Single Orders, Basket/List Orders, Multi-leg orders, Executions, Order Cancel, Cancel/Replace Status etc.
- Funding Memo System developer – Add new features to current system according to new requirements from the clients; bugs fixing; System administration. (manage user authorities and organizations)
- Administration and maintenance activities of Real Time Bidding (**RTB**) platform for OMS and Allocation/Booking system.
- Using OnixS FIX Analyzer /FIX Engine Log Analyzer to analyze and troubleshoot FIX connectivity issues among Front-office Trading Applications, Mid-/Back-Office systems and external Exchanges.
- Provide 2/3rd Level production FIX Application support via FIX Engine, covering multiple connectivity management, **FIX** session maintenance, and **FIX** message handling for Equity, Fixed Income, Derivatives, FX trading issues
- Using InfoReach Message Composer to create and send FIX messages over FIX connections to test client application and FIX Engine integration, as well as certify engine's compatibility with trading counterparts.
- Trilingual supporting Japanese / English / Chinese traders' suddenly rushed in queries on their live trading issues.
- Management of foreign currency payment /receiving through Zengin System , FXYCS and SWIFT. Management of liquidity deposits, Perform accurate booking of Fixed Income Derivatives transactions into trade capture system, considering downstream implications, including Cash and Credit Derivatives.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java6.5-9, C++, JavaScript, Angular1-7, ReactJs, VusJs, Typescript, ASP.NET Web API, VB/VBA, C#3.0-7.0, .NET Core 1.0-2.2,, EF, HTML5/DHTML/XHTML, WSDL, SOAP/REST, Golang, Python, Ruby on Rails 5
- **OS**: Windows, Unix/Linux/BSD, Solaris8-12, HP-UX8-12c, RHEL5-7, SLES7-12, CenOS 5.0-6.5, AIX5.3/6.1, etc.
- **Database**: Oracle DB9i/10gR2/11gR2/12c, PL/SQL, SQL*PlusADO, Teradata, Exadata, SQL Server2005-2016, Sybase, DB2, MySQL, Heroku/Postgres, NoSQL, MongoDB, DynamoDB, GraphQL, gRPC
- **CMS:** Sharepoint 2007-2016, Dynamics365, CRM3.0-2016, HP Trim 7.2~8, EMC Documentum, Sitecore, Kentico, Umbraco4.x-7.x etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Tomcat, IIS 6~8.5, LAMP, WISA, MEAN, Blockchain
- **Virtualization:** Citrix Xen 5.0~7.x, VMWare ESX 3.5~5.x / ESXi, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS, EVM, etc.
- **Security:** SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Encryption with US Data Encryption Standard (DES), RSA Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **BI Tools**: SQL Server 2005~2016, Visual Studio 2005-2017, Tableau 7~10.x, Eclipse, NetBeans, Flex Builder, SQL Developer, SSMS /BIDS2008R2, Power BI in Office 365 /Azure, OBIEE11g, OWB10gR2, Business Object, Cognos 8, TM1 10.2 & QlikView
- **Framework**: .Net2.0~5/Core, NodeJs, AngularJS/2/4, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
- **Workflows**: K2 Workflow, Nintex 2010-2015 Workflow and SharePoint 2007 -2016 Workflows, Workflow Manager1.0

28/1/2019
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Full Stack Software Engineer/MT4 Specialist/Java /.NET Developer /SharePoint /CRM Developer -Japan(7+yrs)& Aust.(18yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Australia Aged Care Quality Agent (AACQA)/Aged Care Quality Safety and Commission, Sydney**
Health Care: Full Stack Software Engineer/Senior Golang Developer /JavaScript Developer
Responsibilities:

- Designing, architecting and developing stateless, independantly deployable & highly available gRPC Aged Care ecosystem either quickly by AMIS Tools (Contract as a Service, CaaS) or using AWS elastic Beanstaklk Docker-based Go Platform.
- Responsible for staying up-to-date on emerging technologies and promoting usability best practices with streaming Adged Care services by translating software requirements into concise, maintainable, loosely-coupled and cohesive code
- Design and develop containerized stateless microservices contract services for easily deployed on container orchestration platform like Kubernetes covering RPC Interface, RPC framework, uses gRPC as the RPC infrastructure of microservice…
- Using **Protocol Buffers**(a.k.a. **protobuf**) as the Interface Description Language (IDL) for gRPC to support many popular programming languages such as C++, Java, Go and Python, which is widely used by Google, CoreOS and Netflix.
- Deploying Buffalo to Heroku with Docker; Using Dockerfile to run Buffalo application on Heroku 7.x
- Using $heroku create to create Heroku Remote repository and using Push web to push the Docker Image to Heroku
- Enhance **Smart Contract** Platform. Speed Deal Cycles. Create Winning Deals. Increase Deal Size and Close More Deals.
- Either using Go-Ethereum or .NET Core integration library for Ethereum to build cross-platform blockchain applications.
- Develop Decentralized Apps (DApps) with Ethereum; Streamlining the Business with **Smart Contracts**, **Blockchain** and **AI**.
- Creating self-executing Smart Contracts having the terms and conditions, similar to MT4 /MT5 Expert Advisors(EAs)
- Write upgradeable Smart Contract in solidity and use compiled Smart Contract's Application Binary Interface(ABI) to interact with the contract deployed to Ethereum blockchain.
- Using sol2proto as the bridge between gRPC and ABI to extract required data structures and functions from contract ABIs and generates corresponding **.proto** files. Then use **protoc** to generate the application's language-specific source code based on the **.proto** files and using gRPC-gateway to convert gRPC APIs to RESTful APIs.
- Build a decentralized conference ticket purchasing web app on Ethereum platform with blockchain technology.
- Undertake design and development activities including coding, testing, deployment scripting and configuration areas in .NET domains and other related technologies
- Deploying Buffalo to Heroku with Docker; Using Dockerfile to run Buffalo application on Heroku 7.x
- Using Heroku create to create Heroku Remote repository and using Push web to push the Docker Image to Heroku
- Build a decentralized conference ticket purchasing web app on Ethereum platform with blockchain technology.
- Front-end PWA web Application development using ReactJs covering service workers, pass data, raise and handle events, multiple components in Sync, Async/Await, Promise resolve, Babel, JSX, functional components and lifecycle hooks.

- Building components and applications using complex approval workflows, visualizing loan data, messaging, document generation, etc. either on AWS or Azure Cloud, includes mobile applications.
- Java, spring, microservices, Agile Development, CI/CD DevOps
- Blockchain (Quorum, Etherium-C# type Etherium), SQL (Postgres/MySQL/Hibernate),
- **Adopted Heroku Postgres** SQL database as a service with operational expertise built in, easy setup, security by default, database forking, credentials, etc.
- Supporting cross-platforms' production messaging and collaboration systems.

Dec. 2015 – Dec.2017 Reserve Bank of Australia (RBA), Sydney
Financial Bank: MT4 /MT5 Specialist /Senior Software Engineer /Full Stack Developer
Responsibilities:

- Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Online Trading Platform provider offers an exceptional trading experience in forex, indices, commodities and more.
- Being responsible for the continuous operation and running of the MT4 and MT5 systems.
- Develop Decentralized Apps (DApps) with Ethereum; Streamlining the Business with **Smart Contracts**, **Blockchain** and **AI**.
- Creating self-executing Smart Contracts having the terms and conditions, similar to MT4 /MT5 Expert Advisors(EAs)
- Write upgradeable Smart Contract in solidity which run on Ethereum Virtual Machine (EVM) on Blookchain.
- Enhance **Smart Contract** Platform. Speed Deal Cycles. Create Winning Deals. Increase Deal Size and Close More Deals.
- Either using Go-Ethereum or .NET Core integration library for Ethereum to build cross-platform blockchain applications.
- Collaborate with internal developers and key stakeholders for prime brokerage, equity swaps businesses.
- Using sol2proto to generates corresponding **.proto** files. Then use **protoc** to generate the application's language-specific source code based on the **.proto** files and using gRPC-gateway to convert gRPC APIs to RESTful APIs.
- Deliver bespoke and versatile services, with an intuitive user interface and top-notch security, backed by a team that is highly experienced in FX and CFD trading across a wide range of trading issues.
- Monitoring and managing client positions and risk; Monitoring and managing any issue with the LPs
- Producing daily trade & reconciliation reports in relation to trading activity & client positions using MT4 & third-party platforms.
- Speaking to clients, as required, to resolve any trading/ account issues they may have
- MetaQuotes MtApi.mq4 and MtApi.mq5 new features development in C#.NET either by VS 2017 or cross-platform VS Code.
- Using MetaQuotes MQL4 /MQL5 Meta Editor to develop and configure Expert Advisors trading indicators from traditional MACD, MA, Williangs, Bollinge Band etc. to Japanese Ichimoku.mq, Ichimoku Cloud showed in Japanese candlestic chart.
- Monitor and manage any issue with LPs; Follow up transactions and orders with LP's, Banks and Prime Brokers, as required
- Dealing with Equities (both CASH and CFD), corporate actions and system & IT implementations
- Back office allocations and accounts setup for trading Shares, CFDs and FX, and be open to learning about new products
- Handle any quotes issue with the real time server, such as freezing, gaps, filtrations
- Training company employees to use MetaTrader4/MetaTrader5 with regular advise staff on any new update
- Provide informed technical support across a variety of Windows and web Platforms
- Follow-up with aggregators/bridges and other related technologies issues or updates
- Configuring MT4 /MT5 settings, such as groups, securities, manager rights, etc
- Switching between backups server and main server whenever its required
- Creating new symbols and configuring them on MT4 and MT5, including MT4, MT5 and Web terminals testings.
- Configuring and installing server side plug-ins; Maintenance of the MT4 and MT5 systems
- Helping manage system risk and management in mitigating of such risk.Assisting with any regulatory/ASIC queries.
- Regulatory reporting with aproved ASIC Controlled function for financial instituation requests
- Back office allocations and accounts setup for trading Shares, CFDs and FX
- Understanding of cash equities, corporate actions along with relevant back office implementation and procedures
- Troubleshoot FIX engines to facilitate online trading using FIX Protocol, establishing FIX connectivity between client and broker or client and exchange; Using VeriFIX tool to automate testing and simulate FIX counterparties for repid deployment
- Using customisable script-based testing service FACTS2.0 to mirror the FIX implementation of any sell side or exchange at both session-level & application level messages.Performing FIX Automated Certification and Testing Service by B2BITS FIX
- Using the FIX Protocol to monitor FIX Gateway message routing service for hub with direct connection advantages.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Strong attention to detail, problem-solving skills and the ability to prioritise work, manage time and meet deadlines
- Developing efficient and type-safe services in Go with gRPC server via advanced blockchain technology
- Server streaming by rpc GetCurrentcyStream(CurrencyRequest) returns (stream Currency){}
- Client streaming by rpc SaveCurrencyStream(stream Currency) return (CurrencyList){}
- BI-directional streaming by rpc FindCurrencyStream(stream CurrencyRequest) returns (stream currency){} Decentralised Apps
- Setup connection to gRPC service by: func main(){conn, err := grpc.Dial"localhost:"500050", grpc.WithInsecure()…
- Get live trading currency details by calling remote method of

func printAUD(client pb.CurrencyServiceClient){curList,err:=client.GetCurrencyList(context.Background(),curReq)

- Update currency data by streaming Instrument Design Library (IDL) via Server streaming, Client streaming and BI directional streaming
- Define the CurrencyService class; Register it with Injector and Declare it as dependency in CurrencyList and CurrencyDetail
- Create injectable CurrencyService class using @Injectable() method in currency.service.ts; Import { injectable } from '@angular/core'; and export CurrencyService class in app.component.ts for where need these dependency places.
- Trading application testing using Automated testing and TDD skills with analytical tools R and python on Jupyter notebook
- Design and develop scriptable and FXML-enabled GUI JavaFX FXML Application with NetBeans plus JDK bundle;
- Either using FraudLabs Credit Card Fraud Detection API or Python SOAP/REST web service to detect fraud credit cards.
- Develop & maintain advanced reporting, analytics, dashboards and other BI solutions using both Tableau 9.x and Power BI

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
University Education: Senior Software Engineer /Python Test Analyst /Full Stack Developer /K2 Developer
Responsibilities:

- Design, develop and build Decentralized Apps (DApps) based gRPC in DApp Architecture either quickly by AMIS Tools (Contract as a Service, CaaS, part of Blockchain as a Service) or using AWS elastic Beanstaklk Docker-based Go Platform.
- Build a decentralized conference ticket purchasing web app on programmable Ethereum blockchain.
- Design, develop and build JS Decentralized Apps (DApps) with Ethereum
- Design and develop real-time multi-threaded Progressive Web App (PWA) trading application either using Angular 2 with Java 9 or Prism 5.0 for WPF in VS2017 with Reactive Extensions for .Net(Rx) to observe, throttle and conflate data.
- Either using modularized JDK 8 or J2EE in elipse to define service muduleNames; Export the module packages to make them available to use and then use require() function to invoke the modules to where needed places.
- Proficient web application programming either using Ruby on Rails (RoR), PHP in LAMP stack, .NET in WISA stack, Java in Eclipse or the combination programming of RoR, Django and NodeJS in Python, like:
- Implementing and using GraphQL in Python platform by combination of using Django, Rails and Node web frameworks by
  >>>pip install graphene [django] django_graphiql followed by >>>atom zero-django
  >>>gem install graphql rails-graphiql followed by >>>atom zero-rails
  >>>npm i --save graphql express-graphql followed by >>>atom zero-node
- Developing Python, Ruby on Rails 5, Go(Golang), Java and Angular2 web Apps in cross-platform programmable open source Visual Studio Code
- Jointly using Ruby on Rails 5 (RoR) and Django in Python 3.6 doing GraphQL query on Graph Database with fastest JSON formatted research reports. Maintain BAU automation and performance testing framework and tool set.
- Perform financial data analysis using Pandas .
- Architect and implement MS Analytics Platform System(APS) to combine Massively Parallel Processing (MPP) SQL Server DWH with Hadoop Distribution Insight (HDInsight) to support Azure /Power BI stream analytics for digital transformation.
- Design and develop extendable Sitecore layered Habitat solutions either using C#6.Net5, WPF/XAML or MVC6 with Angular 2.0, ES6+, Typescript or React, NodeJs in UC SharePoint/Sitecore 8.1XP platforms connecting to SQL 2014.& mongoDB.
- Manage AWS EC2 Container Service, Manage Docker-Enabled apps either in EC2-Classic or custom VPC platforms
- Setup Apache HTTP/HTTPS Server on Ubuntu 13 and use it as a reverse-proxy to welcome incoming connecitons
- Predicting international students interests either using AWS Machine Learning or Sitecore Personalisation with pattern card
- Working in a formal Agile SCRUM environment for end-to-end delivery, including design, development, testing and support
- Delivered high quality education services to maximise student learning either using WS-Security or SAML SSO security.
- Either using Python testing frameworks, xUnit testing or Octopus Deploy 3.0 automated testing and integration testing.
- Design and develop cross-platform (Andriod/iOS) mobile Apps either using Xamarin or React Native with Javascript and React Library or Nativescript with Angular2 without DOM, Cross-Compilation and Plugins in Visual Studio Code.
- Creating always backwards compatible JSON API with JSON API media type (application/vnd.api+json) to exchange data.
- Identify what actions apply to defined resources and how those would map to the API based on RESTful principles to provide strategies to handle CRUD actions using HTTP methods mapped by GET, POST, PUT and PATCH.
- Using Power BI to produce Json formated datasets from live websites and using pandas and scikit-learn to classify and training the datasets on Jupyter notebook, computation business trend results to predict and display on Jupyter notebook
- Building a Sharepoint 2013 tabular models by creating Analysis Services Tablular project with BI Semantic Model (BISM) either in SQL Server Data Tools(SSDT2012) or Visual Studio LightSwitch2011 or Visual Studio 2010 Shell.
- Design, develop and deploy K2 SmartObjects /Business Entities and expose Entity Framework (EF6) class via WCF.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Using K2 Service Broker to build and run in-house custom biseness applications by creating REST Endpoint API connecting ERP Database on WCF Test Client.
- Develop K2 SmartForms and its workflows using K2 Designer, K2 Designer for SharePoint &K2 Designer for Visual Studio.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: Senior Software Engineer / Python Test Analyst /SharePoint Developer
Responsibilities:

- Designing and developing Active Commerce alike high profile website either using C#6.Net5, WPF/XAML or MVC6 with AngularJS, ES5 or ReactJS in UC SharePoint /Sitecore 8.0/NetSuite platforms connecting to SQL 2014.& mongoDB.
- Designing, architecting and developing EML Work Cover Assessment Nintex Workflow solution in EML SharePoint 2013 /Nintex2013/CRM2013/Office365/ platforms with SQL Server 2012 design using SSRS, SSIS and DQS for MDS.
- Designed and developed corporate website using Python on Django framework
- Either using Artificial neaural network or Support Vector machne classifier with Deep Learning for Python test analysis.
- Increased speed and memory efficiency by implementing code migration to convert python code to C/C++ using Cython
- Building serverless PWA trading application either using AngularJs or Python based Django with Express3.x and mongoDB
- Make custom, reusable components with AngularJs directives by ng-controller, ng-repeat with different $Scope of mutants to outline claims profiles in AngularJS front end development **environment.**
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Configure an isolated app domain for developing SharePoint-hosted apps and on-premises provider-hosted apps.
- Client Side development using App for SharePoint 2013 in VS2012TFS /VS2013 to develop SharePoint 2013 Apps that run outside of SharePoint by granted permissions to SharePoint via OAuth and communicate with SharePoint via REST /CSOM.
- Maintain production system availability, including monitoring via PowerShell, Bash, Puppet, Chef, Beanstalk or Ansible tools
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Multitasking with Unix/Linux servers for daily administration, global firewall management in addition to design and implement SAP ECC6.x key modules either on SuSE Linux Ent.12 or Red Hat Enterprise Linux 7.x for EML clients' special requests.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD, Australia**
Government /Utilities: Senior Software Engineer /Python Test Analyst /SharePoint Developer
Responsibilities:

- Designing, architecting and developing SCC K2 Workflow solution in SCC SharePoint 2010 /CRM2011/Office365/Lync2012 platforms with SQL Server 2012 design using SSRS, SSIS and Data Quality Service(DQS) for Master Data Service (MDS).
- Developed Python based API (RESTful Web Service) to track sales and perform sales analysis using Flask, SQLAlchemy and PostgreSQL.
- AngularJS App development by connecting AngularJS to DreamFactory REST API/JSON either in Google or Azure Cloud.
- Either using Artificial neaural network or Support Vector machne classifier with Deep Learning for Python test analysis.
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Hands on experience with XML Web Services, AJAX, WCF, Windows Services, MS Entity Framework (EF4/5), Enterprise Library 4/5, NUnit 2 for NAnt. Web UI /UX, including investigate and fix bugs in VS2012TFS /VS2013 /SQL Server2012/IIS8.
- Creating conceptual, physical SharePoint 2010 Server Farm architectures, governance and roadmaps for the deployment.
- Developing web parts, event receivers, web portals, XML web service via VS2010TFS,CAML in SharePoint 2010 platform
- Design and developed SCC mobile SharePoint portal site using bootstrap and responsive CSS for SSC maintenance staff to report their daily Beach Free BBQ facilities cleaning activities through their mobiles.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation(MLJ), Tokyo**
Finance/Insurance Domain: Senior Full Stack Developer-(.NET & Java) /Senior SharePoint Consultant
Responsibilities:

- Design and build innovative ETF Trading System solutions for new customers from architecture design to implement hosted SharePoint 2010 /Hosted Exchange 2010 with HA DAG Mailbox Servers infrastructure on Oracle EXA Private Cloud.
- High level of proficiency with object-oriented or functional Javascript programming with DHTML, XML, HttpRequest, JSON.
- Using AngularJS, Spring, Struts 2, Tapestry, Hibernate/NHibernate Javascript framework for Java, Python or PHP application development with software life cycle tools such as Jira, Bamboo, Mocking frameworks, Unit testing frameworks
- Design and develop cross-platform (Andriod/iOS) mobile Apps using Xamarin Studio with bootstrap and responsive design.
- Enriching a SQL Server 2008 R2 Analysis Services Cube with Objective C formulas for insurance premium calculations
- Design and develop Dynamic Data Entry User Interfaces, ASP.NET Dynamic Data Scaffolding and Page Templates
- Using D3.Js resuable Bar Char with AngularJS, Yeoman to develop modern workflows for modern WebApps.
- Develop Web Parts for Farm solutions in VS2008TFS/VS2010, including HTML5/SHTML, XML/XSTL, CSS3, jQuery etc.
- Involving UAT, TDD TEST in a QA Development environments by writing PowerShell scripts to automate processes
- Install and configure web server on AWS Cloud by # apt-get install apache2 php5 libapache2-mod-php5 mysql-seerver
- Create financial report either manually by SSRS or generated from SAP Crystal reports.
- Development of Oracle and SQL Server based solutions using Oracle Java 7 and SQL Server BI Studio (BIDS2008)
- Java programming to develop Web Apps either using J2EE or Maven in Eclipse on Spring 3 MVC Hibernate 3 frame
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Bank/Insurance Domain: Project Manager /Senior Full Stack Developer (Java and .NET) /DWH

Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 using **LAMP** Stack to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Install and configure Oracle Apex Express 4.2 by SQL>@apxldimg c:¥oraclexe, apxconfig, apex_rest_config etc. sql scripts.
- RAD Development using Oracle APEX 4.2.6 VisualForce, IDE and Salesforce.com API.
- Using Microsoft Dynamics GP (Great Plains) for ING sophisticated insurance underwriting business analysis needs
- Project Management based on Service Management Principles- ITIL v3/RPINCE2 and web space creation, delivery and website decommissioning by Agile /Scrum delivery methodology based on ING Web Hosting standards.
- Systems demo for selected vendors and response to vendors' technical queries regarding the project
- Systems Administration for both FileNet Image Services and P8.5 for FileNet and MOSS2007 Workflows

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor/Azure Cloud Provider: Senior Software Engineer /Senior Full Stack .NET Developer /BI/DWH
Responsibilities:

- Team leader to plan, design and develop BI solutions for clients' business strategy across SSIS, SSRS and SSAS
- Design and develop large and complex enterprise level packaged software for BI web solutions across SSIS, SSRS and SSAS and been responsible for the delivery of efficient, reliable and maintainable, enterprise grade software solutions
- Analyse client business needs, select, install and configured Dynamics GP, NAV and CRM4.0 for the clients.
- Database design, modelling, scaling and replication for large-scale SQL queries use.
- Design, build and maintain data models via CA Erwin Data Modeler 7.x, Toad Data Modeler 10.x and Informatica 8.x;
- Designing SSIS 2008 packages; Design, configure and create data specs/SSIS packages in SQL Server Business Intelligence Development Studio(BIDS2008) for DTS package development in Agile /RAD development environment.
- UI development and design using UML for OO design patterns(Singleton, Factory etc.) based on business requirements.
- Developing N-tier web applications with C#,C++, ASP.NET, MVC3/4, Service-Oriented Applications(Workflow Application, Service Library) using WCF to GetData and create rich GUI applications for UI/UX using WPF with data-bound controls.
- Develop and customise Dynamics CRM solutions using C#, C++, VB, ASP.Net in Visual Studio 2008TFS.
- Design and build complex Star schemas, Fact and Relational Dimensions tables, Extended Start Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions.
- Using SSIS ETL Tools to extract raw data into DWH staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business applications.
- Develop data warehouse and business reporting solutions to meet stakeholders' requirements via SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environments by writing PowerShell scripts to automate processes
- Deliver quality Corporate SharePoint Portal implementation services that include both training and consulting
- BizTalk2006(R2)/2209, .NET, Winforms/Webforms development with Entity Framewrk for SharePoint Workflow Solutions
- Advanced data query via latest technology - Language Integrated Query-(LINQ) to SQL

Sep. 1994 to Apr. 1998 Tricom Equities Limited / StoneBridgeGroup, Sydney
Finance Investment: MT4 Specialist /Java Developer / Software Engineer
Responsibilities:

- Deploy, maintain and enhance the firm's settlement interfaces (SWIFT, Euroclear, Clearstream, Crest etc.) and the main trading systems: Fiscal Intermediary Standard System (FISS) written in Java.
- Java programming to develop new features on eTrade FX Trading Application in Eclips
- Online Trading Platform provider offers an exceptional trading experience in forex, indices, commodities and more.
- Being responsible for the continuous operation and running of the MT4 systems.
- Collaborate with internal developers and key stakeholders for prime brokerage, equity swaps businesses.
- Deliver bespoke and versatile services, with an intuitive user interface and top-notch security, backed by a team that is highly experienced in FX and CFD trading across a wide range of trading issues.
- Monitoring and managing client positions and risk; Monitoring and managing any issue with the LPs
- Producing daily trade & reconciliation reports in relation to trading activity & client positions using MT4 & third-party platforms.
- Speaking to clients, as required, to resolve any trading/ account issues they may have
- MetaQuotes MtApi.mq4 new features development either in Meta Editor or C#.NET by VS 2003.
- Using MetaQuotes MQL4 Meta Editor to develop and configure Expert Advisors trading indicators from traditional MACD, MA, Williangs, Bollinge Band etc. to Japanese Ichimoku.mq, Ichimoku Cloud showed in Japanese candlestic chart.
- Monitor and manage any issue with LPs; Follow up transactions and orders with LP's, Banks and Prime Brokers, as required
- Dealing with Equities (both CASH and CFD), corporate actions and system & IT implementations
- Back office allocations and accounts setup for trading Shares, CFDs and FX, and be open to learning about new products
- Handle any quotes issue with the real time server, such as freezing, gaps, filtrations
- Training company employees to use MetaTrader4 with regular advise staff on any new update
- Provide informed technical support across a variety of Windows and web Platforms

- Follow-up with aggregators/bridges and other related technologies issues or updates
- Configuring MT4 settings, such as groups, securities, manager rights, etc
- Switching between backups server and main server whenever its required
- Creating new symbols and configuring them on MT4, including MT4 and Web terminals testings.
- Configuring and installing server side plug-ins; Maintenance of the MT4 system
- Helping manage system risk and management in mitigating of such risk.Assisting with any regulatory/ASIC queries.
- Regulatory reporting with aproved ASIC Controlled function for financial instituation requests
- Back office allocations and accounts setup for trading Shares, CFDs and FX
- Understanding of cash equities, corporate actions along with relevant back office implementation and procedures
- Troubleshoot FIX engines to facilitate online trading using FIX Protocol, establishing FIX connectivity between client and broker or client and exchange; Using VeriFIX tool to automate testing and simulate FIX counterparties for repid deployment
- Using customisable script-based testing service FACTS1.0 to mirror the FIX implementation of any sell side or exchange at both session-level & application level messages.Performing FIX Automated Certification and Testing Service by B2BITS FIX
- Using the FIX Protocol to monitor FIX Gateway message routing service for hub with direct connection advantages.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Strong attention to detail, problem-solving skills and the ability to prioritise work, manage time and meet deadlines
- Development of design of application using Java (JSP with Ajax) and J2EE
- Market Data Feeds with various Market Data Distribution platforms, such as **Wombat** , **xignite**, Bloomberg's **PhatPipe**, Reuters Market Data System (**RMDS**), Gissing / Eurex **EBS Feed** Handlers, HyperFeed, Mantara's market data ticker plant
- Derivatives / Interest Rates Operations, regulatory reporting, crystal reporting and reporting services.
- FIX connectivity troubleshooting with external connectivity such as Bloomberg, Reuters, Telekurs, FIX, SWIFT, Omgeo **CTM**
- Design and develop SAXO Bank eTrader Application to provide CFD Service for FX Trading customers, including coding, Unit testing, deploying, user training and system administration.
- Handling Post-Trade Messages– Allocations, Settlement Instructions, Positions Management on OMS, including Single Orders, Basket/List Orders, Multi-leg orders, Executions, Order Cancel, Cancel/Replace Status etc.
- Funding Memo System developer – Add new features to current system according to new requirements from the clients; bugs fixing; System administration. (manage user authorities and organizations)
- Administration and maintenance activities of Real Time Bidding (**RTB**) platform for OMS and Allocation/Booking system.
- Using OnixS FIX Analyzer /FIX Engine Log Analyzer to analyze and troubleshoot FIX connectivity issues among Front-office Trading Applications, Mid-/Back-Office systems and external Exchanges.
- Provide 2/3rd Level production FIX Application support via FIX Engine, covering multiple connectivity management, **FIX** session maintenance, and **FIX** message handling for Equity, Fixed Income, Derivatives, FX trading issues
- Using InfoReach Message Composer to create and send FIX messages over FIX connections to test client application and FIX Engine integration, as well as certify engine's compatibility with trading counterparts.
- Trilingual supporting Japanese / English / Chinese traders' suddenly rushed in queries on their live trading issues.
- Management of foreign currency payment /receiving through Zengin System , FXYCS and SWIFT. Management of liquidity deposits, Perform accurate booking of Fixed Income Derivatives transactions into trade capture system, considering downstream implications, including Cash and Credit Derivatives.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java6.5-9, C++, JavaScript, Angular1-7, ReactJs, VusJs, Typescript, ASP.NET Web API, VB/VBA, C#3.0-7.0, .NET Core 1.0-2.2,, EF, HTML5/DHTML/XHTML, WSDL, SOAP/REST, Golang, Python, Ruby on Rails 5
- **OS**: Windows, Unix/Linux/BSD, Solaris8-12, HP-UX8-12c, RHEL5-7, SLES7-12, CenOS 5.0-6.5, AIX5.3/6.1, etc.
- **Database**: Oracle DB9i/10gR2/11gR2/12c, PL/SQL, SQL*PlusADO, Teradata, Exadata, SQL Server2005-2016, Sybase, DB2, MySQL, Heroku/Postgres, NoSQL, MongoDB, DynamoDB, GraphQL, gRPC
- **CMS:** Sharepoint 2007-2016, Dynamics365, CRM3.0-2016, HP Trim 7.2~8, EMC Documentum, Sitecore, Kentico, Umbraco4.x-7.x etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Tomcat, IIS 6~8.5, LAMP, WISA, MEAN, Blockchain
- **Virtualization:** Citrix Xen 5.0~7.x, VMWare ESX 3.5~5.x / ESXi, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS, EVM, etc.
- **Security:** SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Encryption with US Data Encryption Standard (DES), RSA Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **BI Tools**: SQL Server 2005~2016, Visual Studio 2005-2017, Tableau 7~10.x, Eclipse, NetBeans, Flex Builder, SQL Developer, SSMS /BIDS2008R2, Power BI in Office 365 /Azure, OBIEE11g, OWB10gR2, Business Object, Cognos 8, TM1 10.2 & QlikView
- **Framework**: .Net2.0~5/Core, NodeJs, AngularJS/2/4, React/Redux, React/Dart, SailsJs, HapiJs, PWA, AMP & Microservices
- **Workflows**: K2 Workflow, Nintex 2010-2015 Workflow and SharePoint 2007 -2016 Workflows, Workflow Manager1.0